Math Guide

Mathematical foundations for Halo 2 and Zcash Orchard

m@rek.onl

Abstract

This volume of *The Zcash Arboretum* develops the mathematics underlying the cryptography of Zcash from first principles, assuming only elementary algebra and calculus. It builds a single tower: the notation of sets, functions, and relations in which every later claim is written and spoken; the integers and their remainders modulo a prime, where an extended Euclidean algorithm turns division into multiplication by an inverse; the abstract structures—groups, rings, and fields—that name the laws this arithmetic obeys; polynomials over a field, whose scarcity of roots lets two enormous polynomials be compared at a single random point; the finite fields, their possible sizes, and their cyclic multiplicative structure; the number theory of squares, square roots, and the discrete logarithm; linear algebra and the inner products behind short algebraic receipts for long lists of secrets; the roots of unity that furnish evaluation domains and the fast Fourier transform; elliptic curves, whose point groups have no known subexponential discrete-log attack when chosen appropriately; and finally the probability, asymptotics, and model of computation required to state—and mean—a claim of 128-bit security.

The series consumes three artefacts of this tower: the Pasta curve cycle, constructed in §10; the polynomial machinery on which Halo 2 rests, developed in §5 and §9; and the security-statement language, fixed in §11. A first reading takes the sections in order; a reader after one artefact may enter at its section and follow its backward references down the tower.

Contents

1 Notation: sets, functions, and relations

Open the specification of a modern cryptographic protocol — a signature scheme, say, or a scheme for proving a statement without revealing why it is true — and try to read a single line of it aloud. Before any mathematics has happened, the page presents three alphabets at once: double-struck capitals such as $\mathbb{Z}$ and $\mathbb{F}_q$ naming entire systems of numbers, Greek letters such as λ and ρ naming parameters and challenges, and a thicket of operator symbols — $\oplus$, $\|$, $\xleftarrow{\$}$, $[k]P$ — each with a fixed meaning and a spoken name. A reader who cannot pronounce a line cannot discuss it with a colleague, and a reader who mistakes $\subseteq$ for $\in$, or swaps the order of two quantifiers, has misread the claim itself.

This section fixes the small stock of set-theoretic and logical language in which the whole series is written, and it closes by paying the opening debt in full: three reference tables listing every recurring symbol with its spoken form (§1.5). A reader comfortable with elementary mathematics may skim the prose and keep the tables to hand.

1.1 Sets

A *set* is a collection of distinct objects, its *elements*; one writes $x \in A$ for membership and $x \notin A$ for its negation. A set may be given by listing, $\{a, b, c\}$, or by *set-builder* notation $\{x \in A : P(x)\}$ ("the x in A such that $P(x)$"). One writes $A \subseteq B$ when every element of A lies in B, and $A = B$ exactly when $A \subseteq B$ and $B \subseteq A$ — so proving two sets equal means proving two inclusions. The usual operations are union $A \cup B$, intersection $A \cap B$, difference $A \setminus B$, and the empty set $\varnothing$. The *Cartesian product* is $A \times B = \{(a, b) : a \in A,\ b \in B\}$, and A^n is the set of n-tuples over A. For a finite set A, $|A|$ denotes its number of elements. The standard number systems are the naturals $\mathbb{N}$, the integers $\mathbb{Z}$, the rationals $\mathbb{Q}$, the reals $\mathbb{R}$, and the complex numbers $\mathbb{C}$, with $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$; we construct the underlying arithmetic of the prime field $\mathbb{F}_p$ in §2, certify it as a field—and first call it $\mathbb{F}_p$—in §4, and treat general finite fields $\mathbb{F}_q$ in §6.

1.2 Functions

A *function* $f : A \to B$ assigns to each $a \in A$ a unique value $f(a) \in B$; A is the *domain* and B the *codomain*. It is *injective* (one-to-one) if $f(a) = f(a')$ implies $a = a'$; *surjective* (onto) if every $b \in B$ equals $f(a)$ for some $a \in A$; and *bijective* if both. The *composition* of $f : A \to B$ and $g : B \to C$ is $(g \circ f)(a) = g(f(a))$; composition is associative, $h \circ (g \circ f) = (h \circ g) \circ f$, both sides sending a to $h(g(f(a)))$. The *identity* on A is $\mathrm{id}_A : A \to A$, $\mathrm{id}_A(a) = a$.

> **Theorem 1.1** (Inverse functions). *A function $f : A \to B$ has a two-sided* inverse $f^{-1} : B \to A$, *meaning $f^{-1} \circ f = \mathrm{id}_A$ and $f \circ f^{-1} = \mathrm{id}_B$, if and only if f is bijective. When it exists, the inverse is unique.*

Proof. Suppose a two-sided inverse g exists. If $f(a) = f(a')$, applying g gives $a = g(f(a)) = g(f(a')) = a'$, so f is injective; and every $b \in B$ satisfies $b = f(g(b))$, so f is surjective. Conversely, suppose f is bijective. Each $b \in B$ has at least one preimage by surjectivity and at most one by injectivity, so we may define $g(b)$ to be the unique $a \in A$ with $f(a) = b$; then $g(f(a)) = a$ and $f(g(b)) = b$ hold by construction. Finally, if g and g' are both two-sided inverses, associativity of composition gives $g = g \circ \mathrm{id}_B = g \circ (f \circ g') = (g \circ f) \circ g' = \mathrm{id}_A \circ g' = g'$; the inverse is unique, and we write it f^{-1}. □

For $S \subseteq A$ the *image* is $f(S) = \{f(a) : a \in S\}$, and for $T \subseteq B$ the *preimage* is $f^{-1}(T) = \{a \in A : f(a) \in T\}$, defined whether or not f is invertible. The overloading of f^{-1} is harmless: when the inverse function exists, the preimage of T under f is exactly the image of T under f^{-1}.

1.3 Relations and equivalences

A *binary relation* on A is a subset $R \subseteq A \times A$; one writes $a \sim b$ for $(a, b) \in R$. It is an *equivalence relation* if it is *reflexive* ($a \sim a$), *symmetric* ($a \sim b \Rightarrow b \sim a$), and *transitive* ($a \sim b$ and $b \sim c$ imply $a \sim c$). The *equivalence class* of a is $[a] = \{x \in A : x \sim a\}$, and the set of distinct classes is the *quotient* $A/{\sim}$.

> **Proposition 1.2** (Classes partition the set)**.** *Let $\sim$ be an equivalence relation on a set A. The distinct equivalence classes* partition *A: they are nonempty, pairwise disjoint, and their union is A.*

Proof. Reflexivity gives $a \in [a]$ for every $a \in A$, so each class is nonempty and every element lies in some class; the union of the classes is therefore A. For disjointness, suppose the classes $[a]$ and $[b]$ share an element x, and let $y \in [a]$ be arbitrary. From $y \sim a$ and $a \sim x$ (the latter by symmetry from $x \sim a$), transitivity gives $y \sim x$; combining with $x \sim b$ gives $y \sim b$, so $y \in [b]$. Thus $[a] \subseteq [b]$, and exchanging the roles of a and b gives the reverse inclusion; two classes that meet coincide. □

We use this construction twice in the present volume: to form the integers modulo n (§2) and the cosets of a subgroup (§3).

1.4 Logic and proof conventions

We employ the symbols $\forall$ ("for all"), $\exists$ ("there exists"), $\exists!$ ("there exists a unique"), and " : " or "s.t." for "such that". Negation swaps the quantifiers: $\neg(\forall x\, P(x))$ is $\exists x\, \neg P(x)$, and $\neg(\exists x\, P(x))$ is $\forall x\, \neg P(x)$. The order of unlike quantifiers matters: $\forall x\, \exists y\, P(x, y)$ (the y may depend on x) is weaker than $\exists y\, \forall x\, P(x, y)$ (one y works for all x).

The following conventions are fixed for the entire series. We prove an implication $P \Rightarrow Q$ directly (assume P, derive Q), by contraposition (assume $\neg Q$, derive $\neg P$), or by contradiction; we prove a statement $\forall x\, P(x)$ by taking an arbitrary x, and $\exists x\, P(x)$ by exhibiting a witness. We prove statements about all natural numbers by induction.

1.5 Symbols and how to say them

The debt incurred at the opening of this section falls due here. The series draws on three alphabets: blackboard-bold letters for the standard number systems and structures, Greek letters, and a handful of relational and operator symbols. Tables 1–3 collect the ones the series actually uses and give a spoken form for each — useful when reading aloud, and because several Greek letters are routinely mispronounced. Pronunciations are given in the received-English form (so β is "BEE-ta", not "BAY-ta"); a capitalised syllable carries the stress. Each symbol is defined where it is first used; the notes here are reminders, not definitions.

2 The integers and modular arithmetic

How does one divide in a finite world? The proof systems documented in this series do their arithmetic not with the unbounded integers of school algebra but with the finitely many remainders left upon division by a fixed, enormous prime. Every step a prover or verifier takes is an addition, a multiplication, or—the delicate case—a division of such remainders. Adding and multiplying remainders is easy to imagine: compute in $\mathbb{Z}$, then discard multiples of the modulus. But what can it mean to divide, when "one half" names no remainder? The answer developed in this section is that division is multiplication by a *modular inverse*; that a remainder possesses an inverse exactly when a certain greatest common divisor equals one; and that an algorithm already in Euclid's *Elements*, suitably extended, computes the inverse efficiently.

Symbol	Name	Said	Typical role in the series
$\mathbb{N}$	blackboard N	"natural numbers"	the naturals
$\mathbb{Z}$	blackboard Z	"the integers"	the integers (German *Zahlen*)
$\mathbb{Q}$	blackboard Q	"the rationals"	the rationals (*quotients*)
$\mathbb{R}$	blackboard R	"the reals"	the real numbers
$\mathbb{C}$	blackboard C	"the complexes"	the complex numbers
$\mathbb{F}_q$	blackboard F	"eff-cue"	the finite field of q elements
$\mathbb{G}$	blackboard G	"group G"	an abstract group
$\mathbb{E}$	blackboard E	"expectation"	expectation of a random variable (§11)
$\mathbb{P}$	blackboard P	"the Pallas curve"	the Pallas group (§10; protocol-spec notation)

Table 1: Blackboard-bold letters. "Blackboard bold" names the double-struck style itself; in speech one usually says only the letter or the system it denotes. Elliptic curves are written italic E or calligraphic $\mathcal{E}$, never blackboard bold. The squared $\mathbb{P}^2$ of §10 is unrelated: it denotes the projective plane.

Sym.	Name	Said	Sym.	Name	Said
α	alpha	"AL-fa"	μ	mu	"mew"
β	beta	"BEE-ta"	ν	nu	"new"
γ	gamma	"GAM-a"	ξ	xi	"ksigh" / "zigh"
δ	delta	"DEL-ta"	π	pi	"pie"
ϵ, ε	epsilon	"EP-si-lon"	ρ	rho	"roe"
ζ	zeta	"ZEE-ta"	σ	sigma	"SIG-ma"
η	eta	"EE-ta"	τ	tau	"taw" (rhymes "paw")
θ	theta	"THEE-ta"	ϕ, φ	phi	"fie" or "fee"
ι	iota	"eye-OH-ta"	χ	chi	"kigh" (hard k)
κ	kappa	"KAP-a"	ψ	psi	"sigh" or "psigh"
λ	lambda	"LAM-da"	ω	omega	"OH-mig-a"

Table 2: Lower-case Greek letters used in the series. The commonly muddled ones: ξ, χ, and ψ rhyme with "eye" ("ksigh", "kigh", "sigh" — never "ksee"), χ with a hard k, ψ with a silent or lightly sounded p, and ζ is "ZEE-ta", not "ZAY-ta". The letter τ is also heard rhyming with "cow". Capitals reuse the same names: Δ ("delta"), Σ ("sigma"; the summation sign, and the name of the Σ-protocol family of later volumes), Π ("pi"; the product sign, and a conventional name in those volumes for a protocol or problem), Φ, Ψ, Ω, Θ, Λ.

The integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ are the arithmetical bedrock on which every algebraic and cryptographic structure in this monograph ultimately rests. The finite number systems underlying elliptic curves, the scalar arithmetic of Halo 2, and the polynomial commitments of Orchard (short values that bind a prover to entire polynomials, which a verifier then tests by evaluation) all reduce, in the end, to computations with congruence classes of integers. This section develops the theory of the integers from a small set of assumptions and arrives at the integers modulo n—the set $\mathbb{Z}/n\mathbb{Z}$ of congruence classes—together with its units and the algorithms that make arithmetic with them effective. We keep the treatment constructive throughout: every existence statement comes, wherever possible, with an algorithm that produces the object in question, since cryptography requires not merely that inverses and representatives exist but that one can compute them efficiently.

2.1 Well-ordering and the integers

We take the basic algebraic properties of $\mathbb{Z}$ for granted: addition and multiplication are associative and commutative, and multiplication distributes over addition; the integers 0 and 1 are identities for addition and multiplication respectively; every integer a has a negative $-a$; there are no zero divisors (if $ab = 0$ then $a = 0$ or $b = 0$); and $\mathbb{Z}$ carries a total order $\leq$ compatible with the arithmetic. The single deep property we assume axiomatically is the following.

Symbol	Said	Meaning
$\in$, $\notin$	"in", "not in"	set membership
$\subseteq$, $\subset$	"subset of"	(proper) inclusion
$\cup$, $\cap$	"union", "intersect"	set union and intersection
$A \setminus B$	"A minus B"	set difference
$\times$	"times" / "cross"	Cartesian product
$\mid$, $\nmid$	"divides", "does not divide"	divisibility
$\equiv$	"congruent to"	congruence, qualified $\pmod{n}$
$a \bmod n$	"a mod n"	remainder of division by n
$:=$, $=:$	"is defined as"	definitional equality
$\mapsto$	"maps to"	the action of a function
$g \circ f$	"g after f"	function composition
$\lfloor x \rfloor$, $\lceil x \rceil$	"floor", "ceiling"	round down, round up
$\langle a, b \rangle$	"inner product"	inner product; $\langle g \rangle$ the subgroup generated by g
$\oplus$	"x-or"	bitwise exclusive or
$x \parallel y$	"x concat y"	concatenation of bit or byte strings
$x \leftarrow D$, $x \xleftarrow{\$} S$	"drawn from"	sampling ($ marks uniform); also assignment
$[k]P$	"k times P"	scalar multiplication of a point (§10)
$P^\star$	"P star"	canonical bit-encoding of a point or element
$\cdot$	"dot"	product; argument placeholder, as in $F(\cdot)$
$\lvert A \rvert$	"size of A" / "order of A"	cardinality

Table 3: Relational and operator symbols. The symbol $\oplus$ is read "x-or"; every use in the series is the bitwise exclusive or. The encoding $P^\star$ is protocol-spec notation, defined in later volumes.

Theorem 2.1 (Well-ordering principle)**.** *Every nonempty subset $S \subseteq \mathbb{N} = \{0, 1, 2, \dots\}$ of the natural numbers has a least element: there exists $m \in S$ such that $m \leq s$ for all $s \in S$.*

The well-ordering principle is logically equivalent to the principle of mathematical induction; we use both freely. It is also the engine of this section: nearly every existence result below is, at bottom, an application of well-ordering, in which one exhibits a nonempty set of natural numbers and extracts its minimum. Four instances map the road ahead. The remainder in the division algorithm is the least element of the set $\{\, a - bk \;:\; k \in \mathbb{Z},\; a - bk \geq 0 \,\}$; the Euclidean algorithm terminates because a strictly decreasing sequence of nonnegative integers cannot be infinite; the Bézout identity extracts the least positive element of a set of integer combinations; and existence in the fundamental theorem of arithmetic proceeds by minimal counterexample.

2.2 Divisibility

Definition 2.2. Let $a, b \in \mathbb{Z}$. One says that a *divides* b, written $a \mid b$, if there exists an integer $c \in \mathbb{Z}$ with $b = ac$. In this case a is called a *divisor* or *factor* of b, and b is a *multiple* of a. If no such c exists one writes $a \nmid b$.

Several immediate consequences of the definition deserve to be recorded; we shall use them constantly and usually without explicit citation.

Proposition 2.3. *Let $a, b, c, d \in \mathbb{Z}$. Then:*

1. *$a \mid a$ (reflexivity), $1 \mid a$, and $a \mid 0$.*
2. *If $a \mid b$ and $b \mid c$ then $a \mid c$ (transitivity).*

3. *If $a \mid b$ and $a \mid c$ then $a \mid (bx + cy)$ for all $x, y \in \mathbb{Z}$; in particular $a \mid (b \pm c)$.*

4. *If $a \mid b$ and $b \neq 0$ then $|a| \leq |b|$.*

5. *If $a \mid b$ and $b \mid a$ then $a = \pm b$.*

6. *$0 \mid b$ if and only if $b = 0$.*

Proof. (1) is immediate: $a = a \cdot 1$, $a = 1 \cdot a$, $0 = a \cdot 0$. For (2), write $b = ae$ and $c = bf$; then $c = a(ef)$, so $a \mid c$. For (3), write $b = ae$ and $c = af$; then $bx + cy = a(ex + fy)$. For (4), write $b = ac$ with $b \neq 0$, so $c \neq 0$ and hence $|c| \geq 1$; then $|b| = |a|\,|c| \geq |a|$. For (5), if either is 0 then by (6) so is the other and the claim holds; otherwise (4) gives $|a| \leq |b|$ and $|b| \leq |a|$, so $|a| = |b|$ and $a = \pm b$. For (6), $0 \mid b$ means $b = 0 \cdot c = 0$; conversely $0 \mid 0$. □

The relation $\mid$ is thus reflexive and transitive; it is a partial order when restricted to the natural numbers (where (5) becomes $a = b$), but on all of $\mathbb{Z}$ it fails to be antisymmetric precisely because of signs. The *units* of $\mathbb{Z}$—the integers dividing 1—are exactly ± 1, and two integers that divide each other are called *associates*; in $\mathbb{Z}$ the associates of a are $\pm a$.

2.3 The division algorithm

Divisibility is the special case of exact division. The fundamental fact that makes the arithmetic of $\mathbb{Z}$ tractable is that one can always divide *with remainder*, and that the quotient and remainder are uniquely determined once a convention for the remainder's range is fixed.

Theorem 2.4 (Division algorithm). *Let $a \in \mathbb{Z}$ and $b \in \mathbb{Z}$ with $b > 0$. Then there exist unique integers q (the* quotient*) and r (the* remainder*) such that*

$$a = bq + r, \qquad 0 \leq r < b.$$

Proof. Existence. Consider the set

$$S = \{\, a - bk \,:\, k \in \mathbb{Z},\ a - bk \geq 0 \,\} \subseteq \mathbb{N}.$$

First, $S \neq \varnothing$. If $a \geq 0$ then $a = a - b \cdot 0 \in S$. If $a < 0$ then, since $b \geq 1$, taking $k = a$ gives $a - ba = a(1 - b) \geq 0$ (a product of a nonpositive and a nonpositive factor), so $a - ba \in S$. Thus S is a nonempty subset of $\mathbb{N}$, and by the well-ordering principle (Theorem 2.1) it has a least element $r = a - bq$ for some $q \in \mathbb{Z}$. By construction $r \geq 0$. If $r \geq b$, then $r - b = a - b(q + 1) \geq 0$ would lie in S and be strictly smaller than r, contradicting minimality. Hence $0 \leq r < b$.

Uniqueness. Suppose $a = bq + r = bq' + r'$ with $0 \leq r, r' < b$. Then $b(q - q') = r' - r$, so $b \mid (r' - r)$. But $-b < r' - r < b$, and the only multiple of b strictly between $-b$ and b is 0. Hence $r' = r$, and then $b(q - q') = 0$ with $b \neq 0$ forces $q = q'$. □

Remark 2.5. The hypothesis $b > 0$ is only for definiteness. For general $b \neq 0$ one obtains unique q, r with $a = bq + r$ and $0 \leq r < |b|$, by applying the theorem to $|b|$ and adjusting the sign of q. Note that the remainder convention $0 \leq r < |b|$ differs from the truncation behaviour of many programming languages, where the remainder of a negative dividend may be negative; for the number-theoretic results below the convention above is the appropriate one.

We denote the remainder r produced by the division algorithm by $a \bmod b$, and the quotient q by $\lfloor a/b \rfloor$ when $b > 0$ (the *floor* of the rational number a/b). With this notation, $b \mid a$ holds precisely when $a \bmod b = 0$.

2.4 The greatest common divisor and the Euclidean algorithm

Definition 2.6. A *common divisor* of integers a and b is an integer d with $d \mid a$ and $d \mid b$. The *greatest common divisor* of a and b, not both zero, is the largest such d; it is denoted $\gcd(a, b)$. By convention $\gcd(0, 0) = 0$. Integers a and b are *coprime* (or *relatively prime*) if $\gcd(a, b) = 1$.

The greatest common divisor is well defined when $(a, b) \neq (0, 0)$: the set of common divisors is nonempty (it contains 1) and bounded above (every common divisor of, say, the nonzero a has absolute value at most $|a|$ by Proposition 2.3(4)), so it has a largest element, which is positive since 1 is a common divisor. We establish a much stronger characterisation shortly: the gcd is not merely the numerically largest common divisor but is divisible by every common divisor. The key computational engine is the following observation.

Lemma 2.7. *Let $a, b \in \mathbb{Z}$. For any $q \in \mathbb{Z}$,*

$$\gcd(a, b) = \gcd(b,\ a - qb).$$

In particular, if $a = bq + r$ then $\gcd(a, b) = \gcd(b, r)$*, and* $\gcd(a, 0) = |a|$.

Proof. It suffices to show the two pairs (a, b) and $(b, a - qb)$ have exactly the same common divisors; equal sets of common divisors have equal greatest elements (and the same coprimality status). If $d \mid a$ and $d \mid b$, then $d \mid (a - qb)$ by Proposition 2.3(3), so d is a common divisor of b and $a - qb$. Conversely, if $d \mid b$ and $d \mid (a - qb)$, then $d \mid \big((a - qb) + qb\big) = a$, so d is a common divisor of a and b. The two sets of common divisors coincide. Finally $\gcd(a, 0) = |a|$ because the common divisors of a and 0 are exactly the divisors of a, the largest of which in absolute value is $|a|$ (and $|a| > 0$ when $a \neq 0$; the case $a = 0$ is the convention $\gcd(0, 0) = 0$). □

Iterating the lemma with the division algorithm yields the oldest nontrivial algorithm in mathematics.

Theorem 2.8 (Euclidean algorithm)**.** *Let a, b be integers with $b \neq 0$. Define a sequence of remainders by $r_0 = |a|$, $r_1 = |b|$, and, as long as $r_i \neq 0$, let $r_{i+1} = r_{i-1} \bmod r_i$, so that*

$$r_{i-1} = r_i q_i + r_{i+1}, \qquad 0 \le r_{i+1} < r_i.$$

The sequence $r_1 > r_2 > \cdots$ of positive remainders is strictly decreasing, so some $r_{n+1} = 0$; the last nonzero remainder satisfies $r_n = \gcd(a, b)$.

Proof. The remainders $r_1, r_2, \ldots$ are nonnegative integers and, while nonzero, strictly decreasing because $r_{i+1} = r_{i-1} \bmod r_i < r_i$. A strictly decreasing sequence of nonnegative integers cannot be infinite (again by well-ordering: otherwise the set of values would have no least element), so the algorithm terminates with some $r_{n+1} = 0$ and $r_n > 0$. By Lemma 2.7, each step preserves the gcd:

$$\gcd(a, b) = \gcd(r_0, r_1) = \gcd(r_1, r_2) = \cdots = \gcd(r_n, r_{n+1}) = \gcd(r_n, 0) = r_n,$$

using $\gcd(|a|, |b|) = \gcd(a, b)$ (signs do not affect common divisors). □

Example 2.9. Compute $\gcd(1071, 462)$:

$$\begin{aligned} 1071 &= 2 \cdot 462 + 147, \\ 462 &= 3 \cdot 147 + 21, \\ 147 &= 7 \cdot 21 + 0. \end{aligned}$$

The last nonzero remainder is 21, so $\gcd(1071, 462) = 21$. Indeed $1071 = 21 \cdot 51$ and $462 = 21 \cdot 22$, with $\gcd(51, 22) = 1$.

Remark 2.10. The Euclidean algorithm is efficient. The number of division steps for inputs $a > b > 0$ is $O(\log b)$; the worst case occurs at consecutive Fibonacci numbers (Lamé's theorem), where every quotient q_i equals 1 except the final one (which equals 2) and the remainders shrink as slowly as possible. With schoolbook arithmetic on k-bit inputs the whole computation costs $O(k^2)$ bit operations. This efficiency is what renders gcd-based cryptographic operations—above all modular inversion, below—practical at the sizes used in Halo 2 and Orchard.

2.5 The Bézout identity and the extended Euclidean algorithm

The Euclidean algorithm computes the gcd, but the back-substitution of its equations yields something more powerful: the gcd is an *integer linear combination* of the inputs. This is among the most useful facts in elementary number theory.

> **Theorem 2.11** (Bézout's identity). *Let $a, b \in \mathbb{Z}$, not both zero, and let $d = \gcd(a, b)$. Then there exist integers $u, v \in \mathbb{Z}$ with*
> $$au + bv = d.$$
> *Moreover d is the* smallest *positive integer expressible in the form $ax + by$ with $x, y \in \mathbb{Z}$, and the set of all such combinations is exactly the set of multiples of d:*
> $$\{\, ax + by \,:\, x, y \in \mathbb{Z} \,\} = d\mathbb{Z} = \{\, kd \,:\, k \in \mathbb{Z} \,\}.$$

Proof. Let $I = \{\, ax+by \,:\, x, y \in \mathbb{Z} \,\}$. This set contains a and b and is closed under addition and under multiplication by any integer; concretely, if $s = ax_1 + by_1$ and $t = ax_2 + by_2$ lie in I, then for any $m, n \in \mathbb{Z}$ one has
$$ms + nt = a(mx_1 + nx_2) + b(my_1 + ny_2) \in I.$$
Since a, b are not both zero, I contains a positive element (one of $\pm a, \pm b$), so the set $I \cap \mathbb{N}_{>0}$ of positive elements of I is nonempty. By well-ordering it has a least element $g = au + bv > 0$.

We claim that g divides every element of I. Indeed, let $c \in I$ and write $c = gq + r$ with $0 \le r < g$ by the division algorithm. Then $r = c - gq \in I$ because I is closed under the operations just described. By minimality of g among positive elements of I, and $0 \le r < g$, necessarily $r = 0$. Hence $g \mid c$ for every $c \in I$; in particular $g \mid a$ and $g \mid b$, so g is a common divisor of a and b. Conversely, any common divisor e of a and b divides $au + bv = g$ by Proposition 2.3(3). Therefore g is a common divisor that is divisible by every common divisor; being positive, it is in particular the *greatest* common divisor, so $g = d$. The displayed equation $au + bv = d$ follows, d is the least positive element of I, and $I = g\mathbb{Z} = d\mathbb{Z}$ because every element of I is a multiple of g, as shown, and conversely every multiple $kg = a(ku) + b(kv)$ lies in I. □

Remark 2.12. The proof yields a characterisation of the gcd that is independent of the order $\le$: $d = \gcd(a, b)$ is, up to sign, the unique common divisor of a and b that is divisible by every common divisor. This is the formulation that survives in settings where "largest" may make no sense but "divisible by every common divisor" still does; later sections adopt it when they leave the integers. We call a nonempty subset $I \subseteq \mathbb{Z}$ closed under addition and under multiplication by arbitrary integers an *ideal*; the content of Bézout's identity is that every ideal of $\mathbb{Z}$ has the form $d\mathbb{Z}$ for a single generator d. Ideals recur in section 4, where they drive a general construction whose prototype is the passage from integers to congruence classes carried out later in this section.

The Bézout coefficients u, v are not unique. If $au + bv = d$, then for every $k \in \mathbb{Z}$,
$$a\Big(u + k\frac{b}{d}\Big) + b\Big(v - k\frac{a}{d}\Big) = d,$$
since the added terms $k\frac{ab}{d}$ and $-k\frac{ab}{d}$ cancel; and these are all the solutions. To see this, note first that a/d and b/d are coprime: a common divisor e of both makes de a common divisor of a and b, so $de \mid$

$au+bv = d$ by Proposition 2.3(3), forcing $e = \pm 1$. Now suppose $au'+bv' = d$. Subtracting $au+bv = d$ and dividing by $d > 0$ gives $\frac{a}{d}(u'-u) = -\frac{b}{d}(v'-v)$, so b/d divides $\frac{a}{d}(u'-u)$; by Proposition 2.15(1) below (whose proof rests only on Bézout's identity, already established), b/d divides $u'-u$, say $u' = u + k\frac{b}{d}$ with $k \in \mathbb{Z}$. Substituting back gives $b\big(v' - v + k\frac{a}{d}\big) = 0$, so $v' = v - k\frac{a}{d}$ when $b \neq 0$; and when $b = 0$ we have $u' = u$, $a/d = \pm 1$, and $k = (v - v')\,\frac{a}{d}$ places (u', v') in the family. To compute a particular pair (u, v) efficiently, we augment the Euclidean algorithm so that it carries each remainder as an explicit combination of a and b.

> **Theorem 2.13** (Extended Euclidean algorithm)**.** *There is an algorithm that, on input $a, b \in \mathbb{Z}$ not both zero, outputs $d = \gcd(a, b)$ together with integers u, v satisfying $au + bv = d$, using the same number of division steps as the Euclidean algorithm.*

Proof. Alongside the remainder sequence of Theorem 2.8, maintain two auxiliary sequences (s_i) and (t_i) with the invariant

$$r_i = a\,s_i + b\,t_i \qquad \text{for all } i. \tag{$*$}$$

Initialise (taking $a, b \geq 0$ for clarity; we restore signs at the end)

$$r_0 = a,\ s_0 = 1,\ t_0 = 0; \qquad r_1 = b,\ s_1 = 0,\ t_1 = 1,$$

so $(*)$ holds for $i = 0, 1$. At step $i \geq 1$, having $r_{i-1} = r_i q_i + r_{i+1}$ from the division algorithm, set

$$s_{i+1} = s_{i-1} - q_i s_i, \qquad t_{i+1} = t_{i-1} - q_i t_i.$$

Then

$$a\,s_{i+1} + b\,t_{i+1} = (as_{i-1} + bt_{i-1}) - q_i\,(as_i + bt_i) = r_{i-1} - q_i r_i = r_{i+1},$$

so induction preserves $(*)$. When the algorithm terminates with $r_{n+1} = 0$ and $r_n = d = \gcd(a, b)$, the invariant for $i = n$ reads $d = as_n + bt_n$; output $u = s_n$, $v = t_n$. If $b = 0$, then $a \neq 0$ and termination is immediate with $n = 0$: no division step runs, the last nonzero remainder is $r_0 = a = \gcd(a, 0)$ (Lemma 2.7), and the invariant at $i = 0$ already gives the output $(d, u, v) = (a, 1, 0)$. If $b \neq 0$, termination and the value of d are exactly as in Theorem 2.8. In either case each step does only a constant amount of extra work. For general inputs, run the algorithm on $(|a|, |b|)$ to obtain $|a|\,u' + |b|\,v' = d$ (recall $\gcd(|a|, |b|) = \gcd(a, b)$) and output $u = \operatorname{sgn}(a)\,u'$, $v = \operatorname{sgn}(b)\,v'$, so that $au + bv = d$. □

In practice it is convenient to lay the computation out as a table with columns (r_i, q_i, s_i, t_i), computing each new row from the previous two.

Example 2.14. Run the extended algorithm on $a = 1071$, $b = 462$ (continuing Example 2.9). The quotients are $q_1 = 2, q_2 = 3, q_3 = 7$.

i	r_i	s_i	t_i
0	1071	1	0
1	462	0	1
2	147	1	−2
3	21	−3	7
4	0	22	−51

For instance, row 2 is row 0 minus $q_1 = 2$ times row 1: $s_2 = 1-2\cdot 0 = 1$, $t_2 = 0-2\cdot 1 = -2$, and indeed $1071 \cdot 1 + 462 \cdot (-2) = 1071 - 924 = 147$. The last nonzero remainder is $r_3 = 21 = \gcd(1071, 462)$, with

$$1071 \cdot (-3) + 462 \cdot 7 = -3213 + 3234 = 21.$$

Thus $(u, v) = (-3, 7)$. The final row $(s_4, t_4) = (22, -51) = (b/d, -a/d)$ up to sign records the homogeneous solution: $1071 \cdot 22 + 462 \cdot (-51) = 0$.

A first, decisive application of Bézout's identity is Euclid's lemma, the property of primes that ultimately forces unique factorisation.

Proposition 2.15 (Euclid's lemma)**.** *Let $a, b, c \in \mathbb{Z}$.*

1. *If $c \mid ab$ and $\gcd(c, a) = 1$, then $c \mid b$.*
2. *If $\gcd(a, b) = 1$, $a \mid n$, and $b \mid n$, then $ab \mid n$.*
3. *If $\gcd(a, b) = 1$ and $\gcd(a, c) = 1$, then $\gcd(a, bc) = 1$.*

Proof. (1) By Bézout there are u, v with $cu + av = 1$. Multiply by b: $b = cub + avb = c(ub) + (ab)v$. Since $c \mid ab$, c divides both terms on the right, hence $c \mid b$.

(2) Write $n = ak$. Since $b \mid n = ak$ and $\gcd(b, a) = 1$, part (1) gives $b \mid k$, say $k = bm$. Then $n = a(bm) = (ab)m$, so $ab \mid n$.

(3) By Bézout, $ax_1 + by_1 = 1$ and $ax_2 + cy_2 = 1$ for suitable integers. Multiplying,

$$1 = (ax_1 + by_1)(ax_2 + cy_2) = a\underbrace{(ax_1x_2 + x_1cy_2 + by_1x_2)}_{\in\mathbb{Z}} + bc\,(y_1y_2),$$

which exhibits 1 as an integer combination of a and bc. Hence $\gcd(a, bc)$ divides 1, so it equals 1. □

2.6 Primes and the fundamental theorem of arithmetic

Definition 2.16. An integer $p > 1$ is *prime* if its only positive divisors are 1 and p. An integer $n > 1$ that is not prime is *composite*; thus n is composite iff $n = ab$ for some integers $1 < a, b < n$. The integer 1 is a *unit* and is by convention neither prime nor composite.

The exclusion of 1 from the primes is not arbitrary: it is exactly what makes factorisation into primes unique, since otherwise one could insert arbitrarily many factors of 1. The decisive property of primes is the prime case of Euclid's lemma.

Lemma 2.17 (Euclid's lemma for primes)**.** *Let p be prime and $a, b \in \mathbb{Z}$. If $p \mid ab$ then $p \mid a$ or $p \mid b$. More generally, if $p \mid a_1a_2 \cdots a_k$ then $p \mid a_i$ for some i.*

Proof. The only positive divisors of p are 1 and p, so $\gcd(p, a) \in \{1, p\}$. If $\gcd(p, a) = p$ then $p \mid a$ and the claim holds; otherwise $\gcd(p, a) = 1$, and Proposition 2.15(1) gives $p \mid b$. The general statement follows by induction on k: from $p \mid a_1(a_2 \cdots a_k)$ it follows that $p \mid a_1$ or $p \mid a_2 \cdots a_k$, and the inductive hypothesis handles the latter. □

Theorem 2.18 (Fundamental theorem of arithmetic)**.** *Every integer $n > 1$ can be written as a product of primes,*

$$n = p_1p_2 \cdots p_k,$$

and this factorisation is unique up to the order of the factors. Equivalently, there is a unique expression

$$n = p_1^{e_1}p_2^{e_2} \cdots p_r^{e_r}, \qquad p_1 < p_2 < \cdots < p_r \text{ prime},\ e_i \geq 1.$$

Proof. Existence. Suppose, for contradiction, that some integer > 1 is not a product of primes, and let n be the least such (well-ordering). Then n is not prime (a prime is trivially a product of one prime), so $n = ab$ with $1 < a, b < n$. By minimality, each of a, b is a product of primes; concatenating these factorisations expresses n as a product of primes, a contradiction. Hence every $n > 1$ factors into primes.

Uniqueness. Suppose

$$p_1 p_2 \cdots p_k = q_1 q_2 \cdots q_\ell$$

are two factorisations of the same integer into primes. The argument proceeds by induction on k. If $k = 0$ the product is the empty product 1, forcing $\ell = 0$ as well (a nonempty product of primes is > 1). For the inductive step, p_1 divides the right-hand product, so by Lemma 2.17 $p_1 \mid q_j$ for some j. Since q_j is prime, its only divisor > 1 is itself, so $p_1 = q_j$. Cancelling this common factor (legitimate as $\mathbb{Z}$ has no zero divisors) yields

$$p_2 \cdots p_k = q_1 \cdots \widehat{q_j} \cdots q_\ell,$$

where the hat denotes omission. By the inductive hypothesis these two factorisations agree up to order, with $k - 1 = \ell - 1$; restoring $p_1 = q_j$ shows the original factorisations agree up to order. Collecting equal primes into prime powers gives the canonical form, whose uniqueness is immediate from the uniqueness of the multiset of primes—the factors counted with multiplicity, without regard to order. □

Remark 2.19. Both halves of the theorem are needed and have different characters. Existence is a statement about the order structure (well-ordering) and would hold in any reasonable factorisation; uniqueness rests squarely on Euclid's lemma, hence on Bézout, hence on the division algorithm. There are number systems with a division-like structure failing Euclid's lemma in which factorisation into irreducibles is *not* unique; the integers avoid this pathology precisely because they admit division with remainder, from which the whole chain above flows.

2.7 Congruences and residue classes

The development now passes from divisibility to the arithmetic of remainders. Fix an integer $n \geq 1$, the *modulus*.

Definition 2.20. Let $n \geq 1$ and $a, b \in \mathbb{Z}$. The integer a is *congruent to b modulo n*, written

$$a \equiv b \pmod{n},$$

if $n \mid (a - b)$, equivalently if a and b leave the same remainder upon division by n.

The two formulations agree: if $a = nq_1 + r_1$ and $b = nq_2 + r_2$ with $0 \leq r_1, r_2 < n$, then $a - b = n(q_1 - q_2) + (r_1 - r_2)$, and $n \mid (a - b)$ iff $n \mid (r_1 - r_2)$ iff $r_1 = r_2$ (since $|r_1 - r_2| < n$).

Proposition 2.21. *Fix $n \geq 1$. Congruence modulo n is an equivalence relation on $\mathbb{Z}$ that is moreover compatible with addition and multiplication: if $a \equiv a'$ and $b \equiv b' \pmod{n}$, then*

$$a + b \equiv a' + b' \pmod{n}, \qquad ab \equiv a'b' \pmod{n}.$$

Consequently $a^k \equiv (a')^k \pmod{n}$ for all $k \geq 0$, and one may substitute congruent values freely in any polynomial expression with integer coefficients.

Proof. Reflexivity ($n \mid 0$), symmetry ($n \mid (a - b) \Rightarrow n \mid (b - a)$), and transitivity ($n \mid (a - b)$ and $n \mid (b - c)$ give $n \mid (a - c)$ by Proposition 2.3(3)) are immediate. For compatibility, write $a - a' = ns$ and $b - b' = nt$. Then

$$(a + b) - (a' + b') = n(s + t),$$

and

$$ab - a'b' = ab - a'b + a'b - a'b' = (a - a')b + a'(b - b') = n(sb + a't),$$

so both $n \mid \big((a+b)-(a'+b')\big)$ and $n \mid (ab-a'b')$. The statement about powers and polynomial expressions follows by induction from the multiplicative case. □

Because congruence is an equivalence relation, it partitions $\mathbb{Z}$ into disjoint *congruence classes* (also called *residue classes*).

Definition 2.22. The *congruence class* (or *residue class*) of a modulo n is

$$[a]_n = \{\, b \in \mathbb{Z} : b \equiv a \pmod{n} \,\} = \{\, a + kn : k \in \mathbb{Z} \,\} = a + n\mathbb{Z}.$$

We write the set of all congruence classes as

$$\mathbb{Z}/n\mathbb{Z} = \{\, [a]_n : a \in \mathbb{Z} \,\}.$$

By the division algorithm every integer is congruent modulo n to exactly one of $0, 1, \dots, n-1$, namely its remainder; hence the n classes $[0]_n, [1]_n, \dots, [n-1]_n$ are distinct and exhaust $\mathbb{Z}/n\mathbb{Z}$. The set $\{0, 1, \dots, n-1\}$ is the standard *system of representatives*, called the *least nonnegative residues*.

2.8 Units modulo n and modular inverses

Division is the inverse of multiplication, so the opening problem of the section comes down to a multiplicative question about congruences: for which a does the congruence $ax \equiv 1 \pmod{n}$ admit a solution? The Euclidean machinery characterises the solvable cases completely and computes the solutions.

Definition 2.23. Let $n \geq 1$. A class $[a]_n \in \mathbb{Z}/n\mathbb{Z}$ is a *unit* (or is *invertible*) if there exists an integer x with $ax \equiv 1 \pmod{n}$. Any class $[x]_n$ with this property is the *(modular) inverse* of $[a]_n$, written $[a]_n^{-1}$; one also calls x a modular inverse of a modulo n. We write the set of units as

$$(\mathbb{Z}/n\mathbb{Z})^\times = \{\, [a]_n : [a]_n \text{ is a unit} \,\}.$$

The definition is well posed on classes: by Proposition 2.21, if $a' \equiv a$ and $x' \equiv x \pmod{n}$ then $a'x' \equiv ax \pmod{n}$, so whether $ax \equiv 1 \pmod{n}$ holds depends only on the classes $[a]_n$ and $[x]_n$, never on the representatives chosen. The definite article in "the inverse" is justified by the uniqueness part of the next theorem—the payoff of the section.

Theorem 2.24. *Let $n \geq 1$ and $a \in \mathbb{Z}$. The class $[a]_n$ is a unit in $\mathbb{Z}/n\mathbb{Z}$ if and only if* $\gcd(a, n) = 1$. *When* $\gcd(a, n) = 1$, *the inverse is unique, and the extended Euclidean algorithm computes a representative: if $au + nv = 1$, then $[a]_n^{-1} = [u]_n$.*

Proof. Suppose $\gcd(a, n) = 1$. By Bézout (Theorem 2.11) there are u, v with $au + nv = 1$. Reducing modulo n, $au \equiv 1 \pmod{n}$, so $[u]_n$ is an inverse of $[a]_n$. The extended Euclidean algorithm (Theorem 2.13) produces such a u explicitly.

Conversely, if $[a]_n$ is a unit, say $ax \equiv 1 \pmod{n}$, then $ax - 1 = nk$ for some k, i.e. $ax - nk = 1$. Any common divisor of a and n divides the left side, hence divides 1; thus $\gcd(a, n) = 1$.

Uniqueness: if $ax \equiv 1$ and $ax' \equiv 1 \pmod{n}$, then $x \equiv x(ax') \equiv (xa)x' \equiv x' \pmod{n}$—a sandwich using only the associativity and commutativity of integer multiplication together with Proposition 2.21—so $[x]_n = [x']_n$. □

Example 2.25. Consider the computation of $[17]_{43}^{-1}$. The extended Euclidean algorithm on $(43, 17)$:

$$\begin{aligned} 43 &= 2 \cdot 17 + 9, \\ 17 &= 1 \cdot 9 + 8, \\ 9 &= 1 \cdot 8 + 1, \\ 8 &= 8 \cdot 1 + 0, \end{aligned}$$

so $\gcd(43, 17) = 1$. Back-substituting:

$$1 = 9 - 8 = 9 - (17 - 9) = 2 \cdot 9 - 17 = 2(43 - 2 \cdot 17) - 17 = 2 \cdot 43 - 5 \cdot 17.$$

Hence $-5 \cdot 17 \equiv 1 \pmod{43}$, and $[17]_{43}^{-1} = [-5]_{43} = [38]_{43}$. Check: $17 \cdot 38 = 646 = 15 \cdot 43 + 1$, so $17 \cdot 38 \equiv 1 \pmod{43}$.

Counting the units will matter as much as finding them: the count defined next carries structural weight in later sections, and we fix it here so that those sections can cite a definition rather than re-derive one.

> **Definition 2.26** (Euler totient)**.** *Euler's totient function* $\varphi(n)$ is the number of integers a with $1 \leq a \leq n$ and $\gcd(a, n) = 1$. Equivalently, $\varphi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times|$.

The two counts agree: the classes of $1, 2, \ldots, n$ run exactly once through all of $\mathbb{Z}/n\mathbb{Z}$ (the class of n being $[0]_n$), and by Theorem 2.24 the class $[a]_n$ is a unit precisely when $\gcd(a, n) = 1$.

Example 2.27. We have $\varphi(1) = 1$, and $\varphi(p) = p - 1$ for prime p: for $1 \leq a < p$ the positive divisor $\gcd(a, p)$ of p is 1 or p, and $p \nmid a$ rules out p, so every one of the $p - 1$ nonzero classes is a unit. By direct enumeration, $\varphi(12) = |\{1, 5, 7, 11\}| = 4$.

The cheque presented at the head of the section is now cashed. To divide by a in the finite world of remainders modulo n is to multiply by the inverse class $[a]_n^{-1}$; Theorem 2.24 says exactly when that inverse exists—precisely when $\gcd(a, n) = 1$, hence for every nonzero class when the modulus is prime—and the extended Euclidean algorithm computes it in $O(\log n)$ division steps (Remark 2.10). The laws verified along the way—associativity, commutativity, distributivity, identities, negatives, and now inverses—have been stated concretely, for the integers and their congruence classes alone. They are instances of patterns that pervade mathematics; Sections 3 and 4 isolate those patterns, give them their standard names, and re-read $\mathbb{Z}/n\mathbb{Z}$ in their light. Nothing proved there revises anything proved here.

3 Groups

Two parties who have never met wish to agree on a secret value while every message they exchange is read by an eavesdropper. One solution begins with a public set carrying a single public operation and a public starting element g. Each party privately chooses a count of repetitions—say a and b—combines g with itself that many times, and publishes the result. Each then takes the *other's* published element and applies their own secret count of repetitions to it. If repetition is well behaved, both arrive at the element produced by ab repetitions of g, and so share a secret, while the eavesdropper has seen only g and the two published values. Three demands on the operation are implicit. It must *compose*: "a repetitions, then b more" must be unambiguous however the intermediate steps are bracketed. It must *invert*: repetition counts should obey an arithmetic, with an identity and the possibility of undoing a step, not a mere accumulation. And it must *hide* repetition: recovering the secret count from the published value must be hard. The third demand is a computational matter taken up in later volumes; the first two are algebra, and this section isolates exactly the structure that supplies them.

That structure is the *group*, the foundational abstraction of modern algebra. A group axiomatises the bare minimum needed to discuss symmetry and invertible composition: one associative operation with an identity and inverses. Almost every object met later in this series is a group—the integers modulo a prime under addition, the nonzero residues under multiplication, the points of an elliptic curve, the symmetries of a finite field. This section develops group theory from first principles and proves its central structural results in full: Lagrange's theorem, the classification of cyclic groups, the quotient of an abelian group by a subgroup, and the basic theory of homomorphisms and isomorphisms. The payoff for the opening problem is the cyclic-group theory together with Lagrange's theorem: in any finite group, the arithmetic of repetition counts is precisely the modular arithmetic of section 2, the modulus being the order of the repeated element—a divisor of the group's order.

3.1 The group axioms

The first task is to make "combining two elements" precise.

Definition 3.1 (Binary operation)**.** Let S be a set. A *binary operation* on S is a function

$$*: S \times S \to S, \qquad (a, b) \mapsto a * b.$$

Closure—that $a * b$ again lies in S—is built into the very statement that $*$ maps *into* S. We name it explicitly nonetheless, because for a subset $H \subseteq S$ the question of whether $*$ restricts to a binary operation on H is exactly the question of whether H is *closed* under $*$; this question drives the theory of subgroups below (§3.4).

Definition 3.2 (Group)**.** A *group* is a pair $(G, *)$ consisting of a set G and a binary operation $*$ on G satisfying three axioms.

(G1) *Associativity.* For all $a, b, c \in G$, $(a * b) * c = a * (b * c)$.

(G2) *Identity.* There exists $e \in G$ such that $e * a = a * e = a$ for all $a \in G$.

(G3) *Inverses.* For each $a \in G$ there exists $b \in G$ such that $a * b = b * a = e$, where e is an identity as in (G2).

A set with an operation satisfying only (G1) is a *semigroup*; if it also satisfies (G2) it is a *monoid*. A group is thus a monoid in which every element is invertible.

The number of elements of G, written $|G|$, is the *order* of the group; if $|G|$ is finite, G is a *finite group*, and otherwise an *infinite group*. We frequently say "the group G", leaving the operation understood.

The axioms assert the *existence* of an identity and of inverses, but not, on their face, uniqueness. Uniqueness is a theorem—and a useful one, since it is what licenses the notations e and a^{-1}.

Proposition 3.3. *Let $(G, *)$ be a group.*

1. *The identity element is unique.*
2. *Each $a \in G$ has a unique inverse, denoted a^{-1}.*
3. (Cancellation.) *For all $a, x, y \in G$: if $a * x = a * y$ then $x = y$, and if $x * a = y * a$ then $x = y$.*
4. *For all $a, b \in G$, $(a^{-1})^{-1} = a$ and $(a * b)^{-1} = b^{-1} * a^{-1}$.*

Proof. (1) Suppose e and e' both satisfy (G2). Using that e' is an identity, $e = e * e'$; using that e is an identity, $e * e' = e'$. Hence $e = e'$.

(2) Suppose b and b' are both inverses of a. Then

$$b = b * e = b * (a * b') = (b * a) * b' = e * b' = b',$$

using (G2), the inverse property of b', associativity, and the inverse property of b.

(3) Suppose $a * x = a * y$. Multiplying on the left by a^{-1} and reassociating,

$$x = (a^{-1} * a) * x = a^{-1} * (a * x) = a^{-1} * (a * y) = (a^{-1} * a) * y = y.$$

The right-cancellation law is proved symmetrically, multiplying on the right by a^{-1}.

(4) Since $a * a^{-1} = a^{-1} * a = e$, the element a is an inverse of a^{-1}; by (2) it is *the* inverse, so $(a^{-1})^{-1} = a$. For the product formula, compute

$$(a * b) * (b^{-1} * a^{-1}) = a * (b * b^{-1}) * a^{-1} = a * a^{-1} = e,$$

and symmetrically $(b^{-1} * a^{-1}) * (a * b) = e$; by (2), $(a * b)^{-1} = b^{-1} * a^{-1}$. The order reversal is the "socks–shoes" rule: the inverse of putting on socks, then shoes, is removing shoes, then socks. □

Two notational conventions dominate. *Multiplicative* notation writes the operation as juxtaposition ab (or $a \cdot b$), the identity as 1, the inverse as a^{-1}, and powers as $a^n = \underbrace{a \cdots a}_{n \text{ factors}}$ for $n \geq 1$, with $a^0 = 1$ and $a^{-n} = (a^{-1})^n$. *Additive* notation, reserved almost exclusively for commutative groups, writes $a + b$, the identity as 0, the inverse as $-a$, and na for the n-fold sum. The exponent laws

$$a^m a^n = a^{m+n}, \qquad (a^m)^n = a^{mn}$$

hold for all integers m, n, proved by induction together with Proposition 3.3. Henceforth ab denotes $a * b$, and we drop the symbol $*$ unless clarity demands it.

The power notation presumes something the axioms do not literally grant: that an n-fold product needs no brackets at all. Axiom (G1) speaks only of three factors. The gap is closed once and for all by the next proposition, and thereafter products are written without brackets and powers without comment.

> **Proposition 3.4** (Generalised associativity). *Let $a_1, \dots, a_n$, $n \geq 1$, be elements of a group—or of any set with an associative operation. Every way of bracketing the product $a_1 a_2 \cdots a_n$ yields the same element, namely the* left-normed *product $L_n = (\cdots((a_1 a_2) a_3) \cdots) a_n$. In particular a^n is a single well-defined element, however the n factors are combined.*

Proof. By strong induction on n. For $n \leq 2$ there is only one bracketing, and for $n = 3$ the two bracketings agree by (G1). Let $n \geq 3$ and assume the claim for all shorter products. Any bracketed product P of $a_1, \dots, a_n$ has an outermost multiplication, splitting it as $P = QR$ with Q a bracketed product of $a_1, \dots, a_k$ and R one of $a_{k+1}, \dots, a_n$, for some $1 \leq k < n$. The induction hypothesis evaluates both halves: $Q = L_k$, and R is the left-normed product of $a_{k+1}, \dots, a_n$. If $k = n - 1$, then $R = a_n$ and $P = L_{n-1} a_n = L_n$ directly. If $k < n - 1$, write $R = R' a_n$ with R' the left-normed product of $a_{k+1}, \dots, a_{n-1}$; then (G1) and the induction hypothesis, applied to the bracketed product $L_k R'$ of the $n - 1$ elements $a_1, \dots, a_{n-1}$, give

$$P = L_k(R' a_n) = (L_k R')\, a_n = L_{n-1}\, a_n = L_n.$$

Every bracketing therefore equals L_n. □

> **Definition 3.5** (Abelian group). A group $(G, *)$ is *abelian* (or *commutative*) if $a * b = b * a$ for all $a, b \in G$; otherwise it is *non-abelian*.

The term honours Niels Henrik Abel. In an abelian group products may be freely reordered, and the socks–shoes formula $(ab)^{-1} = b^{-1}a^{-1} = a^{-1}b^{-1}$ loses its order-sensitivity. Convention reserves additive notation for abelian groups precisely because $+$ connotes commutativity.

3.2 First examples

Example 3.6. The pair $(\mathbb{Z}, +)$ is an infinite abelian group: addition is associative and commutative, the identity is 0, and the inverse of n is $-n$. Likewise $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, and $(\mathbb{C}, +)$ are abelian groups. By contrast $(\mathbb{Z}, \cdot)$ is *not* a group: the integer 2 has no multiplicative inverse in $\mathbb{Z}$.

Example 3.7. The nonzero rationals $\mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}$ form an abelian group under multiplication, with identity 1 and inverse $1/a$; likewise $\mathbb{R}^\times$ and $\mathbb{C}^\times$. Zero must be excluded because it has no multiplicative inverse. The pattern is general: for any field F—the reader has met $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$; the abstract notion is defined in section 4—the nonzero elements $F^\times = F \setminus \{0\}$ form an abelian group under multiplication. Indeed, this is part of the very definition of a field.

Congruence classes furnish the central finite examples. Recall from section 2 the set $\mathbb{Z}/n\mathbb{Z}$ of residue classes $[a]_n$ and define operations on classes through representatives,

$$[a]_n + [b]_n = [a + b]_n, \qquad [a]_n\,[b]_n = [ab]_n.$$

By Proposition 2.21 the right-hand sides do not depend on the representatives chosen, so both operations are well defined, and each law of integer arithmetic—associativity, commutativity, the identities $[0]_n$ and $[1]_n$, negatives—passes to classes by computing on representatives. (The full structural statement, that these operations make $\mathbb{Z}/n\mathbb{Z}$ a commutative ring, is Theorem 4.6 in section 4; the present section uses only the fragments just listed.) Addition makes all of $\mathbb{Z}/n\mathbb{Z}$ a group, as Example 3.9 below records. For $n > 1$, multiplication does not—$[0]_n$ has no inverse—but section 2 identified exactly which classes are invertible: the *units*, the classes $[a]_n$ with $\gcd(a, n) = 1$ (Theorem 2.24). The units form a group.

> **Proposition 3.8.** *Under multiplication of classes, the set $(\mathbb{Z}/n\mathbb{Z})^\times$ of units of $\mathbb{Z}/n\mathbb{Z}$ is an abelian group. Its order is $|(\mathbb{Z}/n\mathbb{Z})^\times| = \varphi(n)$, Euler's totient of Definition 2.26: the number of integers a with $1 \le a \le n$ and* $\gcd(a, n) = 1$.

Proof. Multiplication of classes is associative and commutative with identity $[1]_n$, as derived above from Proposition 2.21 by computing on representatives, and $[1]_n$ is a unit (it is its own inverse). The operation is closed on units: if $[a]_n$ and $[b]_n$ are units with inverses $[a]_n^{-1}$ and $[b]_n^{-1}$, then $[ab]_n$ has inverse $[b]_n^{-1}[a]_n^{-1}$, since $(ab)(b^{-1}a^{-1}) \equiv 1 \pmod{n}$ for any representatives of the inverse classes. Every unit has an inverse by definition, and that inverse is itself a unit (it is invertible, with inverse the original class). Hence $(\mathbb{Z}/n\mathbb{Z})^\times$ is an abelian group. By Theorem 2.24 its elements are exactly the classes $[a]_n$ with $\gcd(a, n) = 1$ and $1 \le a \le n$, of which there are $\varphi(n)$ by Definition 2.26. □

Example 3.9. The pair $(\mathbb{Z}/n\mathbb{Z}, +)$ is an abelian group of order n: the discussion above derived associativity, commutativity, the identity $[0]_n$, and negatives from Proposition 2.21, and the classes number exactly n (the least nonnegative residues, Definition 2.22). The invertible classes form the abelian group $(\mathbb{Z}/n\mathbb{Z})^\times$ of order $\varphi(n)$ under multiplication (Theorem 2.24 and Proposition 3.8). When $n = p$ is prime, every nonzero class is a unit (Example 2.27), so $(\mathbb{Z}/p\mathbb{Z})^\times$ has order $p-1$—this is Example 3.7 specialised to $\mathbb{Z}/p\mathbb{Z}$, which section 4 shows to be a field and denotes $\mathbb{F}_p$.

Example 3.10. The one-element set $\{e\}$, with its only possible operation $e * e = e$, is the *trivial group*. The group $\mathbb{Z}/2\mathbb{Z} = \{0, 1\}$ is the smallest nontrivial group. A subtler small example is the *Klein four-group* $V = \{e, a, b, c\}$, in which every element composed with itself gives e and the product of any two distinct nonidentity elements is the third; the verification that this table satisfies the axioms is routine. The group V is abelian of order 4.

Although V and $\mathbb{Z}/4\mathbb{Z}$ both have order 4, they are structurally different: in V every element combined with itself returns the identity, while in $\mathbb{Z}/4\mathbb{Z}$ the class 1 must be added to itself four times to reach 0. The language that makes "structurally different" precise is that of *isomorphism* (§3.8); in its terms, V is not isomorphic to $\mathbb{Z}/4\mathbb{Z}$, because V has no element of order 4 in the sense defined next.

3.3 The order of an element

Definition 3.11 (Order of an element)**.** Let G be a group and $a \in G$. The *order* of a, written $\operatorname{ord}(a)$, is the least positive integer n with $a^n = e$, if such an n exists; a then has *finite order*. If no positive power of a equals e, then a has *infinite order* and one writes $\operatorname{ord}(a) = \infty$.

The word "order" now has two meanings—the order $|G|$ of a group and the order $\operatorname{ord}(a)$ of an element. The clash is deliberate and is reconciled by the cyclic-group theory below: $\operatorname{ord}(a)$ equals the order of the smallest subgroup of G containing a (Proposition 3.21).

Proposition 3.12. *Let a have finite order $d = \operatorname{ord}(a)$, and let $m, k \in \mathbb{Z}$.*

1. $a^m = e$ *if and only if* $d \mid m$.
2. $a^m = a^k$ *if and only if* $m \equiv k \pmod{d}$.
3. *The distinct powers of a are exactly* $e = a^0, a^1, \dots, a^{d-1}$.

Proof. (1) If $d \mid m$, write $m = dq$; then $a^m = (a^d)^q = e^q = e$. Conversely, suppose $a^m = e$ and divide with remainder (Theorem 2.4): $m = dq + r$ with $0 \le r < d$. Then

$$e = a^m = (a^d)^q\, a^r = a^r,$$

so r is a nonnegative integer smaller than d with $a^r = e$. Minimality of d among *positive* exponents forces $r = 0$, whence $d \mid m$.

(2) We have $a^m = a^k$ if and only if $a^{m-k} = e$ (multiply by a^{-k} and use the exponent laws), which by (1) holds if and only if $d \mid (m - k)$, i.e. $m \equiv k \pmod{d}$ (Definition 2.20).

(3) By (2) the powers $a^0, a^1, \dots, a^{d-1}$ are pairwise distinct, their exponents being pairwise incongruent modulo d; and every power a^m equals $a^{m \bmod d}$, again by (2). □

The final step of part (1) deserves to be isolated, because it is the recurring device of this section: to show that every witness of some property is a multiple of the *least positive* witness m, take an arbitrary witness t, divide $t = mq+r$ with $0 \le r < m$, show that the remainder r is itself a witness, and conclude $r = 0$ by minimality. It is the same shape as the proof of Bézout's identity (Theorem 2.11), transplanted from the integers into a group, and it reappears twice below: for the least positive power of g lying in a subgroup (Theorem 3.24) and for the least positive element of a subgroup of $\mathbb{Z}$ (Proposition 3.26).

Example 3.13. In $(\mathbb{Z}/n\mathbb{Z}, +)$, read the powers of Definition 3.11 additively: the k-fold sum of the class 1 is the class k, which is 0 exactly when $n \mid k$, so $\operatorname{ord}(1) = n$. More generally $\operatorname{ord}(a) = n/\gcd(a, n)$ for any class a, by Corollary 3.23 below read additively. In $\mathbb{C}^\times$ the element i has order 4: its successive powers are

$$i^1 = i, \qquad i^2 = -1, \qquad i^3 = -i, \qquad i^4 = 1.$$

In $(\mathbb{Z}, +)$ every nonzero element has infinite order: no nonzero multiple of a nonzero integer is 0.

3.4 Subgroups

Definition 3.14 (Subgroup). A subset $H \subseteq G$ of a group $(G, *)$ is a *subgroup*, written $H \leq G$, if H is itself a group under the restriction of $*$ to $H \times H$: explicitly, H is nonempty, closed under $*$, contains the identity e of G, and contains a^{-1} whenever it contains a. Every group has the *trivial* subgroup $\{e\}$ and G itself; a subgroup other than G is *proper*.

Two small points keep this definition honest. First, the identity of a subgroup necessarily coincides with the identity of G: if $e_H \in H$ satisfies $e_H * e_H = e_H$, then cancelling e_H in G against $e_H * e = e_H$ (Proposition 3.3(3)) gives $e_H = e$. Second, inverses computed in H agree with those computed in G, by uniqueness of inverses in G. Nothing about H, in other words, depends on whether we regard it as a group in its own right or as a subset of G.

Checking all the group axioms for a subset would be wasteful, since associativity is inherited for free. A single condition suffices.

Proposition 3.15 (One-step subgroup test). *Let G be a group and $H \subseteq G$. Then $H \leq G$ if and only if*

1. *H is nonempty, and*
2. *for all $a, b \in H$, the element ab^{-1} lies in H.*

Proof. If $H \leq G$, then H is nonempty and, for $a, b \in H$, it contains b^{-1} and hence ab^{-1}; both conditions hold.

Conversely, suppose (1) and (2) hold. Pick $a \in H$ by (1); taking $b = a$ in (2) gives $e = aa^{-1} \in H$. For any $b \in H$, taking the pair (e, b) in (2) gives $b^{-1} = eb^{-1} \in H$, so H is closed under inverses. Finally, for $a, b \in H$ we now know $b^{-1} \in H$, and applying (2) to the pair (a, b^{-1}) gives $a(b^{-1})^{-1} = ab \in H$, so H is closed under the operation. Associativity is inherited from G, so H is a group. □

Remark 3.16. For a *finite* nonempty subset H, closure under the operation alone forces $H \leq G$. Indeed, given $a \in H$, closure puts all the powers $a, a^2, a^3, \ldots$ in H; since H is finite they cannot all be distinct, so $a^i = a^j$ for some $i < j$, giving $a^{j-i} = e \in H$. Then $a^{-1} = a^{j-i-1} \in H$: this is a positive power of a when $j - i \geq 2$, and when $j - i = 1$ we have $a = e$, whose inverse e already lies in H. Thus for finite subsets, closure under multiplication is the only condition to verify.

Example 3.17. The even integers $2\mathbb{Z} = \{2k : k \in \mathbb{Z}\}$ form a subgroup of $(\mathbb{Z}, +)$: the set is nonempty, and $2k - 2l = 2(k - l)$ is again even (the subgroup test, written additively). More generally $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\} \leq \mathbb{Z}$ for every $n \geq 0$, by the same computation—and these are *all* the subgroups of $\mathbb{Z}$, as Proposition 3.26 will show.

Proposition 3.18. *The intersection $\bigcap_{i \in I} H_i$ of any family of subgroups $H_i \leq G$ is a subgroup of G.*

Proof. The identity e lies in every H_i, so the intersection is nonempty. If a and b lie in the intersection, then for each i both lie in H_i, so $ab^{-1} \in H_i$ by Proposition 3.15; hence ab^{-1} lies in the intersection, and the one-step test applies. □

Proposition 3.18 is what makes "the smallest subgroup containing a given set" well defined: the intersection of all subgroups containing the set is itself a subgroup containing the set, and it is contained in every other one.

Definition 3.19 (Generated subgroup). For a subset $S \subseteq G$, the *subgroup generated by S*, written

$\langle S \rangle$, is the intersection of all subgroups of G containing S. By Proposition 3.18 it is a subgroup, and by construction it is the smallest subgroup of G containing S.

Concretely, $\langle S \rangle$ consists of all finite products $s_1^{\varepsilon_1} s_2^{\varepsilon_2} \cdots s_k^{\varepsilon_k}$ with $s_i \in S$ and $\varepsilon_i \in \{+1, -1\}$, the empty product being e. Indeed, the set of such products contains S, contains e, and is closed under multiplication (concatenate) and under inversion (reverse the factors and negate the exponents, by the socks–shoes rule), so it is a subgroup containing S; conversely, every subgroup containing S is closed under these operations and so contains every such product. Hence the two descriptions coincide.

3.5 Cyclic groups

Definition 3.20 (Cyclic group)**.** A group G is *cyclic* if there exists $g \in G$ with

$$G = \langle g \rangle = \{ g^k : k \in \mathbb{Z} \};$$

such an element g is a *generator* of G.

The displayed description matches Definition 3.19: the set $\{g^k : k \in \mathbb{Z}\}$ contains g, and it is closed under products and inverses by the exponent laws, so it is a subgroup containing g; conversely any subgroup containing g must contain all its powers. Hence $\langle g \rangle = \{g^k : k \in \mathbb{Z}\}$: for a single element g, the general generated-subgroup construction and the naive one agree. Every cyclic group is abelian, since $g^m g^n = g^{m+n} = g^n g^m$.

Proposition 3.21. *Let $G = \langle g \rangle$ be cyclic.*

1. *If g has infinite order, the powers g^k $(k \in \mathbb{Z})$ are pairwise distinct, so G is infinite.*
2. *If g has finite order d, then $G = \{e, g, g^2, \dots, g^{d-1}\}$ has exactly d elements.*

In both cases $|\langle g \rangle| = \mathrm{ord}(g)$.

Proof. (1) Suppose $g^i = g^j$ with $i \neq j$; without loss of generality $i > j$. Then $g^{i-j} = e$ with $i - j > 0$, contradicting infinite order. So the powers are pairwise distinct and G is infinite.

(2) This is Proposition 3.12(3): the distinct powers of g are exactly $g^0, g^1, \dots, g^{d-1}$, and these exhaust $G = \{g^k : k \in \mathbb{Z}\}$. □

Example 3.22. The group $(\mathbb{Z}, +)$ is infinite cyclic, generated by 1 and also by -1. The group $(\mathbb{Z}/n\mathbb{Z}, +)$ is cyclic of order n, generated by 1 (Example 3.13). The unit group $(\mathbb{Z}/8\mathbb{Z})^\times = \{1, 3, 5, 7\}$ is *not* cyclic: each of its four elements squares to 1 ($3^2 = 9 \equiv 1, 5^2 = 25 \equiv 1, 7^2 = 49 \equiv 1 \pmod 8$), so no element has order 4. By contrast $(\mathbb{Z}/5\mathbb{Z})^\times = \{1, 2, 3, 4\}$ is cyclic, generated by 2: its powers run

$$2^1 \equiv 2, \qquad 2^2 \equiv 4, \qquad 2^3 \equiv 3, \qquad 2^4 \equiv 1 \pmod 5,$$

exhausting the group.

The order of a power of g is determined by a gcd, and the formula tells us exactly which powers are again generators.

Corollary 3.23. *Let g have finite order n and let $k \in \mathbb{Z}$. Then*

$$\mathrm{ord}(g^k) = \frac{n}{\gcd(n, k)}.$$

In particular g^k generates $\langle g \rangle$ if and only if $\gcd(n, k) = 1$, so a cyclic group of order n has exactly $\varphi(n)$

generators.

Proof. Set $d = \gcd(n, k)$ and write $k = dk'$, $n = dn'$ with $\gcd(n', k') = 1$. For $m \geq 1$,

$$(g^k)^m = g^{km} = e \iff n \mid km \iff dn' \mid dk'm \iff n' \mid k'm \iff n' \mid m,$$

using Proposition 3.12(1) for the first equivalence and Euclid's lemma (Proposition 2.15(1), with $\gcd(n', k') = 1$) for the last. The least positive such m is $n' = n/d$, which is therefore $\operatorname{ord}(g^k)$.

For the second claim, $\langle g^k \rangle \subseteq \langle g \rangle$ always, and by Proposition 3.21 the subgroup $\langle g^k \rangle$ has $\operatorname{ord}(g^k) = n/\gcd(n, k)$ elements; it equals the n-element set $\langle g \rangle$ exactly when $\gcd(n, k) = 1$. Since the powers $g^1, g^2, \dots, g^n$ run through each element of $\langle g \rangle$ exactly once (Proposition 3.12(3)), the generators correspond to the $k \in \{1, \dots, n\}$ with $\gcd(n, k) = 1$, of which there are $\varphi(n)$ by Definition 2.26. □

The subgroups of a cyclic group can be classified completely. The theorem below is used constantly in the sequel; its part (3) is the origin of the "one subgroup per divisor" picture of $\mathbb{Z}/n\mathbb{Z}$.

Theorem 3.24 (Subgroups of cyclic groups)**.** *Let $G = \langle g \rangle$ be a cyclic group.*

1. *Every subgroup of G is cyclic.*
2. *If G is infinite, its subgroups are exactly $\langle g^m \rangle$ for $m \geq 0$; distinct $m \geq 0$ give distinct subgroups, and for $m \geq 1$ the subgroup $\langle g^m \rangle$ is isomorphic to $(\mathbb{Z}, +)$.*
3. *If $|G| = n$, then for each positive divisor $d \mid n$ there is exactly one subgroup of G of order d, namely $\langle g^{n/d} \rangle$, and these are all the subgroups of G. Subgroups of G thus correspond bijectively to positive divisors of n.*

Proof. (1) Let $H \leq G$. If $H = \{e\}$, then $H = \langle e \rangle$ is cyclic. Otherwise H contains some $g^t \neq e$, so $t \neq 0$; since H also contains $(g^t)^{-1} = g^{-t}$, it contains a power of g with positive exponent. Let m be the least positive integer with $g^m \in H$ (well-ordering, Theorem 2.1). We claim $H = \langle g^m \rangle$. The inclusion $\supseteq$ holds by closure. For $\subseteq$, take any element of H; it is some g^t, since $H \subseteq G = \langle g \rangle$. Divide $t = mq + r$ with $0 \leq r < m$; then

$$g^r = g^{t-mq} = g^t\,(g^m)^{-q} \in H,$$

and minimality of m forces $r = 0$—the division-minimality device again. Hence $g^t = (g^m)^q \in \langle g^m \rangle$.

(2) Let G be infinite. Then g has infinite order, for otherwise G would be finite by Proposition 3.21(2). By part (1) every subgroup is $\{e\} = \langle g^0 \rangle$ or $\langle g^m \rangle$ with $m \geq 1$ least positive, and conversely each $\langle g^m \rangle$ is a subgroup. For distinctness, note that for $m \geq 1$ the positive exponents t with $g^t \in \langle g^m \rangle$ are exactly the positive multiples of m: the elements of $\langle g^m \rangle$ are the powers g^{mk}, and $g^t = g^{mk}$ forces $t = mk$ because distinct exponents give distinct powers (Proposition 3.21(1)). The least such exponent recovers m, so distinct $m \geq 1$ give distinct subgroups, and each differs from $\{e\}$ because $g^m \neq e$. Finally, for $m \geq 1$ the element g^m has infinite order (if $(g^m)^t = g^{mt} = e$ then $mt = 0$, so $t = 0$), so $\langle g^m \rangle$ is an infinite cyclic group and is therefore isomorphic to $(\mathbb{Z}, +)$ by Theorem 3.43(1) below. No circularity arises: the proof of that theorem uses only Propositions 3.12 and 3.21.

(3) Let $|G| = n$, so $\operatorname{ord}(g) = n$ by Proposition 3.21.

Existence. Let $d \mid n$ be a positive divisor. Since n/d divides n, we have $\gcd(n, n/d) = n/d$, so Corollary 3.23 gives

$$\operatorname{ord}\big(g^{n/d}\big) = \frac{n}{\gcd(n,\ n/d)} = \frac{n}{n/d} = d,$$

and $\langle g^{n/d} \rangle$ is a subgroup of order d by Proposition 3.21.

Uniqueness. Let $H \leq G$ with $|H| = d$. If $d = 1$ then $H = \{e\} = \langle g^n \rangle = \langle g^{n/1} \rangle$, since $g^n = e$. Otherwise, as in part (1), $H = \langle g^m \rangle$ for the least positive m with $g^m \in H$, and Propositions 3.12 and 3.21 together with Corollary 3.23 give

$$d = |H| = \operatorname{ord}(g^m) = \frac{n}{\gcd(n, m)}, \qquad \text{so} \qquad \gcd(n, m) = \frac{n}{d}.$$

In particular n/d divides m, say $m = (n/d)\, m'$; then $g^m = (g^{n/d})^{m'}$, so $H = \langle g^m \rangle \subseteq \langle g^{n/d} \rangle$. Both sides have exactly d elements, so $H = \langle g^{n/d} \rangle$.

All subgroups arise. Any $H \le G$ equals some $\langle g^m \rangle$ (or $\{e\}$), whose order $n/\gcd(n,m)$ is a positive divisor d of n; by uniqueness $H = \langle g^{n/d} \rangle$. The assignment $d \mapsto \langle g^{n/d} \rangle$ is therefore surjective onto the set of subgroups, and it is injective because subgroups of different orders are distinct. This is the asserted bijection. $\square$

Remark 3.25. Specialised to $G = \mathbb{Z}/n\mathbb{Z}$ with generator 1 (written additively), Theorem 3.24(3) reads: for each divisor $d \mid n$ there is exactly one subgroup of order d, namely

$$\langle n/d \rangle = \{\, 0,\ n/d,\ 2n/d,\ \dots,\ (d-1)\, n/d \,\},$$

itself cyclic of order d, hence isomorphic to $\mathbb{Z}/d\mathbb{Z}$ by the classification below. For $n = 12$ the six subgroups correspond to the divisors $1, 2, 3, 4, 6, 12$, and inclusion between subgroups mirrors divisibility between the corresponding divisors: the subgroup lattice of $\mathbb{Z}/12\mathbb{Z}$ is the divisor lattice of 12, with chains such as $1 \mid 2 \mid 4$, $1 \mid 2 \mid 6$, and $3 \mid 6$.

The classification promised in Example 3.17 now follows; it is Theorem 3.24(2) read additively for $G = (\mathbb{Z}, +) = \langle 1 \rangle$, but the direct argument is short enough to give in full.

Proposition 3.26. *Every subgroup of $(\mathbb{Z}, +)$ has the form $n\mathbb{Z}$ for a unique integer $n \ge 0$.*

Proof. Let $H \le \mathbb{Z}$. If $H = \{0\}$, then $H = 0\mathbb{Z}$. Otherwise H contains a nonzero integer and, being closed under negation, contains a positive one; let n be its least positive element (Theorem 2.1). Closure under addition and negation gives $n\mathbb{Z} \subseteq H$. Conversely, for any $h \in H$, divide $h = nq + r$ with $0 \le r < n$; then $r = h - nq \in H$, and minimality of n forces $r = 0$, so $h \in n\mathbb{Z}$. Hence $H = n\mathbb{Z}$. For uniqueness, note n is recovered from H as its least positive element (and $n = 0$ exactly for $H = \{0\}$), so distinct $n \ge 0$ give distinct subgroups. $\square$

3.6 Cosets and Lagrange's theorem

A subgroup $H \le G$ slices G into translated copies of itself, and counting the slices is the mechanism underlying Lagrange's theorem: partition the group into blocks, exhibit a bijection between each block and H, and multiply the block count by the block size. The blocks are the cosets.

Definition 3.27 (Cosets and index)**.** Let $H \le G$ and $a \in G$. The *left coset* of H by a is

$$aH = \{\, ah \,:\, h \in H \,\},$$

and the *right coset* is $Ha = \{ha \,:\, h \in H\}$. In additive notation one writes $a + H = \{a + h \,:\, h \in H\}$. The set of left cosets is denoted G/H, and the *index* of H in G is $[G : H] = |G/H|$, the number of left cosets.

Lemma 3.28 (Cosets partition the group)**.** *Let $H \le G$.*

1. *Each $a \in G$ lies in aH; the left cosets cover G.*
2. *Two left cosets are either equal or disjoint: $aH = bH$ if and only if $a^{-1}b \in H$, and otherwise $aH \cap bH = \varnothing$.*
3. *Every left coset has the same cardinality as H: the map $h \mapsto ah$ is a bijection from H onto aH.*

The same statements hold for right cosets, with $ba^{-1} \in H$ as the criterion in (2).

Proof. (1) Since $e \in H$, we have $a = ae \in aH$.

(2) First suppose the cosets meet: $ah_1 = bh_2$ for some $h_1, h_2 \in H$. Then $a^{-1}b = h_1h_2^{-1} \in H$. Conversely, suppose $a^{-1}b = h \in H$, so $b = ah$. Then every element $bh' \in bH$ equals $a(hh') \in aH$, giving $bH \subseteq aH$; and since $b^{-1}a = h^{-1} \in H$, the same argument with the roles of a and b exchanged gives $aH \subseteq bH$. Hence $aH = bH$. Combining the two directions: if two cosets share even one element they coincide, so distinct cosets are disjoint.

(3) The map $\lambda_a : H \to aH$, $h \mapsto ah$, is surjective by the definition of aH and injective by left cancellation (Proposition 3.3(3)): $ah = ah'$ implies $h = h'$. Hence $|aH| = |H|$.

For right cosets, the same three arguments apply with sides exchanged; in (2) the criterion becomes $ba^{-1} \in H$. □

Theorem 3.29 (Lagrange's theorem)**.** *Let G be a finite group and $H \leq G$. Then $|H|$ divides $|G|$, and*

$$|G| = [G : H]\,|H|.$$

Proof. By Lemma 3.28, the distinct left cosets of H partition G: they cover G by part (1) and are pairwise disjoint by part (2). Since G is finite there are finitely many of them, say $[G : H] = k$, and each has exactly $|H|$ elements by part (3). Summing the sizes of the blocks of the partition,

$$|G| = \sum_{\text{distinct cosets } aH} |aH| = k\,|H| = [G : H]\,|H|. \qquad \square$$

Three corollaries follow in quick succession; the third is the theorem this section owes to the rest of the volume.

Corollary 3.30. *In a finite group G, the order of every element divides $|G|$.*

Proof. For $a \in G$, the cyclic subgroup $\langle a \rangle$ has order $\operatorname{ord}(a)$ by Proposition 3.21 (the order is finite, since $\langle a \rangle \subseteq G$ is finite), and Theorem 3.29 makes $|\langle a \rangle|$ a divisor of $|G|$. □

Corollary 3.31. *If G is a finite group of order N, then $a^N = e$ for every $a \in G$.*

Proof. Let $d = \operatorname{ord}(a)$. By Corollary 3.30, $d \mid N$, say $N = dm$; then $a^N = (a^d)^m = e^m = e$. □

Corollary 3.32 (Euler's theorem; Fermat's little theorem)**.** *Let $n \geq 1$ and $a \in \mathbb{Z}$ with* $\gcd(a, n) = 1$. *Then*

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

In particular, for a prime p and a not divisible by p,

$$a^{p-1} \equiv 1 \pmod{p},$$

and $a^p \equiv a \pmod{p}$ holds for all *integers a.*

Proof. Apply Corollary 3.31 to the group $(\mathbb{Z}/n\mathbb{Z})^\times$, which has order $\varphi(n)$ (Proposition 3.8). For $\gcd(a, n) = 1$ the class $[a]_n$ is a unit (Theorem 2.24), so $[a]_n^{\varphi(n)} = [1]_n$, which is the first congruence. For $n = p$ prime, $\varphi(p) = p - 1$ (Example 2.27), and $p \nmid a$ gives $\gcd(a, p) = 1$, yielding the second congruence. Multiplying it by a gives $a^p \equiv a \pmod{p}$; and when $p \mid a$ both sides are congruent to 0, so the last congruence holds for all a. □

Remark 3.33. The congruences of Corollary 3.32 are proved here, once; section 7 restates them under the heading *Fermat's little theorem and Euler's theorem* and develops their cryptographic role there, citing this proof rather than repeating it.

> **Corollary 3.34.** *Every group G of prime order p is cyclic, hence isomorphic to $\mathbb{Z}/p\mathbb{Z}$, and has no proper nontrivial subgroups.*

Proof. Pick $a \in G$ with $a \neq e$ (possible since $p \geq 2$). The subgroup $\langle a\rangle$ has at least two elements, and by Theorem 3.29 its order divides p; a divisor of p exceeding 1 equals p, so $\langle a\rangle = G$ and G is cyclic. The isomorphism with $\mathbb{Z}/p\mathbb{Z}$ is Theorem 3.43 below. Finally, any subgroup has order 1 or p by Lagrange, i.e. is $\{e\}$ or G. ☐

3.7 Quotients of abelian groups

Cosets do more than count: for an abelian group they themselves form a group. This single construction underlies the quotient rings of section 4 and, through them, $\mathbb{Z}/n\mathbb{Z}$ and the polynomial quotients used throughout the series; it is all of quotient theory the volume requires.

> **Proposition 3.35** (Quotient of an abelian group)**.** *Let A be an abelian group, written additively, and let $H \leq A$. The operation*
>
> $$(a + H) + (b + H) = (a + b) + H$$
>
> *on the coset set A/H is well defined and makes A/H an abelian group—the* quotient group *of A by H—with identity $H = 0 + H$ and inverse $-(a + H) = (-a) + H$. If A is finite, then*
>
> $$|A/H| = [A : H] = \frac{|A|}{|H|}.$$

Proof. The only point of substance is that the operation is well defined: its output is stated in terms of chosen representatives a and b, and we must check that replacing them by other representatives of the same cosets leaves the result unchanged. Suppose then that $a + H = a' + H$ and $b + H = b' + H$. By Lemma 3.28(2), written additively, this means $a' - a \in H$ and $b' - b \in H$. Then

$$(a' + b') - (a + b) = (a' - a) + (b' - b) \in H,$$

since H is closed under addition; applying Lemma 3.28(2) once more, $(a' + b') + H = (a + b) + H$. The operation is well defined. This representative-independence check is the template for every coset-wise construction in the volume; it recurs, in the same words, for quotient-ring multiplication in section 4.

The axioms now pass coset-wise from A: for associativity,

$$\big((a + H) + (b + H)\big) + (c + H) = \big((a + b) + c\big) + H = \big(a + (b + c)\big) + H = (a + H) + \big((b + H) + (c + H)\big);$$

commutativity follows the same way from commutativity in A; the coset $0 + H = H$ is an identity; and $(-a) + H$ is an inverse of $a + H$. Hence A/H is an abelian group. When A is finite, the number of cosets is $[A : H] = |A|/|H|$ by Theorem 3.29—the same partition-into-equal-blocks count as before. ☐

The construction is not new to this volume in disguise: for $A = (\mathbb{Z}, +)$ and $H = n\mathbb{Z}$, Definition 2.22 exhibits each residue class as $[a]_n = a + n\mathbb{Z}$, a coset of $n\mathbb{Z} \leq \mathbb{Z}$, and the coset addition above is exactly addition of classes. The notation $\mathbb{Z}/n\mathbb{Z}$, introduced in section 2 as a bare name for the set of classes, is thus an instance of the general quotient notation A/H, and $(\mathbb{Z}/n\mathbb{Z}, +)$ is the quotient group $\mathbb{Z}/n\mathbb{Z}$ in the sense just constructed.

Remark 3.36. For a non-abelian group G the same construction succeeds exactly for those subgroups $N \leq G$ whose left and right cosets coincide, $gN = Ng$ for all $g \in G$; these are the *normal* subgroups. Commutativity makes every subgroup of an abelian group normal, which is why no extra hypothesis was needed in Proposition 3.35. The present volume deliberately needs only the abelian case, and we do not develop the general theory.

3.8 Homomorphisms and isomorphisms

Groups are compared through the maps that respect their operations.

Definition 3.37 (Group homomorphism)**.** Let $(G, *)$ and $(G', \circ)$ be groups. A function $\phi : G \to G'$ is a *group homomorphism* if

$$\phi(a * b) = \phi(a) \circ \phi(b) \qquad \text{for all } a, b \in G.$$

The set of all homomorphisms from G to G' is denoted $\mathrm{Hom}(G, G')$.

Proposition 3.38. *Let $\phi : G \to G'$ be a homomorphism, and write e, e' for the identities of G and G'.*

1. $\phi(e) = e'$.
2. $\phi(a^{-1}) = \phi(a)^{-1}$ *for all* $a \in G$.
3. $\phi(a^n) = \phi(a)^n$ *for all* $a \in G$ *and* $n \in \mathbb{Z}$.
4. *The image* $\phi(G)$ *is a subgroup of* G'.

Proof. (1) From $e = e * e$ we get $\phi(e) = \phi(e) \circ \phi(e)$; cancelling $\phi(e)$ in G' (Proposition 3.3(3)) gives $e' = \phi(e)$.

(2) Applying ϕ to $a * a^{-1} = e = a^{-1} * a$ and using (1),

$$\phi(a) \circ \phi(a^{-1}) = e' = \phi(a^{-1}) \circ \phi(a),$$

so $\phi(a^{-1})$ is an inverse of $\phi(a)$; by uniqueness of inverses, $\phi(a^{-1}) = \phi(a)^{-1}$.

(3) For $n \geq 0$, induct: the case $n = 0$ is (1), and $\phi(a^{n+1}) = \phi(a^n * a) = \phi(a)^n \circ \phi(a) = \phi(a)^{n+1}$. For $n < 0$, write $n = -m$ with $m > 0$ and combine with (2): $\phi(a^{-m}) = \phi((a^{-1})^m) = \phi(a^{-1})^m = (\phi(a)^{-1})^m = \phi(a)^{-m}$.

(4) The image is nonempty, containing $e' = \phi(e)$. For $x = \phi(a)$ and $y = \phi(b)$ in $\phi(G)$,

$$x \circ y^{-1} = \phi(a) \circ \phi(b)^{-1} = \phi(a * b^{-1}) \in \phi(G),$$

using (2); the one-step subgroup test (Proposition 3.15) concludes. □

Definition 3.39 (Kernel and image)**.** The *kernel* and *image* of a homomorphism $\phi : G \to G'$ are

$$\ker(\phi) = \{\, a \in G : \phi(a) = e' \,\} = \phi^{-1}(\{e'\}), \qquad \mathrm{im}(\phi) = \phi(G) = \{\, \phi(a) : a \in G \,\}.$$

Proposition 3.40. *For any homomorphism $\phi : G \to G'$, the kernel is a subgroup,* $\ker(\phi) \leq G$*, and*

> *ϕ is injective if and only if* $\ker(\phi) = \{e\}$.

Proof. The kernel contains e by Proposition 3.38(1), so it is nonempty; and for $a, b \in \ker(\phi)$,

$$\phi(ab^{-1}) = \phi(a) \circ \phi(b)^{-1} = e' \circ (e')^{-1} = e',$$

so $ab^{-1} \in \ker(\phi)$; the one-step test gives $\ker(\phi) \leq G$.

If ϕ is injective and $\phi(a) = e' = \phi(e)$, then $a = e$; the kernel is trivial. Conversely, suppose $\ker(\phi) = \{e\}$ and $\phi(a) = \phi(b)$. Then $\phi(ab^{-1}) = \phi(a) \circ \phi(b)^{-1} = e'$, so $ab^{-1} \in \ker(\phi) = \{e\}$, forcing $ab^{-1} = e$, i.e. $a = b$. □

Kernels are exactly the subgroups by which one may quotient: for abelian G, Proposition 3.35 forms $G/\ker(\phi)$ directly, and in general the kernel satisfies the normality condition of Remark 3.36— for $n \in \ker(\phi)$ and $g \in G$, one computes $\phi(gng^{-1}) = \phi(g) \circ e' \circ \phi(g)^{-1} = e'$, so conjugating the kernel by g lands back in the kernel, whence $g \ker(\phi) = \ker(\phi)\, g$.

> **Definition 3.41** (Isomorphism)**.** An *isomorphism* is a bijective homomorphism $\phi : G \to G'$; when one exists, the groups are *isomorphic*, written $G \cong G'$. An isomorphism from G to itself is an *automorphism*. An injective homomorphism is a *monomorphism*; a surjective one is an *epimorphism*.

> **Proposition 3.42.** *The set-theoretic inverse of an isomorphism is an isomorphism, and $\cong$ is an equivalence relation on any collection of groups.*

Proof. Let $\phi : G \to G'$ be an isomorphism, with inverse function ϕ^{-1} (which exists since ϕ is a bijection, Theorem 1.1). For $x, y \in G'$, write $x = \phi(a)$ and $y = \phi(b)$; then

$$\phi^{-1}(x \circ y) = \phi^{-1}\big(\phi(a * b)\big) = a * b = \phi^{-1}(x) * \phi^{-1}(y),$$

so ϕ^{-1} is a homomorphism, and it is a bijection; hence an isomorphism. For the equivalence relation: reflexivity is witnessed by the identity automorphism id_G; symmetry by the inverse just constructed; and transitivity because a composite of homomorphisms is a homomorphism (apply the defining property twice) and a composite of bijections is a bijection. □

Isomorphic groups are "the same group" for every group-theoretic purpose: any property expressible in terms of the operation alone— orders of elements, the lattice of subgroups, commutativity— transfers across an isomorphism. One therefore classifies groups *up to isomorphism*, and the classification of cyclic groups is the model result of this kind: up to isomorphism there is exactly one cyclic group of each order.

> **Theorem 3.43** (Classification of cyclic groups)**.** *Every cyclic group is isomorphic to exactly one of the following:*
>
> 1. *the infinite cyclic group $(\mathbb{Z}, +)$, if it is infinite;*
> 2. *the finite cyclic group $(\mathbb{Z}/n\mathbb{Z}, +)$, if it has order n.*
>
> *In particular, two cyclic groups are isomorphic if and only if they have the same order.*

Proof. Let $G = \langle g \rangle$ and define

$$\phi : \mathbb{Z} \to G, \qquad \phi(k) = g^k.$$

The exponent law $g^{k+l} = g^k g^l$ says exactly that ϕ is a homomorphism from $(\mathbb{Z}, +)$ to G, and ϕ is surjective because every element of G is a power of g.

Case 1: g has infinite order. Distinct exponents give distinct powers (Proposition 3.21(1)), so ϕ is injective, hence a bijective homomorphism, and $\mathbb{Z} \cong G$.

Case 2: $\operatorname{ord}(g) = n$. By Proposition 3.12(1),

$$\ker(\phi) = \{ k \in \mathbb{Z} : g^k = e \} = n\mathbb{Z}.$$

Define the induced map

$$\bar{\phi} : \mathbb{Z}/n\mathbb{Z} \to G, \qquad \bar{\phi}([k]_n) = g^k.$$

It is well defined: if $[k]_n = [l]_n$ then $n \mid (k-l)$, so $g^k = g^l$ by Proposition 3.12(2). It is a homomorphism: $\bar{\phi}([k]_n + [l]_n) = \bar{\phi}([k+l]_n) = g^{k+l} = \bar{\phi}([k]_n)\,\bar{\phi}([l]_n)$. It is surjective because ϕ is. And it is injective by Proposition 3.40: if $\bar{\phi}([k]_n) = g^k = e$, then $n \mid k$ (Proposition 3.12(1)), so $[k]_n = [0]_n$; the kernel of $\bar{\phi}$ is trivial. Hence $\mathbb{Z}/n\mathbb{Z} \cong G$.

Distinctness. Isomorphic groups have equal order, an isomorphism being in particular a bijection. The group $\mathbb{Z}$ is infinite and the groups $\mathbb{Z}/n\mathbb{Z}$ for $n = 1, 2, 3, \ldots$ have the pairwise distinct finite orders n, so no two of the listed groups are isomorphic, and a cyclic group is isomorphic to exactly one of them—the one matching its order. The final claim follows: cyclic groups of equal order are isomorphic to the same listed group, hence to each other (Proposition 3.42), and isomorphic groups have equal order. $\square$

The cheque presented at the head of the section can now be cashed. Fix any finite group G and a public element $g \in G$, and let $n = \operatorname{ord}(g)$. Associativity makes "$a$ repetitions of g" unambiguous—the element g^a, however bracketed, by generalised associativity (Proposition 3.4)—and the exponent laws give

$$\left(g^a\right)^b = g^{ab} = \left(g^b\right)^a,$$

so the two parties of the opening protocol, each applying a secret repetition count to the other's published value, really do arrive at a common secret element. Repetition counts obey an arithmetic, not a mere accumulation: by the classification theorem the subgroup $\langle g \rangle$ is a copy of $\mathbb{Z}/n\mathbb{Z}$, so exponents add, negate, and matter only modulo n (Proposition 3.12)—the modular arithmetic of section 2, transplanted into the exponent. Lagrange's theorem locates the modulus: n divides $|G|$, and $a^{|G|} = e$ for every $a \in G$ (Corollary 3.31); in the unit groups $(\mathbb{Z}/n\mathbb{Z})^\times$ this specialises to Euler's congruence $a^{\varphi(n)} \equiv 1 \pmod{n}$ (Corollary 3.32). Composition and inversion, the first two demands of the opening problem, are thereby supplied in full generality. The third demand—that publishing g^a should *hide a*—is not a theorem of algebra at all: it is a statement about computation, and it fails in some groups (in $(\mathbb{Z}/n\mathbb{Z}, +)$, recovering a from ag is a single modular division) while appearing to hold in others. Making the demand precise, and building the groups for which it is credible, occupies the sections that follow and the volumes above them.

4 Rings and fields

An arithmetic circuit—the object a proof system ultimately asks a verifier to check—is a network of gates, each of which either *adds* or *multiplies* two previously computed values. A verifier must therefore calculate in a number system where both operations make sense at once and interact through the familiar laws of school algebra; and at certain moments—inverting a random challenge, cancelling a common factor from both sides of an identity—the verifier must *divide* as well. In what kind of structure can one do both, and when can one also divide? The structures of genuine cryptographic interest all support two interacting operations obeying the arithmetic laws of $\mathbb{Z}$: the integers themselves, the rationals, the residue classes modulo n, polynomials, matrices. The abstract capture of this pattern is the *ring*; the key special case in which division by every nonzero element is possible is the *field*. Fields are the natural arena for linear algebra and for the theory of polynomials, and the finite fields $\mathbb{F}_p$ and $\mathbb{F}_q$ are the computational substrate of essentially all the elliptic-curve and proof-system machinery later in the monograph.

The section also discharges a debt. Section 2 verified, concretely and one law at a time, that congruence classes admit an arithmetic, and promised that the verified laws would receive their structural names in due course. They receive them here: $\mathbb{Z}/n\mathbb{Z}$ is re-read as a ring, indeed as the quotient of the ring $\mathbb{Z}$ by the ideal $n\mathbb{Z}$, and the theorem that $\mathbb{Z}/p\mathbb{Z}$ is a field is stated in the language that the rest of the series uses. As a standing convention we assume from §3 only the theory of *abelian* groups; additive structure is written additively with identity 0, multiplicative structure multiplicatively with identity 1.

4.1 Rings

Definition 4.1 (Ring)**.** A *ring* is a set R equipped with two binary operations $+$ and $\cdot$ such that:

1. $(R, +)$ is an abelian group: addition is associative and commutative, there is an element 0 with $a + 0 = a$ for all a, and every a has an *additive inverse* $-a$ with $a + (-a) = 0$;
2. multiplication is associative: $a \cdot (b \cdot c) = (a \cdot b) \cdot c$;
3. there is a *multiplicative identity* 1 with $1 \cdot a = a \cdot 1 = a$ for all a;
4. multiplication distributes over addition on both sides: $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(a + b) \cdot c = a \cdot c + b \cdot c$.

The ring is *commutative* if moreover $a \cdot b = b \cdot a$ for all $a, b \in R$. We write ab for $a \cdot b$, let multiplication bind tighter than addition, and call 0 the *zero* and 1 the *unity* of the ring.

Remark 4.2. Conventions in the literature vary. Some authors omit requirement (3), saying "ring with unity" for the notion above and "rng" for the weaker one. In this monograph *every ring has a* 1, and the structure-preserving maps between rings—the ring homomorphisms of theorem 4.21 below—are required to preserve it. Moreover "ring" frequently means "commutative ring" in the examples that matter to us; commutativity is always stated explicitly when a result depends on it.

The distributive law is the only axiom coupling the two operations, and a surprising amount of everyday algebra follows from it alone. The following identities are used constantly and, after this proposition, without comment.

Proposition 4.3 (Elementary ring arithmetic)**.** *Let R be a ring and $a, b \in R$. Then:*

1. $0 \cdot a = a \cdot 0 = 0$*;*
2. $(-a)b = a(-b) = -(ab)$*;*
3. $(-a)(-b) = ab$*;*
4. *for all integers $m, n \in \mathbb{Z}$, $(ma)(nb) = (mn)(ab)$, where na denotes the n-fold additive multiple ($0a = 0$, $(n + 1)a = na + a$, and $(-n)a = -(na)$, as for any abelian group, §3).*

Proof. (1) From $0 = 0 + 0$ and distributivity, $0 \cdot a = (0 + 0) \cdot a = 0 \cdot a + 0 \cdot a$; adding $-(0 \cdot a)$ to both sides leaves $0 = 0 \cdot a$. The computation for $a \cdot 0$ is symmetric.

(2) By distributivity and (1), $ab + (-a)b = (a + (-a))b = 0 \cdot b = 0$, so $(-a)b$ is the additive inverse of ab, that is, $(-a)b = -(ab)$; likewise $ab + a(-b) = a(b + (-b)) = 0$ gives $a(-b) = -(ab)$.

(3) Applying (2) twice, $(-a)(-b) = -\big(a(-b)\big) = -\big(-(ab)\big) = ab$, the last step because negation is an involution in the abelian group $(R, +)$.

(4) First fix a, b and show $a(nb) = n(ab)$ for $n \geq 0$ by induction: the case $n = 0$ is (1), and $a((n+1)b) = a(nb + b) = a(nb) + ab = n(ab) + ab = (n+1)(ab)$ by distributivity and the inductive

hypothesis. For $n < 0$ write $n = -m$ with $m > 0$; then $a(nb) = a(-(mb)) = -(a(mb)) = -(m(ab)) = n(ab)$ by (2). The same argument in the first factor gives $(ma)c = m(ac)$ for all c, and combining,

$$(ma)(nb) = m\big(a(nb)\big) = m\big(n(ab)\big) = (mn)(ab),$$

the final equality being the usual bookkeeping for iterated multiples in an abelian group, which we omit. $\square$

Remark 4.4 (The zero ring). Nothing in theorem 4.1 forces $1 \neq 0$. If $1 = 0$ in a ring R, then every $a \in R$ satisfies $a = 1 \cdot a = 0 \cdot a = 0$ by theorem 4.3(1), so $R = \{0\}$: the *zero ring* (or *trivial ring*), a legitimate ring in which both operations are the only possible ones. Whenever the zero ring must be excluded we impose the assumption $1 \neq 0$ explicitly; the definition of a field below builds $1 \neq 0$ in as a standing assumption.

Example 4.5 (Basic rings).

1. The integers $\mathbb{Z}$ form the prototype commutative ring; much of ring theory consists of isolating which features of $\mathbb{Z}$ survive in generality.
2. The rationals $\mathbb{Q}$, the reals $\mathbb{R}$, and the complexes $\mathbb{C}$ are commutative rings—indeed fields, in the sense of §4.4.
3. The residue classes $\mathbb{Z}/n\mathbb{Z} = \{\overline{0}, \overline{1}, \dots, \overline{n-1}\}$ for $n \geq 1$—in this section's examples we abbreviate the class $[a]_n$ of Definition 2.22 to $\overline{a}$ when the modulus is clear—under the induced operations form a commutative ring whose additive group is the cyclic group of order n (§3); this is theorem 4.6 below. We write $\mathbb{Z}/n\mathbb{Z}$, or $\mathbb{Z}_n$ when no confusion threatens. The case $n = 1$ gives the zero ring of theorem 4.4.
4. The *polynomial ring* $R[x]$ over a commutative ring R consists of the polynomials $a_0 + a_1x + \cdots + a_dx^d$ with coefficients $a_i \in R$, added coefficient-wise and multiplied by expanding and collecting powers of x; it is a commutative ring with the same 0 and 1 as R (the constant polynomials). The theory of polynomials over a field is developed in its own section later in the volume.
5. The *matrix ring* $\mathrm{M}_n(R)$ of $n \times n$ matrices over a commutative ring R, $n \geq 1$, is a ring under matrix addition and multiplication. It is *not* commutative for $n \geq 2$ and $R \neq \{0\}$, even when R is commutative: in $\mathrm{M}_2(\mathbb{Z})$,
$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \text{but} \qquad \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$
This is the first important noncommutative example.
6. The *product ring* $R \times S$ of two rings carries the componentwise operations, with zero $(0, 0)$ and identity $(1, 1)$; likewise any finite product $R_1 \times \cdots \times R_k$.
7. The zero ring $\{0\}$.

The third example is the one this monograph computes in, and it is the one whose ring structure was promised rather than named in §2. We now discharge that promise.

Theorem 4.6. *Let $n \geq 1$. The operations*

$$[a]_n + [b]_n := [a + b]_n, \qquad [a]_n \cdot [b]_n := [ab]_n$$

are well defined and make $\mathbb{Z}/n\mathbb{Z}$ a commutative ring with additive identity $[0]_n$ and multiplicative identity $[1]_n$. It has exactly n elements.

Proof. The right-hand sides are defined in terms of chosen representatives a, b, so we must check that different choices give the same class. This is exactly the compatibility half of Proposition 2.21: if $a' \equiv a$ and $b' \equiv b \pmod{n}$, then $a' + b' \equiv a + b$ and $a'b' \equiv ab \pmod{n}$, so $[a' + b']_n = [a + b]_n$ and $[a'b']_n = [ab]_n$. The operations are therefore well defined on classes.

Every ring axiom is now inherited from the corresponding law of $\mathbb{Z}$ by passing to classes. For instance, distributivity:

$$\begin{aligned}[a]_n([b]_n + [c]_n) &= [a]_n[b + c]_n = [a(b + c)]_n = [ab + ac]_n \\ &= [ab]_n + [ac]_n = [a]_n[b]_n + [a]_n[c]_n,\end{aligned}$$

using only the definitions and the distributive law of the integers. Associativity and commutativity of both operations follow by the identical pattern. The class $[0]_n$ is an additive identity and $[1]_n$ a multiplicative one, since $[a+0]_n = [a]_n$ and $[1 \cdot a]_n = [a]_n$; and $-[a]_n = [-a]_n$ because $[a]_n + [-a]_n = [a + (-a)]_n = [0]_n$.

Finally, by the division algorithm every integer is congruent modulo n to exactly one of $0, 1, \dots, n-1$ (the least nonnegative residues, Definition 2.22 and the discussion following it), so the classes $[0]_n, \dots, [n-1]_n$ are distinct and exhaust $\mathbb{Z}/n\mathbb{Z}$: the ring has exactly n elements. □

4.2 Units, zero divisors, and integral domains

Two features of the integers now deserve names of their own. First, $\mathbb{Z}$ enjoys *cancellation*: a product of nonzero integers is nonzero, so a nonzero common factor may be struck from both sides of an equation. Second, division *within* $\mathbb{Z}$ is nonetheless rare: only ± 1 divide 1. The next two definitions abstract these phenomena—units capture the elements one can divide by, zero divisors the obstruction to cancelling.

Definition 4.7 (Unit)**.** An element u of a ring R is a *unit* (or is *invertible*) if there exists $v \in R$ with $uv = vu = 1$. Such a v is unique: if $uv = vu = 1$ and $uv' = v'u = 1$, then $v = v \cdot 1 = v(uv') = (vu)v' = 1 \cdot v' = v'$. It is called the *inverse* of u and written u^{-1}. The set of units of R is denoted $R^\times$.

Proposition 4.8. *For any ring R, the set $R^\times$ of units is a group under the multiplication of R, called the* group of units *(or* multiplicative group*) of R.*

Proof. The identity 1 is a unit (it is its own inverse), so $1 \in R^\times$. For closure, let $u, w \in R^\times$; then

$$(uw)(w^{-1}u^{-1}) = u(ww^{-1})u^{-1} = u \cdot 1 \cdot u^{-1} = 1,$$

and symmetrically $(w^{-1}u^{-1})(uw) = 1$, so uw is a unit with $(uw)^{-1} = w^{-1}u^{-1}$. Multiplication in $R^\times$ is associative because it is associative in R. Finally each u^{-1} is itself a unit, its inverse being u; so every element of $R^\times$ has an inverse in $R^\times$, and the group axioms are verified. □

Example 4.9 (Units)**.**

1. The units of the integers are $\mathbb{Z}^\times = \{1, -1\}$: if $uv = 1$ in $\mathbb{Z}$ then $|u|\,|v| = 1$ forces $|u| = 1$.
2. In $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ every nonzero element is a unit: $\mathbb{Q}^\times = \mathbb{Q}\backslash\{0\}$, $\mathbb{R}^\times = \mathbb{R}\backslash\{0\}$, $\mathbb{C}^\times = \mathbb{C}\backslash\{0\}$. Commutative rings with $1 \neq 0$ in which *every* nonzero element is a unit are precisely the fields of §4.4; this is their defining feature. Both qualifiers are needed: the zero ring of theorem 4.4 satisfies the unit condition vacuously, and noncommutative rings satisfying it—the *division rings*—exist but are not fields.

3. In $\mathbb{Z}/n\mathbb{Z}$, the class $\overline{a}$ is a unit if and only if $\gcd(a, n) = 1$, by Theorem 2.24; thus $(\mathbb{Z}/n\mathbb{Z})^\times$ consists of the residue classes coprime to the modulus, and its order is Euler's totient $\varphi(n)$ (Definition 2.26).

4. In the matrix ring, $\mathrm{M}_n(R)^\times$ is the group of invertible $n \times n$ matrices. Over a field it is the *general linear group* GL_n, the matrices of nonzero determinant; the determinant belongs to linear algebra, treated later in the volume.

Definition 4.10 (Zero divisor)**.** A *nonzero* element a of a ring R is a *left zero divisor* if $ab = 0$ for some nonzero $b \in R$, and a *right zero divisor* if $ba = 0$ for some nonzero b. In a commutative ring the two notions coincide and we say simply *zero divisor*. A nonzero element that is neither is called *regular* (or *cancellable*).

The terminology "cancellable" is justified by the following equivalence: zero divisors are exactly the obstruction to dividing out a common factor.

Proposition 4.11. *A nonzero element a of a ring R is not a left zero divisor if and only if it is* left-cancellable*: $ab = ac$ implies $b = c$ for all $b, c \in R$.*

Proof. Suppose a is not a left zero divisor and $ab = ac$. Then $a(b - c) = ab - ac = 0$, using theorem 4.3(2) to expand $a(b + (-c))$; since a is nonzero and not a left zero divisor, the factor $b - c$ must be 0, so $b = c$. Conversely, suppose a is a left zero divisor, say $ab = 0$ with $b \neq 0$. Then $ab = 0 = a \cdot 0$ while $b \neq 0$: cancellation fails. □

Remark 4.12. A unit is never a zero divisor: if $u \in R^\times$ and $ub = 0$, then $b = 1 \cdot b = (u^{-1}u)b = u^{-1}(ub) = u^{-1} \cdot 0 = 0$. The converse fails in general—in $\mathbb{Z}$ every nonzero element is a non-zero-divisor, yet only ± 1 are units—but every nonzero non-zero-divisor is a unit in a finite commutative ring, by the injectivity-implies-surjectivity argument that proves theorem 4.29 below.

Example 4.13 (Zero divisors)**.**

1. In $\mathbb{Z}/6\mathbb{Z}$, $\overline{2} \cdot \overline{3} = \overline{6} = \overline{0}$, so $\overline{2}$ and $\overline{3}$ are zero divisors. In general a nonzero class $\overline{a} \in \mathbb{Z}/n\mathbb{Z}$ is a zero divisor precisely when $d := \gcd(a, n) > 1$: in that case $a \cdot (n/d) = (a/d) \cdot n \equiv 0 \pmod n$ while $\overline{n/d} \neq \overline{0}$ (as $0 < n/d < n$); and when $d = 1$ the class is a unit (Example 4.9(3)), hence no zero divisor by theorem 4.12. Thus in $\mathbb{Z}/n\mathbb{Z}$ every nonzero element is either a unit or a zero divisor—a dichotomy revisited from a structural angle by theorem 4.29.

2. In a product ring $R \times S$ with both factors nonzero, $(1, 0)(0, 1) = (0, 0)$: product rings always have zero divisors.

3. In $\mathrm{M}_2(\mathbb{R})$, the diagonal matrices $\mathrm{diag}(1, 0)$ and $\mathrm{diag}(0, 1)$ multiply to the zero matrix; more generally every nonzero *singular* matrix—the standard name for a square matrix that is not invertible—is a zero divisor there.

Definition 4.14 (Integral domain)**.** An *integral domain* (or simply *domain*) is a commutative ring R with $1 \neq 0$ and no zero divisors: $ab = 0$ implies $a = 0$ or $b = 0$. Equivalently, by theorem 4.11, a domain is a nonzero commutative ring in which every nonzero element is cancellable.

The residue rings sort themselves neatly against this definition, and the sorting is governed by primality.

Proposition 4.15. *Let $n \geq 2$. The ring $\mathbb{Z}/n\mathbb{Z}$ has no zero divisors if and only if n is prime; that is, $\mathbb{Z}/n\mathbb{Z}$ is an integral domain if and only if n is prime.*

Proof. If $n = ab$ is composite with $1 < a, b < n$, then $[a]_n[b]_n = [n]_n = [0]_n$ with both factors nonzero (their representatives lie strictly between 0 and n), so $\mathbb{Z}/n\mathbb{Z}$ has zero divisors. Conversely let $n = p$ be prime and suppose $[a]_p[b]_p = [0]_p$, that is, $p \mid ab$. Euclid's lemma for primes (Lemma 2.17) gives $p \mid a$ or $p \mid b$, that is, $[a]_p = [0]_p$ or $[b]_p = [0]_p$. Since $p \geq 2$ we also have $[1]_p \neq [0]_p$, so $\mathbb{Z}/p\mathbb{Z}$ is an integral domain. □

For prime p the ring $\mathbb{Z}/p\mathbb{Z}$ is in fact far better than a domain: every nonzero class is invertible, so it is a *field*—a term defined in §4.4, where the statement is proved as theorem 4.26.

Example 4.16 (Integral domains)**.** The integers $\mathbb{Z}$ form an integral domain—the example the name commemorates. Every field is a domain (theorem 4.28 below). The ring $\mathbb{Z}/n\mathbb{Z}$ is a domain exactly when n is prime (theorem 4.15; the composite case is witnessed concretely by Example 4.13(1)). If R is a domain, so is the polynomial ring $R[x]$: let f and g be nonzero of degrees m and n—the largest indices carrying nonzero coefficients f_m and g_n, the *leading coefficients*. In fg the coefficient of x^{m+n} is $\sum_{i+j=m+n} f_i g_j$; every term with $i > m$ or $j > n$ has a vanishing factor, and $i \leq m$, $j \leq n$, $i+j = m+n$ force $i = m$, $j = n$, so the coefficient equals $f_m g_n$, nonzero because R is a domain. Hence $fg \neq 0$. Non-domains include $\mathbb{Z}/6\mathbb{Z}$, product rings with two nonzero factors, and the matrix rings $\mathrm{M}_n(R)$ for $n \geq 2$ (Example 4.13).

4.3 Ideals and quotient rings

We next carry quotient formation from groups to rings. For an abelian group and any subgroup, the cosets themselves form a group (Proposition 3.35); the ring-theoretic analogue asks for the subsets of a ring by which one can quotient while keeping *both* operations, and the answer is the notion of an *ideal*. The treatment here is deliberately brief: we need ideals chiefly to construct $\mathbb{Z}/n\mathbb{Z}$ conceptually and to indicate where the finite fields $\mathbb{F}_q$ come from; a fuller development belongs to commutative algebra.

Definition 4.17 (Ideal)**.** Let R be a commutative ring. An *ideal* of R is a subset $I \subseteq R$ such that

1. I is a subgroup of the additive group $(R, +)$: $0 \in I$, and $a - b \in I$ whenever $a, b \in I$; and
2. I *absorbs* multiplication by arbitrary ring elements: $ra \in I$ whenever $r \in R$ and $a \in I$.

In noncommutative rings one distinguishes left, right, and two-sided ideals according to the side on which absorption holds; commutative rings suffice for this volume, and there the three notions coincide.

Despite containing 0 and being closed under the ring's operations in the sense above, an ideal is almost never a subring: if an ideal I contains 1—or, by the same absorption argument, any unit u, since then $1 = u^{-1}u \in I$—then $r = r \cdot 1 \in I$ for every $r \in R$, so $I = R$. The only ideal containing a unit is the whole ring. This little observation will carry real weight: it is the reason a field has no ideals other than the two trivial ones, and hence no interesting quotients.

Example 4.18 (Ideals)**.**

1. In any commutative ring R, the subsets $\{0\}$ and R are ideals—the *trivial* and *improper* ideals respectively.
2. For $a \in R$, the *principal ideal generated by* a is

$$(a) := aR = \{\, ar : r \in R \,\},$$

the set of multiples of a; absorption holds because $r'(ar) = a(r'r)$. In $\mathbb{Z}$, the principal ideal $(n) = n\mathbb{Z}$ is the set of multiples of n. In fact *every* ideal of $\mathbb{Z}$ is principal: an ideal $I \neq \{0\}$ contains a nonzero element and hence (closing under negation) a positive one, so it contains a least positive element n by well-ordering; for any $a \in I$ the division algorithm gives $a = qn + r$ with $0 \leq r < n$, whence $r = a - qn \in I$ by absorption and subgroup closure, forcing $r = 0$ by minimality; so $I = n\mathbb{Z}$. This sharpens the observation of Remark 2.12, where ideals of $\mathbb{Z}$ first appeared as the sets of Bézout combinations.

3. For $a_1, \dots, a_k \in R$, the *ideal generated by* $a_1, \dots, a_k$ is $(a_1, \dots, a_k) = \{ r_1 a_1 + \cdots + r_k a_k : r_i \in R \}$, the smallest ideal containing them all.

An integral domain in which every ideal is principal, as in (2), is called a *principal ideal domain* (PID).

Definition 4.19 (Quotient ring operations)**.** Let I be an ideal of a commutative ring R. Since I is a subgroup of the abelian group $(R, +)$, the additive cosets $a + I$ form the quotient group R/I under coset addition $(a + I) + (b + I) = (a + b) + I$ (Proposition 3.35). Define a multiplication of cosets by

$$(a + I)(b + I) := ab + I.$$

Proposition 4.20. *With the operations of theorem 4.19, the set R/I is a commutative ring—the* quotient ring *of R by I—with zero $0 + I = I$ and identity $1 + I$. In particular the coset multiplication is well defined.*

Proof. Well-definedness is the point at which absorption is needed. Let $a' = a + s$ and $b' = b + t$ be other representatives of the same cosets, with $s, t \in I$. Then

$$a'b' = (a + s)(b + t) = ab + at + sb + st,$$

and each of at, sb, st lies in I by the absorption property; hence $a'b' - ab \in I$ and $a'b' + I = ab + I$. The product coset is therefore independent of the representatives chosen.

The ring axioms now pass to cosets exactly as in the proof of theorem 4.6. For instance, distributivity reads

$$\begin{aligned}(a + I)\big((b + I) + (c + I)\big) &= (a + I)\big((b + c) + I\big) = a(b + c) + I \\ &= (ab + ac) + I = (a + I)(b + I) + (a + I)(c + I),\end{aligned}$$

and associativity and commutativity of both operations follow by the same pattern from the corresponding laws in R. The coset $I = 0 + I$ is the additive identity, $1 + I$ the multiplicative identity, and $-(a + I) = (-a) + I$. □

One more definition names the maps under which ring structure is preserved; we need it to say precisely in what sense the quotient construction generalises the passage from $\mathbb{Z}$ to $\mathbb{Z}/n\mathbb{Z}$.

Definition 4.21 (Ring homomorphism)**.** A *ring homomorphism* between rings R and S is a map $\phi : R \to S$ with

$$\phi(a + b) = \phi(a) + \phi(b), \qquad \phi(ab) = \phi(a)\phi(b), \qquad \phi(1) = 1$$

for all $a, b \in R$. Its *kernel* is $\ker \phi = \{ a \in R : \phi(a) = 0 \}$.

The kernel of a ring homomorphism out of a commutative ring is always an ideal: it is a subgroup of $(R, +)$ because ϕ is in particular a homomorphism of additive groups, and it absorbs because $a \in \ker\phi$ gives $\phi(ra) = \phi(r)\phi(a) = \phi(r) \cdot 0 = 0$.

Example 4.22. Take $R = \mathbb{Z}$ and $I = n\mathbb{Z}$. The cosets $a + n\mathbb{Z}$ are literally the residue classes $[a]_n$ of Definition 2.22, and the coset operations of theorem 4.19 are exactly residue arithmetic: the quotient construction recovers the ring $\mathbb{Z}/n\mathbb{Z}$ of Example 4.5(3), and is the conceptual origin of modular arithmetic. The same construction with a different ring produces the general finite fields: quotients $\mathbb{F}_p[x]/(f)$ of a polynomial ring by the principal ideal of an *irreducible* polynomial f of degree k—irreducibility, the polynomial analogue of primality, is defined in §5—furnish the finite fields $\mathbb{F}_q$ with $q = p^k$ elements. We record the recipe here and take it up in §6; the curves of Halo 2 and Orchard work over *prime* fields, so the prime case $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ carries the main line of development.

Remark 4.23. Structurally, then, $\mathbb{Z}/n\mathbb{Z}$ is the quotient of the ring $\mathbb{Z}$ by the ideal $n\mathbb{Z} = \{\, kn : k \in \mathbb{Z}\,\}$, and the map

$$\pi : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}, \qquad \pi(a) = [a]_n,$$

is a surjective ring homomorphism with kernel $n\mathbb{Z}$. This is the prototype of every quotient construction used in the volume's algebra: one starts from a ring, singles out an ideal of elements to be "declared zero", and computes with cosets. On notation: we abbreviate $[a]_n$ to $\overline{a}$ (as in the examples above) or drop the decoration altogether, writing plain a, whenever the modulus is clear from context; $\mathbb{Z}/n\mathbb{Z}$ and $\mathbb{Z}_n$ are used interchangeably; and $\mathbb{Z}_n$ always means this ring, never the n-adic integers (which do not appear in this series).

4.4 Fields

Definition 4.24 (Field)**.** A *field* is a commutative ring F with $1 \neq 0$ in which every nonzero element is a unit. Equivalently, a field is a set F with two operations $+$ and $\cdot$ such that $(F, +)$ is an abelian group with identity 0, the nonzero elements $(F \setminus \{0\}, \cdot)$ form an abelian group with identity 1, and multiplication distributes over addition. Unwinding the definition: in a field one may add, subtract, multiply, and divide by any nonzero element, with $a/b := ab^{-1}$. The group $F^\times = (F \setminus \{0\}, \cdot)$ is the *multiplicative group* of the field. (Two points of the equivalence are not direct transcriptions. Passing from the first formulation to the second, one must show $F \setminus \{0\}$ is closed under multiplication; that is theorem 4.28 below. Passing back, the ring axioms must be checked for products involving 0: distributivity forces $0 \cdot a = a \cdot 0 = 0$ by the computation of theorem 4.3(1), whereupon every product with a zero factor vanishes and the associativity, commutativity, and identity laws extend from $F \setminus \{0\}$ to all of F; and $1 \neq 0$ holds because the group $F \setminus \{0\}$ contains its identity 1.)

Definition 4.25 (Subfield and extension field)**.** A subset $F \subseteq K$ of a field K is a *subfield* when it contains 0 and 1 and is itself a field under the operations of K restricted to F. When F is a subfield of K, the larger field K is called an *extension field* (or simply an *extension*) of F; the two phrases name one relation viewed from its two ends. More generally, an injective ring homomorphism $F \hookrightarrow K$ between fields exhibits K as an extension of the copy of F inside it, and we permit ourselves the usual abuse of identifying F with that copy.

The finite world of §2 already contains the series' most important fields, and the unit criterion proved there identifies them at once.

Corollary 4.26. *If p is prime, then every nonzero class in $\mathbb{Z}/p\mathbb{Z}$ is a unit, so $\mathbb{Z}/p\mathbb{Z}$ is a field; it is denoted $\mathbb{F}_p$. More generally, $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if n is prime.*

Proof. Let p be prime and $[a]_p \neq [0]_p$, so that $p \nmid a$. The only positive divisors of p are 1 and p, so $\gcd(a, p) = 1$, and Theorem 2.24 makes $[a]_p$ a unit. Since $[1]_p \neq [0]_p$ for $p \geq 2$, the ring $\mathbb{Z}/p\mathbb{Z}$ is a field. Conversely, if $n = ab$ is composite with $1 < a, b < n$, then $[a]_n \neq [0]_n$ but $\gcd(a, n) = a > 1$, so $[a]_n$ is not a unit (Theorem 2.24 again) and the ring is not a field; and $n = 1$ gives the zero ring, which is not a field because a field has $1 \neq 0$ by definition. □

Example 4.27 (Fields)**.**

1. The rationals $\mathbb{Q}$, the reals $\mathbb{R}$, and the complexes $\mathbb{C}$ are fields. The integers are not: 2 has no inverse in $\mathbb{Z}$.
2. For p prime, $\mathbb{F}_p := \mathbb{Z}/p\mathbb{Z}$ is a field (theorem 4.26; theorem 4.29 below recovers the same fact abstractly). It is called the *prime field* of characteristic p—theorem 4.30 names the invariant—and is fundamental to every later cryptographic construction in the series. Its finite extensions $\mathbb{F}_q$ with $q = p^k$ elements exist for every prime power, arising from the quotient recipe of Example 4.22, and are developed in §6.
3. The subset $\mathbb{Q}(i) = \{a + bi : a, b \in \mathbb{Q}\} \subseteq \mathbb{C}$ is a field: the inverse of a nonzero $a + bi$ is $(a - bi)/(a^2 + b^2)$, which again has rational coordinates. More generally the *number fields*, obtained by adjoining to $\mathbb{Q}$ roots of polynomial equations with rational coefficients, are fields.
4. For any field F, the *rational functions* $F(x) = \{f/g : f, g \in F[x],\ g \neq 0\}$ form a field under the usual arithmetic of fractions—the field of fractions of the polynomial ring $F[x]$.

Proposition 4.28. *Every field is an integral domain.*

Proof. Let F be a field. Commutativity and $1 \neq 0$ hold by definition. Suppose $ab = 0$ with $a \neq 0$. Then a is a unit, and multiplying by its inverse,

$$b = 1 \cdot b = (a^{-1}a)b = a^{-1}(ab) = a^{-1} \cdot 0 = 0.$$

Hence F has no zero divisors. □

The converse fails: $\mathbb{Z}$ is a domain but not a field (Example 4.16). Finiteness, however, closes the gap entirely, and does so by a counting device worth naming, for it recurs. Call it the *injectivity-implies-surjectivity argument*: to invert an element a of a finite structure, show that multiplication by a is injective, conclude that it is surjective because an injective map from a finite set to itself must be onto, and read off a preimage of 1. The same device proves the claim of theorem 4.12 that in a finite commutative ring every nonzero non-zero-divisor is a unit.

Theorem 4.29. *Every finite integral domain is a field.*

Proof. Let R be a finite integral domain and $a \in R$ nonzero. Consider the multiplication map

$$\mu_a : R \to R, \qquad \mu_a(x) = ax.$$

It is injective: if $ax = ay$, then $a(x - y) = 0$, and since R is a domain and $a \neq 0$ this forces $x - y = 0$, that is, $x = y$ (equivalently, a is cancellable by theorem 4.11). An injective map from a finite set to itself is surjective: the image $\mu_a(R)$ has $|R|$ distinct elements and sits inside R, so it is all of R. In particular 1 lies in the image, so $ab = 1$ for some $b \in R$; commutativity gives $ba = 1$ as well, so $b = a^{-1}$ and a is a unit. Since R is a domain, $1 \neq 0$, and every nonzero element has just been shown invertible: R is a field. □

Applied to $\mathbb{Z}/p\mathbb{Z}$—a finite domain by theorem 4.15—the theorem recovers theorem 4.26. The two proofs differ instructively. The pigeonhole argument merely asserts that each inverse *exists*; the extended Euclidean algorithm of §2 *computes* it, in $O(\log p)$ division steps. Cryptography needs both: the abstract statement to reason with, and the algorithm to run.

4.5 The characteristic

Every ring receives a canonical map from the integers, and the behaviour of that map is a fundamental invariant: it measures how much of $\mathbb{Z}$ survives inside the ring.

> **Definition 4.30** (Characteristic). Let R be a ring, and for $n \in \mathbb{Z}$ let $n \cdot 1$ denote the n-fold additive multiple of 1 (so $n \cdot 1 = 1 + \cdots + 1$ with n summands for $n > 0$, $0 \cdot 1 = 0$, and $(-n) \cdot 1 = -(n \cdot 1)$). The *characteristic* $\mathrm{char}(R)$ is the least positive integer n with $n \cdot 1 = 0$, if such an n exists, and 0 otherwise.

An equivalent formulation is often more useful. The map

$$\iota \colon \mathbb{Z} \to R, \qquad \iota(n) = n \cdot 1,$$

is a ring homomorphism: additivity is the bookkeeping of iterated multiples in the abelian group $(R, +)$, multiplicativity is theorem 4.3(4) with $a = b = 1$—namely $\iota(m)\iota(n) = (m \cdot 1)(n \cdot 1) = (mn)(1 \cdot 1) = \iota(mn)$—and $\iota(1) = 1$. Its kernel is an ideal of $\mathbb{Z}$, hence of the form $n\mathbb{Z}$ for a unique $n \geq 0$ by Example 4.18(2), and $\mathrm{char}(R)$ is exactly this nonnegative generator: $\mathrm{char}(R) = 0$ precisely when ι is injective, so that a faithful copy of $\mathbb{Z}$ sits inside R, and $\mathrm{char}(R) = n > 0$ when the kernel is $n\mathbb{Z}$.

Example 4.31 (Characteristics). The rings $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ all have characteristic 0: no positive multiple of 1 vanishes. For the residue rings, $\mathrm{char}(\mathbb{Z}/n\mathbb{Z}) = n$: the multiple $n \cdot \overline{1} = \overline{n} = \overline{0}$ vanishes, and no smaller positive multiple does, since $k \cdot \overline{1} = \overline{k} \neq \overline{0}$ for $0 < k < n$. In particular $\mathrm{char}(\mathbb{F}_p) = p$.

> **Theorem 4.32.** *The characteristic of an integral domain—in particular, of a field—is either* 0 *or a prime number.*

Proof. Let R be an integral domain with $\mathrm{char}(R) = n > 0$; we show n is prime. First $n \neq 1$: otherwise $1 \cdot 1 = 1 = 0$, making R the zero ring (theorem 4.4) and contradicting $1 \neq 0$ in a domain. Suppose $n = ab$ with $1 < a, b < n$. By theorem 4.3(4),

$$0 = n \cdot 1 = (ab)(1 \cdot 1) = (a \cdot 1)(b \cdot 1),$$

so $a \cdot 1 = 0$ or $b \cdot 1 = 0$ since R is a domain. Either case exhibits a positive integer smaller than n whose multiple of 1 vanishes, contradicting the minimality of $n = \mathrm{char}(R)$. Hence n admits no such factorisation, and being greater than 1, it is prime. □

Remark 4.33 (The prime subfield). Let F be a field. If $\mathrm{char}(F) = 0$, the homomorphism ι embeds a copy of $\mathbb{Z}$ in F, and dividing—possible in a field—a copy of $\mathbb{Q}$ as well. If $\mathrm{char}(F) = p$, the image of ι is a copy of $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$. The smallest subfield so obtained is called the *prime subfield* of F; every field is thus an extension of $\mathbb{Q}$ or of some $\mathbb{F}_p$, and the dichotomy between characteristic zero and characteristic p pervades the subject. The fields $\mathbb{F}_q$ underlying the cryptography treated in this series all have prime characteristic p. There the "freshman's dream" identity $(a + b)^p = a^p + b^p$ actually holds, and gives rise to the Frobenius endomorphism; both are developed in the finite-fields section (theorem 6.12).

The question that opened the section is now answered in full. A verifier that must add and multiply computes in a ring; the laws verified one at a time for $\mathbb{Z}/n\mathbb{Z}$ in §2 are precisely the ring axioms, and the construction that produced them is the quotient of $\mathbb{Z}$ by the ideal $n\mathbb{Z}$. A verifier that must also divide computes in a field, and theorem 4.26 says exactly which moduli provide one: the primes. The payoff theorem, theorem 4.29, closes the circle with a purely structural guarantee: in a finite world there is no daylight between the absence of zero divisors and the presence of division—any nonzero finite commutative ring in which nonzero elements never multiply to zero is automatically a field. The prime fields $\mathbb{F}_p$ certified by these results are the ground on which the volume now builds: polynomials, linear algebra, roots of unity, and elliptic curves are all developed over them in the sections that follow.

5 Polynomials over a field

Throughout this section F denotes a field. Consider a problem that sounds impossible. Two parties each hold a polynomial of enormous degree—millions of coefficients, say, the encoding of an entire computation—and wish to convince themselves that the two polynomials are equal. Comparing coefficient lists costs as much as reading the computation itself. Could a *single* spot check suffice: pick one value α at random, evaluate both polynomials there, and declare them equal if the two field elements agree? The astonishing answer is yes—with an error probability so small that no adversary can exploit it—and this section develops exactly the algebra that makes the answer honest.

Polynomials are the engine behind finite fields, error-correcting codes, and—most importantly here—the polynomial commitment schemes and *arithmetisation* (the encoding of a computation as identities between polynomials) underlying Halo 2 and Zcash Orchard. The single most exploited fact in that edifice is elementary: a nonzero polynomial of degree d over a field has at most d roots. From this one fact the section extracts Lagrange interpolation and the Schwartz–Zippel principle, the probabilistic heart of every interactive proof we eventually construct. The only prerequisite is the field theory of the preceding section.

5.1 The polynomial ring

One point of principle governs everything that follows, so we state it before the first definition: *a polynomial is not a function*. It is a formal expression—at bottom, a finite list of coefficients—which one may *evaluate* to obtain a function, but which in general carries more information than that function. Over the finite fields of cryptographic interest the distinction is not pedantic but essential, and the closing subsections of this section (§5.9) return to it at length.

Definition 5.1 (Polynomial)**.** A *polynomial* over F in the *indeterminate* X is a formal expression

$$f \;=\; a_0 + a_1X + a_2X^2 + \cdots + a_nX^n \;=\; \sum_{i=0}^{n} a_iX^i,$$

where $n \in \mathbb{N} = \{0, 1, 2, \dots\}$ and the *coefficients* $a_0, \dots, a_n$ lie in F. Formally, f is identified with its coefficient sequence $(a_0, a_1, a_2, \dots)$, in which only finitely many entries are nonzero. Two polynomials are equal precisely when their coefficient sequences agree term by term:

$$\sum_i a_iX^i = \sum_i b_iX^i \quad\Longleftrightarrow\quad a_i = b_i \text{ for every } i \in \mathbb{N}.$$

The symbol X is an *indeterminate* (a formal variable): it is not an element of F and stands for nothing in particular; it is merely a placeholder organising the coefficients. The set of all polynomials over F in X is denoted $F[X]$.

Remark 5.2 (The sequence model). The fully rigorous rendering of theorem 5.1 takes $F[X]$ to be the set of all functions $a : \mathbb{N} \to F$ that are *eventually zero*: there is some N with $a(i) = 0$ for all $i > N$. The expression $\sum_i a_i X^i$ is then merely suggestive notation for the sequence $(a_0, a_1, \ldots)$. We use the two viewpoints interchangeably; the notation is justified because the arithmetic of these sequences, defined next, is exactly the arithmetic the notation suggests.

Definition 5.3 (Ring operations on $F[X]$). Let $f = \sum_i a_i X^i$ and $g = \sum_i b_i X^i$ lie in $F[X]$, with the shorter coefficient sequence zero-padded so that both run over the same index set. Their *sum* and *product* are

$$f + g = \sum_{i \geq 0} (a_i + b_i) X^i, \qquad fg = \sum_{k \geq 0} c_k X^k \quad \text{where} \quad c_k = \sum_{i+j=k} a_i b_j = \sum_{i=0}^{k} a_i b_{k-i}.$$

The coefficient c_k is the *convolution* of the two coefficient sequences. Both defining sums are finite—only finitely many a_i and b_j are nonzero, hence only finitely many c_k are nonzero—so the sum and the product are again polynomials.

The convolution formula is exactly what multiplying out $(\sum_i a_i X^i)(\sum_j b_j X^j)$ and collecting like powers of X gives, using $X^i \cdot X^j = X^{i+j}$. Stated as a convolution, however, it makes no appeal to substituting a value for X: it is a purely formal manipulation of coefficient sequences. The formal/functional distinction announced above is thereby kept intact at the level of the operations themselves.

The verification that these operations obey the ring axioms (theorem 4.1) is a single device applied over and over: to prove an identity between polynomials, compute the coefficient of X^m on each side as a finite sum over index decompositions, then reindex or regroup that sum. No evaluation is ever involved. The associativity step below is the archetype.

Theorem 5.4 ($F[X]$ is a commutative ring). *With the operations of theorem 5.3, the set $F[X]$ is a commutative ring whose additive identity is the zero polynomial* $0 = (0, 0, 0, \ldots)$ *and whose multiplicative identity is the constant polynomial* $1 = (1, 0, 0, \ldots)$. *Moreover the map $c \mapsto (c, 0, 0, \ldots)$ embeds F into $F[X]$ as a subring, the* constant polynomials*; in this sense $F \subseteq F[X]$.*

Proof. Additive structure. Addition is coordinatewise, so its commutativity and associativity are inherited directly from the corresponding laws of the additive group of F, the sequence $(0, 0, \ldots)$ is an additive identity, and $(-a_0, -a_1, \ldots)$ is an additive inverse of $(a_0, a_1, \ldots)$. Hence $(F[X], +)$ is an abelian group.

Commutativity of multiplication. The coefficient of X^k in fg is $\sum_{i+j=k} a_i b_j$; reindexing the sum by $(i, j) \mapsto (j, i)$ and using commutativity of multiplication in F,

$$\sum_{i+j=k} a_i b_j = \sum_{i+j=k} b_i a_j,$$

which is the coefficient of X^k in gf.

Associativity of multiplication. Write $h = \sum_l c_l X^l$. The coefficient of X^m in $(fg)h$ is

$$\sum_{k+l=m} \Big(\sum_{i+j=k} a_i b_j \Big) c_l = \sum_{i+j+l=m} a_i b_j c_l,$$

and expanding $f(gh)$ instead produces $\sum_{i+j+l=m} a_i (b_j c_l)$, the same symmetric triple sum by associativity in F. Hence $(fg)h = f(gh)$ coefficient by coefficient.

Distributivity. The coefficient of X^k in $f(g+h)$ is $\sum_{i+j=k} a_i(b_j + c_j)$, which distributivity in F splits as $\sum_{i+j=k} a_i b_j + \sum_{i+j=k} a_i c_j$: the coefficient of X^k in $fg + fh$. The other distributive law follows from this one and commutativity.

Identity. With $1 = (1, 0, 0, \dots)$, the k-th coefficient of $1 \cdot f$ is $\sum_{i+j=k} 1[i = 0]\, a_j = a_k$, where $1[i = 0]$ denotes the coefficient sequence of 1 (equal to 1 at index 0 and 0 elsewhere); so $1 \cdot f = f$.

Embedding. Let $\iota(c) = (c, 0, 0, \dots)$. The map ι preserves sums coordinatewise, and $\iota(c)\iota(d)$ has k-th coefficient cd for $k = 0$ and 0 otherwise, so $\iota(c)\iota(d) = \iota(cd)$; also $\iota(1) = 1$. Distinct constants give distinct sequences, so ι is an injective ring homomorphism (theorem 4.21) and its image is a subring of $F[X]$ isomorphic to F. □

Remark 5.5 (Scalar multiplication). Multiplying $f = \sum_i a_i X^i$ by a constant $c \in F$, viewed as the constant polynomial $\iota(c)$, gives the *scalar multiple* $cf = \sum_i (ca_i)X^i$. Together with addition, this makes $F[X]$ a *vector space* over F—a structure studied systematically in the linear-algebra section later in the volume; informally, a system closed under addition and under scaling by field elements, in which the monomials $1, X, X^2, X^3, \dots$ play the role of an infinite coordinate system (a *basis*). This linear structure is relied on constantly, for instance whenever we count polynomials of bounded degree.

5.2 Degree

Definition 5.6 (Degree, leading coefficient, monic)**.** Let $f = \sum_i a_i X^i \in F[X]$ be nonzero. The *degree* $\deg f$ is the largest index n with $a_n \neq 0$; the coefficient a_n is the *leading coefficient* $\mathrm{lc}(f)$, the term $a_n X^n$ is the *leading term*, and a_0 is the *constant term*. A nonzero polynomial is *monic* if its leading coefficient is 1. The constant polynomials are the elements of F embedded by theorem 5.4; the nonzero ones have degree 0. The zero polynomial is assigned $\deg 0 = -\infty$, with the conventions $-\infty < n$, $-\infty + n = -\infty$, and $-\infty + (-\infty) = -\infty$ for every $n \in \mathbb{Z}$. Polynomials of degree 1, 2, and 3 are called *linear*, *quadratic*, and *cubic* respectively.

The convention $\deg 0 = -\infty$ is not pedantry: it makes the degree formulas below hold without exceptions, which streamlines every later induction on degree. Induction on degree is thereby set up as a reusable device for the rest of the volume.

Proposition 5.7 (Degree of sums and products)**.** *For all* $f, g \in F[X]$,

$$\deg(f + g) \;\leq\; \max\{\deg f, \deg g\}, \tag{1}$$

with equality whenever $\deg f \neq \deg g$*, and*

$$\deg(fg) \;=\; \deg f + \deg g. \tag{2}$$

Proof. If either polynomial is zero, both statements reduce to the $-\infty$ conventions of theorem 5.6. So let f and g be nonzero, of degrees m and n with leading coefficients a_m and b_n.

Sum. The coefficient of X^k in $f + g$ is $a_k + b_k$, which vanishes for $k > \max\{m, n\}$; this is (1). If $m > n$, the coefficient of X^m is $a_m + 0 = a_m \neq 0$, giving equality (symmetrically for $n > m$). When $m = n$, cancellation $a_m + b_m = 0$ can lower the degree, which is why the sum bound is only an inequality.

Product. The coefficient $c_{m+n} = \sum_{i+j=m+n} a_i b_j$ has only the surviving term $i = m$, $j = n$: terms with $i > m$ have $a_i = 0$, and terms with $i < m$ force $j > n$, so $b_j = 0$. Hence $c_{m+n} = a_m b_n$, and the same argument shows $c_k = 0$ for $k > m + n$. Since F is a field it has no zero divisors (theorem 4.28), so $a_m b_n \neq 0$; therefore $\deg(fg) = m + n$ and $\mathrm{lc}(fg) = \mathrm{lc}(f)\,\mathrm{lc}(g)$. □

The product formula (2) is the first place the field hypothesis does real work: it required $a_m b_n \neq 0$, that is, the absence of zero divisors in F. This single fact has sweeping consequences for everything that follows, beginning with the next three corollaries.

Corollary 5.8 ($F[X]$ is an integral domain). *The ring $F[X]$ has no zero divisors: if $fg = 0$, then $f = 0$ or $g = 0$.*

Proof. If f and g are both nonzero, then $\deg(fg) = \deg f + \deg g \geq 0$ by (2), so $fg \neq 0$. Contrapositively, $fg = 0$ forces $f = 0$ or $g = 0$. □

Corollary 5.9 (Cancellation). *Let $f, g, h \in F[X]$ with $h \neq 0$. If $fh = gh$, then $f = g$.*

Proof. From $fh = gh$ we get $(f - g)h = 0$; since $h \neq 0$, theorem 5.8 forces $f - g = 0$. □

Proposition 5.10 (Units of $F[X]$). *A polynomial $f \in F[X]$ is a unit—has a multiplicative inverse in $F[X]$—if and only if it is a nonzero constant. Thus $F[X]^\times = F^\times$.*

Proof. A nonzero constant c has the inverse $c^{-1} \in F \subseteq F[X]$. Conversely, $fg = 1$ gives $\deg f + \deg g = \deg 1 = 0$ by (2); since degrees of nonzero polynomials are nonnegative, $\deg f = \deg g = 0$, and f is a nonzero constant. □

Definition 5.11 (Associates). Two polynomials $f, g \in F[X]$ are *associates* if $f = ug$ for some unit $u \in F^\times$, that is, if they differ only by a nonzero scalar factor.

Every nonzero polynomial f has exactly one monic associate, obtained by dividing through by its leading coefficient: $\mathrm{lc}(f)^{-1} f$. Choosing the monic representative is the standard way to pin down *the* greatest common divisor or *the* irreducible factor uniquely, and we adopt this normalisation below.

5.3 Division with remainder

The arithmetic of $F[X]$ parallels that of the integers $\mathbb{Z}$ with remarkable fidelity: greatest common divisors, a Euclidean algorithm, a Bézout identity, primes (here called irreducibles), and unique factorisation all reappear. The source of the parallel is a single statement, division with remainder, in which *degree* plays the role that absolute value plays in $\mathbb{Z}$.

Theorem 5.12 (Division algorithm). *Let $f, g \in F[X]$ with $g \neq 0$. There exist* unique *polynomials $q, r \in F[X]$ such that*

$$f = qg + r \qquad \text{and} \qquad \deg r < \deg g.$$

We call q the quotient *and r the* remainder*, and write $q = f \operatorname{div} g$ and $r = f \bmod g$.*

Proof. Existence. Fix g, of degree n with leading coefficient b_n, and induct on $\deg f$. If $\deg f < n$—including $f = 0$ with $\deg f = -\infty$—take $q = 0$ and $r = f$. Otherwise let $m = \deg f \geq n$, with leading coefficient a_m, and form

$$f_1 \;=\; f - \frac{a_m}{b_n} X^{m-n} g.$$

The subtracted term has degree m and leading coefficient $(a_m/b_n)\, b_n = a_m$, so it cancels the leading term of f: either $f_1 = 0$ or $\deg f_1 < m$. By the induction hypothesis $f_1 = q_1 g + r$ with $\deg r < n$, and then $f = \big(q_1 + (a_m/b_n)X^{m-n}\big)g + r$. The field hypothesis enters visibly here: division by the leading coefficient b_n requires b_n^{-1} to exist.

Uniqueness. Suppose $f = qg + r = q'g + r'$ with both remainders of degree $< n$. Then $(q - q')g = r' - r$. If $q \neq q'$, then $\deg\big((q - q')g\big) = \deg(q - q') + n \geq n$ by (2), whereas $\deg(r' - r) \leq \max\{\deg r', \deg r\} < n$ by (1)—a contradiction. Hence $q = q'$, and then $r = f - qg = r'$. □

Remark 5.13 (Polynomial long division). The existence proof is constructive, and it is precisely *polynomial long division*: repeatedly eliminate the current leading term by subtracting an appropriate monomial multiple of g. Each step strictly lowers the degree of the running remainder, so the process terminates after at most $\deg f - \deg g + 1$ steps when $\deg f \geq \deg g$, and takes no steps when $f = 0$ or $\deg f < \deg g$. The device is reusable in both of its guises: as an algorithm, and as the leading-term-elimination pattern for inductions on degree.

For a concrete instance, divide $f = X^3 + 2X + 1$ by $g = X^2 + 1$ over $\mathbb{Q}$. Subtracting $X \cdot g = X^3 + X$ leaves $X + 1$, whose degree 1 is already below $\deg g = 2$; hence $q = X$ and $r = X + 1$, and indeed

$$X^3 + 2X + 1 \;=\; X(X^2 + 1) + (X + 1).$$

One elimination step suffices here.

Definition 5.14 (Divisibility). For $f, g \in F[X]$, we say g *divides* f, written $g \mid f$, if $f = qg$ for some $q \in F[X]$; equivalently, for $g \neq 0$, if the remainder $f \bmod g$ is 0. In that case g is a *divisor* (or *factor*) of f.

5.4 Roots and the factor theorem

The destination of this subsection is the central counting bound of the whole section, stated at once so that the reader knows what the machinery is for: *a nonzero polynomial of degree d over a field has at most d distinct roots* (theorem 5.18 below). Everything the volume later builds on polynomials—the cyclicity of the multiplicative group of a finite field, Lagrange interpolation, the Schwartz–Zippel principle—is load-bearing on this one fact. To reach it we must first say what a root is, and that requires making substitution precise: this is the first place where a polynomial gives rise to a function.

Definition 5.15 (Evaluation and roots). Let $f = \sum_{i=0}^{n} a_i X^i \in F[X]$ and $\alpha \in F$. The *evaluation* of f at α is the field element

$$f(\alpha) \;=\; \sum_{i=0}^{n} a_i \alpha^i \;\in\; F,$$

obtained by substituting α for the indeterminate X. An element $\alpha \in F$ is a *root* (or *zero*) of f if $f(\alpha) = 0$.

Proposition 5.16 (Evaluation is a ring homomorphism). *Fix $\alpha \in F$. The evaluation map $\mathrm{ev}_\alpha : F[X] \to F$, $f \mapsto f(\alpha)$, satisfies*

$$(f + g)(\alpha) = f(\alpha) + g(\alpha), \qquad (fg)(\alpha) = f(\alpha)\, g(\alpha), \qquad 1(\alpha) = 1$$

for all $f, g \in F[X]$; that is, ev_α is a ring homomorphism.

Proof. Additivity is immediate from the coordinatewise definition of addition. For multiplicativity, with $c_k = \sum_{i+j=k} a_i b_j$ as in theorem 5.3,

$$(fg)(\alpha) = \sum_k c_k \alpha^k = \sum_k \sum_{i+j=k} a_i b_j\, \alpha^{i+j} = \Big(\sum_i a_i \alpha^i\Big)\Big(\sum_j b_j \alpha^j\Big) = f(\alpha)\, g(\alpha),$$

the middle equality because expanding the product of the two finite sums and grouping terms by $k = i + j$ produces exactly the double sum. The formal reason evaluation respects the algebra is that the convolution defining polynomial multiplication collapses to ordinary multiplication in F via $\alpha^i \alpha^j = \alpha^{i+j}$. Finally $1(\alpha) = 1$ by direct substitution. □

Theorem 5.17 (Remainder theorem; factor theorem). *Let $f \in F[X]$ and $\alpha \in F$.*

1. (Remainder theorem.) *The remainder of f upon division by the linear polynomial $X - \alpha$ is the constant $f(\alpha)$: there is $q \in F[X]$ with $f = q\,(X - \alpha) + f(\alpha)$.*

2. (Factor theorem.) *The element α is a root of f if and only if $(X - \alpha) \mid f$.*

Proof. (1) Divide f by $X-\alpha$ (theorem 5.12): since $\deg(X-\alpha) = 1$, the remainder has degree < 1, hence is a constant $r \in F$. Evaluating $f = q\,(X-\alpha)+r$ at α via theorem 5.16 gives $f(\alpha) = q(\alpha)(\alpha-\alpha)+r = r$.

(2) By (1) and the uniqueness of the remainder, $(X-\alpha) \mid f$ holds exactly when the remainder $f(\alpha)$ is 0, that is, when α is a root of f. □

The factor theorem is an immediate but powerful consequence of the division algorithm: it converts a statement about *values* (a root) into a statement about *divisibility* (a linear factor). Iterating that conversion proves the promised bound.

Theorem 5.18 (At most d roots). *A nonzero polynomial $f \in F[X]$ of degree d has at most d distinct roots in F.*

Proof. We induct on $d = \deg f$. If $d = 0$, then f is a nonzero constant, which evaluates to itself everywhere and so has $0 \le d$ roots. Let $d \ge 1$. If f has no root in F the claim holds; otherwise let α be a root. By the factor theorem, $f = (X - \alpha)\,q$, and $\deg q = d - 1$ by (2). Now let $\beta \ne \alpha$ be any other root of f. Evaluating with theorem 5.16,

$$0 = f(\beta) = (\beta - \alpha)\,q(\beta);$$

since $\beta - \alpha \ne 0$ and F has no zero divisors (theorem 4.28), we get $q(\beta) = 0$. Every root of f other than α is therefore a root of q, and by the induction hypothesis q has at most $d - 1$ distinct roots. Hence f has at most $(d - 1) + 1 = d$. □

Remark 5.19 (Why a field is needed). Both ingredients of the proof—the absence of zero divisors and the degree bookkeeping—are essential. Over a ring with zero divisors the bound fails outright: in $\mathbb{Z}/8\mathbb{Z}$ the quadratic $X^2 - 1$ has the four roots $\overline{1}, \overline{3}, \overline{5}, \overline{7}$, since $1^2 = 1$, $3^2 = 9 \equiv 1$, $5^2 = 25 \equiv 1$, and $7^2 = 49 \equiv 1 \pmod 8$—a degree-2 polynomial with four roots. This is why the polynomial methods underlying Halo 2 require a field, and indeed a finite field $\mathbb{F}_p$ or $\mathbb{F}_q$, beneath them.

Corollary 5.20 (Few points determine a low-degree polynomial). *Let $f, g \in F[X]$ both have degree at most d. If $f(\alpha) = g(\alpha)$ for $d + 1$ distinct values $\alpha \in F$, then $f = g$ as polynomials.*

Proof. The difference $h = f - g$ has $\deg h \le \max\{\deg f, \deg g\} \le d$ by (1) and vanishes at $d + 1$ distinct points. Were h nonzero, it would be a polynomial of degree at most d with $d+1$ distinct roots, contradicting theorem 5.18. Hence $h = 0$, that is, $f = g$. □

The corollary is the deterministic skeleton of two constructions developed later in this section: Lagrange interpolation (§5.7), its constructive converse, and the Schwartz–Zippel principle (§5.10), its probabilistic refinement. The root bound is thus positioned as the load-bearing fact behind the polynomial identity testing and interpolation machinery used by the proving system.

5.5 Greatest common divisors

Because $F[X]$ supports division with remainder, the entire theory of greatest common divisors from elementary number theory (§2) transfers verbatim from $\mathbb{Z}$ to $F[X]$, with degree replacing absolute value as the measure that decreases. We run through the transfer explicitly, both because the statements are used constantly and because the polynomial Euclidean algorithm is itself a tool of the trade.

Definition 5.21 (Greatest common divisor)**.** Let $f, g \in F[X]$, not both zero. A *greatest common divisor* of f and g is a polynomial d such that

1. $d \mid f$ and $d \mid g$; and
2. every common divisor c of f and g satisfies $c \mid d$.

A greatest common divisor is determined only up to multiplication by a unit; the unique *monic* one (theorem 5.23) is singled out and denoted $\gcd(f, g)$. When $\gcd(f, g) = 1$, the polynomials f and g are called *coprime* (or *relatively prime*). Since every polynomial divides 0, we have $\gcd(f, 0) = \mathrm{lc}(f)^{-1} f$ for $f \neq 0$.

Lemma 5.22 (Euclidean step)**.** *Let $f, g \in F[X]$ with $g \neq 0$, and let $f = qg + r$ be the division of theorem 5.12. Then the pairs (f, g) and (g, r) have exactly the same common divisors; in particular, their greatest common divisors coincide.*

Proof. If $c \mid f$ and $c \mid g$, then $c \mid (f - qg) = r$; so c is a common divisor of (g, r). Conversely, if $c \mid g$ and $c \mid r$, then $c \mid (qg + r) = f$. The two sets of common divisors are therefore equal, hence so are their maximal elements under divisibility. □

Theorem 5.23 (Euclidean algorithm; Bézout's identity)**.** *Let $f, g \in F[X]$, not both zero. Then* $\gcd(f, g)$ *exists and is unique as a monic polynomial, and there exist $s, t \in F[X]$ with*

$$\gcd(f, g) \;=\; sf + tg. \tag{3}$$

Concretely: set $r_0 = f$ and $r_1 = g$, and while $r_i \neq 0$ divide

$$r_{i-1} = q_i r_i + r_{i+1}, \qquad \deg r_{i+1} < \deg r_i.$$

The degrees strictly decrease, so some $r_{N+1} = 0$; the last nonzero remainder r_N, made monic, is $\gcd(f, g)$*.*

Proof. Termination. If $r_1 = 0$, the algorithm stops at once with $N = 0$ and $r_N = f \neq 0$. Otherwise the degrees $\deg r_1 > \deg r_2 > \ldots$ form a strictly decreasing sequence of nonnegative integers, which cannot continue forever; so some $r_{N+1} = 0$ with $r_N \neq 0$.

The gcd property. Theorem 5.22, applied at each division, preserves the *set* of common divisors:

$$\{\text{common divisors of } (f, g)\} = \{\text{common divisors of } (r_1, r_2)\} = \cdots = \{\text{common divisors of } (r_N, 0)\},$$

and the last set is exactly the set of divisors of r_N, since every polynomial divides 0. Thus r_N divides both f and g, and every common divisor of f and g divides r_N: the monic associate $\mathrm{lc}(r_N)^{-1} r_N$ is a monic greatest common divisor.

Uniqueness. Two monic greatest common divisors divide each other, so they differ by a unit $u \in F^{\times}$ (theorem 5.10); comparing leading coefficients of monic polynomials forces $u = 1$.

Bézout. We show by induction on i that each remainder r_i is an $F[X]$-linear combination of f and g. The bases $r_0 = 1 \cdot f + 0 \cdot g$ and $r_1 = 0 \cdot f + 1 \cdot g$ are trivial, and the recurrence $r_{i+1} = r_{i-1} - q_i r_i$ expresses r_{i+1} as a combination once r_{i-1} and r_i are. Hence $r_N = s_0 f + t_0 g$ for some $s_0, t_0 \in F[X]$; dividing through by $\mathrm{lc}(r_N)$ yields (3). □

Remark 5.24 (Extended Euclidean algorithm). Carrying the linear-combination coefficients along with the remainders, exactly as in the integer case (Theorem 2.13), produces s and t explicitly alongside the gcd. The standard application is inversion in a quotient $F[X]/(m)$ for irreducible m: if

$\gcd(f, m) = 1$, then $sf + tm = 1$, so s is the inverse of f modulo m. This is how one divides in the extension fields $\mathbb{F}_q$ constructed in section 6. The polynomial version matters both because it parallels the integer one exactly and because it computes gcds of the polynomials a proof system manipulates. A caveat on deployed practice: the curves of Halo 2 and Orchard are defined over *prime* fields $\mathbb{F}_p$, not extension fields, and field inversion there uses the integer extended Euclidean algorithm (Theorem 2.13) or Fermat's little theorem—not the polynomial algorithm in $F[X]/(m)$.

Corollary 5.25 (Euclid's lemma in $F[X]$). *Let $p, f, g \in F[X]$.*

1. *If $\gcd(p, f) = 1$ and $p \mid fg$, then $p \mid g$.*
2. *If p is irreducible and $p \mid fg$, then $p \mid f$ or $p \mid g$.*

Here irreducible *is the polynomial analogue of prime, made precise in theorem 5.26 below.*

Proof. (1) Multiplying the Bézout identity $sp + tf = 1$ by g gives $spg + tfg = g$. The polynomial p divides both terms on the left—the second because $p \mid fg$—hence divides g.

(2) Suppose $p \nmid f$. The gcd $\gcd(p, f)$ is a monic divisor of p that is not the monic associate of p (otherwise $p \mid f$). By irreducibility the only monic divisors of p are 1 and its monic associate, so $\gcd(p, f) = 1$ and part (1) applies. $\square$

5.6 Unique factorisation

Definition 5.26 (Irreducible polynomial). A polynomial $p \in F[X]$ is *irreducible over F* if $\deg p \geq 1$ and p admits no factorisation $p = gh$ with both $\deg g \geq 1$ and $\deg h \geq 1$; equivalently, p is nonconstant and its only factorisations are the trivial ones, (unit)$\times$(associate of p). A nonconstant polynomial that is not irreducible is *reducible*.

Remark 5.27 (Irreducibility is relative to the base field). The polynomial X^2+1 is irreducible over $\mathbb{R}$ (it has no real root, and a real quadratic factors precisely when it has a root, by the test below), yet splits as $(X-i)(X+i)$ over $\mathbb{C}$, and factors as $(X+1)^2$ over $\mathbb{F}_2$, by direct expansion in characteristic 2. Degree-1 polynomials are irreducible over every field. A useful *rootlessness test*: a polynomial of degree 2 or 3 is irreducible over F if and only if it has no root in F, because any nontrivial factorisation of a quadratic or cubic must include a linear factor, which by the factor theorem (theorem 5.17) supplies a root. The test fails from degree 4 onward: a product of two irreducible quadratics has no roots yet is reducible.

Theorem 5.28 ($F[X]$ is a unique factorisation domain). *Every nonconstant $f \in F[X]$ factors as*

$$f = c\, p_1 p_2 \cdots p_k$$

with $c = \mathrm{lc}(f) \in F^\times$ a constant and each p_i monic irreducible. The factorisation is unique up to order: if also $f = c'\, q_1 \cdots q_l$ with $c' \in F^\times$ and each q_j monic irreducible, then $c = c'$, $k = l$, and the q_j are a reordering of the p_i.

Proof. The proof is templated on the fundamental theorem of arithmetic (Theorem 2.18), with degree in place of absolute value.

Existence, by induction on $\deg f \geq 1$. If f is irreducible, then $f = \mathrm{lc}(f) \cdot p_1$ with p_1 its monic associate. If $f = gh$ is reducible with $1 \leq \deg g, \deg h < \deg f$, the induction hypothesis factors g and h, and multiplying the two factorisations—collecting the two constants into one—factors f.

Uniqueness. Comparing leading coefficients in the two factorisations gives $c = c' = \mathrm{lc}(f)$, since monic factors contribute leading coefficient 1 by (2). Cancelling the constants (theorem 5.9) leaves

$p_1 \cdots p_k = q_1 \cdots q_l$; we induct on k. For $k = 1$ the left side is irreducible, so $l = 1$ and $p_1 = q_1$. For $k \geq 2$: p_1 divides $q_1 \cdots q_l$, so iterating Euclid's lemma (theorem 5.25(2)) gives $p_1 \mid q_j$ for some j. A monic irreducible dividing a monic irreducible equals it: $q_j = p_1 h$ forces $\deg h = 0$ by irreducibility of q_j, and comparing leading coefficients gives $h = 1$. Cancel $p_1 = q_j$ from both sides (theorem 5.9). If $l = 1$, the cancellation leaves $p_2 \cdots p_k = 1$, which is impossible: the left side is a product of $k-1 \geq 1$ nonconstant polynomials and so has degree at least 1 by (2), while $\deg 1 = 0$. Hence $l \geq 2$, and the induction hypothesis, applied to $p_2 \cdots p_k = \prod_{j' \neq j} q_{j'}$, concludes $k = l$ and matches the remaining factors. □

Remark 5.29 (The structural reason: $F[X]$ is a PID). The deeper explanation of unique factorisation is structural. The degree function and theorem 5.12 make $F[X]$ a *Euclidean domain*—an integral domain with division with remainder relative to a size measure—and every Euclidean domain is a *principal ideal domain* (PID): each ideal (theorem 4.17) of $F[X]$ is $(g) = \{ qg : q \in F[X] \}$ for a single generator g, namely any nonzero element of least degree in a nonzero ideal (the zero ideal is generated by 0), by the same division-and-minimality argument that proved every ideal of $\mathbb{Z}$ principal (Example 4.18(2)). The generator of the ideal $\{ sf + tg : s, t \in F[X] \}$ is a gcd of f and g, and Bézout's identity (3) is the concrete shadow of this principality. The standard chain

$$\text{Euclidean} \implies \text{PID} \implies \text{UFD}$$

has thus been walked explicitly for $F[X]$, ending at theorem 5.28.

5.7 Lagrange interpolation

The corollary "few points determine a low-degree polynomial" (theorem 5.20) says that a polynomial of degree at most d is pinned down by its values at $d + 1$ points. The present subsection proves the constructive converse: *any* $d + 1$ prescribed values at distinct points are attained by exactly one polynomial of degree at most d, and there is an explicit formula for it. This is the algebraic backbone of secret sharing, of Reed–Solomon error-correcting codes, and of the polynomial encodings in Halo 2.

> **Theorem 5.30** (Lagrange interpolation). *Let $x_0, \dots, x_d \in F$ be distinct (the* nodes*) and let $y_0, \dots, y_d \in F$ be arbitrary targets. There exists a unique $f \in F[X]$ with $\deg f \leq d$ and $f(x_i) = y_i$ for all i, given explicitly by*
>
> $$f(X) = \sum_{i=0}^{d} y_i \, \ell_i(X), \qquad \textit{where} \qquad \ell_i(X) = \prod_{j \neq i} \frac{X - x_j}{x_i - x_j}$$
>
> *are the* Lagrange basis polynomials *for the nodes.*

Proof. The ℓ_i are genuine polynomials. Each denominator $\prod_{j \neq i}(x_i - x_j)$ is a product of nonzero field elements, the nodes being distinct, hence is an invertible constant; so ℓ_i is a polynomial, of degree exactly d (a product of d linear factors divided by a nonzero constant).

The delta property. Writing δ_{im} for the Kronecker symbol (equal to 1 if $i = m$ and 0 otherwise),

$$\ell_i(x_m) = \delta_{im} \qquad (0 \leq i, m \leq d) : \tag{4}$$

at $m = i$ every factor $(x_i - x_j)/(x_i - x_j)$ is 1, while at $m \neq i$ the factor with $j = m$ has numerator $x_m - x_m = 0$. Consequently $f(x_m) = \sum_i y_i \, \ell_i(x_m) = y_m$ for each m, and $\deg f \leq d$ by (1), each summand having degree at most d.

Uniqueness. Two interpolants of degree at most d agree at the $d+1$ distinct nodes, hence are equal by theorem 5.20. □

Example 5.31 (A worked interpolation over $\mathbb{Q}$)**.** Find the polynomial of degree at most 2 with $f(0) = 1$, $f(1) = 3$, and $f(2) = 7$. The basis polynomials for the nodes $0, 1, 2$ are

$$\begin{aligned} \ell_0 &= \frac{(X-1)(X-2)}{(0-1)(0-2)} = \frac{(X-1)(X-2)}{2}, \\ \ell_1 &= \frac{(X-0)(X-2)}{(1-0)(1-2)} = -X(X-2), \\ \ell_2 &= \frac{(X-0)(X-1)}{(2-0)(2-1)} = \frac{X(X-1)}{2}. \end{aligned}$$

Hence

$$f \;=\; 1\cdot\ell_0 + 3\cdot\ell_1 + 7\cdot\ell_2 \;=\; \tfrac{1}{2}(X^2-3X+2) + 3(2X-X^2) + \tfrac{7}{2}(X^2-X) \;=\; X^2+X+1,$$

and indeed $f(0) = 1$, $f(1) = 3$, $f(2) = 7$.

Remark 5.32 (The evaluation isomorphism). Fix $d+1$ distinct nodes and let $V = \{\, f \in F[X] \,:\, \deg f \le d \,\}$. In the language of the linear-algebra section later in the volume, V is a $(d+1)$-dimensional vector space over F with basis $1, X, \dots, X^d$, and the evaluation map

$$\mathrm{ev}\colon\; V \to F^{d+1}, \qquad f \longmapsto \big(f(x_0), \dots, f(x_d)\big),$$

is F-linear. Its surjectivity is exactly the existence half of theorem 5.30, and its injectivity exactly the uniqueness half; so ev is a linear isomorphism $V \cong F^{d+1}$. The Lagrange basis $\{\ell_i\}$ is precisely the basis of V matched to the standard coordinates of F^{d+1}, by the delta property (4): the polynomial with prescribed values $(y_0, \dots, y_d)$ has coordinates $(y_0, \dots, y_d)$ in the basis $\{\ell_i\}$. This switching between a polynomial's *coefficient representation* and its *evaluation representation* is the conceptual core of fast polynomial arithmetic via the fast Fourier transform, treated with the roots of unity later in the volume, and of the evaluation-domain encodings used throughout Halo 2.

Remark 5.33 (The vanishing polynomial). The companion object to the Lagrange basis is the *vanishing polynomial* of the node set $\{x_0, \dots, x_d\}$,

$$Z(X) \;=\; \prod_{i=0}^{d}(X - x_i),$$

the unique monic polynomial of degree $d+1$ vanishing at every node. A polynomial g vanishes on the whole node set if and only if $Z \mid g$. One direction is immediate from theorem 5.16. For the other, apply the factor theorem repeatedly: $g(x_0) = 0$ gives $g = (X - x_0)\,g_1$; evaluating at $x_1 \ne x_0$ gives $0 = (x_1 - x_0)\,g_1(x_1)$, so $g_1(x_1) = 0$ since F has no zero divisors, and $g_1 = (X - x_1)\,g_2$; continuing through the remaining nodes strips off one linear factor per node. The repetition is legitimate precisely because the nodes are distinct—the linear factors $X - x_i$ are pairwise coprime. The pair (Lagrange basis, vanishing polynomial) is exactly the data a proof system manipulates when arguing that a committed polynomial agrees with prescribed values on an evaluation domain.

5.8 Multiplicity and the formal derivative

A root can be a root "several times over": the quadratic $X^2 - 2X + 1 = (X-1)^2$ vanishes at 1 doubly. Counting roots correctly requires making that idea precise, and detecting it requires a derivative—which we define with no limits at all.

> **Definition 5.34** (Multiplicity of a root)**.** Let $f \in F[X]$ be nonzero and let $\alpha \in F$ be a root of f. The *multiplicity* $\mathrm{mult}_\alpha(f)$ is the largest integer $m \ge 1$ with $(X-\alpha)^m \mid f$. A root of multiplicity 1 is *simple*; a root of multiplicity at least 2 is *repeated* (or *multiple*).

The largest such m exists: the powers $(X-\alpha)^m$ have strictly increasing degrees, so only finitely many can divide the fixed f, and at least one does by the factor theorem. An equivalent characterisation is often handier: $m = \mathrm{mult}_\alpha(f)$ exactly when $f = (X-\alpha)^m g$ with $g(\alpha) \neq 0$. Indeed, writing $f = (X-\alpha)^m g$ with m maximal forces $g(\alpha) \neq 0$, else the factor theorem would split a further factor $X-\alpha$ off g; conversely, if $f = (X-\alpha)^m g$ with $g(\alpha) \neq 0$, then $(X-\alpha)^{m+1} \mid f$ would give $(X-\alpha)^m g = (X-\alpha)^{m+1} h$, and cancelling (theorem 5.9) $g = (X-\alpha)h$, making α a root of g.

The count of roots *with* multiplicity still respects the degree. The proof rests on a small device worth isolating, since it recurs.

Lemma 5.35 (Coprime factors multiply). *Let $f_1, \dots, f_r \in F[X]$ be pairwise coprime* ($\gcd(f_i, f_j) = 1$ *for $i \neq j$) and suppose each f_i divides f. Then $f_1 f_2 \cdots f_r \mid f$.*

Proof. First let $r = 2$, and write $f = f_1 u = f_2 v$. Bézout (3) gives $sf_1 + tf_2 = 1$, so

$$f \;=\; sf_1 f + tf_2 f \;=\; sf_1(f_2 v) + tf_2(f_1 u) \;=\; f_1 f_2\,(sv + tu),$$

and $f_1 f_2 \mid f$. For $r > 2$, induct: by hypothesis $f_1 \cdots f_{r-1} \mid f$, and the product $f_1 \cdots f_{r-1}$ is coprime to f_r—a nonconstant common divisor would have a monic irreducible factor (theorem 5.28), which by Euclid's lemma (theorem 5.25(2)) would divide some f_i with $i < r$ as well as f_r, contradicting $\gcd(f_i, f_r) = 1$. Now apply the case $r = 2$. □

Proposition 5.36 (Counting roots with multiplicity). *Let $f \in F[X]$ be nonzero of degree d, with all its distinct roots $\alpha_1, \dots, \alpha_r \in F$ of multiplicities $m_1, \dots, m_r$. Then*

$$f \;=\; (X-\alpha_1)^{m_1} \cdots (X-\alpha_r)^{m_r}\, g$$

for some $g \in F[X]$ with no roots in F, and $\sum_i m_i \leq d$. The number of roots counted with multiplicity is thus at most d, with equality if and only if f splits into linear factors over F.

Proof. We first check that the powers $(X-\alpha_i)^{m_i}$ are pairwise coprime. Each $X-\alpha_i$ is monic irreducible (degree 1), and by unique factorisation the monic divisors of $(X-\alpha_i)^{m_i}$ are exactly the powers $(X-\alpha_i)^e$ with $0 \leq e \leq m_i$. For $i \neq j$ the only monic polynomial that is simultaneously a power of $X-\alpha_i$ and of $X-\alpha_j$ is 1, since $X-\alpha_i = X-\alpha_j$ would force $\alpha_i = \alpha_j$; so $\gcd\big((X-\alpha_i)^{m_i}, (X-\alpha_j)^{m_j}\big) = 1$.

Each $(X-\alpha_i)^{m_i}$ divides f by the definition of multiplicity, so theorem 5.35 gives $f = (X-\alpha_1)^{m_1} \cdots (X-\alpha_r)^{m_r}\, g$. Maximality of each m_i gives $g(\alpha_i) \neq 0$: a further factor $X-\alpha_i$ inside g would raise the multiplicity. And g has no other roots either: a root β of g is a root of f, hence one of the α_i. Taking degrees with (2),

$$m_1 + \cdots + m_r + \deg g \;=\; d,$$

so $\sum_i m_i \leq d$, with equality precisely when $\deg g = 0$, that is, when f is a constant times a product of linear factors. □

Definition 5.37 (Formal derivative)**.** The *formal derivative* is the map $D\colon F[X] \to F[X]$ defined on $f = \sum_{i=0}^n a_i X^i$ by

$$f' \;=\; Df \;=\; \sum_{i=1}^{n} i\, a_i X^{i-1} \;=\; a_1 + 2a_2 X + \cdots + n a_n X^{n-1},$$

where the coefficient $i\,a_i$ means a_i added to itself i times in F—that is, a_i multiplied by the image of the integer i under the canonical map $\mathbb{Z} \to F$ (theorem 4.30). No limits are involved: the definition

> is purely formal and algebraic, valid over any field, and it is introduced for one purpose—to detect repeated roots without any appeal to analysis.

> **Proposition 5.38** (Rules of formal differentiation)**.** *For all $f, g \in F[X]$, $c \in F$, $\alpha \in F$, and $m \geq 1$:*
>
> 1. (Linearity.) $(f+g)' = f' + g'$ *and* $(cf)' = cf'$.
> 2. (Product rule.) $(fg)' = f'g + fg'$.
> 3. (Power rule.) $\big((X-\alpha)^m\big)' = m\,(X-\alpha)^{m-1}$.

Proof. (1) is immediate from the coordinatewise, visibly linear definition of D.

(2) Both sides are *bilinear* in the pair (f, g): by part (1), each is linear in f with g fixed and linear in g with f fixed. It therefore suffices to verify the identity on monomials $f = X^a$, $g = X^b$, since every polynomial is a linear combination of monomials and the general case follows by expanding $f = \sum_a u_a X^a$, $g = \sum_b v_b X^b$ and applying the monomial case termwise. On monomials,

$$(X^a X^b)' = (X^{a+b})' = (a+b)\,X^{a+b-1} = aX^{a-1}X^b + X^a\,bX^{b-1} = (X^a)'X^b + X^a(X^b)',$$

with the convention that a term carrying the integer coefficient 0 (the case $a = 0$ or $b = 0$) is the zero polynomial.

(3) Induct on m. The base is $(X-\alpha)' = 1$. For the step, the product rule gives

$$\big((X-\alpha)^m\big)' = \big((X-\alpha)(X-\alpha)^{m-1}\big)' = (X-\alpha)^{m-1} + (X-\alpha)\,(m-1)(X-\alpha)^{m-2} = m\,(X-\alpha)^{m-1}. \qquad \square$$

Part (3) of the next theorem must exhibit a repeated root inside an extension field that is not given in advance; the following lemma manufactures the required extension. It is the quotient recipe of Example 4.22 carried out for polynomial rings, and the proof is the argument that made $\mathbb{Z}/p\mathbb{Z}$ a field (theorem 4.26), transposed from $\mathbb{Z}$ to $F[X]$.

> **Lemma 5.39** (Adjoining a root)**.** *Let $h \in F[X]$ be irreducible. Then the quotient ring $K = F[X]/(h)$ is a field; the map $a \mapsto a + (h)$ embeds F into K as a subfield, so that K is an extension of F; and the class $\alpha = X + (h)$ is a root of h in K.*

Proof. The principal ideal $(h) = hF[X]$ is an ideal of the commutative ring $F[X]$ (Example 4.18(2)), so K is a commutative ring under the coset operations (theorem 4.20). Its identity is not zero: $1 + (h) = 0 + (h)$ would say $h \mid 1$, impossible since a factorisation $1 = hk$ has degree $\deg h + \deg k = 0$ by (2), while $\deg h \geq 1$.

Every nonzero class is a unit. Let $f + (h) \neq 0 + (h)$, i.e. $h \nmid f$. The gcd $\gcd(h, f)$ is then a monic divisor of the irreducible h other than its monic associate, hence equals 1, exactly as in the proof of theorem 5.25(2); Bézout's identity (3) supplies $s, t \in F[X]$ with $sf + th = 1$, and reducing modulo (h) gives $(s + (h))(f + (h)) = 1 + (h)$. So K is a field.

The map $\iota(a) = a + (h)$ on constants preserves sums, products, and 1 directly from the coset operations of theorem 4.19, and it is injective: $\iota(a) = 0$ with $a \neq 0$ would make h divide a nonzero constant, again impossible by degrees. We identify F with its image, making K an extension field of F.

Finally, write $h = \sum_i c_i X^i$ and compute in K:

$$h(\alpha) = \sum_i \iota(c_i)\,\big(X + (h)\big)^i = \Big(\sum_i c_i X^i\Big) + (h) = h + (h) = 0 + (h),$$

so α is a root of h in K. $\square$

Theorem 5.40 (Derivative test for repeated roots; separability). *Let $f \in F[X]$ be nonzero and $\alpha \in F$.*

1. *The element α is a repeated root of f—a root of multiplicity at least 2—if and only if $f(\alpha) = 0$ and $f'(\alpha) = 0$.*
2. *If f has a repeated root in F, then $\gcd(f, f') \neq 1$.*
3. *Call f separable if it has no repeated root in any field containing F. Then f is separable if and only if $\gcd(f, f') = 1$ in $F[X]$.*

Proof. (1) Let α be a root of multiplicity $m \geq 1$ and write $f = (X - \alpha)^m g$ with $g(\alpha) \neq 0$. The product and power rules give

$$f' \;=\; m\,(X-\alpha)^{m-1}g + (X-\alpha)^m g' \;=\; (X-\alpha)^{m-1}\left[\, mg + (X-\alpha)\,g' \,\right].$$

If $m = 1$, the first factor is 1 and evaluating at α gives $f'(\alpha) = 1 \cdot g(\alpha) = g(\alpha) \neq 0$: a simple root of f is not a root of f'. If $m \geq 2$, then $(X-\alpha)^{m-1}$ divides f' and vanishes at α, so $f'(\alpha) = 0$. Together with the factor theorem for $f(\alpha) = 0$, this proves the equivalence.

(2) A repeated root $\alpha \in F$ is a common root of f and f' by (1), so the factor theorem puts the common factor $X - \alpha$ into both, and $(X - \alpha) \mid \gcd(f, f')$.

(3) The key observation is that *gcds are stable under field extension*: for an extension field K of F (theorem 4.25), the Euclidean algorithm of theorem 5.23 run on $f, f' \in F[X]$ uses only field operations on their coefficients, which lie in F; run inside $K[X]$ on the same inputs it produces the identical quotients and remainders. Hence the gcd computed in $K[X]$ equals the gcd computed in $F[X]$.

Suppose $\gcd(f, f') = 1$ in $F[X]$, and let $K \supseteq F$ be any extension. By stability, $\gcd(f, f') = 1$ in $K[X]$ as well, so by (2) applied over K, the polynomial f has no repeated root in K: f is separable.

Conversely, suppose $d = \gcd(f, f')$ is nonconstant; we exhibit a repeated root in some extension. Let h be a monic irreducible factor of d (theorem 5.28), and let $K = F[X]/(h)$ be the extension field supplied by theorem 5.39, in which h has a root α. The polynomial h divides d, which divides both f and f'; evaluating the factorisations at α (theorem 5.16) gives $f(\alpha) = f'(\alpha) = 0$. By (1), applied over K, the element α is a repeated root of f in K: f is not separable. □

The stability observation in the proof deserves emphasis as a technique: it licenses testing separability—the absence of repeated roots in *every* extension of the base field at once—by a gcd computation carried out entirely in the base field.

Remark 5.41 (The pitfall of positive characteristic). The integer coefficient i in $f' = \sum_i i\, a_i X^{i-1}$ is interpreted *in* F, and may vanish there. Over a field of characteristic p—where $p = 0$ in F, p prime, the setting of $\mathbb{F}_p$ and $\mathbb{F}_q$ (theorem 4.30)—the derivative of X^p is $pX^{p-1} = 0$. Thus $f = X^p - 1$ over $\mathbb{F}_p$ has $f' = 0$ and $\gcd(f, f') = \gcd(f, 0) = f \neq 1$; and indeed

$$X^p - 1 \;=\; (X-1)^p \quad \text{in } \mathbb{F}_p[X],$$

a factorisation the finite-fields section establishes through the freshman's-dream identity $(a + b)^p = a^p + b^p$ of characteristic p (theorem 6.12, applied to $(X + (-1))^p$ together with $(-1)^p = -1$ in every $\mathbb{F}_p$): a single root 1 of multiplicity p. This characteristic-p behaviour is a structural feature to respect when working over the finite fields of Orchard, not a pathology to avoid.

5.9 Polynomials versus polynomial functions

The opening of this section insisted that a polynomial is not a function. The debt of that insistence now falls due: we make the relationship precise and show exactly when it is safe—and when it is not—to conflate the two.

Definition 5.42 (Polynomial function)**.** Every $f \in F[X]$ induces a *polynomial function*

$$\tilde{f} : F \to F, \qquad \alpha \longmapsto f(\alpha).$$

The assignment $f \mapsto \tilde{f}$ is a ring homomorphism from $F[X]$ to the ring F^F of *all* functions $F \to F$ under pointwise operations ($(u+v)(\alpha) = u(\alpha) + v(\alpha)$ and $(uv)(\alpha) = u(\alpha)v(\alpha)$), by theorem 5.16 applied at every point.

Proposition 5.43 (When formal equals functional)**.** *Let $f, g \in F[X]$.*

1. *If F is infinite, then $\tilde{f} = \tilde{g}$ as functions implies $f = g$ as polynomials. Equivalently, $f \mapsto \tilde{f}$ is injective, and polynomials over an infinite field may be identified with polynomial functions.*
2. *If F is finite, the map is* not *injective. Over $\mathbb{F}_p$, the nonzero polynomial $X^p - X$ induces the zero function, since Fermat's little theorem (Corollary 3.32) gives $\alpha^p = \alpha$ for every $\alpha \in \mathbb{F}_p$.*

Proof. (1) Let $h = f - g$, so that $\tilde{h}$ is the zero function: h vanishes at every point of the infinite set F. Were h nonzero of degree d, it would have at most d roots (theorem 5.18), yet it has infinitely many—a contradiction. Hence $h = 0$.

(2) Let F be finite, with $q = |F|$ elements, and consider the *vanishing-polynomial witness*

$$w(X) = \prod_{\alpha \in F} (X - \alpha),$$

a monic polynomial of degree q—in particular nonzero as a formal object—which vanishes at every point of F by construction. Thus $\tilde{w} = \tilde{0}$ while $w \neq 0$: the map $f \mapsto \tilde{f}$ identifies distinct polynomials. Over $F = \mathbb{F}_p$ the witness specialises to $X^p - X$: by Fermat's little theorem every element of $\mathbb{F}_p$ is a root of the monic degree-p polynomial $X^p - X$, so theorem 5.36 factors it as $\prod_{\alpha \in \mathbb{F}_p} (X - \alpha)$—the p distinct linear factors exhaust the degree—and $X^p - X$ induces the zero function despite having degree $p \geq 1$. □

Remark 5.44 (Why the formal viewpoint is the correct one). Part (2) of theorem 5.43 is precisely why polynomials are defined as coefficient sequences and not as functions. Cryptography works over finite fields, where the functional view loses information: the polynomials X and X^p induce the same function on $\mathbb{F}_p$, yet have degrees 1 and p. And the *degree* of a polynomial—a property of the formal object, invisible to the induced function over a finite field—is what the security of polynomial commitment schemes rests on. A prover commits to a low-degree *formal* polynomial; the verifier tests it by *evaluation*; and the gap between agreement-as-function at the tested points and identity to the claimed formal polynomial is exactly what the Schwartz–Zippel lemma, next, quantifies.

5.10 The Schwartz–Zippel lemma

Only one preliminary is needed to state the payoff theorem. To draw α *uniformly at random* from a finite nonempty set S means to select each element with equal likelihood $1/|S|$; the *probability* of an event is then the fraction of elements of S producing it, so that $\Pr[f(\alpha) = 0] = |\{\alpha \in S : f(\alpha) = 0\}|/|S|$. The systematic language of probability is developed later in the volume; this counting case is all the theorem needs.

Theorem 5.45 (Univariate Schwartz–Zippel)**.** *Let $f \in F[X]$ be nonzero of degree at most d, let*

> *$S \subseteq F$ be finite and nonempty, and let α be drawn uniformly at random from S. Then*
>
> $$\Pr\big[f(\alpha) = 0\big] \le \frac{d}{|S|}. \tag{5}$$
>
> *More generally, if $f, g \in F[X]$ are distinct, each of degree at most d, then* $\Pr\big[f(\alpha) = g(\alpha)\big] \le d/|S|$.

Proof. By theorem 5.18, the nonzero f has at most d roots in F, hence at most d in the subset S. Under the uniform draw,

$$\Pr\big[f(\alpha) = 0\big] = \frac{|\{\alpha \in S : f(\alpha) = 0\}|}{|S|} \le \frac{d}{|S|}.$$

For the general form, apply this to $h = f - g$, which is nonzero and has degree at most d by (1), noting that $f(\alpha) = g(\alpha)$ exactly when $h(\alpha) = 0$. □

Remark 5.46 (The lemma stated plainly, and used). Inequality (5) formalises the slogan: *two low-degree polynomials that agree at many points must be equal.* The deterministic version is theorem 5.20—agreement at $d + 1$ points forces equality—and Schwartz–Zippel is its probabilistic refinement: a *single* random evaluation point detects unequal polynomials of degree at most d, except with probability at most $d/|S|$. With $|S|$ exponentially large—taking $S = \mathbb{F}_q$ with $q \approx 2^{254}$, the 255-bit fields underlying Orchard—and d polynomially bounded, the failure probability is astronomically small; the term *negligible* receives its precise asymptotic meaning later in the volume. This is the foundational soundness argument for polynomial identity testing, and hence for the polynomial commitment schemes and arithmetisation checks of Halo 2: to verify a purported polynomial identity, evaluate both sides at a random challenge and compare.

Remark 5.47 (Multivariate generalisation). The full Schwartz–Zippel lemma extends to polynomials in several indeterminates $f \in F[X_1, \dots, X_n]$: a nonzero polynomial of total degree d—the largest sum of exponents $i_1 + \cdots + i_n$ over the monomials with nonzero coefficient—vanishes at a uniformly random point of S^n with probability at most $d/|S|$. The univariate case proved here is both the base case of the multivariate inductive proof and the version most directly used: the later volumes invoke only the univariate lemma directly. Where an identity in several random challenges arises—for instance a permutation argument drawing a pair of challenges (β, γ)—the multivariate bound applies and follows from the univariate lemma applied once per variable.

The cheque drawn in the section's opening can now be cashed in full. Two parties hold polynomials f and g of degree at most d over a 255-bit field $\mathbb{F}_q$—even with d in the millions, they agree on a uniformly random challenge $\alpha \in \mathbb{F}_q$ with probability at most $d/q \approx d \cdot 2^{-254}$ unless they are literally the same formal polynomial. One evaluation apiece, one comparison of field elements, and the enormous coefficient lists never change hands: that is the trade the proving system makes on every identity it checks, and the entire cost of the bargain is the elementary fact with which the section began—a nonzero polynomial of degree d over a field has at most d roots.

6 Finite fields

A processor stores a number in a word of fixed width. A 64-bit word can hold exactly 2^{64} distinct values, and the hardware's arithmetic wraps around on overflow: what the machine natively computes is addition and multiplication modulo 2^{64}. For a verifier that must add, multiply, *and divide*, this built-in arithmetic falls at the last hurdle: in $\mathbb{Z}/2^{64}\mathbb{Z}$ no even class has an inverse, since a class is a unit only when its representative is coprime to the modulus (Theorem 2.24). The question this section answers is the natural one to ask next. Among all finite arithmetics—finite sets closed under an addition and a multiplication obeying the ring laws—which ones allow division by *every* nonzero element? In the language of §4: what are the finite fields, what sizes do they come in, and what structure do they carry?

Part of the answer is already in hand. The residue ring $\mathbb{Z}/n\mathbb{Z}$ is a field exactly when n is prime (theorem 4.26), giving one finite field $\mathbb{F}_p$ for every prime p; and the quotient recipe of Example 4.22 promised further examples of prime-power size p^k. This section completes the picture with two theorems. The first, theorem 6.9, is a prohibition: *no other sizes occur*—every finite field has exactly p^k elements for some prime p, so there is no field with 6, 10, or 12 elements. This constrains cardinalities, not binary representation widths: fields of size 2^k exist for every positive k. The second, theorem 6.16, is the section's payoff and one of the most consequential facts in the volume: the multiplicative group $\mathbb{F}_q^\times$ is *cyclic*—the powers of a single well-chosen element sweep out every nonzero value. Cyclicity is what makes discrete logarithms meaningful and what guarantees the roots of unity on which the polynomial machinery of later sections runs.

The route is as follows. We first make the prime field $\mathbb{F}_p$ computational, with two inversion algorithms and the square-and-multiply technique that powers one of them. We then determine the possible sizes of a finite field through the characteristic and the prime subfield, meet the Frobenius map $x \mapsto x^p$—the symmetry peculiar to characteristic p—and finally prove cyclicity by a counting argument of Gauss.

6.1 The prime field and two inversion routes

The field $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ certified by theorem 4.26 is the workhorse of the entire series, and the first order of business is to compute in it. Addition, subtraction, and multiplication are residue arithmetic as in §2; the operation that needs an algorithm is inversion. As §4 stressed, an existence proof and an algorithm are different assets, and cryptography needs both. For inversion in $\mathbb{F}_p$ there are two standard routes, with different characters.

Construction 6.1 (Inversion by the extended Euclidean algorithm)**.** Given a with $0 < a < p$, run the extended Euclidean algorithm (Theorem 2.13) on the pair (a, p). Since p is prime and $p \nmid a$, the gcd is 1, and the algorithm outputs integers u, v with $au + pv = 1$. Reducing modulo p gives $au \equiv 1 \pmod p$, that is,

$$a^{-1} = [u]_p \quad \text{in } \mathbb{F}_p.$$

The cost is $O(\log p)$ division steps (Theorem 2.13). This is the standard general-purpose method; a worked instance is Example 2.25, which computed $[17]_{43}^{-1} = [38]_{43}$ by exactly this route.

The second route rests on Fermat's little theorem and on a technique for fast exponentiation that is used throughout cryptography, so we record the technique first, in the generality it deserves.

Construction 6.2 (Square-and-multiply)**.** Let a be an element of any structure with an associative multiplication, and let $e \geq 1$ have binary expansion $e = \sum_i e_i 2^i$ with digits $e_i \in \{0, 1\}$. Squaring repeatedly produces the values

$$a^{2^0} = a, \qquad a^{2^{i+1}} = \left(a^{2^i}\right)^2,$$

and multiplying together the factors indexed by the nonzero digits gives

$$a^e = \prod_{i\,:\,e_i = 1} a^{2^i}.$$

The computation uses $\lfloor \log_2 e \rfloor$ squarings and at most $\lfloor \log_2 e \rfloor$ further multiplications—$O(\log e)$ multiplications in all, against the $e - 1$ multiplications of the naive product $a \cdot a \cdots a$. For the 255-bit exponents of later sections the difference is that between a few hundred multiplications and a count that no conceivable computer could finish.

Remark 6.3. Written additively, the same algorithm computes the multiple $e \cdot P$ of a group element P by repeated doubling and selective adding. Under the name *double-and-add* it is the algorithm for scalar multiplication on elliptic curves (theorem 10.17), where the group operation is written $+$; the two names denote one algorithm in two notations.

Construction 6.4 (Inversion by Fermat's little theorem)**.** Let $a \in \mathbb{F}_p$ be nonzero. Fermat's little theorem (Corollary 3.32) gives $a^{p-1} = 1$, hence $a \cdot a^{p-2} = 1$, and so

$$a^{-1} = a^{p-2},$$

computed by square-and-multiply (theorem 6.2) with exponent $p-2$, at a cost of $O(\log p)$ modular multiplications when $p > 2$. For $p = 2$, the only nonzero element is 1, whose inverse is 1; the exponent is zero and no multiplication is needed.

Remark 6.5 (Constant-time inversion). The two routes are asymptotically comparable, but they differ in a respect invisible to asymptotics and central to practice. In theorem 6.4 the sequence of operations performed—which steps square, which multiply—depends only on the binary digits of the *public* exponent $p - 2$, never on the secret input a; the method is *constant-time-friendly*. The extended Euclidean route, by contrast, performs a sequence of divisions whose lengths and quotients depend on a itself, so its running time and memory-access pattern can leak information about a to an observer timing the computation. Side-channel-resistant implementations therefore often prefer the Fermat route despite its slightly larger constant factor.

6.2 Characteristic, prime subfield, and prime-power order

We turn to the sizes question: which cardinalities can a finite field have? The tool is the characteristic of section 4.5. Recall the canonical homomorphism

$$\iota\colon \mathbb{Z} \to F, \qquad \iota(n) = n \cdot 1_F,$$

whose kernel is an ideal $n\mathbb{Z}$ of $\mathbb{Z}$; the characteristic $\mathrm{char}(F)$ is its nonnegative generator (Definition 4.30).

Theorem 6.6. *Every finite field F has prime characteristic:* $\mathrm{char}(F) = p$ *for some prime p. In particular* $\mathrm{char}(\mathbb{F}_p) = p$.

Proof. The map ι cannot be injective, since its domain $\mathbb{Z}$ is infinite and its codomain F is finite; two distinct integers $m < n$ must satisfy $\iota(m) = \iota(n)$, and then $n - m$ is a nonzero element of $\ker \iota$. Hence $\ker \iota = n\mathbb{Z}$ with $n > 0$, that is, $\mathrm{char}(F) > 0$; and the characteristic of a field, being nonzero, is prime by Theorem 4.32. For $F = \mathbb{F}_p$ itself, Example 4.31 computed $\mathrm{char}(\mathbb{Z}/p\mathbb{Z}) = p$ directly. □

The characteristic locates a canonical copy of $\mathbb{F}_p$ inside every field of characteristic p. Remark 4.33 described this copy informally; we now make it precise, beginning with the notion of a subfield.

Definition 6.7 (Subfield; prime subfield)**.** A *subfield* of a field F is a subset $K \subseteq F$ containing 0 and 1 and closed under addition, negation, multiplication, and inversion of nonzero elements; with the restricted operations, K is itself a field. The *prime subfield* of F is the intersection of all subfields of F—equivalently, the smallest subfield of F, and equivalently again the subfield generated by 1_F.

Two words on the equivalences. The intersection of any family of subfields is again a subfield, since each defining closure property survives intersection; and the intersection of *all* subfields is contained in every subfield, hence is the smallest one. That this smallest subfield is exactly what

1_F generates—the closure of $\{1_F\}$ under the field operations—follows from the classification below, whose proof exhibits the prime subfield inside the closure of $\{1_F\}$.

> **Proposition 6.8** (Classification of prime subfields)**.** *Let F be a field. If* $\operatorname{char}(F) = p > 0$*, the prime subfield of F is isomorphic to $\mathbb{F}_p$. If* $\operatorname{char}(F) = 0$*, it is isomorphic to $\mathbb{Q}$.*

Proof. Suppose first that $\operatorname{char}(F) = p > 0$, so that $\ker \iota = p\mathbb{Z}$. Define

$$\bar{\iota}\colon\ \mathbb{F}_p = \mathbb{Z}/p\mathbb{Z} \to F, \qquad \bar{\iota}([a]_p) = a \cdot 1_F.$$

The map is well defined and injective at a stroke: $a \cdot 1_F = b \cdot 1_F$ holds if and only if $(a - b) \cdot 1_F = 0$, i.e. $a - b \in \ker \iota = p\mathbb{Z}$, i.e. $[a]_p = [b]_p$. It is a ring homomorphism because ι is one (section 4.5) and $\bar{\iota}$ merely re-reads ι on residue classes. Its image $K = \{\, a \cdot 1_F \,:\, a \in \mathbb{Z}\,\}$ is therefore a subring of F isomorphic to the field $\mathbb{F}_p$, and K is in fact a subfield: for $[a]_p \neq [0]_p$ the computation $\bar{\iota}([a]_p)\,\bar{\iota}([a]_p^{-1}) = \bar{\iota}([1]_p) = 1_F$ exhibits the inverse of each nonzero element of K inside K. Finally, K lies in every subfield: a subfield contains 1_F, hence all its additive multiples $a \cdot 1_F$ by closure under addition and negation. So K is contained in the intersection of all subfields and is itself a subfield, forcing K to *be* that intersection: the prime subfield is $K \cong \mathbb{F}_p$.

Now suppose $\operatorname{char}(F) = 0$, so that ι is injective and $b \cdot 1_F \neq 0$ for every nonzero integer b. Define

$$j\colon\ \mathbb{Q} \to F, \qquad j\Big(\frac{a}{b}\Big) = (a \cdot 1_F)(b \cdot 1_F)^{-1}.$$

The map is well defined: if $a/b = c/d$ with $b, d \neq 0$, then $ad = bc$ in $\mathbb{Z}$, so $(a\cdot 1_F)(d\cdot 1_F) = (b\cdot 1_F)(c\cdot 1_F)$ by applying ι, and multiplying both sides by $(b \cdot 1_F)^{-1}(d \cdot 1_F)^{-1}$ gives $(a \cdot 1_F)(b \cdot 1_F)^{-1} = (c \cdot 1_F)(d \cdot 1_F)^{-1}$. That j preserves sums, products, and 1 is the usual arithmetic of fractions, transported by the homomorphism ι and commutativity. Injectivity is immediate: $j(a/b) = 0$ forces $a \cdot 1_F = 0$, hence $a = 0$. The image of j is thus a subfield of F isomorphic to $\mathbb{Q}$; and it lies in every subfield K, since K contains 1_F, hence every $a \cdot 1_F$, hence—being closed under inversion and multiplication—every $(a \cdot 1_F)(b \cdot 1_F)^{-1}$. As before the image must equal the intersection of all subfields, and the prime subfield is a copy of $\mathbb{Q}$. □

The size constraint now falls out of a counting device worth naming, for it recurs whenever a field sits over a subfield: *vector-space counting over the prime subfield.* Every field F of characteristic p is a vector space over its prime subfield $\mathbb{F}_p$: vectors are the elements of F, vector addition is the field addition, and scalar multiplication $c \cdot x$ for $c \in \mathbb{F}_p \subseteq F$ is the field multiplication, so the vector-space axioms are instances of the ring axioms of F. The proof below uses exactly two facts about vector spaces: a space with a finite spanning set has a basis, and coordinates with respect to a basis are unique. Both are stated and proved in the linear-algebra section (section 8), whose development is independent of the present theorem, so no circularity arises.

> **Theorem 6.9** (Finite fields have prime-power order)**.** *Every finite field F has cardinality $|F| = p^k$ for some prime p and integer $k \geq 1$, where $p =$* $\operatorname{char}(F)$*.*

Proof. By theorem 6.6 the characteristic of F is a prime p, and by theorem 6.8 the prime subfield of F is a copy of $\mathbb{F}_p$, with which we identify it. As explained above, F is then a vector space over $\mathbb{F}_p$. It is spanned by the finite set F itself, so it has a finite basis $e_1, \dots, e_k$; and $k \geq 1$ because $1 \neq 0$ makes F nonzero. Every element of F is thus

$$c_1 e_1 + c_2 e_2 + \cdots + c_k e_k$$

for a *unique* tuple $(c_1, \dots, c_k)$ with each $c_i \in \mathbb{F}_p$: existence because the e_i span, uniqueness because coordinates with respect to a basis are unique (section 8). The assignment $(c_1, \dots, c_k) \mapsto \sum_i c_i e_i$ is therefore a bijection from $\mathbb{F}_p^k$ to F, and counting tuples gives $|F| = p \cdot p \cdots p = p^k$. □

Fields of every prime-power size do exist, and are unique; we state the fact in full, but prove only the part the series stands on.

> **Theorem 6.10** (Existence and uniqueness of $\mathbb{F}_q$). *For every prime power $q = p^k$ there exists a field with exactly q elements, and any two finite fields of the same cardinality are isomorphic. One constructs such a field as the quotient $\mathbb{F}_p[x]/(f)$ of the polynomial ring by the ideal generated by an irreducible polynomial f of degree k—the recipe recorded in Example 4.22.*

Remark 6.11. The theorem is deliberately left unproved, and the omission costs the series nothing. The fields on the critical path of Halo 2 and Orchard are *prime* fields, the case $q = p$ with $k = 1$, and there the theorem is already proved: existence is theorem 4.26, and uniqueness is immediate from the classification, since a field with p elements has order $p^k = p$ with $k = 1$ by theorem 6.9, so it equals its own prime subfield, a copy of $\mathbb{F}_p$ (theorem 6.8). The general case $k \geq 2$ would require a development of polynomial factorisation over field extensions on which nothing later in the series relies, so we state it for orientation only. *Convention, fixed henceforth*: the letter $q = p^k$ denotes a prime power, and $\mathbb{F}_q$ a field with q elements, of characteristic p; results are stated for general $\mathbb{F}_q$ whenever the proof is identical to the prime case, and the reader focused on Halo 2 may read $q = p$ throughout.

6.3 The Frobenius endomorphism

Characteristic p is not merely a bookkeeping invariant; it changes what algebra is true. The most famous instance is the schoolroom error $(a+b)^p = a^p + b^p$, false over $\mathbb{Q}$ and $\mathbb{R}$ for every $p \geq 2$, which in characteristic p becomes a theorem.

> **Lemma 6.12** (Freshman's dream). *Let R be a commutative ring of prime characteristic p. Then for all $a, b \in R$,*
> $$(a + b)^p = a^p + b^p,$$
> *and more generally $(a + b)^{p^n} = a^{p^n} + b^{p^n}$ for every $n \geq 1$.*

Proof. The binomial theorem holds in any commutative ring: the usual induction on the exponent uses only distributivity and commutativity, so

$$(a + b)^p \;=\; \sum_{i=0}^{p} \binom{p}{i} a^i b^{p-i},$$

where the integer coefficient $\binom{p}{i}$ acts as the additive multiple $\binom{p}{i} \cdot x = x + \cdots + x$. We claim that $p \mid \binom{p}{i}$ for $0 < i < p$. From the factorial identity

$$\binom{p}{i}\, i!\,(p-i)! \;=\; p!$$

the prime p divides the right-hand side. It does not divide $i!\,(p-i)!$: that product is a product of integers all smaller than p, and if p divided the product it would divide one of the factors by Euclid's lemma for primes (Lemma 2.17), which is impossible. Applying Euclid's lemma to the left-hand side instead, p must divide $\binom{p}{i}$, say $\binom{p}{i} = pm$.

Now let $0 < i < p$ and consider the corresponding term. Writing $\iota(n) = n \cdot 1_R$ for the canonical homomorphism of section 4.5, the coefficient acts as multiplication by $\iota(\binom{p}{i}) = \iota(pm) = \iota(p)\,\iota(m) = 0 \cdot \iota(m) = 0$, since $\operatorname{char}(R) = p$ means exactly $\iota(p) = p \cdot 1_R = 0$. Every middle term of the binomial expansion therefore vanishes, and the terms $i = p$ and $i = 0$ leave $(a + b)^p = a^p + b^p$.

The general case follows by induction on n, the base case $n = 1$ being the identity just proved. For $n \geq 2$, apply the base case to the elements $a^{p^{n-1}}$ and $b^{p^{n-1}}$:

$$(a + b)^{p^n} = \left((a + b)^{p^{n-1}}\right)^p = \left(a^{p^{n-1}} + b^{p^{n-1}}\right)^p = a^{p^n} + b^{p^n},$$

using the inductive hypothesis in the middle step. □

The identity says that in characteristic p, raising to the p-th power respects addition—and it obviously respects multiplication. A map respecting both operations is a homomorphism, and a homomorphism from a structure to itself is called an *endomorphism*; this one has a name of its own.

Definition 6.13 (Frobenius endomorphism). Let F be a field of characteristic $p > 0$. The *Frobenius map* of F is

$$\mathrm{Frob} : F \to F, \qquad \mathrm{Frob}(x) = x^p.$$

Proposition 6.14. *Let F be a field of characteristic p. The Frobenius map is a ring homomorphism $F \to F$ fixing the prime subfield pointwise. If F is finite,* Frob *is an automorphism of F—a bijective ring homomorphism of F onto itself.*

Proof. Multiplicativity is commutativity and associativity alone: $(xy)^p = x^p y^p$. Additivity is the freshman's dream (Lemma 6.12), and $1^p = 1$; so Frob is a ring homomorphism.

Injectivity holds for a reason worth isolating, since it recurs: *every ring homomorphism between fields is injective*. Indeed, if $\phi : F \to F'$ had $\phi(x) = 0$ for some $x \neq 0$, then

$$1 = \phi(1) = \phi(x^{-1}x) = \phi(x^{-1})\,\phi(x) = \phi(x^{-1}) \cdot 0 = 0$$

in F', contradicting $1 \neq 0$ in a field. In particular Frob is injective.

The prime subfield is fixed pointwise. By theorem 6.8 its elements are the values $\bar{\iota}([a]_p) = a \cdot 1_F$, and since $\bar{\iota}$ is a ring homomorphism,

$$\mathrm{Frob}\big(\bar{\iota}([a]_p)\big) = \bar{\iota}([a]_p)^p = \bar{\iota}\big([a]_p^p\big) = \bar{\iota}\big([a^p]_p\big) = \bar{\iota}([a]_p),$$

the last step because $a^p \equiv a \pmod p$ for *all* integers a (Corollary 3.32).

Finally, let F be finite. An injective map from a finite set to itself is surjective—the injectivity-implies-surjectivity device of section 4.4—so Frob is a bijective ring homomorphism of F onto itself, an automorphism. □

The iterates $\mathrm{Frob}^n(x) = x^{p^n}$ are again homomorphisms, as the general case of the freshman's dream states directly. On the prime field $\mathbb{F}_p$ itself the Frobenius map is the identity; on larger fields of characteristic p it is a genuine, structure-preserving symmetry, and theorem 6.23 below records where it re-enters the cryptographic story.

6.4 Cyclicity of the multiplicative group

The multiplicative group $\mathbb{F}_q^\times$ of a finite field is a finite abelian group of order $q - 1$: every one of the $q - 1$ nonzero elements is invertible, precisely because $\mathbb{F}_q$ is a field. Finite abelian groups can in general be quite far from cyclic—the unit group $(\mathbb{Z}/8\mathbb{Z})^\times$ of Example 3.22 has order 4 with no element of order 4—so it is a genuinely strong fact that $\mathbb{F}_q^\times$ is always cyclic. The proof is a counting argument of Gauss, and its arithmetic engine is an identity about the totient function φ. Everything we use about φ is its *counting definition* (Definition 2.26): $\varphi(n)$ is the number of integers a with $1 \leq a \leq n$ and $\gcd(a, n) = 1$. No further totient theory is needed or assumed.

Lemma 6.15 (Gauss's divisor-sum identity). *For every integer $n \geq 1$,*

$$\sum_{d \mid n} \varphi(d) = n,$$

> *the sum running over the positive divisors d of n.*

Proof. Partition the set $\{1, 2, \dots, n\}$ according to the value of $\gcd(a, n)$. For each a the gcd is a positive divisor d of n, so the classes

$$C_d = \{\, a : 1 \le a \le n,\ \gcd(a, n) = d \,\}, \qquad d \mid n,$$

are disjoint and cover $\{1, \dots, n\}$; hence $n = \sum_{d \mid n} |C_d|$.

We claim $|C_d| = \varphi(n/d)$. A member of C_d is divisible by d, so it has the form $a = db$ with $1 \le b \le n/d$; we show that, for such a,

$$\gcd(db, n) = d \quad \Longleftrightarrow \quad \gcd(b, n/d) = 1.$$

For the forward direction, suppose $e = \gcd(b, n/d) > 1$; then de divides both $db = a$ and $d \cdot (n/d) = n$, so de is a common divisor exceeding d and $\gcd(a, n) \ge de > d$. For the converse, suppose $\gcd(b, n/d) = 1$. The integer d is a common divisor of a and n, so d divides $g = \gcd(a, n)$ by the strongest-common-divisor property (Remark 2.12); write $g = dc$ with $c \ge 1$. From $dc \mid db$ we get $c \mid b$, and from $dc \mid n = d(n/d)$ we get $c \mid n/d$ (cancel d in each divisibility); so c divides $\gcd(b, n/d) = 1$, forcing $c = 1$ and $g = d$.

The members of C_d are therefore exactly the integers db with $1 \le b \le n/d$ and $\gcd(b, n/d) = 1$, and by the counting definition of the totient (Definition 2.26) there are precisely $\varphi(n/d)$ of them. Summing,

$$n = \sum_{d \mid n} \varphi\Big(\frac{n}{d}\Big) = \sum_{d \mid n} \varphi(d),$$

the second equality because $d \mapsto n/d$ is a bijection of the set of positive divisors of n onto itself. □

The theorem now follows from a beautiful squeeze. We count the elements of each possible order twice over—once inside the group, once through the totient—and Gauss's identity leaves the two counts no room to differ.

> **Theorem 6.16** (Finite subgroups of $F^\times$ are cyclic)**.** *Let F be any field and let $G \le F^\times$ be a finite subgroup of the multiplicative group, of order N. Then G is cyclic. In particular, for a finite field $\mathbb{F}_q$ the whole multiplicative group $\mathbb{F}_q^\times$ is cyclic of order $q - 1$.*

Proof. For each divisor $d \mid N$, let

$$\psi(d) = |\{\, a \in G : \operatorname{ord}(a) = d \,\}|$$

count the elements of G of order exactly d. Every element of the finite group G has finite order dividing $N = |G|$ (Corollary 3.30), so the sets counted by the $\psi(d)$ partition G and

$$\sum_{d \mid N} \psi(d) = N. \tag{6}$$

The key claim is that *for each $d \mid N$*, either $\psi(d) = 0$ or $\psi(d) = \varphi(d)$. Suppose $\psi(d) > 0$ and choose $a \in G$ of order d. The powers $a^0, a^1, \dots, a^{d-1}$ are d distinct elements of G (Proposition 3.12(3)), and each satisfies

$$\left(a^j\right)^d = \left(a^d\right)^j = 1^j = 1,$$

so each is a root in F of the polynomial $X^d - 1 \in F[X]$. That polynomial has degree d, so it has *at most* d roots in F (Theorem 5.18); the d distinct powers of a therefore account for *all* of its roots. Now take any $b \in G$ of order d. Then $b^d = 1$, so b is a root of $X^d - 1$ and hence $b = a^j$ for some $0 \le j < d$. By Corollary 3.23,

$$\operatorname{ord}(a^j) = \frac{d}{\gcd(d, j)},$$

which equals d exactly when $\gcd(j,d) = 1$. The elements of order d in G are thus precisely the powers a^j with $0 \le j < d$ and $\gcd(j,d) = 1$, and by the counting definition of the totient there are $\varphi(d)$ of them: $\psi(d) = \varphi(d)$. This proves the claim, and with it the inequality $\psi(d) \le \varphi(d)$ for every $d \mid N$.

Summing the inequality over the divisors of N and comparing (6) with Gauss's identity (Lemma 6.15),

$$N = \sum_{d\mid N} \psi(d) \;\le\; \sum_{d\mid N} \varphi(d) = N.$$

The two ends are equal, so the inequality is an equality term by term: $\psi(d) = \varphi(d)$ for *every* divisor $d \mid N$. In particular

$$\psi(N) = \varphi(N) \ge 1,$$

since $a = 1$ always witnesses $\gcd(1,N) = 1$ in the totient count. Hence G contains an element g of order N; its powers form a subgroup of G with N distinct elements (Proposition 3.12(3)), which must be all of G. So $G = \langle g\rangle$ is cyclic.

For the final statement, let $F = \mathbb{F}_q$ be finite. The multiplicative group $\mathbb{F}_q^\times$ has $q-1$ elements—every nonzero element of a field is a unit—and is a finite subgroup of itself, so it is cyclic of order $q-1$. □

The equality $\psi(d) = \varphi(d)$ established along the way is worth extracting: it was proved for subgroups of a field's multiplicative group, but it holds in any cyclic group, by the same gcd computation and with no field in sight.

Corollary 6.17 (Element-order census in cyclic groups)**.** *A cyclic group G of order N contains, for each divisor $d \mid N$, exactly $\varphi(d)$ elements of order d, and no elements of any other order. In particular G has exactly $\varphi(N)$ generators.*

Proof. Write $G = \langle g\rangle$ with $\mathrm{ord}(g) = N$; the elements of G are the powers g^j, $0 \le j < N$, each appearing once (Proposition 3.12(3)). By Corollary 3.23, $\mathrm{ord}(g^j) = N/\gcd(N,j)$, which is always a divisor of N; and $\mathrm{ord}(g^j) = d$ if and only if $\gcd(j,N) = N/d$. By the gcd computation in the proof of Lemma 6.15 (applied with $n = N$ and divisor N/d), the exponents $j \in \{1,\dots,N\}$ with $\gcd(j,N) = N/d$ are exactly $j = (N/d)\,t$ with $1 \le t \le d$ and $\gcd(t,d) = 1$, of which there are $\varphi(d)$ by Definition 2.26. The generators are the elements of order N, numbering $\varphi(N)$. □

Definition 6.18 (Primitive root)**.** A generator of the cyclic group $\mathbb{F}_q^\times$ is called a *primitive root* (or *primitive element*) of $\mathbb{F}_q$. Equivalently, an element $g \in \mathbb{F}_q^\times$ is primitive if and only if $\mathrm{ord}(g) = q-1$.

Existence is theorem 6.16, and the census refines it: $\mathbb{F}_q$ has exactly $\varphi(q-1)$ primitive roots (theorem 6.17), always at least one.

Remark 6.19. The name has a second reading, developed later in the volume: a primitive root of $\mathbb{F}_q$ is precisely a primitive $(q-1)$-th root of unity in the sense of theorem 9.3, where the identification is restated once roots of unity have been defined in general. The two vocabularies—group-theoretic "generator of $\mathbb{F}_q^\times$" and polynomial-flavoured "primitive $(q-1)$-th root of unity"—name the same elements.

Example 6.20 (Primitive roots of $\mathbb{F}_7$)**.** The group $\mathbb{F}_7^\times$ has order 6. Testing $g = 3$, the successive powers modulo 7 are

$$3^1 = 3, \quad 3^2 = 2, \quad 3^3 = 6, \quad 3^4 = 4, \quad 3^5 = 5, \quad 3^6 = 1,$$

which run through all of $\{1,\dots,6\}$: the element 3 is a primitive root of $\mathbb{F}_7$. By contrast 2 is not: its powers are $2, 4, 1$, so $\mathrm{ord}(2) = 3 < 6$. The full roster of primitive roots consists of the powers 3^j with $\gcd(j,6) = 1$ (Corollary 3.23), namely $j = 1$ and $j = 5$: the primitive roots of $\mathbb{F}_7$ are $3^1 = 3$ and $3^5 = 5$, and there are $\varphi(6) = 2$ of them, as the census predicts.

Cyclicity settles, in one stroke, how many solutions the equation $x^n = 1$ has in a finite field—the question on which the evaluation domains of later sections turn.

> **Corollary 6.21** (Roots of unity in $\mathbb{F}_q$). *Let $\mathbb{F}_q$ be a finite field and $n \geq 1$. The solutions of $x^n = 1$ in $\mathbb{F}_q^\times$ number exactly* $\gcd(n, q-1)$*, and they form the cyclic subgroup of $\mathbb{F}_q^\times$ of that order. In particular, when $n \mid q-1$ the equation has exactly n solutions.*

Proof. Fix a primitive root g (theorem 6.16) and write each element of $\mathbb{F}_q^\times$ uniquely as $x = g^j$ with $0 \leq j \leq q-2$ (Proposition 3.12(3)). Then $x^n = g^{jn}$, and $g^{jn} = 1$ holds if and only if $(q-1) \mid jn$ (Proposition 3.12(1)). Set $d = \gcd(n, q-1)$ and write $q-1 = dm$, $n = dn'$ with $\gcd(m, n') = 1$. Then

$$(q-1) \mid jn \iff dm \mid j\,dn' \iff m \mid jn' \iff m \mid j,$$

the last step by Euclid's lemma (Proposition 2.15(1), with $\gcd(m, n') = 1$). The qualifying exponents in $\{0, \dots, q-2\}$ are therefore the multiples of $m = (q-1)/d$, namely $j = tm$ for $t = 0, 1, \dots, d-1$: exactly d of them. The corresponding solutions $x = (g^m)^t$ are exactly the powers of g^m, which by Corollary 3.23 has order $(q-1)/\gcd(q-1, m) = (q-1)/m = d$; so the solution set is the cyclic subgroup $\langle g^m \rangle$ of order $d = \gcd(n, q-1)$. When $n \mid q-1$ we have $d = n$, giving exactly n solutions. □

Remark 6.22 (Two-adicity and proof-system fields). The corollary is precisely what renders a curve such as Pallas or Vesta suitable for Halo 2: one chooses the field size q so that $q-1$ is divisible by a large power of 2, and the corollary then *guarantees* a multiplicative subgroup of 2-power order—the evaluation domain on which the number-theoretic transform, and hence efficient polynomial arithmetic, operates. The construction of these domains and the transform itself are the business of section 9; the curves are introduced in section 10.

Remark 6.23 (Cyclicity and Frobenius in cryptography). The section's two structural discoveries each carry cryptographic weight. First, the cyclicity of $\mathbb{F}_q^\times$ (theorem 6.16) underlies the discrete-logarithm problem—writing every nonzero element as g^j invites the question of recovering j, and the presumed hardness of doing so is a foundational assumption of later volumes—and, through theorem 6.21, guarantees the existence of the roots of unity used in polynomial commitment schemes. Second, the Frobenius endomorphism (theorem 6.13) is not merely abstract: on an elliptic curve over $\mathbb{F}_p$ it induces an endomorphism of the curve whose characteristic polynomial $T^2 - tT + p$ encodes the number of curve points as $N = p + 1 - t$, where t is the Hasse trace—tying the size of the cryptographic group directly to the field-theoretic Frobenius. Both connections are taken up in section 10.

The question that opened the section is now answered in full. A finite arithmetic in which every nonzero element inverts exists only at prime-power sizes—theorem 6.9 forbids every other cardinality—and at each admissible size $q = p^k$ there is exactly one field, up to isomorphism (theorem 6.10). The machine's own wrap-around arithmetic is instructive here: 2^{64} is a prime-power size, so a field with 2^{64} elements exists, but it is *not* the ring $\mathbb{Z}/2^{64}\mathbb{Z}$ of word arithmetic—whose even classes have no inverses—but the polynomial-quotient construction of theorem 6.10, with a different multiplication altogether. The proof systems of this series instead take the prime route: a prime p of roughly 255 bits, the field $\mathbb{F}_p$, inversion by theorem 6.1 or theorem 6.4. The payoff theorem then delivers the structure that everything later rides on: the multiplicative group of the chosen field is a single cycle, generated by one element, with exactly $\varphi(d)$ elements of each order $d \mid q-1$ and a guaranteed subgroup of every size dividing $q-1$. Polynomials over these fields, the subject that the volume develops next into evaluation domains, fast transforms, and commitment schemes, will draw on that guarantee constantly.

7 Number theory for cryptography

An implementation that transmits a point of an elliptic curve—a pair (x, y) of field elements satisfying an equation $y^2 = x^3 + ax + b$, as constructed in section 10—need not send both coordinates. For a given x the equation determines y^2 exactly, hence determines y up to sign, so wire formats send x together with a single bit selecting one of the two candidates, and the transmission halves in size. The receiver is then left with two number-theoretic tasks: decide whether $x^3 + ax + b$ is a square modulo p at all—if not, no point has this x-coordinate—and, if it is, *compute* a square root of it. Both tasks must run in the time of a few modular exponentiations, on numbers hundreds of bits long. The same constructions make a complementary demand in the opposite direction: they publish powers g^x of a known group element while the exponent x is a secret key, so exponentiation must be easy to perform and yet infeasible to *invert*. This section assembles the classical number theory behind both demands: the totient and the Chinese Remainder Theorem, Fermat's little theorem and Euler's theorem in their role as workhorses of modular exponentiation, multiplicative orders and primitive roots, quadratic residues with the Euler criterion, square-root extraction modulo a prime, and finally the discrete logarithm problem together with the reason *prime* group order is so important for cryptographic hardness.

Throughout, n and m denote positive integers and p denotes a prime. The section builds directly on the arithmetic of section 2, the group theory of section 3, and the cyclicity theorem of section 6; several statements proved there reappear here, restated in the arithmetic language in which cryptography uses them, with proofs reduced to citations.

7.1 Euler's totient function and the Chinese Remainder Theorem

Recall from section 2 that a class $[a]_n \in \mathbb{Z}/n\mathbb{Z}$ is a unit if and only if $\gcd(a, n) = 1$ (Theorem 2.24), that the units form a finite abelian group $(\mathbb{Z}/n\mathbb{Z})^\times$ (Proposition 3.8), and that Euler's totient $\varphi(n)$ counts them: $\varphi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times|$ (Definition 2.26). In particular $\varphi(p) = p-1$ for a prime p (Example 2.27), so $\mathbb{F}_p^\times$ has order $p-1$. These counts govern everything that follows: the totient is the exponent in Euler's theorem, the order of the group whose cyclic structure the next subsections exploit, and—through the orders of elliptic-curve groups—a quantity that the parameter choices of Halo 2 and Orchard are built around. What section 2 did not provide is a way to *compute* $\varphi(n)$ beyond direct enumeration. Two facts close the gap: the totient's value on prime powers, and its behaviour on coprime factors.

> **Proposition 7.1** (Totient of a prime power). *For a prime p and an integer $k \geq 1$,*
>
> $$\varphi(p^k) = p^k - p^{k-1} = p^k\Big(1 - \frac{1}{p}\Big).$$

Proof. Among $1, \dots, p^k$, we claim the integers *not* coprime to p^k are exactly the multiples of p. If $p \mid a$ then $p \mid \gcd(a, p^k)$, so the gcd exceeds 1. Conversely, if $g = \gcd(a, p^k) > 1$, then g has a prime factor ℓ; from $\ell \mid g \mid p^k$ and Euclid's lemma for primes (Lemma 2.17) we get $\ell = p$, and $\ell \mid g \mid a$ gives $p \mid a$. The multiples of p in the range are $p, 2p, \dots, p^{k-1} \cdot p$, of which there are p^{k-1}; subtracting them from the p^k integers in the range leaves $\varphi(p^k) = p^k - p^{k-1}$. □

The behaviour on coprime factors is a consequence of a structural theorem that will be used again, in a quite different role, at the end of the section: a residue modulo a product of two coprime moduli carries exactly the same information as a *pair* of residues, one modulo each factor. The natural home for the statement is the language of rings (section 4): recall the product ring $R \times S$ with componentwise operations (Example 4.5), and call a bijective ring homomorphism (Definition 4.21) a *ring isomorphism*, mirroring Definition 3.41 for groups.

> **Theorem 7.2** (Chinese Remainder Theorem). *If $\gcd(m, n) = 1$, then the map*
>
> $$\mathbb{Z}/mn\mathbb{Z} \longrightarrow \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}, \qquad [a]_{mn} \longmapsto \big([a]_m, [a]_n\big)$$

is a ring isomorphism. It restricts to a group isomorphism

$$(\mathbb{Z}/mn\mathbb{Z})^\times \;\cong\; (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times.$$

Proof. The map is well defined: if $[a]_{mn} = [a']_{mn}$ then $mn \mid a - a'$, hence $m \mid a - a'$ and $n \mid a - a'$, so the two components agree. It is a ring homomorphism because each component is: reduction modulo m (and modulo n) sends sums to sums, products to products, and 1 to 1 (Theorem 4.6), and the product ring's operations are componentwise.

For injectivity it suffices, since the map is in particular a homomorphism of additive groups, to check that the kernel is trivial (Proposition 3.40). Suppose $[a]_m = [0]_m$ and $[a]_n = [0]_n$, that is, $m \mid a$ and $n \mid a$. Since $\gcd(m, n) = 1$, Euclid's lemma (Proposition 2.15(2)) gives $mn \mid a$, i.e. $[a]_{mn} = [0]_{mn}$.

Both sides have exactly mn elements (Theorem 4.6 for each factor), and an injective map between finite sets of equal size is bijective: the mn distinct inputs have mn distinct images, which must exhaust the codomain. Hence the map is a ring isomorphism.

For the multiplicative statement, a ring isomorphism ϕ matches units with units: if $uv = 1$ then $\phi(u)\phi(v) = \phi(1) = 1$, and the same applies to ϕ^{-1}. The units of the product ring are exactly the pairs of units, $(u, v)^{-1} = (u^{-1}, v^{-1})$ existing precisely when both components are invertible. Restricting ϕ to unit groups therefore gives a bijection $(\mathbb{Z}/mn\mathbb{Z})^\times \to (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times$ that respects multiplication: a group isomorphism. □

Corollary 7.3 (Multiplicativity and the product formula). *If* $\gcd(m, n) = 1$ *then* $\varphi(mn) = \varphi(m)\,\varphi(n)$. *Hence if* $n = \prod_i p_i^{k_i}$ *is the prime factorisation of n, then*

$$\varphi(n) = \prod_i \left(p_i^{k_i} - p_i^{k_i - 1}\right) = n \prod_{p \mid n} \Big(1 - \frac{1}{p}\Big),$$

the last product running over the distinct primes dividing n.

Proof. The unit-group isomorphism of Theorem 7.2 is in particular a bijection, and the cardinality of a product of two sets is the product of the cardinalities, so $\varphi(mn) = \varphi(m)\,\varphi(n)$.

For the product formula, the prime powers $p_i^{k_i}$ are pairwise coprime: a common divisor exceeding 1 of two of them would have a prime factor dividing both, which by Euclid's lemma for primes (Lemma 2.17) would equal both p_i and p_j. Moreover each $p_i^{k_i}$ is coprime to the product of the others, by repeated application of Proposition 2.15(3). Induction on the number of factors therefore extends multiplicativity to $\varphi(n) = \prod_i \varphi\big(p_i^{k_i}\big)$, and Proposition 7.1 evaluates each factor:

$$\varphi(n) = \prod_i \left(p_i^{k_i} - p_i^{k_i - 1}\right) = \prod_i p_i^{k_i}\Big(1 - \frac{1}{p_i}\Big) = n \prod_{p \mid n} \Big(1 - \frac{1}{p}\Big). \qquad \square$$

As a sanity check, the product formula gives $\varphi(12) = 12\big(1 - \frac{1}{2}\big)\big(1 - \frac{1}{3}\big) = 4$, agreeing with the direct enumeration $\{1, 5, 7, 11\}$ of Example 2.27. Together with Gauss's divisor-sum identity $\sum_{d \mid n} \varphi(d) = n$, proved as Lemma 6.15, this completes the totient's basic theory.

7.2 Fermat's little theorem and Euler's theorem

The congruences of this subsection were proved once, as Corollary 3.32, where they fell out of Lagrange's theorem in three lines. We restate them here in the arithmetic form in which cryptography uses them; the proofs are citations, not repetitions.

Theorem 7.4 (Euler's theorem). *If* $\gcd(a, n) = 1$*, then*

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Proof. The hypothesis says that $[a]_n$ is a unit (Theorem 2.24), i.e. an element of the finite group $(\mathbb{Z}/n\mathbb{Z})^\times$, whose order is $\varphi(n)$ (Definition 2.26). By Corollary 3.31—every element g of a finite group of order N satisfies $g^N = e$, a consequence of Lagrange's theorem—we get $[a]_n^{\varphi(n)} = [1]_n$, which is the stated congruence. This is Corollary 3.32 verbatim. □

Corollary 7.5 (Fermat's little theorem). *Let p be prime. If $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$. Moreover $a^p \equiv a \pmod{p}$ for* every *integer a.*

Proof. For $p \nmid a$ we have $\gcd(a, p) = 1$, so Theorem 7.4 with $n = p$ and $\varphi(p) = p - 1$ (Example 2.27) gives the first congruence; multiplying it by a gives $a^p \equiv a \pmod{p}$. When $p \mid a$ both a^p and a are congruent to 0, so the second congruence holds for all a. □

Remark 7.6 (Cryptographic use of Euler and Fermat). Euler's theorem underlies RSA: if $ed \equiv 1 \pmod{\varphi(n)}$, say $ed = 1 + k\varphi(n)$, then

$$(a^e)^d = a^{1+k\varphi(n)} = a \cdot \left(a^{\varphi(n)}\right)^k \equiv a \pmod{n}$$

whenever $\gcd(a, n) = 1$: raising to the e-th power is undone by raising to the d-th, which is what makes one exponent a public encryption key and the other its private inverse. The elliptic-curve cryptography of Orchard instead exploits Fermat's little theorem in the field $\mathbb{F}_p$: every nonzero x satisfies $x^{p-1} = 1$, so inversion is computable as $x^{-1} = x^{p-2}$ (theorem 6.4), and Fermat is the engine of the Euler criterion for square roots developed below.

7.3 Multiplicative order and primitive roots

Definition 7.7 (Multiplicative order). Let $\gcd(a, n) = 1$. The *multiplicative order* of a modulo n, written $\mathrm{ord}_n(a)$, is the least positive integer k with $a^k \equiv 1 \pmod{n}$. Such a k exists because $a^{\varphi(n)} \equiv 1$ by Theorem 7.4.

The quantity $\mathrm{ord}_n(a)$ is exactly the order of $[a]_n$ as an element of the group $(\mathbb{Z}/n\mathbb{Z})^\times$, so the order theory of section 3 applies verbatim: $a^m \equiv 1 \pmod{n}$ if and only if $\mathrm{ord}_n(a) \mid m$ (Proposition 3.12(1))—in particular $\mathrm{ord}_n(a) \mid \varphi(n)$, by Euler's theorem—and

$$\mathrm{ord}_n\left(a^j\right) = \frac{\mathrm{ord}_n(a)}{\gcd\left(j, \mathrm{ord}_n(a)\right)}$$

(Corollary 3.23).

Definition 7.8 (Primitive root). A *primitive root* modulo n is an element g with $\mathrm{ord}_n(g) = \varphi(n)$; equivalently, g generates $(\mathbb{Z}/n\mathbb{Z})^\times$ as a cyclic group, so that $\{g^0, g^1, \ldots, g^{\varphi(n)-1}\}$ lists all the units.

Primitive roots need not exist for every modulus. The unit group $(\mathbb{Z}/8\mathbb{Z})^\times = \{1, 3, 5, 7\}$ has order $\varphi(8) = 4$, yet every one of its elements squares to 1 (Example 3.22): the element orders are 1 for the identity and 2 for each of 3, 5, 7, so no element has order 4 and no primitive root modulo 8 exists. Indeed the unit group is a copy of the Klein four-group of Example 3.10: every element is its own inverse, and the product of two distinct non-identity elements a and b is neither e (which would force $b = a^{-1} = a$) nor a nor b (which would force $b = e$ or $a = e$), so it is the third. The decisive case for cryptography is $n = p$ prime, where the multiplicative group is always cyclic.

Corollary 7.9 ($\mathbb{F}_p^\times$ is cyclic). *For every prime p, the group $\mathbb{F}_p^\times = (\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic of order $p-1$. In particular primitive roots modulo p exist, and there are exactly $\varphi(p-1)$ of them; Example 6.20 exhibits the two primitive roots 3 and 5 of $\mathbb{F}_7$.*

Proof. The group $\mathbb{F}_p^\times$ is a finite subgroup, of order $p-1$, of the multiplicative group of the field $\mathbb{F}_p$, hence cyclic by Theorem 6.16. A generator g has order $p-1$, and the powers g^k that again generate are those with $\gcd(k, p-1) = 1$ (Corollary 3.23), numbering $\varphi(p-1)$. □

Remark 7.10 (Cyclicity and prime-order groups). Cyclicity is why the prime-order groups of elliptic-curve cryptography behave exactly like $\mathbb{Z}/q\mathbb{Z}$ under addition once a generator is fixed: a cyclic group of order q is isomorphic to $(\mathbb{Z}/q\mathbb{Z}, +)$ via $g^k \leftrightarrow k$ (Theorem 3.43). For completeness we record, without proof and without needing it later, the full classification: primitive roots modulo n exist exactly when $n \in \{1, 2, 4, p^k, 2p^k\}$ for an odd prime p.

7.4 Quadratic residues and the Euler criterion

We turn to squares modulo a prime. They control whether a curve point with a given x-coordinate exists—the first of the two tasks posed at the head of the section—and they are the objects from which square roots in $\mathbb{F}_p$ will be extracted.

Definition 7.11 (Quadratic residue). Let p be an odd prime and a an integer with $p \nmid a$. We say a is a *quadratic residue* modulo p (a QR) if the congruence $x^2 \equiv a \pmod{p}$ has a solution, and a *quadratic non-residue* (a QNR) otherwise.

Proposition 7.12 (Exactly half are residues). *For an odd prime p, exactly $\frac{p-1}{2}$ of the nonzero residues modulo p are quadratic residues and $\frac{p-1}{2}$ are non-residues. The squaring map $x \mapsto x^2$ is exactly two-to-one on $\mathbb{F}_p^\times$.*

Proof. Consider $\sigma : \mathbb{F}_p^\times \to \mathbb{F}_p^\times$, $\sigma(x) = x^2$; its image is exactly the set of quadratic residues. For $x, y \in \mathbb{F}_p^\times$,

$$\sigma(x) = \sigma(y) \iff x^2 - y^2 = 0 \iff (x-y)(x+y) = 0 \iff y = \pm x,$$

the middle equivalence because a field has no zero divisors (Proposition 4.28). Since p is odd, $x \neq -x$ for $x \neq 0$ (otherwise $2x = 0$ with 2 and x nonzero), so every fibre of σ over its image has exactly two elements. Counting, $|\operatorname{im}\sigma| = \frac{p-1}{2}$, and the complement—the non-residues—has the same size. □

Theorem 7.13 (Euler's criterion). *Let p be an odd prime and $p \nmid a$. Then*

$$a^{\frac{p-1}{2}} \equiv \begin{cases} 1 \pmod{p} & \text{if } a \text{ is a quadratic residue,} \\ -1 \pmod{p} & \text{if } a \text{ is a quadratic non-residue.} \end{cases}$$

Proof. First, the value is always ± 1. Fermat's little theorem (Corollary 7.5) gives $\left(a^{(p-1)/2}\right)^2 = a^{p-1} \equiv 1$, and the only square roots of 1 modulo a prime are ± 1: from $x^2 \equiv 1$ we get $p \mid (x-1)(x+1)$, so $p \mid x-1$ or $p \mid x+1$ by Euclid's lemma for primes (Lemma 2.17), i.e. $x \equiv \pm 1$. Hence $a^{(p-1)/2} \equiv \pm 1 \pmod{p}$.

If a is a residue, say $a \equiv x^2$ with $p \nmid x$, then $a^{(p-1)/2} \equiv x^{p-1} \equiv 1$, again by Fermat. Thus every one of the $\frac{p-1}{2}$ quadratic residues (Proposition 7.12) is a root of the polynomial $t^{(p-1)/2} - 1$ over $\mathbb{F}_p$.

That polynomial has degree $\frac{p-1}{2}$, so it has *at most* $\frac{p-1}{2}$ roots (Theorem 5.18); the residues therefore account for all of them. A non-residue a is consequently not a root, yet still satisfies $a^{(p-1)/2} \equiv \pm 1$; the remaining possibility is $a^{(p-1)/2} \equiv -1$. $\square$

The standard notation for quadratic-residue status is the Legendre symbol.

Definition 7.14 (Legendre symbol)**.** For an odd prime p and an integer a, the *Legendre symbol* is

$$\left(\frac{a}{p}\right) = \begin{cases} 0 & \text{if } p \mid a, \\ 1 & \text{if } a \text{ is a quadratic residue modulo } p, \\ -1 & \text{if } a \text{ is a quadratic non-residue modulo } p. \end{cases}$$

Euler's criterion (Theorem 7.13) then reads, uniformly,

$$\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p},$$

so the symbol is computable by a single modular exponentiation—an $O(\log p)$ affair by square-and-multiply (theorem 6.2). The criterion also shows that the symbol is *completely multiplicative* in a:

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right), \qquad \text{since} \qquad (ab)^{\frac{p-1}{2}} = a^{\frac{p-1}{2}} b^{\frac{p-1}{2}}.$$

Both observations carry weight in the worked example that follows, which verifies a constant deployed in Orchard.

Example 7.15 (13 is a non-square in both Pasta fields)**.** The Pasta hash-to-curve construction used by Orchard—a procedure, taken up in later volumes, that maps arbitrary byte strings to curve points and requires a fixed non-square constant in the base field—fixes $Z = -13$ as a verifiably non-square constant in both base fields. Let p and q be the Pasta primes of section 10; both satisfy $p \equiv q \equiv 1 \pmod{4}$. We verify the non-squareness here, working modulo p; the computation modulo q is identical.

First, -1 is a square modulo any prime $\equiv 1 \pmod{4}$: the exponent $\frac{p-1}{2}$ is even, so Euler's criterion gives

$$\left(\frac{-1}{p}\right) \equiv (-1)^{\frac{p-1}{2}} = 1.$$

By complete multiplicativity,

$$\left(\frac{-13}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{13}{p}\right) = \left(\frac{13}{p}\right),$$

so -13 is a non-square precisely when 13 is. Whether 13 is a square is settled by one Euler-criterion exponentiation in each field, a computation any computer-algebra system reproduces in milliseconds:

$$13^{\frac{p-1}{2}} \equiv p - 1 \equiv -1 \pmod{p}, \qquad 13^{\frac{q-1}{2}} \equiv q - 1 \equiv -1 \pmod{q},$$

whence $\left(\frac{13}{p}\right) = \left(\frac{13}{q}\right) = -1$. Thus 13, and with it $Z = -13$, is a quadratic non-residue modulo both Pasta primes, exactly as the hash-to-curve design requires.

7.5 Square roots modulo a prime

Deciding that a is a residue is one matter; *producing* an x with $x^2 \equiv a$ is another, and the point decompression of the section's opening needs the production, not just the decision: the receiver must reconstruct the coordinate y from y^2. How hard this is depends on $p \bmod 4$.

Proposition 7.16 (The $p \equiv 3 \pmod 4$ shortcut). *Let $p \equiv 3 \pmod 4$ be prime and let a be a quadratic residue modulo p with $p \nmid a$. Then*

$$x \equiv a^{\frac{p+1}{4}} \pmod p$$

is a square root of a, and the two square roots of a are $\pm x$.

Proof. Since $p \equiv 3 \pmod 4$, the quantity $\frac{p+1}{4}$ is a positive integer. Compute

$$x^2 = a^{\frac{p+1}{2}} = a^{\frac{p-1}{2}} \cdot a \equiv \left(\frac{a}{p}\right) \cdot a = a \pmod p,$$

using Euler's criterion (Theorem 7.13) and the hypothesis that a is a residue, so $\left(\frac{a}{p}\right) = 1$. The roots come in the pair $\pm x$ by the two-to-one property of squaring (Proposition 7.12). □

Remark 7.17 (Cost and the Pasta primes). The single-exponentiation shortcut requires $p \equiv 3 \pmod 4$, which is one reason cryptographers often favour such primes. For $p \equiv 5 \pmod 8$ a closed-form variant (Atkin's formula) still exists. In general the cost of square-root extraction grows with the 2-adic valuation s of $p-1$: writing $p - 1 = 2^s d$ with d odd, the Tonelli–Shanks algorithm below performs up to s corrective iterations. The Pasta primes have $s = 32$, a value chosen deliberately for FFT-friendliness (theorem 9.10), so their implementations use Tonelli–Shanks.

When $p \equiv 1 \pmod 4$ one writes $p - 1 = 2^s d$ with d odd and $s \geq 2$. The algorithm operates inside the subgroup of $\mathbb{F}_p^\times$ of order 2^s—the unique such subgroup, since $\mathbb{F}_p^\times$ is cyclic (Corollary 7.9 and Theorem 3.24(3))—where the obstruction to the naive exponentiation trick lives, and clears that obstruction one power of two at a time.

Theorem 7.18 (Tonelli–Shanks). *Let p be an odd prime, $p - 1 = 2^s d$ with d odd, and let a be a quadratic residue with $p \nmid a$. Given any quadratic non-residue z modulo p, one can compute a square root of a modulo p using $O(s^2 + \log p)$ modular multiplications.*

Proof. Write $H \leq \mathbb{F}_p^\times$ for the unique cyclic subgroup of order 2^s. Set $c = z^d$. We claim c generates H. By Euler's criterion (Theorem 7.13) $z^{(p-1)/2} \equiv -1$, so

$$c^{2^{s-1}} = z^{2^{s-1}d} = z^{\frac{p-1}{2}} \equiv -1 \neq 1, \qquad c^{2^s} = z^{p-1} \equiv 1$$

by Fermat (Corollary 7.5); hence $\mathrm{ord}_p(c)$ divides 2^s but not 2^{s-1}, forcing $\mathrm{ord}_p(c) = 2^s$ and $\langle c \rangle = H$.

Initialise

$$x = a^{\frac{d+1}{2}}, \qquad t = a^d, \qquad m = s$$

(the exponent $\frac{d+1}{2}$ is an integer because d is odd). We maintain three invariants:

1. $x^2 = a\,t$;
2. the current c has order exactly 2^m;
3. the order of t divides 2^{m-1}.

Initially (1) holds since $x^2 = a^{d+1} = a \cdot a^d$; (2) was just proved; and (3) holds because $t^{2^{s-1}} = a^{(p-1)/2} \equiv 1$ by Euler's criterion (Theorem 7.13), a being a residue.

Now iterate. If $t = 1$, then $x^2 = a$ by (1) and we are done. Otherwise let $2^i = \mathrm{ord}_p(t)$, found as the least $i \geq 1$ with $t^{2^i} = 1$ by repeatedly squaring t; invariant (3) gives $1 \leq i \leq m - 1$. Set

$$b = c^{2^{m-i-1}},$$

and update

$$x \leftarrow xb, \qquad t \leftarrow tb^2, \qquad c \leftarrow b^2, \qquad m \leftarrow i.$$

The invariants survive. For (1), $(xb)^2 = x^2b^2 = a\,t\,b^2$, which is a times the new t. For (2), the old c has order 2^m, so $\mathrm{ord}_p(b) = 2^m/\gcd\big(2^m, 2^{m-i-1}\big) = 2^{i+1}$ (Corollary 3.23), and the new $c = b^2$ has order 2^i—exactly 2 to the new m. For (3), both the old t and b^2 have order exactly 2^i, so $t^{2^{i-1}}$ and $(b^2)^{2^{i-1}}$ each have order 2; the only element of order 2 in $\mathbb{F}_p^\times$ is -1, because the square roots of 1 are ± 1 (as in the proof of Theorem 7.13); hence

$$\big(t\,b^2\big)^{2^{i-1}} = t^{2^{i-1}}\big(b^2\big)^{2^{i-1}} = (-1)(-1) = 1,$$

so the new t has order dividing 2^{i-1}, which is 2^{m-1} for the new m.

Each iteration strictly decreases m, from s through a descending chain of nonnegative integers, so after at most s iterations the order of t reaches $2^0 = 1$, i.e. $t = 1$, and x is the desired root. For the cost: locating i and computing b each take at most $m \le s$ squarings, so the loop costs $O(s)$ multiplications per iteration and $O(s^2)$ in total; the three initial exponentiations, with exponents smaller than p, cost $O(\log p)$ multiplications each by square-and-multiply (theorem 6.2). □

The one ingredient the theorem presupposes is a quadratic non-residue z, and here—for the only time in this section—randomness enters. A uniformly random element of $\mathbb{F}_p^\times$ is a non-residue with probability exactly $\frac{1}{2}$, since precisely half the elements qualify (Proposition 7.12), and each candidate is checked by one Euler-criterion exponentiation (Definition 7.14), i.e. $O(\log p)$ modular multiplications. A handful of random trials therefore finds z almost immediately, and implementations simply hard-code one non-residue per field, verified once—as the constant $Z = -13$ of Example 7.15 is. No deterministic polynomial-time method for finding a non-residue is invoked anywhere in this volume.

Remark 7.19 (Which root, and detecting failure). In every case the square roots of a residue come in a pair $\pm x$ (Proposition 7.12), so a *canonical* root must be pinned down by a convention—for instance "choose the root whose least nonnegative representative is even", as elliptic-curve point encodings do; the compressed point's extra bit selects between the two. If a is *not* a residue, both algorithms above detect the failure, in different ways. The shortcut of Proposition 7.16 still outputs a candidate $a^{(p+1)/4}$, and that candidate fails the final check $x^2 \equiv a$—one squaring suffices to detect failure. The variable-time Tonelli–Shanks algorithm just given cannot complete its update on a non-residue: Euler's criterion gives $a^{(p-1)/2} \equiv -1$, so $t = a^d$ satisfies $t^{2^{s-1}} \equiv -1$ and has order exactly 2^s, and already the first search for the least $i \ge 1$ with $t^{2^i} = 1$ returns $i = m$, which invariant (3) of the proof of Theorem 7.18 forbids and for which the update exponent 2^{m-i-1} is undefined. This variant can report "not a residue" the moment $i = m$ occurs; equivalently it may verify $a^{(p-1)/2} \equiv 1$ up front, at the same $O(\log p)$ cost as the initial exponentiations.

7.6 The discrete logarithm problem

The second demand of the section's opening—exponentiation easy forward, infeasible backward—now takes centre stage. We phrase it for a generic cyclic group, written multiplicatively, because in Orchard the group in question is the group of points of an elliptic curve, written additively, of large prime order; every statement below transfers by renaming the operation.

> **Definition 7.20** (Discrete logarithm problem). Let $G = \langle g \rangle$ be a finite cyclic group of order N with generator g. Given $h \in G$, the *discrete logarithm* of h to base g, written $\log_g h$, is the unique residue $x \in \mathbb{Z}/N\mathbb{Z}$ with $g^x = h$. The *discrete logarithm problem* (DLP) is to compute x from (G, g, h).

Existence and uniqueness of $\log_g h$ are the content of the classification of cyclic groups: the map $x \mapsto g^x$ is a group isomorphism $(\mathbb{Z}/N\mathbb{Z}, +) \xrightarrow{\sim} G$ (Theorem 3.43, whose map $\bar{\phi}$ this is), so it has a well-defined inverse, and $\log_g$ is that inverse. The two directions could not differ more in cost. Computing

g^x from x takes $O(\log N)$ group operations by square-and-multiply (theorem 6.2); computing x from g^x is the DLP, for which no efficient algorithm is known in well-chosen groups. This asymmetry—a bijection cheap in one direction and expensive in the other—is the one-wayness on which elliptic-curve cryptography rests.

Remark 7.21 (Generic complexity and group choice). Two *generic* algorithms solve the DLP in any group of order N, using nothing but the group operation. *Baby-step giant-step* is deterministic: set $m = \lceil\sqrt{N}\rceil$ and write the unknown as $x = im + j$ with $0 \le i, j < m$; tabulate the "baby steps" g^j for all j, then walk the "giant steps" $h(g^{-m})^i$ for $i = 0, 1, 2, \dots$ until one equals a tabulated g^j, at which point $hg^{-im} = g^j$ gives $x = im + j$. Both the table and the walk are $O(\sqrt{N})$, so the cost is $O(\sqrt{N})$ time *and* space. *Pollard's rho* (not developed here) is a randomised alternative with the same $O(\sqrt{N})$ expected running time—on a heuristic analysis, made explicit in section 10—but only $O(1)$ space, and is what attackers would run in practice. In the other direction, Shoup's theorem shows that $\Omega(\sqrt{N})$ group operations are *necessary* for any generic algorithm when N is prime, so for generic attacks the square-root cost is exact. Reaching a λ-bit security level—no attack cheaper than 2^λ operations—therefore requires $N \approx 2^{2\lambda}$. This is why the Pallas and Vesta scalar fields have prime order roughly 255 bits long, targeting roughly 128-bit security (section 10). The qualifier *generic* matters: in $(\mathbb{Z}/p\mathbb{Z})^\times$ sub-exponential index-calculus attacks exist, which exploit the ring structure of the integers behind the group, and this is why elliptic-curve groups—where no such attack is known—permit much smaller parameters than multiplicative groups of comparable security.

The next theorem makes the phrase "well-chosen" precise: if the group order N is composite with small prime factors, the DLP falls apart into small pieces, and the Chinese Remainder Theorem—proved in this section for a purely arithmetic purpose—reassembles the pieces into the secret.

Theorem 7.22 (Pohlig–Hellman reduction). *Let $G = \langle g\rangle$ be cyclic of order $N = \prod_{i=1}^r q_i^{e_i}$, the q_i distinct primes. The discrete logarithm in G is computable by solving discrete logarithms in subgroups of prime order q_i—e_i of them for each i—and combining the answers with the Chinese Remainder Theorem, at a total cost of*

$$O\Big(\sum_{i=1}^{r} e_i\big(\log N + \sqrt{q_i}\,\big)\Big)$$

group operations. When N has a large prime factor $q_{\max}$ the $\sqrt{q_{\max}}$ term dominates; when N is smooth*—a product of primes bounded polynomially in* $\log N$*—every term is polynomial in* $\log N$ *and the DLP is easy. Corollary 7.23 draws the design consequence.*

Proof. Let $h = g^x$ with x the unknown. The reduction has two stages.

Stage (a): splitting across coprime factors. For each i set $N_i = N/q_i^{e_i}$ and project:

$$g_i = g^{N_i}, \qquad h_i = h^{N_i}.$$

By Corollary 3.23, $\operatorname{ord}(g_i) = N/\gcd(N, N_i) = N/N_i = q_i^{e_i}$, so g_i generates the unique subgroup of G of order $q_i^{e_i}$ (Theorem 3.24(3)). Moreover $h_i = g^{xN_i} = g_i^x$, and since g_i has order $q_i^{e_i}$, the exponent matters only modulo $q_i^{e_i}$ (Proposition 3.12(2)): writing $x_i = x \bmod q_i^{e_i}$,

$$\log_{g_i} h_i = x_i.$$

Solving the r smaller DLPs yields $x \bmod q_i^{e_i}$ for every i; since the prime powers $q_i^{e_i}$ are pairwise coprime, the Chinese Remainder Theorem (Theorem 7.2, iterated across the factors as in the proof of Corollary 7.3) determines x modulo N uniquely. Each projection costs $O(\log N)$ operations by square-and-multiply.

Stage (b): lifting a prime power digit by digit. Fix one prime power q^e (dropping the subscript i), with $\gamma = g_i$ of order q^e and $\eta = h_i = \gamma^{x'}$, $x' = x_i$. Write x' in base q:

$$x' \equiv x_0 + x_1 q + \cdots + x_{e-1}q^{e-1} \pmod{q^e}, \qquad 0 \le x_j < q.$$

The element $\delta = \gamma^{q^{e-1}}$ has order q (Corollary 3.23 again). Raise η to the power q^{e-1}: every digit beyond the first is multiplied by a multiple of q^e in the exponent and vanishes, leaving

$$\eta^{q^{e-1}} = \gamma^{x' q^{e-1}} = \delta^{x_0},$$

a single DLP in the group $\langle\delta\rangle$ of prime order q, which baby-step giant-step (Remark 7.21) solves in $O(\sqrt{q})$ operations. With $x_0, \dots, x_{j-1}$ in hand, strip them off and repeat: the element $\eta_j = \eta\,\gamma^{-(x_0 + x_1 q + \cdots + x_{j-1} q^{j-1})}$ equals γ raised to $x_j q^j$ + (multiples of q^{j+1}), so

$$\eta_j^{q^{e-1-j}} = \delta^{x_j},$$

and digit x_j is again one order-q DLP. Each of the e digits costs one such DLP plus $O(\log N)$ operations for the exponentiations.

Summing over the digits of all the prime-power factors gives $\sum_i e_i$ order-q_i DLPs at $O(\sqrt{q_i})$ apiece plus $O(\log N)$ bookkeeping for each, which is the stated bound. □

> **Corollary 7.23** (Why prime order matters). *If N has a small prime factor q, an attacker can recover the discrete logarithm modulo q in $O(\log N + \sqrt{q})$ group operations—$O(\log N)$ to project into the order-q subgroup and $O(\sqrt{q})$ for the small DLP there—leaking part of the secret x. Security therefore requires N to have a large prime factor; the safest and simplest choice is to make N itself a large prime, so that G has no proper nontrivial subgroups and the Pohlig–Hellman reduction offers no purchase at all. This is precisely why the elliptic-curve groups used in Halo 2 and Orchard have large prime order.*

Proof. For the first claim, run stage (a) of Theorem 7.22 for the single factor q: the projections $g^{N/q}$, $h^{N/q}$ cost $O(\log N)$ operations, and the resulting DLP in a group of order q costs $O(\sqrt{q})$ by baby-step giant-step—yielding $x \bmod q$. For the second, if N is prime then G has no subgroups besides $\{e\}$ and G (Corollary 3.34), so the reduction's only output is the original problem, and the generic $\Omega(\sqrt{N})$ floor of Remark 7.21 stands at full height. □

Remark 7.24 (Cofactors and small-subgroup attacks). In practice an elliptic curve over $\mathbb{F}_p$ has group order $h \cdot q$, where q is a large prime and h is a small *cofactor*, and cryptographic operations stay confined to the subgroup of prime order q. If a protocol neglects to check that incoming points lie in that subgroup (or fails to clear the cofactor), an adversary can supply a point of low order and run Pohlig–Hellman in the small factor to extract information about the secret modulo h—a real-world *small-subgroup attack*. The Pasta curves of Halo 2 have cofactor $h = 1$: their group orders are themselves prime (the primes recorded in section 10), which eliminates this class of attacks at the level of the group itself rather than by protocol-level checks.

The two demands with which the section opened are now both met. A receiver handed a compressed point (x, bit) decides in one modular exponentiation whether $x^3 + ax + b$ is a square—Euler's criterion—and, when it is, extracts a root by Proposition 7.16 or by Tonelli–Shanks (Theorem 7.18), the transmitted bit selecting between the pair $\pm y$ (Remark 7.19); for the Pasta fields, with their deliberately large 2-adic valuation $s = 32$, Tonelli–Shanks is the deployed route. On the security side, exponentiation runs in $O(\log N)$ operations while its inverse, the discrete logarithm, costs any generic attacker $\Omega(\sqrt{N})$—a floor that composite group orders would undermine through Pohlig–Hellman, and that the prime, roughly 255-bit orders of the Pasta groups preserve in full, with no small subgroup left to attack. The structure of $\mathbb{Z}/q\mathbb{Z}$ for a large prime q thus supplies both the functionality—a field of scalars acting on the group—and the security of the constructions built on it. The curves themselves, and the primes p and q that this section has treated as black boxes, are the business of section 10.

8 Linear algebra over a field

Begin with the engineering problem that shapes the whole section. A prover holds a list of n secret numbers—the intermediate values of some large computation, the *wire values* of an *arithmetic circuit* in the vocabulary of later volumes, with n in the millions, say—and must hand a verifier a *receipt* for the list: a single short object that pins the list down, to be opened or argued about later. Two demands make the problem interesting. The receipt must be *one* element of some fixed algebraic structure, however large n grows; and receipts must *add*—whoever holds the receipts for two lists, together with two scaling factors, must be able to compute the receipt for the scaled sum of the lists without ever seeing the lists themselves. The construction that meets both demands is disarmingly brief: fix n public elements $G_1, \dots, G_n$ of a suitable group, and issue for the list $(a_1, \dots, a_n)$ the single group element $[a_1]G_1 + \cdots + [a_n]G_n$. Whether this can possibly work—what "receipts add" should mean exactly, how one group element can absorb n secrets, and what price in ambiguity the compression exacts—are questions of *linear algebra*, and answering them is the business of this section.

Linear algebra is the study of *vector spaces*—collections of objects that can be added together and scaled by elements of a field—together with the structure-preserving maps between them. It is the computational backbone of the cryptography this monograph serves: commitments to vectors, the linear constraints that encode arithmetic circuits, the polynomial-evaluation arguments at the heart of the inner-product argument, and the Pedersen-style multiscalar multiplications of Halo 2 and Orchard are all, at bottom, statements about linear maps and (bi)linear forms over a finite field. The section develops the theory from the ground up, assuming only the notion of a field (section 4), and then specialises to the two settings that matter here: the coordinate space F^n, in practice taken over a finite field $\mathbb{F}_q$, and the "mixed" pairing between vectors of scalars and vectors of group elements.

The section also settles two debts contracted below it. Remark 5.5 promised that the linear structure of $F[X]$—closure under addition and scaling, with the monomials as an infinite coordinate system—would be studied systematically in due course; the study happens here. And the proof of Theorem 6.9 borrowed, against a promissory note, exactly two facts about vector spaces: that a space with a finite spanning set has a basis, and that coordinates with respect to a basis are unique. The first is proved in Theorem 8.17; the second follows from linear independence, as recorded in Definition 8.14. These arguments use nothing from the theory of finite fields, so the note is honoured without circularity. Throughout, F denotes an arbitrary field with identities 0 and 1; the field axioms of Definition 4.24 are used freely and without citation.

8.1 Vector spaces and subspaces

Definition 8.1 (Vector space)**.** Let F be a field. A *vector space over F* (an *F-vector space*) is a set V equipped with two operations, *vector addition* $+ : V \times V \to V$ and *scalar multiplication* $\cdot : F \times V \to V$, such that for all $u, v, w \in V$ and all $a, b \in F$:

1. $(V, +)$ is an abelian group: addition is associative and commutative, there is a *zero vector* $0 \in V$ with $v + 0 = v$, and every v has an *additive inverse* $-v$ with $v + (-v) = 0$;
2. scalar multiplication is compatible with the multiplication of F: $a \cdot (b \cdot v) = (ab) \cdot v$;
3. the identity of F acts as the identity: $1 \cdot v = v$;
4. scalar multiplication distributes over vector addition: $a \cdot (u + v) = a \cdot u + a \cdot v$;
5. scalar multiplication distributes over scalar addition: $(a + b) \cdot v = a \cdot v + b \cdot v$.

The elements of V are called *vectors*, the elements of F *scalars*. We write av for $a \cdot v$, and we allow the scalar $0 \in F$ and the vector $0 \in V$ to share a symbol whenever context distinguishes them.

A few consequences of the axioms recur so often that we record them once and, thereafter, use them without comment.

Proposition 8.2. *Let V be an F-vector space, $v \in V$, and $a \in F$. Then:*

1. $0 \cdot v = 0$;
2. $a \cdot 0 = 0$;
3. $(-1) \cdot v = -v$;
4. *if* $av = 0$, *then* $a = 0$ *or* $v = 0$.

Proof. (1) By distributivity over scalar addition, $0 \cdot v = (0+0) \cdot v = 0 \cdot v + 0 \cdot v$; adding $-(0 \cdot v)$ to both sides leaves $0 = 0 \cdot v$.

(2) Likewise $a \cdot 0 = a \cdot (0+0) = a \cdot 0 + a \cdot 0$, and cancelling in the abelian group $(V, +)$ gives $a \cdot 0 = 0$.

(3) Using the identity axiom, distributivity, and (1),

$$v + (-1) \cdot v = 1 \cdot v + (-1) \cdot v = \big(1 + (-1)\big) \cdot v = 0 \cdot v = 0,$$

so $(-1) \cdot v$ is the additive inverse of v; by uniqueness of inverses in a group, $(-1) \cdot v = -v$.

(4) Suppose $av = 0$ and $a \neq 0$. Since F is a field, a^{-1} exists, and by the identity and compatibility axioms

$$v = 1 \cdot v = (a^{-1}a) \cdot v = a^{-1} \cdot (av) = a^{-1} \cdot 0 = 0,$$

the last step by (2). □

Example 8.3 (The coordinate space F^n)**.** The single most important example is the *coordinate space*

$$F^n = \{ (x_1, \dots, x_n) : x_i \in F \},$$

the set of n-tuples of scalars, under componentwise operations:

$$(x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n), \qquad a \cdot (x_1, \dots, x_n) = (ax_1, \dots, ax_n).$$

The zero vector is $(0, \dots, 0)$, and every axiom of theorem 8.1 follows from the corresponding field axiom applied in each coordinate. The degenerate case $n = 0$ is the *trivial* (or *zero*) vector space $\{0\}$, consisting of the empty tuple alone. As a standing convention we write vectors of F^n in boldface and think of them as columns, $\mathbf{x} = (x_1, \dots, x_n)^\mathsf{T}$.

Example 8.4 (Further examples)**.**

1. *Polynomials.* The polynomials $F[X]$ form an F-vector space under addition and the scalar multiplication of Remark 5.5. So does the subset

$$F[X]_{<n} = \{ f \in F[X] : f = 0 \text{ or } \deg f < n \}$$

of polynomials of degree less than n: sums and scalar multiples of such polynomials again have degree less than n (or are zero), by Proposition 5.7, so the operations restrict and the axioms are inherited. This space is exactly the setting in which polynomial commitment schemes operate.

2. *Functions.* For any set S, the set F^S of all functions $S \to F$ is an F-vector space under pointwise operations: $(f+g)(s) = f(s) + g(s)$ and $(af)(s) = a\,f(s)$. Example 8.3 is the special case $S = \{1, \dots, n\}$, a tuple being a function of its index.

3. *Matrices.* The set $F^{m \times n}$ of $m \times n$ arrays of scalars is an F-vector space under entrywise addition and scaling—the case of (2) in which S is the grid of index pairs (i, j).

4. *Field extensions.* If F is a subfield of a larger field K (§4), then K is an F-vector space: vector addition is the addition of K, and scalar multiplication is the multiplication of K restricted to pairs in $F \times K$, so the axioms are instances of the field axioms of K. The dimension of this vector space—in the sense of theorem 8.17 below—is called the *degree* of the extension, written $[K : F]$. For instance $\mathbb{C}$ is an $\mathbb{R}$-vector space of dimension 2, with basis $1, i$, so $[\mathbb{C} : \mathbb{R}] = 2$; and the finite field $\mathbb{F}_q$ with $q = p^k$ is a k-dimensional $\mathbb{F}_p$-vector space, $[\mathbb{F}_q : \mathbb{F}_p] = k$: this is precisely the structure on which the proof of Theorem 6.9 rested.

A vector space often contains smaller vector spaces sitting inside it, sharing its operations; the solution sets of homogeneous linear equations, the spans of the next subsection, and the kernels and images of linear maps are all of this kind.

Definition 8.5 (Subspace)**.** A subset $W \subseteq V$ of an F-vector space V is a *(linear) subspace* if W is itself a vector space under the operations inherited from V; equivalently—as the next proposition makes precise—if and only if W is nonempty and closed under the vector-space operations.

Proposition 8.6 (Subspace criterion)**.** *A subset $W \subseteq V$ is a subspace if and only if*

1. $0 \in W$;
2. *W is closed under addition: $u, v \in W$ implies $u + v \in W$; and*
3. *W is closed under scalar multiplication: $a \in F$ and $v \in W$ imply $av \in W$.*

Equivalently, W is a subspace if and only if $W \neq \varnothing$ and $au + bv \in W$ for all $a, b \in F$ and $u, v \in W$.

Proof. If W is a subspace, then (1)–(3) hold: closure (2) and (3) is what it means for the inherited operations to be operations *on* W, and the zero vector of W is the zero vector of V, since a zero vector 0_W of W satisfies $0_W + 0_W = 0_W$, and cancelling in V gives $0_W = 0$.

Conversely, suppose (1)–(3) hold. Closure makes the two operations well defined on W. The additive inverse of any $v \in W$ is $(-1)v \in W$, by (3) and Proposition 8.2(3). Every remaining axiom of theorem 8.1 is a universally quantified identity—associativity, commutativity, compatibility, the two distributive laws, the action of 1—and each holds for all elements of V, in particular for all elements of W. Hence W is a vector space.

For the two-scalar form: if W is a subspace, then $au+bv \in W$ by (2) and (3) combined, and $W \ni 0$ is nonempty. Conversely, choosing $(a, b) = (1, 1)$ recovers (2), choosing $(a, b) = (a, 0)$ recovers (3) (using Proposition 8.2(1) to identify $0v$), and $0 = 0v \in W$ for any v in the nonempty W, which is (1). □

Example 8.7. In F^3 the set $W = \{(x_1, x_2, x_3) : x_1 + x_2 + x_3 = 0\}$ is a subspace: it contains 0, and if the coordinates of x and y each sum to 0, then those of $a\mathrm{x} + b\mathrm{y}$ sum to $a \cdot 0 + b \cdot 0 = 0$. By contrast $\{(x_1, x_2, x_3) : x_1 + x_2 + x_3 = 1\}$ is *not* a subspace: it does not contain 0. The same computation shows that the solution set of any single homogeneous linear equation $c_1x_1 + \cdots + c_nx_n = 0$ (the c_i fixed scalars) is a subspace of F^n, and the solution set of a finite *system* of such equations, being an intersection of subspaces, is a subspace by the next proposition. These solution sets return shortly in structural guise, as the kernels of linear maps (theorem 8.18).

Proposition 8.8 (Intersections and sums)**.** *Let V be an F-vector space.*

1. *The intersection $\bigcap_{i \in I} W_i$ of an arbitrary family of subspaces of V is a subspace.*

2. *The* sum $W_1 + \cdots + W_k = \{ w_1 + \cdots + w_k : w_j \in W_j \}$ *of finitely many subspaces is a subspace, and it is the smallest subspace of V containing every W_j.*

The union *of two subspaces is in general* not *a subspace.*

Proof. (1) Each W_i contains 0, so the intersection does. If u, v lie in every W_i and $a, b \in F$, then $au + bv \in W_i$ for every i by Proposition 8.6, hence $au + bv$ lies in the intersection, which is therefore a subspace by the same criterion.

(2) The sum contains $0 = 0 + \cdots + 0$, and it contains each W_j (take every other summand 0). For closure, take two elements $\sum_j w_j$ and $\sum_j w'_j$ with $w_j, w'_j \in W_j$, and scalars a, b; regrouping,

$$a \sum_j w_j + b \sum_j w'_j = \sum_j (aw_j + bw'_j),$$

and $aw_j + bw'_j \in W_j$ since W_j is a subspace, so the combination lies in the sum. Finally, any subspace U containing every W_j contains each summand w_j of any element of the sum, hence contains the element itself by closure under addition; so the sum is contained in U, and is the smallest such subspace.

For the union, take $V = F^2$ (any field has $|F| > 1$, since $1 \neq 0$) and the two axes $\{(x, 0) : x \in F\}$ and $\{(0, y) : y \in F\}$—subspaces by the criterion. Their union contains $(1, 0)$ and $(0, 1)$ but not the sum $(1, 1) = (1, 0) + (0, 1)$, since $(1, 1)$ lies on neither axis; closure under addition fails. □

8.2 Linear combinations, span, and linear independence

The subspaces met so far were carved out by equations. The complementary way to produce a subspace is to build one up from chosen vectors, taking everything their repeated addition and scaling can reach.

Definition 8.9 (Linear combination and span)**.** Let $S \subseteq V$. A *linear combination* of vectors in S is any finite sum

$$a_1 v_1 + a_2 v_2 + \cdots + a_k v_k, \qquad a_i \in F,\ v_i \in S.$$

The *span* of S, written $\operatorname{span}(S)$ or $\langle S \rangle$, is the set of all linear combinations of finitely many vectors of S; by convention $\operatorname{span}(\varnothing) = \{0\}$. The set S *spans* (or *generates*) V if $\operatorname{span}(S) = V$, and V is *finitely generated* (synonymously, *finite-dimensional*) if some finite set spans it.

Proposition 8.10. *For any $S \subseteq V$, the span* $\operatorname{span}(S)$ *is the smallest subspace of V containing S: it is a subspace, it contains S, and it is contained in every subspace of V that contains S.*

Proof. The zero vector is the empty linear combination (or, for nonempty S, the combination $0v$), and a sum of two linear combinations of vectors in S, or a scalar multiple of one, is again such a linear combination—concatenate the lists and rescale the coefficients. By Proposition 8.6, $\operatorname{span}(S)$ is a subspace, and it contains each $v \in S$ as the combination $1 \cdot v$. If W is any subspace containing S, then closure under addition and scaling forces every linear combination of elements of S into W, by induction on the number of summands; hence $\operatorname{span}(S) \subseteq W$. □

Definition 8.11 (Linear independence)**.** A finite list of vectors $v_1, \ldots, v_k \in V$ is *linearly independent* if the only way to express the zero vector as a linear combination of the list is with all coefficients zero:

$$a_1 v_1 + \cdots + a_k v_k = 0 \quad \Longrightarrow \quad a_1 = \cdots = a_k = 0.$$

Otherwise the list is *linearly dependent*, and a relation $\sum_i a_i v_i = 0$ with some $a_i \neq 0$ is called a *dependence relation*. An arbitrary—possibly infinite—set $S \subseteq V$ is linearly independent if every finite list of distinct vectors from S is linearly independent.

Remark 8.12. Linear independence captures *non-redundancy*: no vector of an independent list contributes anything the others already provide. In the smallest cases the definition unwinds to familiar statements. A single vector v is independent if and only if $v \neq 0$ (Proposition 8.2(4)). Two vectors are dependent if and only if one is a scalar multiple of the other. In general, a list is dependent exactly when some vector in it lies in the span of the others—the content of the next lemma.

Lemma 8.13 (Dependence and redundancy)**.** *A list $v_1, \dots, v_k$ with $k \geq 1$ is linearly dependent if and only if some v_j is a linear combination of the others. Moreover, if $v_1, \dots, v_k$ is independent but $v_1, \dots, v_k, w$ is dependent, then $w \in \operatorname{span}(v_1, \dots, v_k)$, and the coefficients expressing w are unique.*

Proof. Suppose first that $v_j = \sum_{i \neq j} a_i v_i$. Then

$$\sum_{i \neq j} a_i v_i + (-1)\, v_j = 0$$

is a dependence relation, its coefficient -1 on v_j being nonzero (as $1 \neq 0$ in a field). Conversely, given a dependence relation $\sum_i a_i v_i = 0$ with $a_j \neq 0$, the field supplies a_j^{-1}, and solving for v_j gives $v_j = -a_j^{-1} \sum_{i \neq j} a_i v_i$, a linear combination of the others.

For the second statement, take a dependence relation $\sum_i a_i v_i + bw = 0$ for the extended list. The coefficient b cannot be 0: otherwise the relation would be a nontrivial relation among $v_1, \dots, v_k$ alone, contradicting their independence. Hence $w = -b^{-1} \sum_i a_i v_i \in \operatorname{span}(v_1, \dots, v_k)$. For uniqueness, suppose $w = \sum_i c_i v_i = \sum_i c_i' v_i$; subtracting, $\sum_i (c_i - c_i') v_i = 0$, and independence forces $c_i = c_i'$ for every i. □

8.3 Bases, dimension, and linear maps

Spanning says a list reaches everything; independence says it carries no dead weight. A list with both properties is a coordinate system.

Definition 8.14 (Basis)**.** A *basis* of V is a linearly independent set B that spans V. Every $v \in V$ is then a *unique finite* linear combination of elements of B: existence because B spans, uniqueness because two expressions subtract to a relation among finitely many distinct elements of the independent set (as in Lemma 8.13). For a finite basis $B = (b_1, \dots, b_n)$, write

$$v = c_1 b_1 + \cdots + c_n b_n, \qquad [v]_B = (c_1, \dots, c_n),$$

and call the scalars c_i the *coordinates* of v in B.

Example 8.15 (The two bases in constant use)**.** The *standard basis* of F^n is $e_1, \dots, e_n$, where e_i has a 1 in position i and 0 elsewhere. Both properties reduce to reading off coordinates: the combination $\sum_i c_i e_i$ is the tuple $(c_1, \dots, c_n)$, so every vector is such a combination (spanning) and only the zero combination gives 0 (independence). The *monomial basis* of $F[X]_{<n}$ is $1, X, \dots, X^{n-1}$: a polynomial of degree less than n is, by construction, a linear combination of these monomials, and independence is the fact that polynomials are equal exactly when their coefficient sequences agree (Definition 5.1).

That all bases of a given space have the same length is not obvious; it rests on the following exchange principle, which lets an independent list consume a spanning list one slot at a time.

Lemma 8.16 (Steinitz exchange). *If $v_1, \dots, v_m$ are linearly independent in V and $w_1, \dots, w_n$ span V, then $m \le n$.*

Proof. We prove by induction on $k = 0, 1, \dots, m$ that, after a suitable renumbering of the w_j, the hybrid list

$$v_1, \dots, v_k, w_{k+1}, \dots, w_n$$

spans V. The case $k = 0$ is the hypothesis on the w_j.

For the induction step, suppose the hybrid list at stage $k < m$ spans V. In particular it expresses v_{k+1}:

$$v_{k+1} = \sum_{i \le k} a_i v_i + \sum_{j > k} c_j w_j.$$

Some coefficient c_j with $j > k$ must be nonzero: were they all zero, v_{k+1} would be a linear combination of $v_1, \dots, v_k$, and Lemma 8.13 would make the list $v_1, \dots, v_{k+1}$ dependent, contradicting the independence of the v_i. (This step silently requires a slot $j > k$ to exist, i.e. $k < n$; if instead $k = n$, the hybrid list is $v_1, \dots, v_n$ and the displayed equation already yields the same contradiction.) Renumber so that $c_{k+1} \ne 0$, and solve for the displaced vector:

$$w_{k+1} = c_{k+1}^{-1}\Big(v_{k+1} - \sum_{i \le k} a_i v_i - \sum_{j > k+1} c_j w_j\Big).$$

Every vector of the stage-k list therefore lies in the span of the stage-$(k+1)$ list $v_1, \dots, v_{k+1}, w_{k+2}, \dots, w_n$ —the v_i and the remaining w_j trivially, w_{k+1} by the display—so by Proposition 8.10 the stage-$(k+1)$ list spans V, completing the induction.

Each step trades one w for one v, so the process can run to $k = m$ only if there are at least m slots to trade: $m \le n$. ☐

Theorem 8.17 (Dimension). *Let V be a finitely generated F-vector space. Then V has a basis, and any two bases of V have the same number of elements. This common number is the* dimension $\dim_F V$ *(written* $\dim V$ *when the field is clear). In particular* $\dim_F F^n = n$ *and* $\dim_F F[X]_{<n} = n$.

Proof. Existence. Let $w_1, \dots, w_n$ be a finite spanning list. While the list is dependent, Lemma 8.13 exhibits some w_j in the span of the others; discarding it leaves the span unchanged, since any combination using w_j can be rewritten without it (substitute and regroup, Proposition 8.10). Each discard shortens the list, so after at most n discards the process halts at an independent spanning list—a basis. (If V is the zero space, the process discards everything and halts at the empty list, a basis of $\{0\}$ with $\operatorname{span}(\varnothing) = \{0\}$; the dimension is 0.)

Invariance. Every basis is finite: if a basis contained more than the n elements of a finite spanning list, any $n + 1$ of its elements would contradict Lemma 8.16. Let $b_1, \dots, b_m$ and $b'_1, \dots, b'_n$ be bases. Applying Lemma 8.16 with the first list as the independent one and the second as the spanning one gives $m \le n$; exchanging the roles gives $n \le m$. Hence $m = n$.

The two named spaces have the explicit bases of Example 8.15, each of length n. ☐

With coordinates in hand, the maps that respect the linear structure come next; they are the section's real subject, for the Pedersen-style commitments developed in the series are such maps.

Definition 8.18 (Linear map, kernel, image). A function $T : V \to W$ between F-vector spaces is *linear* (an *F-linear map*) if

$$T(au + bv) = a\,T(u) + b\,T(v) \qquad \text{for all } a, b \in F \text{ and } u, v \in V.$$

Its *kernel* and *image* are

$$\ker T = \{ v \in V \,:\, T(v) = 0 \} \subseteq V, \qquad \operatorname{im} T = \{ T(v) \,:\, v \in V \} \subseteq W.$$

A bijective linear map is an *isomorphism*.

Three basic facts follow at once from the subspace criterion and linearity, and we verify them here once. First, taking $a = b = 0$ gives $T(0) = 0$, so $0 \in \ker T$ and $0 \in \operatorname{im} T$. Second, both kernel and image are subspaces: if $u, v \in \ker T$ then $T(au + bv) = a\,T(u) + b\,T(v) = 0$, and if $T(u), T(v) \in \operatorname{im} T$ then $a\,T(u) + b\,T(v) = T(au + bv) \in \operatorname{im} T$; Proposition 8.6 applies to each. Third, T is injective if and only if $\ker T = \{0\}$: if $T(u) = T(v)$, then $T(u - v) = 0$ by linearity, so a trivial kernel forces $u = v$; conversely an injective T sends only 0 to $T(0) = 0$. We also note that for fixed V and W the set $\operatorname{Hom}(V, W)$ of all linear maps $V \to W$ is itself an F-vector space under the pointwise operations of Example 8.4(2)—a sum or scalar multiple of linear maps is linear, so $\operatorname{Hom}(V, W)$ is a subspace of the space of all functions $V \to W$.

Theorem 8.19 (Rank–nullity). *Let $T : V \to W$ be a linear map with V finite-dimensional. Then*

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

Proof. The kernel is a subspace of the finite-dimensional V, and is itself finite-dimensional: an independent list in $\ker T$ has length at most $\dim V$ by Lemma 8.16 (tested against a basis of V, which spans), so a longest independent list $u_1, \dots, u_k$ in $\ker T$ exists, and it spans $\ker T$ —any $u \in \ker T$ outside its span would extend it to a longer independent list by Lemma 8.13. Thus $u_1, \dots, u_k$ is a basis of $\ker T$.

Extend it to a basis of V: while the current independent list fails to span V, adjoin any vector outside its span—independence is preserved by Lemma 8.13—and the process terminates, again because independent lists cannot exceed $\dim V$ in length (Lemma 8.16 against any basis of V). Write the resulting basis as $u_1, \dots, u_k, v_1, \dots, v_r$, so that $\dim V = k + r$.

We claim $T(v_1), \dots, T(v_r)$ is a basis of $\operatorname{im} T$; the theorem follows, since then $\dim \operatorname{im} T = r$ and $\dim V = k + r = \dim \ker T + \dim \operatorname{im} T$.

Spanning. Any element of $\operatorname{im} T$ is $T(v)$ for some $v = \sum_i a_i u_i + \sum_j c_j v_j$, and linearity with $T(u_i) = 0$ gives $T(v) = \sum_j c_j\, T(v_j)$.

Independence. Suppose $\sum_j c_j\, T(v_j) = 0$. By linearity $T\big(\sum_j c_j v_j\big) = 0$, so $\sum_j c_j v_j \in \ker T$, and expressing this kernel element in the kernel basis gives $\sum_j c_j v_j = \sum_i a_i u_i$ for some scalars a_i. Rearranged, $\sum_i a_i u_i - \sum_j c_j v_j = 0$ is a linear relation among the combined basis of V, whose independence forces every coefficient—in particular every c_j—to vanish. □

Rank–nullity is the dimension-counting engine of the section. It shows at once that a linear map from a higher-dimensional space into a lower-dimensional one must have a nontrivial kernel—the image cannot have dimension exceeding the target's, so the kernel absorbs the difference—and distinct inputs must therefore collide. That forced collision is exactly what makes vector commitments compressing and only *computationally* binding (Remark 8.29).

8.4 Bilinear forms and the inner product on F^n

The maps considered so far are linear in a single argument. Commitment and proof systems are pervaded by expressions linear in *each* of two arguments—above all the inner product between two vectors. Over $\mathbb{R}$ such products come wrapped in geometry: lengths, angles, positivity. Over the finite fields of cryptographic interest there is no ordering, hence no “positive” and no length; what survives, and what the applications actually use, is the bare algebra. The appropriate level of generality is therefore the *bilinear form*.

Definition 8.20 (Bilinear form)**.** A *bilinear form* on an F-vector space V is a function $\beta : V \times V \to F$ that is linear in each argument separately:

$$\beta(au + a'u',\ v) = a\,\beta(u,v) + a'\,\beta(u',v), \qquad \beta(u,\ bv + b'v') = b\,\beta(u,v) + b'\,\beta(u,v'),$$

for all $u, u', v, v' \in V$ and $a, a', b, b' \in F$. The form is *symmetric* if $\beta(u,v) = \beta(v,u)$ for all u, v, and *nondegenerate* if $\beta(u,v) = 0$ for all v implies $u = 0$.

Relative to a fixed basis $b_1, \dots, b_n$ of V, a bilinear form is captured by finitely many scalars: expanding both arguments in the basis and applying bilinearity termwise, $\beta(u,v) = \sum_{i,j} \beta(b_i, b_j)\,[u]_i[v]_j$, so β is determined by—and represented by—the unique $n \times n$ array M with entries $M_{ij} = \beta(b_i, b_j)$, via $\beta(u,v) = [u]_B^{\mathsf{T}} M\,[v]_B$ in matrix shorthand. We record the representation for orientation but make no further use of it; only the special case with M the identity array matters below.

Definition 8.21 (Standard inner product on F^n)**.** The *standard inner product* (or *dot product*) on F^n is

$$\langle \mathrm{a},\ \mathrm{b}\rangle = \sum_{i=1}^{n} a_i b_i = \mathrm{a}^{\mathsf{T}}\mathrm{b}, \qquad \mathrm{a} = (a_1, \dots, a_n),\ \mathrm{b} = (b_1, \dots, b_n).$$

Bilinearity holds because each term $a_i b_i$ is linear in each factor, and symmetry because multiplication in F commutes; the representing array in the standard basis is the identity ($\langle e_i,\ e_j\rangle$ is 1 if $i = j$ and 0 otherwise). The form is nondegenerate: if $\langle \mathrm{a},\ \mathrm{b}\rangle = 0$ for every b, then in particular $a_i = \langle \mathrm{a},\ e_i\rangle = 0$ for each i, so $\mathrm{a} = 0$.

Remark 8.22 (No positivity over finite fields). Over $\mathbb{R}$ the standard inner product is *positive definite*: $\langle \mathrm{a},\ \mathrm{a}\rangle = \sum_i a_i^2 \geq 0$, with equality only at 0. Positivity is what yields the Euclidean norm $\|\mathrm{a}\| = \sqrt{\langle \mathrm{a},\ \mathrm{a}\rangle}$ and the Cauchy–Schwarz inequality $|\langle \mathrm{a},\ \mathrm{b}\rangle| \leq \|\mathrm{a}\|\,\|\mathrm{b}\|$—the geometric superstructure of real inner products. Over a finite field $\mathbb{F}_q$ there is no ordering compatible with the arithmetic, so "≥ 0" is meaningless, and the superstructure collapses: there exist nonzero *isotropic* vectors $\mathrm{a} \neq 0$ with $\langle \mathrm{a},\ \mathrm{a}\rangle = 0$. Over $\mathbb{F}_5$, for instance, the vector $\mathrm{a} = (1, 2)$ has

$$\langle \mathrm{a},\ \mathrm{a}\rangle = 1^2 + 2^2 = 1 + 4 = 5 = 0$$

in $\mathbb{F}_5$. What survives the collapse are the algebraically robust properties: bilinearity, symmetry, and nondegeneracy. These are all that the arguments in proof systems rely on, which is why the loss of positivity costs the applications nothing.

Nondegeneracy is not a consolation prize; it powers a dictionary between vectors and the scalar-valued linear maps on them.

Proposition 8.23 (Functionals via the inner product)**.** *For each* $\mathrm{a} \in F^n$*, the map* $\mathrm{b} \mapsto \langle \mathrm{a},\ \mathrm{b}\rangle$ *is a linear map* $F^n \to F$ *(a* linear functional*). The assignment*

$$F^n \longrightarrow \mathrm{Hom}(F^n, F), \qquad \mathrm{a} \longmapsto \langle \mathrm{a},\ \cdot\rangle,$$

is an isomorphism of F-vector spaces onto the dual space $\mathrm{Hom}(F^n, F)$ *of all linear functionals on* F^n*. In particular both spaces have dimension n.*

Proof. Each $\langle \mathrm{a},\ \cdot\rangle$ is linear in b by bilinearity, and the assignment itself is linear in a for the same reason, mapping into the vector space $\mathrm{Hom}(F^n, F)$ of theorem 8.18. It is injective: if $\langle \mathrm{a},\ \cdot\rangle$ is the zero functional, then $\langle \mathrm{a},\ \mathrm{b}\rangle = 0$ for every b, and nondegeneracy (theorem 8.21) gives $\mathrm{a} = 0$, so the kernel is trivial. It is surjective: given a linear functional φ, set $a_i = \varphi(e_i)$ and $\mathrm{a} = (a_1, \dots, a_n)$; then for every

$\mathrm{b} = \sum_i b_i e_i$, linearity of φ gives

$$\varphi(\mathrm{b}) = \sum_i b_i\, \varphi(e_i) = \sum_i a_i b_i = \langle \mathrm{a},\, \mathrm{b}\rangle,$$

so $\varphi = \langle \mathrm{a},\, \cdot\rangle$. A bijective linear map is an isomorphism, and $\dim \mathrm{Hom}(F^n, F) = \dim F^n = n$ follows since an isomorphism carries a basis to a basis (its inverse is linear, and both directions preserve spanning and independence). □

Remark 8.24 (The dictionary in use). Proposition 8.23 says that "a linear functional on F^n" and "a vector in F^n" are interchangeable: every linear way of producing one scalar from a vector *is* an inner product against a fixed vector, and conversely. The instance to keep in mind is polynomial evaluation. Fix a point $z \in F$; the map sending a polynomial to its value at z is linear in the coefficients, so it must be an inner product against some fixed vector—and indeed, for $p \in F[X]_{<n}$ with coefficient vector $\mathrm{c} = (c_0, c_1, \dots, c_{n-1})$,

$$p(z) = c_0 + c_1 z + \cdots + c_{n-1} z^{n-1} = \langle \mathrm{c},\, (1, z, z^2, \dots, z^{n-1})\rangle.$$

Evaluating a *committed* polynomial at a challenge point is therefore an inner-product claim about the committed coefficient vector; this is precisely the shape of statement an inner-product argument proves (theorem 8.32).

8.5 The mixed inner product with group elements

The commitments used in Halo 2 and Orchard pair a vector of *scalars* with a vector of *group elements*. We isolate the algebra here, working over an abstract group; the group of actual cryptographic interest—the points of an elliptic curve—is constructed in the elliptic-curve section later in the volume, and only its abelian-group structure is needed for everything proved below, so the results transfer verbatim.

Construction 8.25 (Scalar multiplication by $\mathbb{F}_r$ on a prime-order group)**.** Let $(\mathbb{G}, +)$ be a finite abelian group of prime order r, written additively with identity $\mathcal{O}$. For an integer $a \in \mathbb{Z}$ and $G \in \mathbb{G}$, write $[a]G$ for the a-fold multiple of G—the additive reading of the powers of §3: $[0]G = \mathcal{O}$, $[a]G = G + \cdots + G$ with a summands for $a > 0$, and $[-a]G = -([a]G)$. The brackets keep the scalar visually separate from the group element. The multiple $[a]G$ depends only on the residue of a modulo r: the order of G divides $|\mathbb{G}| = r$ (Corollary 3.30), and $[a]G = [b]G$ whenever $a \equiv b \pmod{\mathrm{ord}(G)}$ (Proposition 3.12(2), read additively), so congruence modulo r suffices. The operation therefore *descends* to scalars in the field $\mathbb{F}_r = \mathbb{Z}/r\mathbb{Z}$ (theorem 4.26): for $a \in \mathbb{F}_r$ and $G \in \mathbb{G}$, define $[a]G$ using any integer representative of a. It is this descended operation—group elements scaled by elements of a *field*—that the rest of the section uses.

Proposition 8.26 ($\mathbb{G}$ as an $\mathbb{F}_r$-vector space)**.** *Let $\mathbb{G}$ be an abelian group of prime order r, with the scalar multiplication of theorem 8.25. Then $\mathbb{G}$ is an $\mathbb{F}_r$-vector space of dimension* 1*; any element $G \neq \mathcal{O}$ is a basis, and every element of $\mathbb{G}$ is $[a]G$ for a unique $a \in \mathbb{F}_r$.*

Proof. The vector-space axioms are the standard laws of multiples. For integers a, b, the identities $[a+b]G = [a]G + [b]G$ and $[a]\big([b]G\big) = [ab]G$ are the exponent laws of §3 in additive notation, $[1]G = G$ is the definition, and $[a](G+H) = [a]G + [a]H$ holds in any *abelian* group: for $a > 0$ it is the rearrangement of a summands $G + H$ into a summands G followed by a summands H, legitimate by commutativity and associativity (formally, induction on a), and the cases $a \leq 0$ follow by negating. Each law respects reduction modulo r by theorem 8.25, so the four scalar axioms of theorem 8.1 hold with scalars in $\mathbb{F}_r$, and $(\mathbb{G}, +)$ is an abelian group by hypothesis: $\mathbb{G}$ is an $\mathbb{F}_r$-vector space.

Now let $G \neq \mathcal{O}$. By Lagrange's theorem (Theorem 3.29, via Corollary 3.30) the order of G divides the prime r, and it exceeds 1 since G is not the identity; hence $\operatorname{ord}(G) = r$. The r multiples $[0]G, [1]G, \dots, [r-1]G$ are then pairwise distinct (Proposition 3.12(2), additively: $[a]G = [b]G$ forces $a \equiv b \pmod{r}$), and $\mathbb{G}$ has exactly r elements, so the multiples exhaust $\mathbb{G}$. Thus $\{G\}$ spans; and it is independent, since $[a]G = \mathcal{O}$ forces $a = 0$ in $\mathbb{F}_r$, again because $\operatorname{ord}(G) = r$. A one-element basis gives $\dim_{\mathbb{F}_r} \mathbb{G} = 1$, and the uniqueness of a in $[a]G$ is the uniqueness of coordinates in a basis (theorem 8.14). □

Definition 8.27 (Mixed inner product, multiscalar multiplication)**.** Let $\mathrm{G} = (G_1, \dots, G_n) \in \mathbb{G}^n$ be a fixed tuple of group elements and $\mathrm{a} = (a_1, \dots, a_n) \in \mathbb{F}_r^n$ a vector of scalars. Their *mixed inner product*—in computational contexts a *multiscalar multiplication* (MSM), in cryptographic ones a *Pedersen-style inner product*—is the group element

$$\langle \mathrm{a}, \mathrm{G} \rangle = \sum_{i=1}^{n} [a_i]\, G_i \in \mathbb{G}.$$

We reuse the angle-bracket notation of theorem 8.21 deliberately; the type of the second argument—scalar vector or vector of group elements—determines which product is meant.

Proposition 8.28 (Bilinearity of the mixed product)**.** *Fix* $\mathrm{G} \in \mathbb{G}^n$. *The map* $\mathrm{a} \mapsto \langle \mathrm{a}, \mathrm{G} \rangle$ *is* $\mathbb{F}_r$*-linear from* $\mathbb{F}_r^n$ *to* $\mathbb{G}$*:*

$$\langle c\,\mathrm{a} + c'\,\mathrm{a}', \mathrm{G} \rangle = [c]\langle \mathrm{a}, \mathrm{G} \rangle + [c']\langle \mathrm{a}', \mathrm{G} \rangle \qquad \textit{for all } c, c' \in \mathbb{F}_r \textit{ and } \mathrm{a}, \mathrm{a}' \in \mathbb{F}_r^n.$$

Symmetrically, for fixed $\mathrm{a} \in \mathbb{F}_r^n$ *the map* $\mathrm{G} \mapsto \langle \mathrm{a}, \mathrm{G} \rangle$ *is a group homomorphism* $\mathbb{G}^n \to \mathbb{G}$*, and is* $\mathbb{F}_r$*-linear when* $\mathbb{G}^n$ *is viewed as an* $\mathbb{F}_r$*-vector space (componentwise, as in Example 8.3 with* $\mathbb{G}$ *in place of* F*). Thus* $\langle \cdot, \cdot \rangle : \mathbb{F}_r^n \times \mathbb{G}^n \to \mathbb{G}$ *is bilinear over* $\mathbb{F}_r$*.*

Proof. Both statements are the vector-space laws of $\mathbb{G}$ (Proposition 8.26) applied componentwise. For the first,

$$\langle c\,\mathrm{a} + c'\,\mathrm{a}', \mathrm{G} \rangle = \sum_i [\, ca_i + c'a_i' \,]\, G_i = \sum_i \big([c][a_i]G_i + [c'][a_i']G_i\big) = [c] \sum_i [a_i]G_i + [c'] \sum_i [a_i']G_i,$$

using $[a+b]G = [a]G + [b]G$ and $[ab]G = [a]([b]G)$ in each component and then regrouping the sum in the abelian group $\mathbb{G}$. For the second, linearity in G follows by the symmetric computation on the other slot of the defining sum, using $[a](G + H) = [a]G + [a]H$ componentwise. □

Remark 8.29 (Why the mixed product is the right abstraction). Read $\langle \mathrm{a}, \mathrm{G} \rangle$ as a *commitment* to the scalar vector a under the public "basis" G—the receipt of the section's opening problem. Proposition 8.28 is then exactly the *homomorphic* property that the problem demanded: the commitment to $c\,\mathrm{a} + c'\,\mathrm{a}'$ is the $[c]$-, $[c']$-scaled combination of the individual commitments, computable by anyone holding those commitments and the scalars c, c', without knowledge of a or a'. Provided the G_i are chosen so that no party can exhibit a nontrivial relation $\sum_i [a_i]G_i = \mathcal{O}$—a computational hypothesis, the *discrete-logarithm hardness assumption*, made precise in later volumes of this series—the commitment is *binding*: the map $\mathrm{a} \mapsto \langle \mathrm{a}, \mathrm{G} \rangle$ is computationally injective, in the sense that no feasible computation produces two vectors with the same image. Genuinely injective it cannot be once $n \geq 2$: the map is linear from the n-dimensional $\mathbb{F}_r^n$ into the one-dimensional $\mathbb{G}$ (Proposition 8.26), so by rank–nullity (Theorem 8.19) its kernel has dimension at least $n - 1$. Collisions exist in abundance; binding is the claim that they cannot be *found*, a computational rather than information-theoretic statement. The classical Pedersen commitment to a single value a with randomness ρ is the special case $n = 2$, with $H \neq \mathcal{O}$,

$$[a]G + [\rho]H = \langle (a, \rho),\ (G, H) \rangle,$$

which is moreover *perfectly hiding*: for fixed a, as ρ ranges uniformly over $\mathbb{F}_r$ the term $[\rho]H$ ranges uniformly over $\mathbb{G}$ (it is a bijection of $\mathbb{G}$, by unique representation in the basis $\{H\}$), so the commitment's distribution is uniform and carries no information about a.

8.6 How this machinery appears in commitment and proof systems

We close by drawing the threads together. Each point below is developed rigorously in later volumes of this series; the remarks here are descriptive, but the linear-algebraic content behind them is already proved in full.

Remark 8.30 (Witnesses, constraints, and assignments). An arithmetic circuit over $\mathbb{F}_q$ becomes, after *arithmetisation*—the encoding of a computation as a system of polynomial constraints on the wire values—a collection of equations that the vector $\mathrm{z} \in \mathbb{F}_q^N$ of all wire values must satisfy. The *linear* part of such a system—the wiring (or "copy") constraints of the PLONK family, which assert that certain wires carry equal values—consists of homogeneous linear equations in the entries of z, organised in practice as arrays of coefficients (one row per equation) acting on the assignment vector. Satisfying the linear constraints confines z to a subspace of $\mathbb{F}_q^N$ (Example 8.7), or, when public inputs fix some entries, to a translate $\mathrm{z}_0 + W$ of one—an *affine* subspace. The multiplicative constraints are handled separately, by the polynomial machinery of the surrounding sections. Rank–nullity (Theorem 8.19) quantifies the degrees of freedom that remain after the linear constraints are imposed: the dimension of the witness space.

Remark 8.31 (Commitments as linear maps). A Pedersen vector commitment $\mathrm{a} \mapsto \langle \mathrm{a},\ \mathrm{G} \rangle$ is a linear map $\mathbb{F}_r^n \to \mathbb{G}$ (Proposition 8.28), and its linearity is the source of every algebraic manipulation a verifier performs: taking known linear combinations of commitments, folding two commitment keys G, G' into $\mathrm{G} + [x]\mathrm{G}'$ under a challenge x, and checking claimed openings by re-evaluating the same linear map. A polynomial commitment is the identical map applied to a coefficient vector, and an *evaluation* of the committed polynomial at a point z is the scalar inner product $\langle \mathrm{c},\ (1, z, \dots, z^{n-1}) \rangle$ (Remark 8.24).

Remark 8.32 (The inner-product relation). The core statement an inner-product argument proves has the shape: "I know vectors $\mathrm{a}, \mathrm{b} \in \mathbb{F}_r^n$ such that

$$P = \langle \mathrm{a},\ \mathrm{G} \rangle + \langle \mathrm{b},\ \mathrm{H} \rangle + \big[\, \langle \mathrm{a},\ \mathrm{b} \rangle \,\big] U$$

for public $\mathrm{G}, \mathrm{H} \in \mathbb{G}^n$ and $U \in \mathbb{G}$." Three of this section's constructions meet in the one equation: two mixed inner products $\langle \mathrm{a},\ \mathrm{G} \rangle$ and $\langle \mathrm{b},\ \mathrm{H} \rangle$, committing to the witness vectors; one scalar inner product $\langle \mathrm{a},\ \mathrm{b} \rangle$, the value being argued; and the bilinearity of all three, which is what permits prover and verifier to *fold* a length-n instance into a length-$n/2$ one and recurse. The argument's soundness ultimately reduces to the nondegeneracy of these forms together with the hardness of finding kernel elements of the commitment map—exactly the linear-algebraic and computational facts assembled above.

The cheque drawn at the head of the section is now cashed. The receipt for a list $(a_1, \dots, a_n)$ of secrets is the mixed inner product $\langle \mathrm{a},\ \mathrm{G} \rangle = \sum_i [a_i] G_i$: one group element, whatever the size of n. "Receipts add" means precisely that the receipt map is linear, and that is the payoff theorem, Proposition 8.28. One element can absorb n secrets only because the map compresses an n-dimensional space into a one-dimensional one, and rank–nullity prices the compression exactly: a kernel of dimension at least $n - 1$, an unavoidable reservoir of collisions that Remark 8.29 tames computationally rather than absolutely.

In summary: vector spaces, bases and dimension, linear maps with their kernels and images governed by rank–nullity, bilinear forms, and the two flavours of inner product—scalar with scalar, scalar with group—constitute the linear-algebraic vocabulary for the cryptography ahead. Pedersen-style commitments are linear maps, and their polynomial evaluation openings assert inner products; non-degeneracy and dimension counting supply the relevant algebra. Other commitment constructions, such as hash-based commitments, need not be linear. The next section applies the machinery to the polynomial spaces $F[X]_{<n}$ over structured evaluation domains, and the section on elliptic curves constructs the group $\mathbb{G}$ that the mixed inner product has so far taken as abstract.

9 Roots of unity and evaluation domains

Run a computation for n steps and record, at each step, the value it holds in each of its registers. The record is a table: one column per register, one row per step, every entry an element of a finite field. A proof system of the kind underlying Halo 2 and Zcash Orchard convinces a verifier that such a table obeys its rules—that every row follows from the one before it—without the verifier replaying the computation. Its basic move is to treat each column as the value table of a *polynomial*: a function given by its values at n fixed points is, by Lagrange interpolation (theorem 5.30), the same data as a polynomial of degree less than n, and statements about the table become statements about polynomials—checkable all at once, at a single random point, by the Schwartz–Zippel principle (theorem 5.45). For this plan to be workable two demands must be met. The n evaluation points must be *structured*, so that "every row" conditions, "next row" shifts, and the bookkeeping polynomials attached to the point set stay cheap; and converting between a polynomial's values and its coefficients—and multiplying two polynomials—must cost far less than the n^2 field operations of the schoolbook methods, or provers could never handle tables of practical size. Both demands are met by one choice of point set: the powers of a *root of unity*. Nearly every operation such a proof system performs—committing to a polynomial, checking that two polynomials agree, enforcing that a constraint vanishes everywhere it should, multiplying or interpolating efficiently—is ultimately a computation indexed by the finite multiplicative subgroup those powers form. This section develops the theory of these subgroups and delivers, in the fast Fourier transform, the $O(n \log n)$ arithmetic promised above.

Throughout, $\mathbb{F}$ denotes a field. When $\mathbb{F}$ is to be finite, the notation $\mathbb{F}_q$ denotes the field with q elements (so q is a prime power, theorem 6.9); when we intend a prime field specifically we write $\mathbb{F}_p$. The multiplicative group of nonzero elements is $\mathbb{F}^\times = \mathbb{F} \setminus \{0\}$ (Example 3.7). We reserve the letter n for the order of the domain we shall construct; throughout, $n \geq 1$.

The route is as follows. We first define roots of unity and the primitive ones among them, and settle exactly when a finite field contains them: the condition is $n \mid q - 1$, a direct consequence of the cyclicity of $\mathbb{F}_q^\times$ proved in §6. We then fix the *evaluation domain* H, the subgroup those roots form, and work out its toolkit: the vanishing polynomial $X^n - 1$ and its cosets, the Lagrange basis in closed form, and the evaluation/interpolation isomorphism in matrix form. The isomorphism, read as a matrix, is the number-theoretic transform; the convolution theorem turns polynomial multiplication into entrywise multiplication of value vectors; and the fast Fourier transform computes the transform in $O(n \log n)$ operations. A closing synthesis assembles the pieces into the reason these domains form the substrate of polynomial proof systems.

9.1 Roots of unity

The starting point is the equation $X^n = 1$. Its solutions in a field are the roots of unity of order dividing n.

Definition 9.1 (Root of unity). Let $\mathbb{F}$ be a field and $n \in \mathbb{N}_{>0}$. An element $\omega \in \mathbb{F}$ is an *n-th root of unity* if $\omega^n = 1$. We write

$$\mu_n(\mathbb{F}) = \{\, \omega \in \mathbb{F} : \omega^n = 1 \,\}$$

for the set of all n-th roots of unity in $\mathbb{F}$.

Note that $0 \notin \mu_n(\mathbb{F})$, since $0^n = 0 \neq 1$; every n-th root of unity is therefore a unit, i.e. lies in $\mathbb{F}^\times$. The set $\mu_n(\mathbb{F})$ is not merely a set; it is a group.

Proposition 9.2. *The set $\mu_n(\mathbb{F})$ is a finite subgroup of $\mathbb{F}^\times$ of order at most n.*

Proof. The identity 1 satisfies $1^n = 1$, so $1 \in \mu_n(\mathbb{F})$. If $\omega, \zeta \in \mu_n(\mathbb{F})$, then $(\omega\zeta)^n = \omega^n\zeta^n = 1 \cdot 1 = 1$ because $\mathbb{F}^\times$ is abelian, so $\omega\zeta \in \mu_n(\mathbb{F})$; and $(\omega^{-1})^n = (\omega^n)^{-1} = 1^{-1} = 1$, so $\omega^{-1} \in \mu_n(\mathbb{F})$. Hence $\mu_n(\mathbb{F})$ is a subgroup of $\mathbb{F}^\times$ (theorem 3.14). For the order bound, every element of $\mu_n(\mathbb{F})$ is a root of the polynomial $X^n - 1 \in \mathbb{F}[X]$, which is nonzero of degree n and so has at most n roots in $\mathbb{F}$ (theorem 5.18). Thus $|\mu_n(\mathbb{F})| \leq n$. □

The bound $|\mu_n(\mathbb{F})| \leq n$ need not be attained. Whether it is attained is precisely the question of the existence of a *primitive* n-th root of unity.

Definition 9.3 (Primitive root of unity)**.** An element $\omega \in \mathbb{F}$ is a *primitive n-th root of unity* if $\omega^n = 1$ and $\omega^k \neq 1$ for every integer k with $1 \leq k < n$. Equivalently, ω is a primitive n-th root of unity if its multiplicative order $\operatorname{ord}(\omega)$ (theorem 3.11) equals n.

Here $\operatorname{ord}(\omega)$ is the least positive integer d with $\omega^d = 1$; it exists for any $\omega \in \mu_n(\mathbb{F})$ because n itself is such an exponent and the positive integers are well-ordered. The two formulations in theorem 9.3 agree: $\omega^k \neq 1$ for $1 \leq k < n$ together with $\omega^n = 1$ states exactly that the least positive exponent killing ω is n.

Remark 9.4 (Primitive roots revisited). The identification promised in theorem 6.19 can now be stated. A primitive root of a finite field $\mathbb{F}_q$—a generator of the cyclic group $\mathbb{F}_q^\times$, in the sense of theorem 6.18—is an element of multiplicative order $q-1$, which by theorem 9.3 is precisely a primitive $(q-1)$-th root of unity. The group-theoretic vocabulary "generator of $\mathbb{F}_q^\times$" and the polynomial-flavoured vocabulary "primitive $(q-1)$-th root of unity" name the same elements.

Lemma 9.5 (Order divides exponent)**.** *Let $\omega \in \mathbb{F}^\times$ have finite order $d = \operatorname{ord}(\omega)$. Then for any integer m we have $\omega^m = 1$ if and only if $d \mid m$. In particular, ω is an n-th root of unity if and only if $d \mid n$, and ω is a* primitive *n-th root of unity if and only if $d = n$.*

Proof. The first claim is theorem 3.12(1), read in the abelian group $\mathbb{F}^\times$: if $d \mid m$, write $m = dk$, so $\omega^m = (\omega^d)^k = 1$; conversely, division with remainder $m = qd + r$, $0 \leq r < d$, gives $1 = \omega^m = \omega^r$, and minimality of d forces $r = 0$. The remaining clauses instantiate it: $\omega^n = 1 \iff d \mid n$ is the case $m = n$, and primitivity is the case $d = n$ by theorem 9.3. □

The lemma's constant companion is the order-of-a-power formula in a cyclic group, $\operatorname{ord}(g^k) = \operatorname{ord}(g)/\gcd(k, \operatorname{ord}(g))$, proved as Corollary 3.23; we use the pair repeatedly below as the standard device for computing orders.

Proposition 9.6. *If $\omega \in \mathbb{F}$ is a primitive n-th root of unity, then*

$$\mu_n(\mathbb{F}) = \langle\omega\rangle = \{1, \omega, \omega^2, \dots, \omega^{n-1}\}$$

is a cyclic group of order exactly n, and these n powers are pairwise distinct.

Proof. The powers $1, \omega, \dots, \omega^{n-1}$ are pairwise distinct: if $\omega^i = \omega^j$ with $0 \leq i \leq j \leq n-1$, then $\omega^{j-i} = 1$ with $0 \leq j - i < n$, so primitivity (theorem 9.5, with $d = n$) forces $j - i = 0$. Thus $\langle\omega\rangle$ has exactly n elements, all of which lie in $\mu_n(\mathbb{F})$ since $(\omega^k)^n = (\omega^n)^k = 1$. But $|\mu_n(\mathbb{F})| \leq n$ by theorem 9.2, so the inclusion $\langle\omega\rangle \subseteq \mu_n(\mathbb{F})$ is an equality of n-element sets. □

Over the complex numbers a primitive n-th root of unity always exists, namely $\omega = e^{2\pi i/n}$: this is the classical picture of n equally spaced points on the unit circle, with the primitive roots sitting at the angles $2\pi k/n$ for $\gcd(k, n) = 1$. Over a finite field existence is more delicate, and it is exactly the finite case that matters for cryptography.

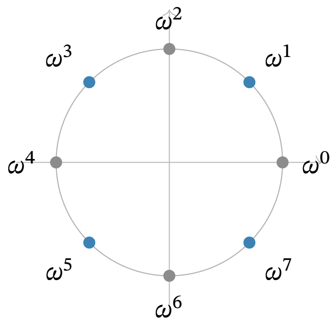

Figure 1: The eight 8-th roots of unity in $\mathbb{C}$, equally spaced on the unit circle at angles $2\pi k/8$, with $\omega = e^{2\pi i/8}$. The four primitive ones ($k = 1, 3, 5, 7$, exactly the k coprime to 8; marked in colour) each generate the whole cycle; the grey points $\omega^0 = 1$, $\omega^2 = i$, $\omega^4 = -1$, $\omega^6 = -i$ have orders $1, 4, 2, 4$ and generate proper subgroups. Over a finite field the same group structure survives with the geometry stripped away—the picture to keep is the cycle, not the circle.

9.2 Existence over finite fields: the condition $n \mid q - 1$

The decisive structural fact about finite fields is that their multiplicative group $\mathbb{F}_q^\times$ is cyclic of order $q - 1$ (theorem 6.16); the existence theorem for roots of unity rests on it.

> **Theorem 9.7** (Existence and count of primitive n-th roots of unity). *Let $\mathbb{F}_q$ be a finite field and $n \in \mathbb{N}_{>0}$ with $\gcd(n, q) = 1$ (automatic when $n < p$, where $p = \operatorname{char} \mathbb{F}_q$; equivalently, in characteristic p, exactly the condition $p \nmid n$). Then:*
>
> 1. *$\mathbb{F}_q$ contains a primitive n-th root of unity if and only if $n \mid q - 1$.*
> 2. *When $n \mid q - 1$, the group $\mu_n(\mathbb{F}_q)$ is cyclic of order exactly n, and the number of primitive n-th roots of unity in $\mathbb{F}_q$ is $\varphi(n)$, where φ is Euler's totient function (theorem 2.26).*

Proof. Let g be a generator of $\mathbb{F}_q^\times$, which is cyclic of order $q - 1$ by theorem 6.16.

(⇒) Suppose ω is a primitive n-th root of unity in $\mathbb{F}_q$. Then $\omega \in \mathbb{F}_q^\times$ has order n by theorem 9.3, and by Lagrange's theorem the order of an element divides the group order $q - 1$ (Corollary 3.30). Hence $n \mid q - 1$.

(⇐) Suppose $n \mid q - 1$, say $q - 1 = nm$. Set $\omega = g^m$. Then $\omega^n = g^{mn} = g^{q-1} = 1$, so $\omega \in \mu_n(\mathbb{F}_q)$. Its order is

$$\operatorname{ord}(\omega) = \operatorname{ord}(g^m) = \frac{\operatorname{ord}(g)}{\gcd(m, \operatorname{ord}(g))} = \frac{q-1}{\gcd(m, q-1)} = \frac{q-1}{m} = n,$$

using the order-of-a-power formula (Corollary 3.23) together with $m \mid q - 1$ (so $\gcd(m, q - 1) = m$). Thus ω is a primitive n-th root of unity, and by theorem 9.6, $\mu_n(\mathbb{F}_q) = \langle\omega\rangle$ is cyclic of order exactly n.

Count. In the cyclic group $\mu_n(\mathbb{F}_q) = \langle\omega\rangle$ of order n, the element ω^k has order $n/\gcd(k, n)$, again by Corollary 3.23. This equals n precisely when $\gcd(k, n) = 1$, and the number of $k \in \{0, 1, \dots, n-1\}$ with $\gcd(k, n) = 1$ is by definition $\varphi(n)$ (theorem 2.26). Hence there are exactly $\varphi(n)$ primitive n-th roots of unity.

(A side note on the coprimality hypothesis. The condition $\gcd(n, q) = 1$ is what guarantees that $X^n - 1$ is separable—$\gcd(X^n - 1, nX^{n-1}) = 1$ when $n \neq 0$ in $\mathbb{F}_q$—hence has n distinct roots in a splitting field over $\mathbb{F}_q$ (theorem 5.40); that all n of them lie in $\mathbb{F}_q$ itself is exactly the condition $n \mid q-1$. If $p = \operatorname{char} \mathbb{F}_q$ divided n, then $X^n - 1$ would have repeated roots and $\mu_n(\mathbb{F}_q)$ would be strictly smaller. But $n \mid q-1$ already forces $\gcd(n, q) = 1$, since $\gcd(q-1, q) = 1$, so the hypothesis is in fact subsumed.) □

Remark 9.8. The proof shows that for a finite field the clean statement is simply: $\mathbb{F}_q$ *has a primitive n-th root of unity* $\iff$ $n \mid q-1$, with no separate coprimality side-condition, because $n \mid q-1$ automatically makes n coprime to q. We stated the condition $\gcd(n, q) = 1$ explicitly only to flag the role of the characteristic.

Example 9.9. Take $\mathbb{F}_q = \mathbb{F}_p$ with $p = 7$. Here $q - 1 = 6$, and 3 is a generator of $\mathbb{F}_7^\times$: its successive powers are $3, 2, 6, 4, 5, 1$, all six nonzero residues (Example 6.20). For $n = 3 \mid 6$, set $m = 6/3 = 2$ and $\omega = 3^2 = 2$. Indeed $2^1 = 2$, $2^2 = 4$, $2^3 = 8 \equiv 1$, so $\omega = 2$ has order 3 and $\mu_3(\mathbb{F}_7) = \{1, 2, 4\}$. There are $\varphi(3) = 2$ primitive cube roots of unity, namely 2 and 4. By contrast $4 \nmid 6$, so $\mathbb{F}_7$ has *no* primitive 4-th root of unity: $X^4 - 1 = (X^2 - 1)(X^2 + 1)$, and $X^2 + 1$ is irreducible over $\mathbb{F}_7$ because it has no root there, -1 not being a square modulo 7.

Remark 9.10 (Two-adicity and FFT-friendly primes). For the fast algorithms below it is desirable that n be a power of two, or at least *smooth*—a product of small primes, ideally with many factors of two. Theorem 9.7 shows this is possible exactly when $2^k \mid q-1$. The largest k with $2^k \mid q-1$ is called the *two-adicity* of the field. Designers deliberately choose primes p with high two-adicity—so that $p - 1 = 2^k \cdot c$ for large k—as scalar fields in proof systems precisely so that large smooth evaluation domains exist. Both Pasta primes—those underlying the Pallas and Vesta curves of section 10—satisfy $2^{32} \mid p-1$, with two-adicity exactly 32 (deployed as the constant $S = 32$ in the `pasta_curves` crate, `src/fields/fp.rs` and `src/fields/fq.rs`; the primes are specified in §5.4.9.6, "Pallas and Vesta", of the Zcash protocol specification), so each Pasta field contains primitive 2^k-th roots of unity for every $k \leq 32$, furnishing power-of-two evaluation domains of every size up to 2^{32}; this is what makes the entire machinery of this section available to Halo 2 and Orchard.

9.3 The evaluation domain H and its vanishing polynomial

We now fix the central object of this section.

Definition 9.11 (Evaluation domain). Let $\mathbb{F}$ be a field containing a primitive n-th root of unity ω. The *evaluation domain* of order n is the multiplicative subgroup

$$H = \langle \omega \rangle = \{ 1, \omega, \omega^2, \dots, \omega^{n-1} \} = \mu_n(\mathbb{F}) \subseteq \mathbb{F}^\times,$$

a cyclic group of order n by theorem 9.6. We index its elements as $\omega^0, \omega^1, \dots, \omega^{n-1}$, and abbreviate ω^i by h_i when convenient. Both $\mathbb{F}$ and ω remain fixed for the rest of the section.

Because H is a multiplicative *group*, it is closed under products and inverses, and $1 \in H$. These elementary facts are exactly what make H substantially more useful than an arbitrary set of n interpolation points: cosets, products, and shifts of H interact algebraically. The single most important polynomial attached to H is the one that vanishes precisely on it.

Definition 9.12 (Vanishing polynomial). The *vanishing polynomial* (or *zerofier*) of H is

$$Z_H(X) = \prod_{h \in H} (X - h) \in \mathbb{F}[X].$$

On a generic n-point set the vanishing polynomial (theorem 5.33) is an unstructured product of n linear factors. On the group H it collapses.

Theorem 9.13 (Closed form and factorisation of the vanishing polynomial). *With $H = \mu_n(\mathbb{F})$ of order n generated by a primitive n-th root of unity ω,*

$$Z_H(X) \;=\; X^n - 1 \;=\; \prod_{i=0}^{n-1}(X - \omega^i).$$

Moreover Z_H is separable: it has n distinct roots and $\gcd(Z_H, Z_H') = 1$.

Proof. Every $h \in H$ satisfies $h^n = 1$, hence is a root of $X^n - 1$. The n elements of H are distinct (theorem 9.6), so $X^n - 1$ has n distinct roots, namely all of H, and factors as $X^n - 1 = c\prod_{h\in H}(X - h)$ for some constant $c \in \mathbb{F}^\times$ by iterating the factor theorem (theorem 5.36, with the degree leaving no room for further factors). Comparing leading coefficients—both $X^n - 1$ and the product are monic—gives $c = 1$, whence $X^n - 1 = \prod_{h\in H}(X - h) = Z_H(X)$.

For separability, the formal derivative is $Z_H'(X) = nX^{n-1}$ (theorem 5.37). The existence of the n distinct roots already forces $\operatorname{char}\mathbb{F} \nmid n$: if $\operatorname{char}\mathbb{F} = \ell$ divided n, say $n = \ell m$, the freshman's dream (theorem 6.12) would give $X^n - 1 = (X^m - 1)^\ell$, a polynomial with at most $m < n$ distinct roots—a contradiction. Hence $n \neq 0$ in $\mathbb{F}$, so nX^{n-1} is nonzero at every element of H, since each is nonzero. Thus it has no root in common with $X^n - 1$, so $\gcd(Z_H, Z_H') = 1$ and Z_H is separable (theorem 5.40). □

The contraposition through the freshman's dream is worth isolating as a device: it converts a *root count* (n distinct n-th roots of unity exist) into a *characteristic condition* ($\operatorname{char}\mathbb{F} \nmid n$), and thereby licenses dividing by n in $\mathbb{F}$—which the closed forms below do constantly.

The closed form $Z_H(X) = X^n - 1$ is the workhorse of polynomial proof systems: testing whether a polynomial f vanishes on all of H reduces to checking divisibility by $X^n - 1$, which one can verify by exhibiting a quotient $t(X)$ with $f(X) = t(X)(X^n - 1)$. Moreover, evaluating Z_H at any point costs a single exponentiation X^n, not n multiplications. We record two structural consequences.

Proposition 9.14 (Cosets and shifts). *Let $\gamma \in \mathbb{F}^\times$ and let $\gamma H = \{\gamma h : h \in H\}$ be the corresponding coset (theorem 3.27). Then the polynomial vanishing on γH is*

$$Z_{\gamma H}(X) = \prod_{h\in H}(X - \gamma h) = X^n - \gamma^n.$$

In particular, distinct cosets of H in $\mathbb{F}^\times$ have vanishing polynomials $X^n - c$ for distinct constants $c = \gamma^n$, and the cosets of H partition $\mathbb{F}^\times$ (theorem 3.28) into translates of H, on each of which X^n is constant.

Proof. Substitute $Y = X/\gamma$:

$$\prod_{h\in H}(X - \gamma h) = \gamma^n \prod_{h\in H}(Y - h) = \gamma^n(Y^n - 1) = \gamma^n\big((X/\gamma)^n - 1\big) = X^n - \gamma^n,$$

using theorem 9.13 for the middle equality. The constant is $c = \gamma^n$, and

$$\gamma_1 H = \gamma_2 H \iff \gamma_1/\gamma_2 \in H \iff (\gamma_1/\gamma_2)^n = 1 \iff \gamma_1^n = \gamma_2^n,$$

so distinct cosets give distinct constants c. □

Proposition 9.15 (Sum of roots of unity). *If $n > 1$ then $\sum_{i=0}^{n-1}\omega^i = 0$, and more generally for any* *integer k,*

$$\sum_{i=0}^{n-1} \omega^{ki} = \begin{cases} n & \text{if } n \mid k, \\ 0 & \text{if } n \nmid k. \end{cases}$$

Proof. If $n \mid k$, then $\omega^{ki} = (\omega^n)^{ki/n} = 1$ for each i, so the sum is n. If $n \nmid k$, set $\zeta = \omega^k \neq 1$ (since ω has order n, we have $\omega^k = 1 \iff n \mid k$ by theorem 9.5), and the finite geometric series—valid over any field: multiplying $S = \sum_{i=0}^{n-1} \zeta^i$ by ζ shifts each term one slot, so $(\zeta - 1)\,S = \zeta S - S$ telescopes to $\zeta^n - 1$, and for $\zeta \neq 1$ one may divide—gives

$$\sum_{i=0}^{n-1} \zeta^i = \frac{\zeta^n - 1}{\zeta - 1} = \frac{(\omega^n)^k - 1}{\zeta - 1} = \frac{1 - 1}{\zeta - 1} = 0,$$

the denominator being nonzero because $\zeta \neq 1$. The first claim is the case $k = 1$, valid for $n > 1$ since then $n \nmid 1$. ☐

Remark 9.16 (Why "orthogonality"). Call two vectors in $\mathbb{F}^n$ *orthogonal* when their standard inner product (theorem 8.21) vanishes. For each frequency k let $v_k = (1, \omega^k, \omega^{2k}, \dots, \omega^{(n-1)k})$ be its vector of powers, and let $\bar{v}_k$ be the mirror vector built from ω^{-1}. Then $\langle \bar{v}_k, v_{k'} \rangle = \sum_i \omega^{i(k'-k)}$, which theorem 9.15 evaluates to 0 for $k \neq k'$ and to n for $k = k'$: distinct frequency vectors are orthogonal, and each pairs with its own mirror to n. This is the finite-field counterpart of the orthogonality of the exponentials $\theta \mapsto e^{2\pi i k\theta}$ in classical Fourier analysis, the balanced trip around the cycle replacing the integral over the circle. It is the engine of the inverse-transform proof below (theorem 9.27): there the v_k appear as the columns of the transform matrix, so in the inverse sum every wrong frequency's contribution rides on one of these vanishing pairings and cancels, while the right one survives n-fold and the $1/n$ rescales it.

9.4 The Lagrange basis on H

Attention now passes from the group H to the polynomials we study on it. Since H has exactly n points, values on H pin down polynomials of degree less than n (theorem 5.20); the basis adapted to this situation is the Lagrange basis of §5.7, which on the group H acquires a closed form it has on no generic point set.

Definition 9.17. Let $\mathbb{F}[X]_{<n}$ denote the $\mathbb{F}$-vector space of polynomials of degree strictly less than n, together with the zero polynomial. It has the monomial basis $\{1, X, \dots, X^{n-1}\}$ (theorem 8.14), so $\dim_{\mathbb{F}} \mathbb{F}[X]_{<n} = n$ (theorem 8.17).

Definition 9.18. For each $i \in \{0, 1, \dots, n-1\}$, the i-th *Lagrange basis polynomial* for $H = \{\omega^0, \dots, \omega^{n-1}\}$ is

$$L_i(X) \;=\; \prod_{\substack{j=0 \\ j \neq i}}^{n-1} \frac{X - \omega^j}{\omega^i - \omega^j} \;\in\; \mathbb{F}[X]_{<n},$$

the basis polynomial ℓ_i of theorem 5.30 for the nodes $\omega^0, \dots, \omega^{n-1}$.

Proposition 9.19 (Interpolation property)**.** *The polynomials $L_0, \dots, L_{n-1}$ satisfy the Kronecker-delta conditions*

$$L_i(\omega^j) = \delta_{ij} = \begin{cases} 1 & i = j, \\ 0 & i \neq j, \end{cases}$$

they form a basis of $\mathbb{F}[X]_{<n}$*, and they give the unique interpolant: for any data* $(v_0, \dots, v_{n-1}) \in \mathbb{F}^n$*, the polynomial*

$$f(X) = \sum_{i=0}^{n-1} v_i\, L_i(X)$$

is the unique element of $\mathbb{F}[X]_{<n}$ *with* $f(\omega^i) = v_i$ *for all i.*

Proof. Each L_i is well defined and of degree $n-1$ because the nodes ω^j are distinct (theorem 9.6), so the denominators $\omega^i - \omega^j$ with $j \neq i$ are nonzero. The delta conditions, the interpolation $f(\omega^j) = v_j$, and the uniqueness are theorem 5.30 (with (4)) applied to these nodes with $d = n-1$; uniqueness rests, as there, on the root count: two interpolants in $\mathbb{F}[X]_{<n}$ differ by a polynomial of degree $< n$ with n distinct roots, which is the zero polynomial (theorem 5.18). It remains to see that the L_i form a basis of $\mathbb{F}[X]_{<n}$. They are linearly independent: if $\sum_i c_i L_i = 0$, evaluation at ω^j yields $c_j = 0$ for every j (theorem 8.11). Being n independent vectors in the n-dimensional space $\mathbb{F}[X]_{<n}$ (theorem 9.17), they form a basis (theorem 8.17). □

The root-count step deserves a name, for it recurs: to prove two polynomials of degree $< n$ equal, show they agree on all n points of H. We use this *root-count device* for the uniqueness above, for theorem 9.22, and for Example 9.23 below.

On a generic point set the Lagrange polynomials admit no closed form simpler than the defining product. On the *group H* they collapse to a compact expression involving the vanishing polynomial.

Theorem 9.20 (Closed form of the Lagrange basis on H)**.** *For* $H = \mu_n(\mathbb{F})$ *with primitive root* ω*, and for each i,*

$$L_i(X) \;=\; \frac{\omega^i}{n} \cdot \frac{X^n - 1}{X - \omega^i} \;=\; \frac{\omega^i\,(X^n - 1)}{n\,(X - \omega^i)}.$$

In particular, dividing out one linear factor and scaling yields all n Lagrange polynomials from the single *polynomial* $Z_H(X) = X^n - 1$.

Proof. Fix i. From theorem 9.13, $Z_H(X) = \prod_{j=0}^{n-1}(X - \omega^j)$, so

$$\frac{Z_H(X)}{X - \omega^i} = \prod_{\substack{j=0 \\ j \neq i}}^{n-1} (X - \omega^j),$$

which is exactly the numerator of L_i in theorem 9.18. It remains to evaluate the denominator $\prod_{j \neq i}(\omega^i - \omega^j)$. Writing $Z_H = (X - \omega^i)\, g$ with $g = \prod_{j \neq i}(X - \omega^j)$, the product rule (theorem 5.38) gives $Z_H' = g + (X - \omega^i)\, g'$, so

$$Z_H'(\omega^i) = g(\omega^i) = \prod_{\substack{j=0 \\ j \neq i}}^{n-1} (\omega^i - \omega^j)$$

—the derivative at a simple root equals the product of the differences to the other roots, the computation already made for simple roots in the proof of theorem 5.40. Concretely $Z_H'(X) = nX^{n-1}$, so

$$Z_H'(\omega^i) = n\,(\omega^i)^{n-1} = n\,\omega^{in}\,\omega^{-i} = n \cdot 1 \cdot \omega^{-i} = \frac{n}{\omega^i},$$

using $\omega^{in} = (\omega^n)^i = 1$. Therefore

$$L_i(X) = \frac{Z_H(X)/(X - \omega^i)}{Z_H'(\omega^i)} = \frac{(X^n - 1)/(X - \omega^i)}{n/\omega^i} = \frac{\omega^i\,(X^n - 1)}{n\,(X - \omega^i)},$$

as claimed. □

Remark 9.21 (Barycentric form). The closed form renders the *barycentric* interpolation formula on H especially clean. Writing $f(X) = \sum_i v_i L_i(X)$ and substituting theorem 9.20,

$$f(X) = \frac{X^n - 1}{n} \sum_{i=0}^{n-1} \frac{\omega^i v_i}{X - \omega^i}.$$

Evaluating f at a point $z \notin H$ thus costs one evaluation of $z^n - 1$ and n terms of the sum, with no per-point polynomial reconstruction. A verifier can use this formula when the values v_i are known, as for public-input polynomials; a commitment to hidden values does not by itself permit this evaluation.

Proposition 9.22 (Lagrange polynomials sum to one). *The Lagrange basis polynomials for H satisfy* $\sum_{i=0}^{n-1} L_i(X) = 1$ *identically.*

Proof. The polynomial $g(X) = \sum_i L_i(X) - 1$ lies in $\mathbb{F}[X]_{<n}$ and vanishes at every ω^j, because $\sum_i L_i(\omega^j) = \sum_i \delta_{ij} = 1$ (theorem 9.19). By the root-count device—a polynomial of degree $< n$ with n distinct roots is zero (theorem 5.18)—$g \equiv 0$. □

Example 9.23 (The all-ones value vector). The constant polynomial 1 interpolates the data $v_i = 1$ for all i; theorem 9.22 is the statement $\sum_i 1 \cdot L_i = 1$. Likewise the values of X on H are $v_i = \omega^i$, and, for $n \geq 2$, theorem 9.19 gives $\sum_i \omega^i L_i(X) = X$: both sides have degree $< n$ and agree on all of H, so the root-count device applies. We invoke these identities constantly when rewriting monomials in the Lagrange basis.

9.5 Evaluation and interpolation as mutually inverse linear maps

An element of $\mathbb{F}[X]_{<n}$ admits two natural descriptions: by its n coefficients, or by its n values on H. Both are coordinate systems on the same n-dimensional vector space, and passage between them is a linear isomorphism—the specialisation to H of the evaluation isomorphism met for general nodes in theorem 5.32.

Definition 9.24 (Evaluation map). The *evaluation map* on H is the $\mathbb{F}$-linear map (theorem 8.18)

$$\mathrm{ev}_H : \mathbb{F}[X]_{<n} \longrightarrow \mathbb{F}^n, \qquad \mathrm{ev}_H(f) = \big(f(\omega^0), f(\omega^1), \ldots, f(\omega^{n-1})\big).$$

Linearity is immediate from $(af + bg)(\omega^i) = af(\omega^i) + bg(\omega^i)$, an instance of theorem 5.16.

Theorem 9.25 (Evaluation and interpolation are mutually inverse). *The evaluation map* ev_H *is a vector-space isomorphism. Its inverse is the* interpolation map

$$\mathrm{int}_H : \mathbb{F}^n \longrightarrow \mathbb{F}[X]_{<n}, \qquad \mathrm{int}_H(v_0, \ldots, v_{n-1}) = \sum_{i=0}^{n-1} v_i L_i(X).$$

That is, $\mathrm{int}_H \circ \mathrm{ev}_H = \mathrm{id}_{\mathbb{F}[X]_{<n}}$ *and* $\mathrm{ev}_H \circ \mathrm{int}_H = \mathrm{id}_{\mathbb{F}^n}$.

Proof. Both maps are linear and act between spaces of the same finite dimension n, so verifying one composition would suffice (a one-sided inverse makes ev_H injective, hence surjective by rank–nullity, theorem 8.19); we check both for clarity. For $v \in \mathbb{F}^n$, the polynomial $\mathrm{int}_H(v) = \sum_i v_i L_i$ has value $\sum_i v_i L_i(\omega^j) = v_j$ at ω^j by theorem 9.19, so $\mathrm{ev}_H(\mathrm{int}_H(v)) = v$. For $f \in \mathbb{F}[X]_{<n}$, the polynomial $\mathrm{int}_H(\mathrm{ev}_H(f)) = \sum_i f(\omega^i) L_i$ is, by the uniqueness clause of theorem 9.19, the unique degree-$< n$ polynomial taking the values $f(\omega^i)$ on H—which is f itself. Hence the two maps are inverse; in particular ev_H is bijective and linear, i.e. an isomorphism (theorem 8.18). □

In coordinates, ev_H is given by a matrix, which we name here because it is the bridge to the Fourier transform. The notation is worth fixing once, since §8 did not need it: an *$n\times n$ matrix* over $\mathbb{F}$ is a doubly indexed family $A = (A_{ik})_{0\le i,k\le n-1}$ of elements of $\mathbb{F}$, acting on a vector $c \in \mathbb{F}^n$ by $(Ac)_i = \sum_k A_{ik}c_k$—the general form, written out through the standard basis (Example 8.15), of a linear map $\mathbb{F}^n \to \mathbb{F}^n$. The product of two matrices is defined so that $(AB)\,c = A(Bc)$, which forces $(AB)_{ij} = \sum_k A_{ik}B_{kj}$; the *identity matrix* I, with entries δ_{ik}, acts as the identity map; and A is *invertible* if some matrix A^{-1} satisfies $A^{-1}A = AA^{-1} = I$.

Definition 9.26 (Vandermonde / DFT matrix)**.** Let $f(X) = \sum_{k=0}^{n-1} c_kX^k$ have coefficient vector $c = (c_0, \dots, c_{n-1})$. Then $\mathrm{ev}_H(f)$ has i-th entry $f(\omega^i) = \sum_k c_k\,\omega^{ik}$. Thus, relative to the monomial basis on $\mathbb{F}[X]_{<n}$ and the standard basis on $\mathbb{F}^n$, the *Vandermonde matrix*

$$V = \left(\omega^{ik}\right)_{0\le i,k\le n-1} = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \cdots & \omega^{n-1} \\ 1 & \omega^2 & \omega^4 & \cdots & \omega^{2(n-1)} \\ \vdots & & & & \vdots \\ 1 & \omega^{n-1} & \omega^{2(n-1)} & \dots & \omega^{(n-1)^2} \end{pmatrix}$$

—rows indexed by the evaluation point ω^i, columns by the power k—represents ev_H, so that $\mathrm{ev}_H(f) = Vc$. We call V the (discrete Fourier) transform matrix on H.

Theorem 9.27 (Invertibility and the inverse transform)**.** *The matrix V of theorem 9.26 is invertible, with*

$$(V^{-1})_{k,i} = \frac{1}{n}\,\omega^{-ik}.$$

Equivalently, $V^{-1} = \frac{1}{n}\overline{V}$, where $\overline{V} = (\omega^{-ik})_{i,k}$ is the transform matrix built from ω^{-1} (itself a primitive n-th root of unity). Consequently the coefficient and value representations are related by the mutually inverse formulas

$$c_k = \frac{1}{n}\sum_{i=0}^{n-1} f(\omega^i)\,\omega^{-ik}, \qquad f(\omega^i) = \sum_{k=0}^{n-1} c_k\,\omega^{ik}.$$

Proof. First note that $\omega^{-1} = \omega^{n-1}$ has order n by Corollary 3.23 (with $\gcd(n-1, n) = 1$), so $\overline{V}$ is again a matrix of the shape in theorem 9.26. We show $\overline{V}V = nI$. The (k, k') entry of $\overline{V}V$ is

$$\sum_{i=0}^{n-1} \omega^{-ik}\,\omega^{ik'} = \sum_{i=0}^{n-1} \omega^{i(k'-k)}.$$

By theorem 9.15 this sum is n if $n \mid (k' - k)$ and 0 otherwise; for $k, k' \in \{0, \dots, n-1\}$ we have $n \mid (k' - k) \iff k = k'$. Hence $\overline{V}V = nI$, and the same computation with the roles of the two factors exchanged gives $V\overline{V} = nI$. Since $n \ne 0$ in $\mathbb{F}$ (theorem 9.13), we may divide by n: $V^{-1} = \frac{1}{n}\overline{V}$. The component formulas write out the matrix identities $\mathrm{ev}_H(f) = Vc$ and $c = V^{-1}\,\mathrm{ev}_H(f)$. □

Remark 9.28. Theorem 9.25 and theorem 9.27 are two faces of one statement. The abstract face: evaluation on H is an isomorphism whose inverse is Lagrange interpolation. The concrete face: that isomorphism is the matrix V, and its inverse is, up to the scalar $1/n$, the same matrix with ω replaced by ω^{-1}. The near-symmetry between a transform and its inverse—differing only by the substitution $\omega \mapsto \omega^{-1}$ and a scale $1/n$—is the algebraic reason the fast inverse transform of theorem 9.35 costs exactly as much as the forward one.

9.6 The number-theoretic transform and convolution

The map ev_H, read as the matrix V, is the discrete Fourier transform, but over a finite field rather than over $\mathbb{C}$. Because it involves no analytic structure—no complex exponentials, no convergence—it is called the *number-theoretic transform*.

Definition 9.29 (Number-theoretic transform). Let ω be a primitive n-th root of unity in $\mathbb{F}$. The *number-theoretic transform* (NTT) sends the coefficient vector of a polynomial to its value vector on H: it is the evaluation map ev_H written in coordinates. Explicitly, the NTT of $a = (a_0, \dots, a_{n-1}) \in \mathbb{F}^n$ is $\hat{a} = (\hat{a}_0, \dots, \hat{a}_{n-1})$ with

$$\hat{a}_j = \sum_{k=0}^{n-1} a_k\, \omega^{jk}, \qquad j = 0, \dots, n-1,$$

each entry being the value at ω^j of the polynomial with coefficients a. The *inverse number-theoretic transform* is interpolation—the recovery of the coefficients from the values—and by theorem 9.27 it too is a single explicit sum:

$$a_k = \frac{1}{n} \sum_{j=0}^{n-1} \hat{a}_j\, \omega^{-jk}.$$

The NTT enjoys the same convolution property as the classical DFT, and this is the mechanism underlying fast polynomial multiplication. Recall that the coefficients of a product are a *convolution*: with $f = \sum_i a_i X^i$ and $g = \sum_j b_j X^j$, the coefficient of X^k in fg is $\sum_{i+j=k} a_i b_j$ (theorem 5.3). The *cyclic* variant folds indices modulo n: for $a, b \in \mathbb{F}^n$, the cyclic convolution $a * b \in \mathbb{F}^n$ has entries

$$(a * b)_k \;=\; \sum_{\substack{0 \le i, j \le n-1 \\ i+j \equiv k \pmod{n}}} a_i\, b_j, \qquad k = 0, \dots, n-1.$$

Theorem 9.30 (Convolution theorem). *For $f, g \in \mathbb{F}[X]_{<n}$ with $\deg f + \deg g < n$, let $c = fg$ be their product (of degree $< n$). Then for every $h \in H$,*

$$c(h) = f(h)\, g(h),$$

i.e. pointwise multiplication of value vectors corresponds to polynomial multiplication. Equivalently, writing $a, b \in \mathbb{F}^n$ for the coefficient vectors of f, g (padded with zeros up to length n),

$$\widehat{a * b}_j = \hat{a}_j\, \hat{b}_j \qquad (j = 0, \dots, n-1),$$

so the NTT turns convolution into entrywise multiplication.

Proof. The first identity is the statement that evaluation at h is a ring homomorphism $\mathbb{F}[X] \to \mathbb{F}$ (theorem 5.16): $(fg)(h) = f(h)\, g(h)$. Since $\deg(fg) < n$, interpolation recovers the product $c = fg$ from its values on the n-point set H (theorem 9.19), so the entrywise product of the value vectors uniquely determines c.

For the convolution statement the plan is: (i) multiplication in $\mathbb{F}[X]/(X^n - 1)$ *is* cyclic convolution of coefficient vectors; (ii) evaluation on H is a ring isomorphism from that quotient onto $\mathbb{F}^n$ with entrywise operations; (iii) the degree bound rules out wrap-around, so the cyclic product is the genuine one. For (i), work in the quotient ring $\mathbb{F}[X]/(X^n - 1)$ (theorem 4.19 and theorem 4.20, applied to the ideal of multiples of $X^n - 1$, theorem 4.17). Every class has a unique representative of degree $< n$, by division with remainder (theorem 5.12); and multiplying two such representatives and reducing replaces each X^k with $k \geq n$ by X^{k-n}, since $X^n \equiv 1$. The coefficient of X^k in the reduced product of the representatives with coefficient vectors a and b is therefore $\sum_{i+j\equiv k \pmod n} a_i b_j = (a * b)_k$: multiplication in $\mathbb{F}[X]/(X^n - 1)$ *is* cyclic convolution of coefficient vectors.

Now consider the map $[p] \mapsto \mathrm{ev}_H(p)$ from $\mathbb{F}[X]/(X^n - 1)$ to $\mathbb{F}^n$, where $\mathbb{F}^n$ is a commutative ring under entrywise addition and multiplication (the ring laws hold in each coordinate because they hold in $\mathbb{F}$), with multiplicative identity $(1, \dots, 1)$. The map is well defined: two representatives of a class differ by a multiple of $X^n - 1$, which vanishes on H (theorem 9.13). It is a ring homomorphism (theorem 4.21), because evaluation at each point of H is one (theorem 5.16) and the operations on $\mathbb{F}^n$ are entrywise. And it is bijective: classes correspond to their degree-$<n$ representatives, on which ev_H is a bijection by theorem 9.25. Hence it is a ring isomorphism $\mathbb{F}[X]/(X^n-1) \xrightarrow{\sim} \mathbb{F}^n$, and it sends the product of the classes of f and g—coefficient vector $a * b$—to the entrywise product of $\hat{a}$ and $\hat{b}$. This is the displayed identity. Finally, when $\deg f + \deg g < n$, the genuine product fg already has degree $< n$, so no folding occurs, the cyclic and ordinary products coincide, and the statement computes fg itself. □

Remark 9.31 (Computational significance). Theorem 9.30 reduces multiplying two degree-$< n/2$ polynomials—a quadratic-cost operation by the schoolbook method—to: transform both (NTT), multiply the value vectors entrywise (n multiplications), and transform back (inverse NTT). If the NTT runs in $O(n \log n)$ operations, the whole multiplication costs $O(n \log n)$. The fast Fourier transform supplies that conditional.

Example 9.32 (The convolution theorem in $\mathbb{F}_{17}$). In $\mathbb{F}_{17}$, the element $\omega = 4$ has $\omega^2 = 16 = -1$, so it is a primitive 4-th root of unity with domain $H = \{1, 4, 16, 13\}$. To multiply $f = 1+2X$ and $g = 3+X$, pad both coefficient vectors to length 4 and transform: $\hat{a} = (3, 9, 16, 10)$ and $\hat{b} = (4, 7, 2, 16)$, the values of f and of g on H. Multiplying entrywise gives $(12, 12, 15, 7)$, the values of fg on H, and the inverse transform returns $(3, 7, 2, 0)$: the coefficients of $fg = 3 + 7X + 2X^2$, agreeing with the schoolbook product. At $n = 4$ nothing is saved; the point is the shape of the pipeline, which becomes decisive once the transforms themselves cost $O(n \log n)$ (§9.7).

9.7 The fast Fourier transform

Computing $\hat{a} = Va$ directly from theorem 9.26 costs $\Theta(n^2)$ field multiplications: each of the n outputs is a sum of n products. The fast Fourier transform (FFT)—here a number-theoretic FFT—computes the same vector in $O(n \log n)$ operations by a divide-and-conquer recursion, provided n is a power of two (or, more generally, smooth). This is where the requirement that H have smooth order, theorem 9.10, becomes indispensable.

Theorem 9.33 (Cooley–Tukey radix-2 step). *Suppose $n = 2m$ and ω is a primitive n-th root of unity in $\mathbb{F}$. Given $f(X) = \sum_{k=0}^{n-1} a_k X^k$, split its coefficients by parity into*

$$f_{\mathrm{even}}(Y) = \sum_{k=0}^{m-1} a_{2k}\, Y^k, \qquad f_{\mathrm{odd}}(Y) = \sum_{k=0}^{m-1} a_{2k+1}\, Y^k,$$

so that $f(X) = f_{\mathrm{even}}(X^2) + X f_{\mathrm{odd}}(X^2)$. Then $\eta := \omega^2$ is a primitive m-th root of unity, and for $j = 0, \dots, m-1$,

$$f(\omega^j) = f_{\mathrm{even}}(\eta^j) + \omega^j f_{\mathrm{odd}}(\eta^j), \tag{7}$$
$$f(\omega^{j+m}) = f_{\mathrm{even}}(\eta^j) - \omega^j f_{\mathrm{odd}}(\eta^j). \tag{8}$$

Thus the length-n NTT of a reduces to two length-m NTTs (of the even and odd coefficient subsequences, over the domain $\langle \eta \rangle$) plus $O(n)$ extra operations.

Proof. The identity $f(X) = f_{\text{even}}(X^2) + X f_{\text{odd}}(X^2)$ regroups the sum $\sum_k a_k X^k$ into even-index and odd-index terms; in the odd terms one factor of X is pulled out, leaving powers X^{2k}. That $\eta = \omega^2$ is a primitive m-th root of unity follows from theorem 9.5 and the order-of-a-power formula (Corollary 3.23): $\operatorname{ord}(\omega^2) = n/\gcd(2,n) = n/2 = m$, using $2 \mid n$.

It remains to evaluate. For (7), put $X = \omega^j$; then $X^2 = \omega^{2j} = \eta^j$, so $f(\omega^j) = f_{\text{even}}(\eta^j) + \omega^j f_{\text{odd}}(\eta^j)$. For (8), put $X = \omega^{j+m}$; then $X^2 = \omega^{2j+2m} = \omega^{2j}\,\omega^n = \eta^j \cdot 1 = \eta^j$ (since $2m = n$ and $\omega^n = 1$), while the linear factor is $\omega^{j+m} = \omega^j\,\omega^m = -\omega^j$. Here $\omega^m = -1$, because ω^m has order $n/\gcd(m,n) = n/m = 2$ (Corollary 3.23), and -1 is the unique element of order 2 in a field: $x^2 = 1$ with $x \neq 1$ forces $(x-1)(x+1) = 0$ and so $x = -1$; and $-1 \neq 1$, since $\operatorname{char}\mathbb{F} \nmid n$ (theorem 9.13) with n even forces $\operatorname{char}\mathbb{F} \neq 2$. Substituting, $f(\omega^{j+m}) = f_{\text{even}}(\eta^j) - \omega^j f_{\text{odd}}(\eta^j)$. □

The half-domain trick just used—$\omega^m = -1$ whenever $n = 2m$, so the second half of the outputs differs from the first only by a sign on the odd part—recurs in every radix-2 argument over an evaluation domain; it is worth remembering alongside the theorem.

Remark 9.34 (The butterfly). Equations (7)–(8) constitute the *butterfly*: from the two subtransform outputs $E_j = f_{\text{even}}(\eta^j)$ and $O_j = f_{\text{odd}}(\eta^j)$ one forms the *twiddle* $t_j = \omega^j O_j$ and reads off the two outputs $E_j + t_j$ and $E_j - t_j$. Pictorially, E_j and O_j sit as two nodes on the left; each is carried to both outputs on the right, the two E_j-edges with weight 1 and the two O_j-edges with weights $+\omega^j$ and $-\omega^j$; the pair of crossing edges forms the × that names the operation. Each butterfly costs one multiplication (the twiddle) and two additions, and there are $n/2$ of them per level.

> **Theorem 9.35** ($O(n \log n)$ cost of the FFT)**.** *Let $n = 2^t$ and let $T(n)$ be the number of field additions and multiplications the radix-2 recursion of theorem 9.33 uses to compute a length-n NTT. Then*
>
> $$T(n) = 2\,T(n/2) + cn \quad (n > 1), \qquad T(1) = O(1),$$
>
> *for a constant c, and therefore $T(n) = O(n \log n)$. The inverse NTT has the same cost, since it is the forward NTT with ω replaced by ω^{-1}, followed by scaling each entry by $1/n$ (theorem 9.27).*

Proof. The recursion is exactly theorem 9.33: a length-n transform calls two length-$n/2$ transforms (on the even and odd coefficient subsequences) and combines them with $n/2$ butterflies of $O(1)$ operations each, for the cn overhead. Unrolling, with $n = 2^t$,

$$T(2^t) = 2\,T(2^{t-1}) + c\,2^t.$$

Divide by 2^t and set $S(t) = T(2^t)/2^t$: then $S(t) = S(t-1) + c$, so $S(t) = S(0) + ct = O(1) + ct$, and $T(2^t) = 2^t S(t) = O(2^t) + ct\,2^t = O(t\,2^t)$. With $t = \log_2 n$ this is $T(n) = O(n \log n)$. (Equivalently: the recursion tree has $\log_2 n$ levels, each performing $\Theta(n)$ total work across all subproblems at that level.) The inverse-transform claim is theorem 9.27: the inverse is the same Cooley–Tukey recursion with ω^{-1} in place of ω (also a primitive n-th root of unity), plus a final pass of n divisions by n, all within $O(n \log n)$. □

Remark 9.36 (Mixed radix and smoothness). The hypothesis that enables fast transforms is that n be *smooth*—a product of small primes—and, crucially, that the field actually *contain* a primitive n-th root of unity, i.e. $n \mid q - 1$ (theorem 9.7). A power-of-two n dividing $q - 1$ is the optimal case: pure radix-2 with the cleanest butterfly.

Example 9.37 (A length-4 transform)**.** Take $n = 4$ with primitive 4-th root ω (so $\omega^2 = -1$ and $\omega^4 = 1$, and $\eta = \omega^2 = -1$ is the primitive 2nd root). For $f = a_0 + a_1X + a_2X^2 + a_3X^3$, the two length-2 subtransforms over the domain $\{1, -1\}$—each itself a single butterfly, with twiddle $\eta^0 = 1$—are

$$E = (a_0 + a_2,\ a_0 - a_2), \qquad O = (a_1 + a_3,\ a_1 - a_3),$$

and the butterflies give

$$\hat{a}_0 = E_0 + \omega^0 O_0, \quad \hat{a}_2 = E_0 - \omega^0 O_0, \quad \hat{a}_1 = E_1 + \omega^1 O_1, \quad \hat{a}_3 = E_1 - \omega^1 O_1.$$

(Note the output indexing: outputs j and $j + m$ pair up, per (7)–(8).) This uses 2 subtransforms of 2 additions each plus 2 butterflies of 2 additions each ($n/2 = 2$ per level, theorem 9.34)—together 2 twiddle multiplications and 8 additions—versus 16 coefficient-products and 12 additions for the direct 4×4 matrix multiply; and the gap widens as n grows.

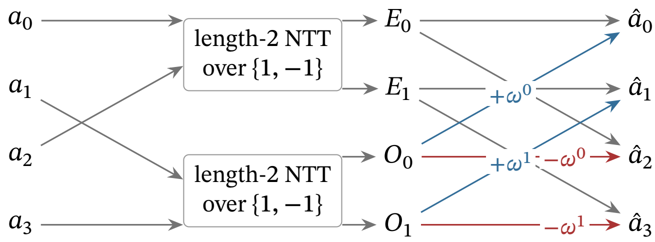

Figure 2: The length-4 transform of theorem 9.37 as a circuit. The inputs split by parity into two length-2 subtransforms, whose outputs E_0, E_1 and O_0, O_1 feed the two butterflies of theorem 9.34: each E_j is carried to outputs $\hat{a}_j$ and $\hat{a}_{j+2}$ with weight 1, each O_j with weights $+\omega^j$ and $-\omega^j$ (eq. (7)–eq. (8)); the crossing pair of weighted edges is the $\times$ that names the butterfly.

9.8 The role of smooth multiplicative domains in polynomial proof systems

This section closes by assembling the pieces into the reason these domains form the substrate of Halo 2 and Orchard. A proof system of the polynomial type encodes a computation as a system of polynomial identities and convinces a verifier that those identities hold. The evaluation domain $H = \mu_n$ is where the encoding lives—the n-step table of the section's opening is laid out on the n points of H—and every property proved above is load-bearing.

Remark 9.38 (The roles played by H, Z_H, the Lagrange basis, and the FFT).

- **Witnesses as evaluations on H.** A prover lays out the trace of a computation as the values of one or more polynomials on the n points of H. By theorem 9.25 this is the same data as a polynomial of degree $< n$; the prover may move freely between the value (Lagrange) representation, natural for stating constraints row by row, and the coefficient representation, natural for committing and for low-degree testing. The change of representation is the (inverse) NTT.

- **Constraints as divisibility by Z_H.** "Constraint polynomial C holds on every row" means C vanishes on all of H, which by theorem 9.13 means exactly that $Z_H = X^n - 1$ divides C. The prover demonstrates this by exhibiting a quotient t with $C(X) = t(X)\,(X^n - 1)$. The inexpensive closed form $Z_H(X) = X^n - 1$—one exponentiation to evaluate—is what makes this check cheap at a random point. Theorem 9.14 extends the same technique to cosets γH, used to separate quotient and remainder domains.

- **Lagrange selectors.** The Lagrange polynomials L_i of theorem 9.18 are *selectors*: L_i is 1 on row i and 0 elsewhere (theorem 9.19), so boundary and instance constraints (e.g. "the first row equals the public input") become $L_0 \cdot (\dots)$. The closed form and the barycentric formula (theorem 9.20, theorem 9.21) let the verifier evaluate these at a random challenge in $O(n)$ or even $O(\log n)$ field operations rather than reconstructing polynomials.

- **Group structure for permutations and shifts.** Because H is a cyclic *group* (theorem 9.11), multiplication by ω is the cyclic "next-row" shift $\omega^i \mapsto \omega^{i+1}$. Constraints relating a row to its neighbour become $f(\omega X)$ versus $f(X)$; permutation arguments—the heart of Halo 2's copy constraints—rest on the action of the group on itself and on its cosets. An arbitrary set of n points offers none of this.

- **Smoothness for speed.** Provers manipulate polynomials of size n (often after blowing the domain up by a small constant factor for low-degree testing). Every transform between representations, every multiplication, every coset evaluation is an NTT. By theorem 9.35 these cost $O(n \log n)$ instead of $O(n^2)$—the difference between feasible and infeasible at the sizes used in practice—*provided* n is smooth and $\mathbb{F}_q$ contains a primitive n-th root of unity. By theorem 9.7 that requires $n \mid q-1$; by theorem 9.10 this is why high-two-adicity primes are chosen for the scalar fields of the Pallas and Vesta curves (section 10).

In summary: the existence condition $n \mid q-1$ (theorem 9.7) supplies the group H; the group supplies the clean vanishing polynomial (theorem 9.13), the closed-form Lagrange basis (theorem 9.20), and the evaluation/interpolation isomorphism (theorem 9.25); and smoothness of n supplies the $O(n \log n)$ FFT (theorem 9.35). Together these discharge both demands of the section's opening—a structured point set on which an execution trace can live, and $O(n \log n)$ multiplication and interpolation on it—and they are exactly the ingredients a polynomial proof system requires, which is why such systems rest on smooth multiplicative evaluation domains and on the fields that contain them.

10 Elliptic curves

The multiplicative groups of the preceding sections harbour a structural weakness. As theorem 7.21 recorded, the discrete logarithm in $\mathbb{F}_p^\times$ does not resist attack at the generic square-root cost: the *index-calculus* family of algorithms exploits the fact that residues come from ordinary integers, and integers factor into primes, to solve the problem in subexponential time. The practical consequence is severe. To push the cheapest known attack on $\mathbb{F}_p^\times$ beyond 2^{128} operations, the prime p must run to *thousands* of bits, and every exponentiation drags that bulk along. What cryptography wants instead is a group of roughly 2^{255} elements for which no attack faster than the square-root bound is known—a group without known exploitable arithmetic behind its elements, where the generic $O(\sqrt{N})$ attacks of theorem 7.21 remain the best known. This section constructs the standard supplier of such groups.

An *elliptic curve* is, concretely, the solution set of a certain cubic equation in two variables, together with one extra "point at infinity". Its notable feature is that this solution set carries the structure of an abelian group, defined by a strikingly geometric rule; and that inverting scalar multiplication in this group is hard—the discrete logarithm problem in a well-chosen elliptic-curve group appears to require exponential time. The combination of clean algebraic structure with a hard computational problem is exactly what cryptography requires.

Throughout the section we work over a field K whose characteristic (theorem 4.30) is neither 2 nor 3. In the applications K is a finite prime field $\mathbb{F}_p$ with p a large prime, but the basic theory is cleaner over a general field, and we specialise when needed. The reader should keep in mind the two *Pasta curves*

$$E_p : y^2 = x^3 + 5 \quad \text{over } \mathbb{F}_p, \qquad E_q : y^2 = x^3 + 5 \quad \text{over } \mathbb{F}_q,$$

where p and q are the two 255-bit primes

$$p = 2^{254} + 0x224698fc094cf91b992d30ed00000001,$$
$$q = 2^{254} + 0x224698fc0994a8dd8c46eb2100000001.$$

These are the primes that theorem 6.22 and theorem 7.17 handled as black boxes; here they finally acquire their curves. The constants are the deployed ones: the primes are the `MODULUS` values of the `pasta_curves` crate (`src/fields/fp.rs` and `src/fields/fq.rs`), the curve equations $y^2 = x^3 + 5$ are defined in `src/curves.rs`, and both are specified in §5.4.9.6 ("Pallas and Vesta") of the Zcash protocol specification. The curve E_p, defined over $\mathbb{F}_p$, is called *Pallas*; the curve E_q, defined over $\mathbb{F}_q$, is called *Vesta*. Together they form a *cycle*: the number of $\mathbb{F}_p$-points of Pallas equals q, and the number of $\mathbb{F}_q$-points of Vesta equals p—a coincidence of design, not chance, whose meaning unfolds across the section. The equation $y^2 = x^3 + 5$ serves as the running example throughout, with a toy curve over $\mathbb{F}_{11}$ developed in parallel for hand computation.

10.1 Weierstrass equations

Definition 10.1 (Short Weierstrass equation)**.** Let K be a field of characteristic $\neq 2, 3$, and let $a, b \in K$. The *short Weierstrass equation* with coefficients a, b is the equation

$$y^2 = x^3 + ax + b. \tag{9}$$

The elements a and b are the *Weierstrass coefficients*.

Over a field of characteristic $\neq 2, 3$, an invertible affine change of variables always reduces the most general cubic of this shape, $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$, to the form (9): completing the square in y (possible since $2 \neq 0$) removes the xy and y terms but changes the quadratic coefficient to $a_2 + a_1^2/4$; translating with the substitution $x = X - (a_2 + a_1^2/4)/3$ (possible since $3 \neq 0$) removes the X^2 term. This is precisely why characteristics 2 and 3 are excluded: those cases require the longer Weierstrass form, and they are not relevant to the Pasta curves, which live over large prime fields.

Not every choice of a, b gives a usable curve. We must exclude the cubics whose graph has a self-intersection or a cusp, because the group law defined below breaks down at such a singular point.

Definition 10.2 (Discriminant and nonsingularity)**.** The *discriminant* of the short Weierstrass equation (9) is

$$\Delta = -16\,(4a^3 + 27b^2) \in K. \tag{10}$$

The equation, and the cubic $f(x) = x^3 + ax + b$, is *nonsingular* (or *smooth*) if $\Delta \neq 0$, equivalently $4a^3 + 27b^2 \neq 0$. (The factor -16 is a convention inherited from the general Weierstrass form; over our fields it is nonzero and affects nothing.)

The quantity $4a^3 + 27b^2$ is exactly the obstruction to f having distinct roots. The following lemma makes this explicit by a computation reusable for any depressed cubic—a cubic with no square term—and is the engine of the smoothness criterion.

Lemma 10.3 (Discriminant of a depressed cubic)**.** *Let $f(x) = x^3 + ax + b = (x - r_1)(x - r_2)(x - r_3)$ with r_1, r_2, r_3 in some field containing K. Then*

$$\prod_{i<j}(r_i - r_j)^2 = -(4a^3 + 27b^2).$$

Proof. Expanding $(x - r_1)(x - r_2)(x - r_3)$ and comparing coefficients with $x^3 + ax + b$ gives Vieta's relations for the depressed cubic:

$$r_1 + r_2 + r_3 = 0, \qquad r_1r_2 + r_1r_3 + r_2r_3 = a, \qquad r_1r_2r_3 = -b.$$

Differentiating $f(x) = \prod_i (x - r_i)$ by the product rule and evaluating at a root kills every term but one: $f'(r_i) = \prod_{j\neq i}(r_i - r_j)$. Multiplying the three evaluations pairs each factor $(r_i - r_j)$ with $(r_j - r_i)$:

$$\prod_i f'(r_i) = \prod_{i<j}(r_i - r_j)(r_j - r_i) = (-1)^3 \prod_{i<j}(r_i - r_j)^2.$$

It remains to compute $\prod_i f'(r_i) = \prod_i (3r_i^2 + a)$. Expanding the product and collecting the elementary symmetric functions of the squares r_i^2,

$$\prod_i (3r_i^2 + a) = 27\prod_i r_i^2 + 9a\sum_{i<j} r_i^2 r_j^2 + 3a^2 \sum_i r_i^2 + a^3.$$

Vieta's relations evaluate each symmetric function: with $e_1 = 0$, $e_2 = a$, $e_3 = -b$,

$$\sum_i r_i^2 = e_1^2 - 2e_2 = -2a, \qquad \sum_{i<j} r_i^2 r_j^2 = e_2^2 - 2e_1e_3 = a^2, \qquad \prod_i r_i^2 = e_3^2 = b^2.$$

Substituting, $\prod_i f'(r_i) = 27b^2 + 9a^3 - 6a^3 + a^3 = 4a^3 + 27b^2$, and the two displays together give $\prod_{i<j}(r_i - r_j)^2 = -(4a^3 + 27b^2)$. □

Proposition 10.4 (Smoothness, four ways)**.** *Let $f(x) = x^3 + ax + b \in K[x]$. The following are equivalent.*

1. *The quantity $4a^3 + 27b^2$ of theorem 10.2 is nonzero.*
2. *The cubic f has no repeated root in the algebraic closure $\overline{K}$—the smallest extension field of K in which every nonconstant polynomial over K factors into linear factors.*
3. *The polynomials f and f' are coprime in $\overline{K}[x]$.*
4. *There is no point $(x_0, y_0) \in \overline{K}^2$ satisfying the curve equation $y^2 = f(x)$ together with the two partial-derivative equations $2y = 0$ and $f'(x) = 0$.*

Proof. Equivalence (1) ⇔ (2): over $\overline{K}$ the monic cubic factors as $f = (x - r_1)(x - r_2)(x - r_3)$, and theorem 10.3 gives $\prod_{i<j}(r_i - r_j)^2 = -(4a^3 + 27b^2)$. The left side vanishes exactly when two roots coincide, so f has a repeated root if and only if $4a^3 + 27b^2 = 0$.

Equivalence (2) ⇔ (3): this is the derivative test of theorem 5.40, applied over $\overline{K}$ where f splits: a repeated root of f is a common root of f and f', hence a common factor $(x - \alpha)$, and conversely a nontrivial $\gcd(f, f')$ has a root in $\overline{K}$ which is then a repeated root of f.

Equivalence (4) ⇔ (2): a singular point of the affine curve $y^2 = f(x)$ is a point where both partial derivatives of $g(x, y) = y^2 - f(x)$ vanish, namely $\partial g/\partial y = 2y = 0$ and $\partial g/\partial x = -f'(x) = 0$. Since $\mathrm{char}(K) \neq 2$, the first equation forces $y_0 = 0$; the curve equation then gives $f(x_0) = 0$, which combined with $f'(x_0) = 0$ says x_0 is a repeated root of f (theorem 5.40(1)). Conversely a repeated root x_0 of f yields the singular point $(x_0, 0)$. □

Definition 10.5 (Elliptic curve and rational points)**.** An *elliptic curve* over K is a short Weierstrass equation (9) with nonzero discriminant, $4a^3 + 27b^2 \neq 0$. For any field extension $L \supseteq K$, the set of *L-rational affine points* is

$$E^{\mathrm{aff}}(L) = \{\, (x, y) \in L \times L \,:\, y^2 = x^3 + ax + b \,\},$$

and the set of *L-rational points* is

$$E(L) = E^{\mathrm{aff}}(L) \cup \{\mathcal{O}\},$$

where $\mathcal{O}$ is a single formal symbol called the *point at infinity*. When $L = K$ one writes simply E for the curve and $E(K)$ for its rational points.

Remark 10.6 (The Pasta curves are smooth). For the Pasta curves $a = 0$ and $b = 5$, so $4a^3 + 27b^2 = 27 \cdot 25 = 675 = 3^3 \cdot 5^2$. The curve is therefore nonsingular whenever the characteristic does not divide 675, i.e. whenever the prime is neither 3 nor 5; the Pasta primes are enormous, so this is automatic. More generally, any curve $y^2 = x^3 + b$ with $b \neq 0$ is nonsingular away from characteristics 2 and 3.

10.2 The point at infinity, geometrically

The extra point $\mathcal{O}$ is not decoration; it is the identity element of the group to come, and it has an honest geometric home. To see it, view the affine plane as sitting inside the *projective plane*. A point of the projective plane $\mathbb{P}^2(K)$ is an equivalence class $[X : Y : Z]$ of nonzero triples $(X, Y, Z) \in K^3 \setminus \{0\}$, where $(X, Y, Z) \sim (\lambda X, \lambda Y, \lambda Z)$ for every $\lambda \in K^\times$. The affine points (x, y) embed as $[x : y : 1]$; the points with $Z = 0$ form the *line at infinity*, the extra "directions". Homogenising the Weierstrass equation (9) by setting $x = X/Z$, $y = Y/Z$ and clearing denominators gives the *projective Weierstrass equation*

$$Y^2Z = X^3 + aXZ^2 + bZ^3. \tag{11}$$

Setting $Z = 0$ in (11) forces $X^3 = 0$, hence $X = 0$, leaving the *single* projective point $[0 : 1 : 0]$. This is the point at infinity $\mathcal{O}$. It lies infinitely far up in the vertical direction, and every vertical line $x =$ const passes through it. The reader may safely picture $\mathcal{O}$ as one point pinned at the top (and bottom) of the y-axis, where all vertical lines meet; the projective equation above is the licence for that picture.

10.3 The chord-and-tangent group law: geometry

We now describe the procedure for "adding" two points of $E(K)$. The construction is purely geometric, and it rests on a single fact: a line meets a cubic curve in exactly three points, counted with multiplicity. The point at infinity is engineered precisely so that the counting is always exact.

Definition 10.7 (The group law, geometric form)**.** Let $P, Q \in E(K)$. Define $P + Q$ as follows.

- If $P = \mathcal{O}$, set $P + Q = Q$; if $Q = \mathcal{O}$, set $P + Q = P$.
- Otherwise $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ are affine. Draw the line ℓ through P and Q; if $P = Q$, let ℓ be the tangent line to the curve at P. The line ℓ meets the curve in a third point R', counted with multiplicity, where a vertical ℓ has third point $\mathcal{O}$. Define $P + Q$ to be the *reflection of R' across the x-axis*: if $R' = (x_3, -y_3)$ then $P + Q = (x_3, y_3)$, and if $R' = \mathcal{O}$ then $P + Q = \mathcal{O}$.

The reflection step deserves a comment, because it is what converts "the three intersection points of a line sum to a fixed thing" into a genuine group law with $\mathcal{O}$ as identity. The underlying rule is this: *the three intersection points of any line with the curve add up to $\mathcal{O}$.* (Stated here as a guiding picture; every use of it below is discharged by an explicit computation with the formulas of §10.4.) Reflection across the x-axis sends an affine point (x, y) to $(x, -y)$, which—as the next subsection confirms—is its additive inverse; and it sends $\mathcal{O}$ to $\mathcal{O}$. To convert the geometry into formulas one must (i) find the third intersection point of a line with the cubic and (ii) handle the degenerate cases. The key algebraic device for (i) is Vieta's relation between the roots of a cubic and its coefficients, already deployed in theorem 10.3.

10.4 Explicit affine formulas

Fix an elliptic curve $E : y^2 = x^3 + ax + b$ over K, and let $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ denote affine points.

Definition 10.8 (Negation)**.** Define

$$-\mathcal{O} = \mathcal{O}, \qquad -(x_1, y_1) = (x_1, -y_1).$$

The reflected point $(x_1, -y_1)$ is again on the curve, since the curve equation depends on y only through y^2. Geometrically, the line through P and $-P$ is the vertical line $x = x_1$, which meets the curve at P, $-P$, and $\mathcal{O}$.

Construction 10.9 (Chord addition). Suppose $P, Q \neq \mathcal{O}$ and $x_1 \neq x_2$. The line through P and Q is $y = \lambda x + \mu$ with

$$\lambda = \frac{y_2 - y_1}{x_2 - x_1}, \qquad \mu = y_1 - \lambda x_1.$$

Substituting the line into the curve equation, $(\lambda x + \mu)^2 = x^3 + ax + b$, and collecting terms gives

$$g(x) := x^3 - \lambda^2 x^2 + (a - 2\lambda\mu)\,x + (b - \mu^2) = 0.$$

Both x_1 and x_2 are roots of the monic cubic g: each point lies on the line and on the curve, so $g(x_i) = f(x_i) - (\lambda x_i + \mu)^2 = f(x_i) - y_i^2 = 0$. Since $x_1 \neq x_2$, factoring out $(x - x_1)(x - x_2)$ (theorem 5.17, applied twice) leaves a monic linear factor $x - x_3$ with $x_3 \in K$: the x-coordinate of the third intersection point R'. By Vieta, the sum of the roots equals the negative of the x^2-coefficient—so $x_1 + x_2 + x_3 = \lambda^2$, and the unknown root is read off with no factoring at all. Hence

$$x_3 = \lambda^2 - x_1 - x_2, \qquad y_3 = \lambda\,(x_1 - x_3) - y_1, \tag{12}$$

and $P + Q = (x_3, y_3)$. The y-formula already incorporates the reflection across the x-axis: the third intersection point is $(x_3, \lambda x_3 + \mu)$, and reflecting gives $y_3 = -(\lambda x_3 + \mu) = -(\lambda x_3 + y_1 - \lambda x_1) = \lambda(x_1 - x_3) - y_1$.

Construction 10.10 (Tangent doubling). Suppose $P = Q = (x_1, y_1)$ with $y_1 \neq 0$. Implicit differentiation of $y^2 = x^3 + ax + b$ yields the tangent slope: from $2y\,dy = (3x^2 + a)\,dx$,

$$\lambda = \frac{3x_1^2 + a}{2y_1}.$$

With $\mu = y_1 - \lambda x_1$, substituting $y = \lambda x + \mu$ into the curve equation produces the same monic cubic $g(x) = x^3 + ax + b - (\lambda x + \mu)^2$ as in theorem 10.9, and here x_1 is a *double* root. The slogan "tangency counts the intersection twice" is a one-line check: $g(x_1) = f(x_1) - y_1^2 = 0$, and

$$g'(x_1) = 3x_1^2 + a - 2\lambda\,(\lambda x_1 + \mu) = 3x_1^2 + a - 2\lambda y_1 = 0$$

by the choice of λ, so x_1 is a repeated root of g (theorem 5.40(1)). Factoring $(x - x_1)^2$ out of g leaves a monic linear factor $x - x_3$ with $x_3 \in K$, and Vieta now reads $x_1 + x_1 + x_3 = \lambda^2$, hence

$$x_3 = \lambda^2 - 2x_1, \qquad y_3 = \lambda\,(x_1 - x_3) - y_1, \qquad [2]P = (x_3, y_3). \tag{13}$$

The remaining cases are exactly those in which the line is vertical.

- If $P = \mathcal{O}$, then $P + Q = Q$; symmetrically if $Q = \mathcal{O}$.
- If $x_1 = x_2$ but $y_1 = -y_2$ (so $Q = -P$, including the subcase $y_1 = y_2 = 0$), the line is vertical, the third point is $\mathcal{O}$, and $P + Q = \mathcal{O}$.
- If $P = Q$ and $y_1 = 0$, the tangent is vertical (the slope formula in theorem 10.10 has zero denominator), so $[2]P = \mathcal{O}$. These are exactly the points of order two, studied in section 10.9.

Every case is covered: either an input is $\mathcal{O}$; or both inputs are affine with $x_1 \neq x_2$ (use (12)); or both are affine with $x_1 = x_2$, in which case either $y_1 = -y_2$ (result $\mathcal{O}$) or $y_1 = y_2 \neq 0$ (use (13)).

Proposition 10.11 (Closure). *For all $P, Q \in E(K)$, the point $P + Q$ computed by the case analysis above lies in $E(K)$. In particular $E(K)$ is closed under $+$.*

Proof. If either input is $\mathcal{O}$, the result is the other input, which lies in $E(K)$ by hypothesis; if the result is $\mathcal{O}$, it lies in $E(K)$ by definition. Otherwise (x_3, y_3) is produced by (12) or (13), and in both constructions x_3 was exhibited as a root of the cubic $g(x) = x^3 + ax + b - (\lambda x + \mu)^2$—as the third root beside x_1, x_2 in theorem 10.9, and beside the double root x_1 in theorem 10.10. The equation $g(x_3) = 0$ states precisely that the third intersection point $(x_3, \lambda x_3 + \mu) = (x_3, -y_3)$ satisfies the curve equation; and since the curve equation is invariant under $y \mapsto -y$, the reflected point (x_3, y_3) is on the curve too, so $P + Q \in E^{\mathrm{aff}}(K) \subseteq E(K)$. For rationality, observe that all coordinates are rational expressions in x_1, x_2, y_1, y_2, a whose denominators are $x_2 - x_1$ or $2y_1$, nonzero in the cases where each formula is invoked; so $x_3, y_3 \in K$. □

Remark 10.12 (Complete versus incomplete addition). The chord formula (12) is *incomplete*: it fails whenever $x_1 = x_2$—the doubling and inverse cases—and an implementation must branch into the case analysis above. *Complete* addition formulas, single rational expressions valid for *all* input pairs including $P = Q$ and $P = -Q$, exist for the curves used here, at the cost of more field multiplications per addition. The distinction is load-bearing in later volumes. Halo 2's elliptic-curve gadgets provide complete addition (the `halo2_gadgets` crate, `src/ecc/chip/add.rs`) and use incomplete gates only where their callers justify the exclusions; the Sinsemilla hash of Orchard deliberately uses the *incomplete* formulas nondeterministically (`halo2_gadgets`, `src/sinsemilla/chip/hash_to_point.rs`)—an exceptional case does not violate the constraints but leaves the affected output unconstrained (the prover may choose the intermediate slope freely), which the specification models by letting the hash return $\perp$ and weakening the proved statement accordingly; its Merkle path validity check is even permitted to pass on the resulting sentinel values. Safety rests not on the proof system but on Theorem 5.4.4 of the Zcash protocol specification: any input reaching an exceptional case of the incomplete addition immediately yields a nontrivial discrete-logarithm relation among the Sinsemilla generators, so under discrete-log hardness on Pallas (theorem 10.28) such inputs are infeasible to find, for honest and adversarial provers alike.

10.5 Identity, inverses, commutativity, and associativity

The destination of the whole construction is the following theorem.

Theorem 10.13 (Group law). *Let E be an elliptic curve over a field K. The set $E(K)$ with the chord-and-tangent operation $+$ of theorem 10.7 is an abelian group with identity $\mathcal{O}$.*

Closure is theorem 10.11. The next proposition supplies every remaining axiom except one.

Proposition 10.14 (Identity, inverses, commutativity). *The operation $+$ on $E(K)$ has the following properties.*

1. *The point $\mathcal{O}$ is a two-sided identity: $P + \mathcal{O} = \mathcal{O} + P = P$ for all P.*
2. *Every P has the two-sided inverse $-P$ of theorem 10.8: $P + (-P) = \mathcal{O}$.*
3. *The operation is commutative: $P + Q = Q + P$.*

Proof. (1) is the first clause of theorem 10.7. (2): for $P = (x, y)$ with $y \neq 0$, the line through P and $-P = (x, -y)$ is vertical, so its third point is $\mathcal{O}$ and $P + (-P) = \mathcal{O}$ by the vertical-line case; for $y = 0$ one has $P = -P$ and $[2]P = \mathcal{O}$ directly by the vertical-tangent case; and $\mathcal{O} + (-\mathcal{O}) = \mathcal{O} + \mathcal{O} = \mathcal{O}$. (3): the chord through P and Q does not depend on the order in which the two points are named, and neither does the tangent in the doubling case; algebraically, the formulas (12) are symmetric under the swap $(x_1, y_1) \leftrightarrow (x_2, y_2)$—the slope λ is unchanged, and $x_3 = \lambda^2 - x_1 - x_2$ is symmetric in x_1, x_2. □

The one axiom not yet established is *associativity*, $(P + Q) + R = P + (Q + R)$, and it is the deep and surprising part of the theory: the definition gives no reason why reflecting third intersection points should be an associative operation, and a brute-force verification through the explicit formulas splinters into a large number of coordinate cases, each demanding lengthy rational-function identities. We sketch the conceptual proof, which exhibits the reason associativity holds.

Proof of associativity (sketch). The key tool is the classical *Cayley–Bacharach theorem* for plane cubics: if two cubics with no common component meet in nine points counted with intersection multiplicity, then any third cubic passing through eight of them with the required multiplicities also passes through the ninth. To prove $(P + Q) + R = P + (Q + R)$, one writes both sides via the chord-and-tangent construction and arranges the relevant intersections among the curve E and two degenerate cubics, each a product of three suitably chosen lines. Cayley–Bacharach forces the two candidate sums to coincide; its intersection-multiplicity formulation includes tangencies and coincident points. Translating "reflect across the x-axis" into "the three points of any line sum to $\mathcal{O}$" renders the bookkeeping uniform.

A complete, case-free treatment identifies $E(K)$ with the degree-zero part $\mathrm{Pic}^0(E)$ of the divisor class group of the curve via the map $P \mapsto [P] - [\mathcal{O}]$, a bijection that intertwines the two operations; the group law on $\mathrm{Pic}^0(E)$ is manifestly associative, because it is inherited from addition of divisors modulo principal divisors. Either route yields a genuine proof; both lie beyond the elementary scope of this volume, so we take associativity as established. □

Remark 10.15 (The notation $[n]P$). For $n \in \mathbb{Z}$ and $P \in E(K)$, we write $[n]P$ for the n-fold sum $P + \cdots + P$, with $[0]P = \mathcal{O}$ and $[-n]P = -([n]P)$. This makes the abelian group $E(K)$ a $\mathbb{Z}$-module: integers act on points. The map $n \mapsto [n]P$ is the *scalar multiplication* of theorem 10.17—not any coordinate-wise product.

10.6 Projective and Jacobian coordinates

The affine formulas (12) and (13) each require a field *inversion*, by $x_2 - x_1$ or $2y_1$. In a large prime field an inversion costs roughly as much as dozens of multiplications, whether computed by the extended Euclidean algorithm (theorem 6.1) or as z^{p-2} by Fermat (theorem 6.4). When a long chain of group operations arises, as in the scalar multiplication of the next subsection, it pays to defer all inversions to the very end by carrying points in a redundant coordinate system whose group law uses only additions, subtractions, and multiplications.

Remark 10.16 (Inversion-free coordinate systems). Two redundant systems serve this purpose. In *(homogeneous) projective coordinates*, a triple $[X : Y : Z]$ with $Z \neq 0$ represents the affine point $(X/Z, Y/Z)$ on the projective curve (11), with $\mathcal{O} = [0 : 1 : 0]$. In *Jacobian coordinates*, a triple $(X : Y : Z)$ with $Z \neq 0$ represents $(X/Z^2, Y/Z^3)$, under the equivalence $(X, Y, Z) \sim (\lambda^2 X, \lambda^3 Y, \lambda Z)$; the curve equation becomes $Y^2 = X^3 + aXZ^4 + bZ^6$. In either system the addition and doubling formulas can be rewritten with no divisions at all, at the cost of more multiplications per operation; a long chain of group operations then defers every inversion to a single final recovery of affine coordinates—compute Z^{-1} once, then the needed powers of it by multiplication. Jacobian coordinates usually yield the faster doubling and are the standard choice in cryptographic libraries, including the implementation Halo 2 uses: the `pasta_curves` crate carries its curve points in Jacobian coordinates (`src/curves.rs`). For curves $y^2 = x^3 + b$ with $a = 0$, the term aZ^4 in the Jacobian doubling formulas vanishes, making doubling especially cheap—one reason the Pasta curves take $a = 0$.

10.7 Scalar multiplication and double-and-add

The fundamental operation of elliptic-curve cryptography is *scalar multiplication*: given a point P and an integer $n \geq 0$, compute

$$[n]P = \underbrace{P + P + \cdots + P}_{n \text{ times}}.$$

Computing this by $n - 1$ successive additions is infeasible when n has hundreds of bits. Instead one uses the binary expansion of n, exactly as in the square-and-multiply technique of theorem 6.2; as theorem 6.3 promised, the additive rendition of that algorithm is the following.

Definition 10.17 (Double-and-add). If $n = 0$, return $\mathcal{O}$. Otherwise write $n = \sum_{i=0}^{k-1} n_i 2^i$ with digits $n_i \in \{0, 1\}$ and top digit $n_{k-1} = 1$. The *left-to-right double-and-add* algorithm computes $[n]P$ as follows: set $R \leftarrow P$; for $i = k - 2$ down to 0, set $R \leftarrow [2]R$, and if $n_i = 1$ set $R \leftarrow R + P$; return R.

Proposition 10.18. *For $n \geq 1$, double-and-add returns $[n]P$ using exactly $k - 1$ doublings and at most $k - 1$ additions, where $k = \lfloor \log_2 n \rfloor + 1$ is the bit length of n. For $n = 0$, it returns $\mathcal{O}$ without a group operation.*

Proof. The case $n = 0$ is immediate. Let $n \geq 1$, and let R_j denote the value of R after bits $n_{k-1}, \dots, n_j$ have been incorporated. We claim $R_j = [\lfloor n/2^j \rfloor]P$, by downward induction on j. At initialization, $R_{k-1} = P = [\lfloor n/2^{k-1} \rfloor]P$ because the top bit is 1. Assuming $R_{j+1} = [\lfloor n/2^{j+1} \rfloor]P$, the next iteration computes

$$[2]R_{j+1} + [n_j]P = [\, 2\lfloor n/2^{j+1} \rfloor + n_j \,]P = [\lfloor n/2^j \rfloor]P,$$

where the last step is the digit identity $2\lfloor n/2^{j+1} \rfloor + n_j = \lfloor n/2^j \rfloor$: stripping the last bit of $\lfloor n/2^j \rfloor$ and re-appending it recovers the number. At $j = 0$ the invariant reads $R_0 = [n]P$, which is what the algorithm returns. For the counts, the loop has exactly $k-1$ iterations, each performing one doubling and at most one addition. □

Scalar multiplication by an n-bit scalar therefore costs $O(n)$ group operations—$O(n)$ field operations up to constant factors, or $O(n \log^2 p)$ bit operations with schoolbook field arithmetic. This efficiency of the *forward* map, contrasted with the apparent hardness of the inverse problem (§10.11), is the foundation of the cryptography.

10.8 The group structure of $E(\mathbb{F}_p)$

Now specialise $K = \mathbb{F}_p$, the prime field with p elements (p prime, $p > 3$). Since $\mathbb{F}_p$ is finite, so is $E(\mathbb{F}_p)$: each of the p values of x admits at most two values of y (theorem 5.18, applied to $y^2 - f(x)$ as a polynomial in y), so with $\mathcal{O}$ there are at most $2p + 1$ points. By theorem 10.13, the set $E(\mathbb{F}_p)$ is thus a *finite abelian group*. One of the central theorems of the subject governs its order.

Theorem 10.19 (Hasse). *Let E be an elliptic curve over $\mathbb{F}_p$. Write*

$$\#E(\mathbb{F}_p) = p + 1 - t,$$

which defines *the integer t, called the* trace of Frobenius. *Then*

$$|t| \leq 2\sqrt{p};$$

equivalently, $\#E(\mathbb{F}_p)$ lies in the Hasse interval $[\, p + 1 - 2\sqrt{p},\ p + 1 + 2\sqrt{p}\,]$.

Proof (sketch). The count has an exact character-sum expression. For fixed x, the number of $y \in \mathbb{F}_p$ with $y^2 = f(x)$ is $1+\left(\frac{f(x)}{p}\right)$ when $f(x) \neq 0$—two solutions for a nonzero square, none for a non-square (theorem 7.12)—and exactly $1 = 1 + \left(\frac{0}{p}\right)$ when $f(x) = 0$, using the Legendre symbol of theorem 7.14 with the convention $\left(\frac{0}{p}\right) = 0$. Summing over x,

$$\#E^{\mathrm{aff}}(\mathbb{F}_p) = \sum_{x\in\mathbb{F}_p}\left(1 + \left(\frac{x^3 + ax + b}{p}\right)\right) = p + \sum_{x\in\mathbb{F}_p}\left(\frac{x^3 + ax + b}{p}\right),$$

so $t = -\sum_x \left(\frac{x^3+ax+b}{p}\right)$. Heuristically, as x ranges over $\mathbb{F}_p$ the value $f(x)$ is a nonzero square about half the time (two points), a non-square about half the time (none), and zero occasionally (one point); the p summands ± 1 behave like a random walk, and Hasse's bound $|t| \leq 2\sqrt{p}$ is exactly square-root cancellation in this character sum.

The conceptual proof realises t as the trace of the p-power *Frobenius endomorphism* $\pi : (x, y) \mapsto (x^p, y^p)$ of the curve—the curve-level shadow of the field automorphism of theorem 6.13. One shows that π satisfies the characteristic equation $\pi^2 - [t]\pi + [p] = 0$ in the endomorphism ring of E, and that the associated quadratic form (the degree) is positive definite; the discriminant condition $t^2 - 4p \leq 0$ for the complex roots of the characteristic polynomial then yields $|t| \leq 2\sqrt{p}$. The full argument requires the theory of isogenies and the Weil pairing, beyond the present scope. □

Hasse's theorem says the order of $E(\mathbb{F}_p)$ is *very close* to p: the deviation from $p + 1$ is at most $2\sqrt{p}$, a relative error of order $p^{-1/2}$. For cryptography this is crucial twice over. It guarantees with no brute-force count that the group is large—about p elements, so about 255 bits for the Pasta curves—and it anchors the point-counting algorithms (Schoof–Elkies–Atkin) that compute the *exact* order in polynomial time, which is how curve designers certify their group orders.

For the cryptographic application one wants $E(\mathbb{F}_p)$ to contain a large subgroup of *prime* order. The general structure theory of $E(\mathbb{F}_p)$ as an abelian group is not needed in this series, because the Pasta curves are designed so that $\#E(\mathbb{F}_p)$ is itself prime: a group of prime order is cyclic with no proper nontrivial subgroups (theorem 3.34), so $E(\mathbb{F}_p)$ is cyclic of prime order—the optimal case.

10.9 Torsion, the cofactor, and prime-order subgroups

Definition 10.20 (Order of a point and torsion)**.** The *order* of a point $P \in E(\mathbb{F}_p)$ is its order as a group element (theorem 3.11): the least integer $m \geq 1$ with $[m]P = \mathcal{O}$. It exists and divides $\#E(\mathbb{F}_p)$, by Lagrange's theorem (theorem 3.29 via theorem 3.30). For an integer $m \geq 1$, the *m-torsion subgroup* is

$$E[m] = \{ P \in E(\overline{\mathbb{F}_p}) : [m]P = \mathcal{O} \},$$

the points over the algebraic closure killed by m; its rational part is $E(\mathbb{F}_p)[m] = E[m] \cap E(\mathbb{F}_p)$.

Definition 10.21 (Cofactor)**.** Suppose $\#E(\mathbb{F}_p) = h \cdot r$, where r is the (large) prime to be used and h is the remaining factor. The integer h is the *cofactor*, and r is the order of the *prime-order subgroup*: a point P of order r generates a cyclic subgroup $\langle P\rangle \cong \mathbb{Z}/r\mathbb{Z}$, and this subgroup is where the cryptography lives.

A nontrivial cofactor $h > 1$ is undesirable: it admits *small-subgroup* points, of order dividing h, which can leak information or break protocol assumptions—the small-subgroup attacks of theorem 7.24. Implementations therefore either *clear the cofactor*, multiplying incoming points by h, or use prime-order curves with $h = 1$. The Pasta curves have cofactor 1: the orders $\#E_p(\mathbb{F}_p) = q$ and

$\#E_q(\mathbb{F}_q) = p$ are themselves prime, so $E_p(\mathbb{F}_p) \cong \mathbb{Z}/q\mathbb{Z}$ and $E_q(\mathbb{F}_q) \cong \mathbb{Z}/p\mathbb{Z}$ are cyclic of prime order and no cofactor clearing is ever needed. This is one of the principal design advantages of the Pasta cycle for proof systems.

Proposition 10.22 (Two-torsion)**.** *The affine points of order* 2 *in* $E(\mathbb{F}_p)$ *are exactly the points* $(x_0, 0)$ *with* $f(x_0) = x_0^3 + ax_0 + b = 0$. *Hence* $E(\mathbb{F}_p)$ *has* 0, 1, *or* 3 *points of order* 2 *according as* f *has* 0, 1, *or* 3 *roots in* $\mathbb{F}_p$*; and* $\#E(\mathbb{F}_p)$ *is even if and only if* f *has a root in* $\mathbb{F}_p$.

Proof. A point $P = (x, y)$ has order 2 iff $P = -P$ and $P \neq \mathcal{O}$, i.e. $y = -y$, i.e. $y = 0$ (since $\mathrm{char}(\mathbb{F}_p) \neq 2$); the curve equation then forces $f(x) = 0$. Conversely each root x_0 of f in $\mathbb{F}_p$ gives the order-2 point $(x_0, 0)$.

Next, the number N of roots of f in $\mathbb{F}_p$ can only be 0, 1, or 3: two roots force a third. Suppose f has distinct roots $r_1 \neq r_2$ in $\mathbb{F}_p$. The factor theorem (theorem 5.17) gives $f = (x - r_1)h$; evaluating at r_2 yields $0 = (r_2 - r_1)\, h(r_2)$, so $h(r_2) = 0$, and a second application factors the monic cubic completely, $f = (x-r_1)(x-r_2)(x-r_3)$ with $r_3 \in \mathbb{F}_p$—explicitly $r_3 = -r_1-r_2$ by Vieta, the depressed cubic having no x^2 term. Nonsingularity says f has no repeated root even over $\overline{\mathbb{F}_p}$ (theorem 10.4(2)), so $r_3 \notin \{r_1, r_2\}$, and f has exactly three roots (theorem 5.18 caps the count). Hence $N \in \{0, 1, 3\}$, matching the trichotomy of the statement. Together with $\mathcal{O}$ the order-2 points form the rational 2-torsion subgroup $E(\mathbb{F}_p)[2]$, which is accordingly trivial, $\mathbb{Z}/2\mathbb{Z}$, or $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

For the parity claim, consider negation $P \mapsto -P$, an *involution*—a self-inverse map—of the finite set $E(\mathbb{F}_p)$. Its fixed points are exactly $\mathcal{O}$ and the points $(x_0, 0)$; every other point pairs off with its distinct negative into a 2-element orbit. Counting the set by orbits,

$$\#E(\mathbb{F}_p) \equiv 1 + N \pmod 2,$$

where $N \in \{0, 1, 3\}$ is the number of roots of f in $\mathbb{F}_p$. Hence $\#E(\mathbb{F}_p)$ is even iff N is odd, i.e. iff $N \geq 1$, i.e. iff f has a root in $\mathbb{F}_p$. (The device is worth remembering: an involution reads off a group's cardinality modulo 2 from its fixed points alone.) □

Remark 10.23 (No two-torsion on the Pasta curves). For $y^2 = x^3 + 5$ over $\mathbb{F}_p$, the curve has a point of order 2 iff $x^3 + 5 = 0$ has a solution, i.e. iff -5 is a cube in $\mathbb{F}_p$. For the Pasta primes the order $\#E_p(\mathbb{F}_p) = q$ is odd (it is prime), so there are *no* rational 2-torsion points; consistently, -5 is not a cube in $\mathbb{F}_p$.

10.10 Base fields, scalar fields, and the Pasta cycle

Step back from the geometry and ask what a machine actually does when it works on $E(\mathbb{F}_p)$. Every operation of the group law—the slope $\lambda = (y_2 - y_1)(x_2 - x_1)^{-1}$, the squarings, and the inversions needed to add and to double—is arithmetic in $\mathbb{F}_p$, performed with exactly the prime-field algorithms of section 6.1 (extended-Euclid or Fermat inversion). The field $\mathbb{F}_p$ is called the *base field* (or coordinate field) of the curve: it is where the coordinates live.

Scalars live elsewhere. By theorem 10.19 the group $E(\mathbb{F}_p)$ is a finite abelian group of order $N = \#E(\mathbb{F}_p)$ close to p; for cryptography one selects a curve whose order has a large prime factor r, and keys and signatures reside in the subgroup of order r (theorem 10.21). A scalar multiplication $P \mapsto [s]P$ on a point of order r depends only on s mod r, since $[r]P = \mathcal{O}$ (theorem 3.12); scalars therefore live modulo r—and since r is prime, they form a *field* (theorem 4.26).

Definition 10.24 (Base field and scalar field)**.** For an elliptic curve used in cryptography, with a prime-order subgroup of order r:

- the *base field* $\mathbb{F}_p$ is the field over which the curve is defined and in which point coordinates lie;

- the *scalar field* $\mathbb{F}_r = \mathbb{Z}/r\mathbb{Z}$ is the field of scalars by which one multiplies points, i.e. the integers modulo the subgroup order r.

Both are prime fields, but in general $p \neq r$.

Remark 10.25 (Two fields at once, and the Pasta cycle). A typical elliptic-curve operation manipulates two distinct prime fields simultaneously: the exponent arithmetic runs in $\mathbb{F}_r$ while the coordinate arithmetic runs in $\mathbb{F}_p$. Halo 2's Pallas and Vesta curves (the "Pasta" pair) exploit this deliberately. They form a *cycle of curves*: the base field of one is the scalar field of the other and vice versa,

$$\#E_p(\mathbb{F}_p) = q, \qquad \#E_q(\mathbb{F}_q) = p,$$

so the scalar field of Pallas is $\mathbb{F}_q$, the base field of Vesta, and symmetrically (the `pasta_curves` crate, `src/curves.rs`; §5.4.9.6 of the Zcash protocol specification). This lets a proof system express the verifier of one curve's arithmetic *natively* in the field of the other, enabling the recursive proof composition at the heart of Halo 2. Moreover both primes are chosen $\equiv 1 \pmod{2^{32}}$, so by theorem 6.21 each field contains a multiplicative subgroup of order 2^{32}, furnishing the high-order roots of unity that the number-theoretic transform of section 9 needs for fast polynomial multiplication.

10.11 The elliptic-curve discrete logarithm problem

The section's opening problem now returns in its proper habitat. The discrete logarithm problem of theorem 7.20 was stated for an abstract cyclic group; the elliptic-curve instance reads as follows, in additive notation.

Definition 10.26 (ECDLP). Let $P \in E(\mathbb{F}_p)$ be a point of large prime order r, generating $G = \langle P \rangle$. The *elliptic-curve discrete logarithm problem* (ECDLP) is: given P and a point $Q \in G$, find the unique integer $n \in \{0, 1, \ldots, r-1\}$ with $Q = [n]P$. One writes $n = \log_P Q$.

The map $n \mapsto [n]P$ is a group isomorphism $\mathbb{Z}/r\mathbb{Z} \to G$ (theorem 3.43), so a discrete logarithm always exists and is unique; the entire content of the problem is *computational*. The forward map is fast by double-and-add (theorem 10.18); no efficient algorithm is known to invert it on a well-chosen curve. What is known is the generic pair of attacks, which transfer verbatim from theorem 7.21 to the additive setting.

Proposition 10.27 (Square-root attacks). *Baby-step giant-step solves the ECDLP in a group of prime order r in $O(\sqrt{r})$ group operations and $O(\sqrt{r})$ space. Pollard's rho method matches the $O(\sqrt{r})$ bound in expectation with only $O(1)$ space, under a modelling assumption on its pseudo-random walk that is stated explicitly in the proof and is unproved for every concrete walk.*

Proof. For baby-step giant-step, set $m = \lceil \sqrt{r} \rceil$ and write the unknown logarithm as $n = im + j$ with $0 \le i, j < m$; since $m^2 \ge r$, every $n \in \{0, \ldots, r-1\}$ has such a decomposition. Tabulate the *baby steps* $[j]P$ for $j = 0, \ldots, m-1$ (a table of m points, built with $m-1$ additions), compute $S = [m]P$ once, and walk the *giant steps* $Q - [i]S$ for $i = 0, 1, 2, \ldots$, testing each against the table. At $i = \lfloor n/m \rfloor$ the walk hits $Q - [im]P = [n - im]P = [j]P$, a table entry; the collision reveals i and j, hence $n = im + j$. Both the table and the walk cost $O(m) = O(\sqrt{r})$ group operations, and the table holds $O(\sqrt{r})$ points.

Pollard's rho eliminates the table, at the price of an explicitly heuristic analysis. The algorithm takes a pseudo-random walk on G whose every position is maintained in the form $[a]P + [b]Q$ with known coefficients a, b, started from such a combination with random coefficients; the walk's next step depends only on the current point, so once the walk revisits a point it cycles forever. A repeat is therefore detected with $O(1)$ memory: run a second copy of the walk at double speed, and the two copies meet at a common point inside the cycle within a constant factor of the first repeat time.

Modelling assumption: the successive positions of the walk behave like independent uniform samples from G. No proof of this is known for any concrete walk—the assumption is what "well-designed" means, and experiment supports it—and the rest of the argument is conditional on it. Under the assumption, the first k positions are pairwise distinct with probability at most $e^{-k(k-1)/(2r)}$ by the birthday bound—the forward reference theorem 11.18 is quantitative machinery from the closing section—so the number T of steps to the first repeat satisfies $\Pr[T > k] \le e^{-k(k-1)/(2r)}$. The tail-sum formula (theorem 11.33) converts the tail bound into an expectation bound: with $m = \lceil\sqrt{r}\rceil$ and $r \ge 4$,

$$\mathbb{E}[T] = \sum_{k\ge1} \Pr[T \ge k] \le m \sum_{j\ge0} \Pr[T > jm] \le m\Big(1 + \sum_{j\ge1} e^{-j/4}\Big) = O(\sqrt{r}),$$

where the middle step groups the indices k into blocks of length m, bounding each block by its first term, and the last uses $jm(jm-1) \ge j^2 m(m-1) \ge j^2(r-\sqrt{r}) \ge j^2 r/2$ for $j \ge 1$, so that $\Pr[T > jm] \le e^{-j^2/4} \le e^{-j/4}$, a convergent geometric tail. A repeat presents one group element with two representations, $[a]P + [b]Q = [a']P + [b']Q$. If $b' \not\equiv b \pmod r$, this rearranges to $[a-a']P = [b'-b]Q = [(b'-b)\,n]P$, whence

$$n \equiv (a-a')(b'-b)^{-1} \pmod r,$$

the inverse existing because r is prime. If instead $b' \equiv b$, then $[a]P = [a']P$ forces $a \equiv a'$: the two representations coincide, the collision reveals nothing, and the walk is restarted with fresh random starting coefficients; that such degenerate collisions are rare is a further part of the heuristic, borne out in practice. Both algorithms run in time exponential in the bit length $\log_2 r$. □

Remark 10.28 (Best known attacks on well-chosen curves). For well-chosen curves, no unrestricted classical attack faster than $O(\sqrt{r})$ is known; this is not an unconditional lower bound for all classical algorithms. In the classical generic-group model, however, Shoup's matching lower bound is proved (theorem 7.21). For $r \approx 2^{254}$—the 255-bit Pasta orders—this is about 2^{127} operations, infeasible. Curves with special structure can be weaker, and the known weak classes are avoided by design. *Anomalous* curves, those with $\#E(\mathbb{F}_p) = p$, admit a polynomial-time attack through the formal group. The MOV/Frey–Rück attack transfers the ECDLP into a finite-field discrete logarithm when the *embedding degree*—the least $k \ge 1$ with $r \mid p^k - 1$, i.e. the degree (theorem 8.4; the dimension of the extension over its base) of the smallest extension of $\mathbb{F}_p$ whose multiplicative group contains a subgroup of order r—is small, where index calculus applies. Cryptographically secure curves, the Pasta curves included, are chosen with a large prime subgroup order, a large embedding degree, and $\#E(\mathbb{F}_p) \ne p$, defeating every known subexponential attack.

Remark 10.29 (The index-calculus contrast). The contrast with the multiplicative groups $\mathbb{F}_p^\times$ is the economic heart of the subject. In $\mathbb{F}_p^\times$ the index-calculus method solves the discrete logarithm in subexponential time $L_p[1/3]$—shorthand for $2^{\Theta(\lambda^{1/3}(\log\lambda)^{2/3})}$ operations in the bit length $\lambda = \log_2 p$, a notation defined precisely in theorem 11.41 of the closing section—exploiting the integer arithmetic behind the residues (theorem 7.21); those groups must therefore be made very large. On a well-chosen elliptic curve no index calculus is known—the points offer no analogue of "factoring into small primes"—so the generic $O(\sqrt{r})$ bound stands, and much smaller groups suffice: about 256 bits for 128-bit security, versus thousands of bits for $\mathbb{F}_p^\times$. Smaller groups mean faster arithmetic and shorter keys, which is why elliptic curves dominate modern protocols, including Halo 2.

10.12 The j-invariant and the GLV endomorphism

Two final pieces of structure round out the theory: a single quantity that classifies curves up to isomorphism, and an "extra symmetry" available exactly when that quantity vanishes—which the Pasta curves arrange on purpose.

Definition 10.30 (j-invariant)**.** The *j-invariant* of the curve $y^2 = x^3 + ax + b$ (with $4a^3 + 27b^2 \neq 0$) is

$$j = 1728\,\frac{4a^3}{4a^3 + 27b^2} \in K,$$

where $1728 = 12^3$ is a normalising constant of historical origin.

Proposition 10.31. *Two elliptic curves over $\overline{K}$ are isomorphic over $\overline{K}$ if and only if they have the same j-invariant. The admissible coordinate changes preserving short Weierstrass form are $(x, y) \mapsto (u^2x, u^3y)$ for $u \in \overline{K}^\times$, under which $(a, b) \mapsto (u^4a, u^6b)$; the combination j is invariant.*

Proof (sketch). A direct computation shows that the only isomorphisms between short Weierstrass curves fixing $\mathcal{O}$ are the scalings $(x, y) \mapsto (u^2x, u^3y)$. Substituting $x = u^{-2}x'$, $y = u^{-3}y'$ into $y^2 = x^3 + ax + b$ and clearing u^6 gives $y'^2 = x'^3 + u^4a\,x' + u^6b$, which is the stated action on coefficients. Under it, numerator and denominator of $4a^3/(4a^3 + 27b^2)$ both scale by u^{12}, so j is unchanged. Conversely, given two curves with equal j-invariants one solves for a suitable u transforming one coefficient pair into the other, with separate easy cases $j = 0$ (both a vanish) and $j = 1728$ (both b vanish). □

Remark 10.32 (The Pasta curves have $j = 0$). For $y^2 = x^3 + b$ one has $a = 0$, so $j = 0$; both Pasta curves have $j = 0$. Curves with $j = 0$ are special: they admit an extra automorphism of order 3, which underlies the efficient endomorphism described next. (At the other extreme, $b = 0$ gives $j = 1728$ and an order-4 automorphism.) The vanishing $j = 0$ is deliberate in the Pasta design: it is what makes the Gallant–Lambert–Vanstone (GLV) speedup available, roughly halving the doubling count in scalar multiplication.

Proposition 10.33 (The GLV endomorphism for $y^2 = x^3 + b$)**.** *Suppose $\mathbb{F}_p$ contains a primitive cube root of unity ω, i.e. $p \equiv 1 \pmod 3$ (theorem 6.21), so $\omega \in \mathbb{F}_p$ satisfies $\omega^2 + \omega + 1 = 0$. On $E : y^2 = x^3 + b$, the map*

$$\phi(x, y) = (\omega x,\ y), \qquad \phi(\mathcal{O}) = \mathcal{O},$$

is a group endomorphism of $E(\mathbb{F}_p)$. Let $G = \langle P \rangle$ be a subgroup of prime order r with $r \equiv 1 \pmod 3$ and $\phi(G) \subseteq G$. Then ϕ acts on G as multiplication by a scalar: there is $\lambda \in \mathbb{Z}/r\mathbb{Z}$ with $\lambda^2 + \lambda + 1 \equiv 0 \pmod r$ and

$$\phi(Q) = [\lambda]\,Q \quad \textit{for all } Q \in G.$$

For the Pasta curves the hypothesis $\phi(G) \subseteq G$ is automatic, since $E(\mathbb{F}_p)$ itself is the prime-order group G; and $r \equiv 1 \pmod 3$ holds for both orders.

Proof. The map lands on the curve. If $y^2 = x^3 + b$ then $y^2 = (\omega x)^3 + b$, because $\omega^3 = 1$; so $(\omega x, y) \in E(\mathbb{F}_p)$.

The map is a homomorphism. We check $\phi(P_1 + P_2) = \phi(P_1) + \phi(P_2)$ against the case analysis of §10.4. The $\mathcal{O}$ cases are trivial, and $\phi(-P) = -\phi(P)$ since negation touches only y; in particular the inverse case $\phi(P) + \phi(-P) = \mathcal{O} = \phi(\mathcal{O})$ is respected, and ϕ preserves the case split, since it preserves equality of x-coordinates. For a chord ($x_1 \neq x_2$), the slope transforms as

$$\lambda' = \frac{y_2 - y_1}{\omega x_2 - \omega x_1} = \omega^{-1}\lambda = \omega^2\lambda,$$

using $\omega^{-1} = \omega^2$. Then, by (12) and $\omega^4 = \omega$,

$$x_3' = \lambda'^2 - \omega x_1 - \omega x_2 = \omega^4\lambda^2 - \omega x_1 - \omega x_2 = \omega\,(\lambda^2 - x_1 - x_2) = \omega x_3,$$

$$y_3' = \lambda'(\omega x_1 - x_3') - y_1 = \omega^2\lambda \cdot \omega(x_1 - x_3) - y_1 = \lambda(x_1 - x_3) - y_1 = y_3,$$

so $\phi(P_1) + \phi(P_2) = (\omega x_3, y_3) = \phi(P_1 + P_2)$. The tangent case runs identically with $\lambda' = 3(\omega x_1)^2/(2y_1) = \omega^2\lambda$ in (13).

The relation $\phi^2 + \phi + \mathrm{id} = 0$. Let $P = (x, y)$ be affine; we show $P + \phi(P) + \phi^2(P) = \mathcal{O}$ directly from the formulas of §10.4.

First let $x \neq 0$. The points $P = (x, y)$, $\phi(P) = (\omega x, y)$, $\phi^2(P) = (\omega^2 x, y)$ share their y-coordinate and have pairwise distinct x-coordinates, since $1, \omega, \omega^2$ are distinct and $x \neq 0$. The chord formula (12) applies to $P + \phi(P)$, with slope $\lambda = (y - y)/(\omega x - x) = 0$:

$$x_3 = 0 - x - \omega x = -(1 + \omega)\,x = \omega^2 x, \qquad y_3 = 0 \cdot (x - x_3) - y = -y,$$

using $1+\omega+\omega^2 = 0$. Hence $P+\phi(P) = (\omega^2 x, -y) = -\phi^2(P)$ (theorem 10.8), and $P+\phi(P)+\phi^2(P) = \mathcal{O}$.

If instead $x = 0$, the three points coincide: $\phi(P) = \phi^2(P) = P$. Here $y \neq 0$, since $x = y = 0$ on the curve would force $b = 0$, which nonsingularity $27b^2 \neq 0$ forbids; so the doubling formula (13) applies, with $a = 0$:

$$\lambda = \frac{3 \cdot 0^2 + 0}{2y} = 0, \qquad x_3 = 0 - 2 \cdot 0 = 0, \qquad y_3 = 0 \cdot (0 - 0) - y = -y,$$

so $[2]P = (0, -y) = -P$, hence $P + \phi(P) + \phi^2(P) = [3]P = \mathcal{O}$ in this case too. (The three points P, $\phi(P)$, $\phi^2(P)$ are exactly the intersections of the horizontal line through P with the curve, the roots of $X^3 = y^2 - b$ being $x, \omega x, \omega^2 x$; the computation just performed is the "three points of a line sum to $\mathcal{O}$" picture made rigorous, including the triple intersection at $x = 0$.)

With $\phi(\mathcal{O}) = \mathcal{O}$ the identity holds on all of $E(\mathbb{F}_p)$; as an identity of endomorphisms, $\phi^2+\phi+\mathrm{id} = 0$. Incidentally $\phi^3 = \mathrm{id}$, since $\omega^3 = 1$; and whenever $\#E(\mathbb{F}_p) > 3$—as always at cryptographic sizes, the Pasta curves included—ϕ has order exactly 3, because at most two affine points, $(0, \pm y_0)$ with $y_0^2 = b$, share the coordinate $x = 0$, so some point of $E(\mathbb{F}_p)$ has $x \neq 0$ and ϕ moves it. The proviso is not vacuous: $y^2 = x^3 + 4$ over $\mathbb{F}_7$ has only the three points $\mathcal{O}$, $(0, 2)$, $(0, 5)$, and there $\phi = \mathrm{id}$.

Action on G as a scalar. The subgroup $G \cong \mathbb{Z}/r\mathbb{Z}$ is cyclic with generator P, and $\phi(G) \subseteq G$ by hypothesis, so $\phi(P) = [\lambda]P$ for some $\lambda \in \mathbb{Z}/r\mathbb{Z}$; then for any $Q = [m]P \in G$, the homomorphism property gives $\phi(Q) = [m]\phi(P) = [m\lambda]P = [\lambda]Q$. (This is the statement $\mathrm{Hom}(\mathbb{Z}/r\mathbb{Z}, \mathbb{Z}/r\mathbb{Z}) \cong \mathbb{Z}/r\mathbb{Z}$: an endomorphism of a cyclic group is determined by, and equal to multiplication by, the image of a generator.) Applying the endomorphism relation to P,

$$[\lambda^2 + \lambda + 1]P = \mathcal{O} \quad\Longrightarrow\quad \lambda^2 + \lambda + 1 \equiv 0 \pmod{r},$$

since P has order r. Consistently, this congruence is solvable exactly when $\mathbb{F}_r$ contains a primitive cube root of unity, i.e. when $r \equiv 1 \pmod 3$ (theorem 6.21 again)—the hypothesis of the statement. □

Remark 10.34 (How GLV halves the doubling count). Computing ϕ costs a single field multiplication $x \mapsto \omega x$, negligible against a group operation. One therefore decomposes a scalar $n \in \mathbb{Z}/r\mathbb{Z}$ as

$$[n]P = [n_1]P + [n_2]\,\phi(P), \qquad n \equiv n_1 + n_2\lambda \pmod{r},$$

where n_1, n_2 can each be taken only about $\frac{1}{2}\log_2 r$ bits long—such a pair is found by lattice reduction in the lattice of pairs (c_1, c_2) with $c_1 + c_2\lambda \equiv 0 \pmod{r}$. The two half-length scalar multiplications are then run *simultaneously*, sharing their doublings (Straus–Shamir's trick): one double-and-add loop of half the length, adding P, $\phi(P)$, or both according to the bits of n_1 and n_2. The result costs about half the doublings of a single full-length double-and-add—a substantial saving, and its availability is one reason the Pasta curves take the form $y^2 = x^3 + b$ over primes $p \equiv 1 \pmod 3$.

10.13 A worked toy example

The entire machinery now runs on a curve small enough for hand computation.

Example 10.35 (The curve $y^2 = x^3 + 5$ over $\mathbb{F}_{11}$). Take $p = 11$: then $p > 3$, and $4 \cdot 0 + 27 \cdot 5^2 = 675 \equiv 4 \not\equiv 0 \pmod{11}$, so the curve $E : y^2 = x^3 + 5$ is nonsingular over $\mathbb{F}_{11}$ (theorem 10.5). Squaring the residues $0, 1, \ldots, 10$ yields the squares $\{0, 1, 3, 4, 5, 9\}$, so the nonzero quadratic residues are $\{1, 3, 4, 5, 9\}$. Tabulating $f(x) = x^3 + 5 \bmod 11$ and reading off the y-values:

x	$x^3 \bmod 11$	$f(x) = x^3 + 5$	y with $y^2 = f(x)$
0	0	5	± 4 ($4^2 = 16 \equiv 5$)
1	1	6	none ($6 \notin$ QR)
2	8	2	none
3	5	10	none
4	9	3	± 5 ($5^2 = 25 \equiv 3$)
5	4	9	± 3
6	7	1	± 1
7	2	7	none
8	6	0	0
9	3	8	none
10	10	4	± 2

Counting: the five values $x \in \{0, 4, 5, 6, 10\}$ contribute two points each (10 points), $x = 8$ contributes the single point $(8, 0)$, and $\mathcal{O}$ completes the census:

$$\#E(\mathbb{F}_{11}) = 10 + 1 + 1 = 12.$$

Hasse's bound checks out: $p + 1 = 12$, so the trace is $t = 0$, and indeed $|t| = 0 \le 2\sqrt{11} \approx 6.63$. The point $(8, 0)$ has order 2, consistent with theorem 10.22 and the even order 12; here $-5 \equiv 6$ *is* a cube, namely $8^3 = 512 \equiv 6$.

An addition. Let $P = (0, 4)$ and $Q = (5, 3)$. Since $x_1 = 0 \ne 5 = x_2$, the chord formula (12) applies:

$$\lambda = \frac{3 - 4}{5 - 0} = (-1) \cdot 5^{-1} = (-1) \cdot 9 = -9 \equiv 2 \pmod{11},$$

using $5^{-1} = 9$ (check: $5 \cdot 9 = 45 \equiv 1$). Then

$$x_3 = \lambda^2 - x_1 - x_2 = 4 - 0 - 5 = -1 \equiv 10, \qquad y_3 = \lambda(x_1 - x_3) - y_1 = 2\,(0 - 10) - 4 = -24 \equiv 9,$$

so $P + Q = (10, 9)$. The result is on the curve: $9^2 = 81 \equiv 4$ and $10^3 + 5 = 1005 \equiv 4 \pmod{11}$, as required.

A doubling. Take $P = (0, 4)$. With $a = 0$, formula (13) gives

$$\lambda = \frac{3x_1^2 + a}{2y_1} = \frac{0}{8} = 0, \qquad x_3 = \lambda^2 - 2x_1 = 0, \qquad y_3 = \lambda(x_1 - x_3) - y_1 = -4 \equiv 7,$$

so $[2](0, 4) = (0, 7) = -(0, 4)$. Hence $[3](0, 4) = \mathcal{O}$: the point $(0, 4)$ has order 3.

The GLV endomorphism. Since $11 \equiv 2 \pmod 3$, the field $\mathbb{F}_{11}$ contains *no* primitive cube root of unity (theorem 6.21 requires $3 \mid p - 1$), so the GLV map is not available over $\mathbb{F}_{11}$. The Pasta primes do satisfy $p \equiv q \equiv 1 \pmod 3$, so GLV applies there. For a small GLV-friendly illustration take instead $p = 13 \equiv 1 \pmod 3$, where $\omega = 3$ satisfies $3^2 + 3 + 1 = 13 \equiv 0 \pmod{13}$; then $\phi(x, y) = (3x, y)$ is a nontrivial endomorphism of $y^2 = x^3 + b$ over $\mathbb{F}_{13}$.

10.14 Pallas and Vesta assembled

Every property developed in this section was chosen with the closing example in mind. Here the pieces assemble.

Example 10.36 (The Pasta curves). For the running example $y^2 = x^3 + 5$, with p and q the 255-bit primes of the section opening:

- *Pallas*, over $\mathbb{F}_p$: the order is $\#E_p(\mathbb{F}_p) = q$, a prime, so the group is cyclic, $E_p(\mathbb{F}_p) \cong \mathbb{Z}/q\mathbb{Z}$ (theorem 3.34), with cofactor $h = 1$ (theorem 10.21). The trace is $t = p + 1 - q$, an 87-bit (negative) integer, far inside the Hasse bound $2\sqrt{p} \approx 2^{128}$ of theorem 10.19.
- *Vesta*, over $\mathbb{F}_q$: the order is $\#E_q(\mathbb{F}_q) = p$, a prime, so $E_q(\mathbb{F}_q) \cong \mathbb{Z}/p\mathbb{Z}$, again with cofactor 1.
- Both curves have $j = 0$ (theorem 10.32), and both primes satisfy $p \equiv q \equiv 1 \pmod 3$, so each curve admits the order-3 endomorphism $\phi(x, y) = (\omega x, y)$ of theorem 10.33 and supports the GLV speedup of theorem 10.34.
- Both fields have two-adicity 32 in the sense of theorem 9.10: 2^{32} divides $p - 1$ and $q - 1$, so each field carries the multiplicative subgroups of order 2^{32} promised by theorem 10.25—the evaluation domains of the fast polynomial arithmetic of section 9.
- The cycle property $\#E_p(\mathbb{F}_p) = q$ and $\#E_q(\mathbb{F}_q) = p$ means the scalar field of each curve is the base field of the other (theorem 10.24): the base field of Pallas is $\mathbb{F}_p$ and its scalar field is $\mathbb{F}_q$, matching the base field of Vesta, and symmetrically. This is exactly what lets recursive proof systems chain proofs across the two curves without expensive non-native field arithmetic (theorem 10.25). The cycle is a deployed fact: the `pasta_curves` crate wires the scalar field of Pallas to be $\mathbb{F}_q$ and that of Vesta to be $\mathbb{F}_p$ (`src/curves.rs`), as specified in §5.4.9.6 of the Zcash protocol specification.

The discrete logarithm problem in $E_p(\mathbb{F}_p)$ and $E_q(\mathbb{F}_q)$ is believed to require about 2^{127} operations—$\sqrt{r}$ for the 255-bit orders $r \approx 2^{254}$ (theorem 10.27)—with no faster attack known (theorem 10.28), just below the 128-bit security level targeted by Halo 2 and Zcash Orchard.

The cheque from the section's opening is hereby cashed. The demand was a roughly 255-bit group for which no attack faster than the square-root bound is known, and each Pasta curve supplies one: a cyclic group of prime 255-bit order (theorem 10.19 and theorem 3.34), carrying a fast forward map (theorem 10.18, accelerated by theorem 10.34) whose inversion—the ECDLP—resists everything known except the generic $O(\sqrt{r})$ attacks of theorem 10.27. Where $\mathbb{F}_p^\times$ needs thousands of bits to outrun index calculus, the curve group needs 255; and the two groups even interlock as a cycle, each one's scalars living in the other's coordinates. Later volumes build on precisely this pair.

11 Probability, asymptotics, and computation

A deployed signature scheme is typically advertised as offering "128-bit security". The question this section exists to answer is: what, precisely, does that phrase assert, and about whom? It cannot assert that forgery is impossible. A forger may simply guess the secret key: keys are finite strings, so the guess succeeds with some positive probability, and a forger with astronomical luck wins outright. Any honest guarantee is therefore a statement of *probability* — a bound on the chance that the forger wins. Nor can it speak of one forger only: a guarantee against yesterday's attack says nothing about tomorrow's, so the bound must quantify over *every* attacker whose resources fall within stated limits. Making the sentence "every efficient forger succeeds with at most a stated probability" precise takes three intertwined languages: the language of *probability*, which permits reasoning about random keys, random challenges, and the probability that an attack succeeds; the language of *asymptotics*, which describes how resources and success probabilities scale; and the language of *computation*, which fixes what an efficient algorithm is and what it means for a problem to be hard. This section develops all three from elementary foundations and closes by cashing the opening cheque: the final subsection states exactly what "128-bit security" asserts, and about whom. The only prerequisites are familiarity with finite sets, sums, and elementary calculus — limits, the exponential function, and, for the closing subsection's density-defined laws, the improper Riemann integral; everything else is constructed here.

Until the closing subsection, all probability spaces are finite. This is no real restriction for the cryptographic material: keys, messages, group elements, bounded-length messages, and the random coins of an algorithm with a fixed running-time bound live in finite sets, and finite spaces need no measure theory. Unbounded retry loops need the extension below even when they terminate with probability one. The closing subsection (§11.7) then extends the theory by exactly the rungs later volumes need — countable additivity, continuity along monotone events, a waiting-time toolkit, and laws defined by densities — declaring its one measure-theoretic debt in place (Remark 11.31).

11.1 Finite probability spaces and events

The basic object is a finite set of outcomes together with an assignment of weights summing to one.

Definition 11.1 (Finite probability space)**.** A *finite probability space* is a pair $(\Omega, \Pr)$ where Ω is a nonempty finite set, called the *sample space*, and $\Pr : \Omega \to [0,1]$ is a function, called the *probability mass function*, satisfying the normalisation condition

$$\sum_{\omega\in\Omega} \Pr[\omega] = 1.$$

An element $\omega \in \Omega$ is an *outcome* (or *elementary event*).

Definition 11.2 (Event)**.** An *event* is a subset $A \subseteq \Omega$. Its *probability* is

$$\Pr[A] = \sum_{\omega\in A} \Pr[\omega],$$

with the convention $\Pr[\varnothing] = 0$ (an empty sum). The event A *occurs* on outcome ω if $\omega \in A$.

Because Ω is finite, every subset is an event. This convenience is worth flagging: on an uncountable sample space the general theory specifies a family of admissible events together with its probability assignment, rather than automatically admitting all subsets; finiteness spares us that machinery, and Remark 11.31 marks the one place this series touches it. The following elementary facts hold, and we use them freely.

Proposition 11.3 (Basic axioms)**.** *For any finite probability space* $(\Omega, \Pr)$ *and events* $A, B \subseteq \Omega$*:*

1. $0 \le \Pr[A] \le 1$, $\Pr[\varnothing] = 0$, *and* $\Pr[\Omega] = 1$.
2. (Complement.) $\Pr[\Omega \setminus A] = 1 - \Pr[A]$.
3. (Monotonicity.) *If* $A \subseteq B$ *then* $\Pr[A] \le \Pr[B]$.
4. (Inclusion–exclusion for two events.) $\Pr[A \cup B] = \Pr[A] + \Pr[B] - \Pr[A \cap B]$.
5. (Finite additivity.) *If* $A_1, \dots, A_n$ *are pairwise disjoint then* $\Pr\left[\bigcup_{i=1}^n A_i\right] = \sum_{i=1}^n \Pr[A_i]$.

Proof. Every claim follows by manipulating the defining sums. For (1), each summand $\Pr[\omega]$ lies in $[0,1]$ and the total is 1 by normalisation; an event's probability is a partial sum of nonnegative terms, hence lies in $[0,1]$. For (2), the sets A and $\Omega \setminus A$ partition Ω, so $\Pr[A] + \Pr[\Omega \setminus A] = \sum_{\omega\in\Omega} \Pr[\omega] = 1$.

For (3), splitting the sum over B gives $\Pr[B] = \sum_{\omega\in A}\Pr[\omega] + \sum_{\omega\in B\setminus A}\Pr[\omega] \geq \Pr[A]$, the second term being a sum of nonnegative numbers. For (5), disjointness means each ω in the union lies in exactly one A_i, so the double sum $\sum_i\sum_{\omega\in A_i}\Pr[\omega]$ counts each $\omega \in \bigcup_i A_i$ exactly once. For (4), write $A \cup B = A \sqcup (B \setminus A)$ (disjoint), so by (5) $\Pr[A \cup B] = \Pr[A] + \Pr[B \setminus A]$; also $B = (A \cap B) \sqcup (B \setminus A)$, giving $\Pr[B \setminus A] = \Pr[B] - \Pr[A \cap B]$. Substituting yields the claim. □

The single most important example, ubiquitous in cryptography, is the uniform distribution.

Definition 11.4 (Uniform distribution)**.** Let S be a nonempty finite set. The *uniform distribution* on S is the probability space $(S, \Pr)$ with $\Pr[\omega] = 1/|S|$ for every $\omega \in S$. The notation $x \leftarrow S$ (or $x \xleftarrow{\$} S$) means that x is sampled according to the uniform distribution on S. Under the uniform distribution, for any event $A \subseteq S$,

$$\Pr[A] = \frac{|A|}{|S|}.$$

Thus, for uniform sampling, probability is exactly the ratio of favourable outcomes to total outcomes, recovering the classical counting notion of probability.

11.2 Conditional probability and independence

Frequently we learn that one event has occurred and must update the probability of another accordingly. Conditioning captures this.

Definition 11.5 (Conditional probability)**.** Let A, B be events with $\Pr[B] > 0$. The *conditional probability of A given B* is

$$\Pr[A \mid B] = \frac{\Pr[A \cap B]}{\Pr[B]}.$$

When $\Pr[B] = 0$ the quantity is left undefined.

Conditioning on B restricts the sample space to B and renormalises so that B has probability 1: the map $\omega \mapsto \Pr[\{\omega\} \mid B]$ is itself a probability mass function on B, and $\Pr[A \mid B]$ is the probability it assigns to $A \cap B$. Two immediate and indispensable consequences follow.

Proposition 11.6 (Chain rule and total probability)**.** *Let $A_1, \dots, A_n$ be events with $\Pr[A_1 \cap \cdots \cap A_{n-1}] > 0$. Then*

$$\Pr[A_1 \cap \cdots \cap A_n] = \Pr[A_1]\ \Pr[A_2 \mid A_1]\ \Pr[A_3 \mid A_1 \cap A_2] \cdots \Pr[A_n \mid A_1 \cap \cdots \cap A_{n-1}].$$

Moreover, if $B_1, \dots, B_k$ partition Ω (pairwise disjoint with union Ω) and each $\Pr[B_i] > 0$, then for any event A,

$$\Pr[A] = \sum_{i=1}^{k} \Pr[A \mid B_i]\ \Pr[B_i].$$

Proof. The chain rule follows by telescoping: each factor $\Pr[A_j \mid A_1 \cap \cdots \cap A_{j-1}]$ equals $\Pr[A_1 \cap \cdots \cap A_j]/\Pr[A_1 \cap \cdots \cap A_{j-1}]$ by theorem 11.5, and the product collapses to $\Pr[A_1 \cap \cdots \cap A_n]/1$. (Monotonicity ensures that all the intermediate intersections have positive probability, so every factor is defined.) For total probability, the events $A \cap B_i$ are pairwise disjoint with union A (since the B_i partition Ω), so finite additivity and $\Pr[A \cap B_i] = \Pr[A \mid B_i]\ \Pr[B_i]$ give $\Pr[A] = \sum_i \Pr[A \cap B_i] = \sum_i \Pr[A \mid B_i]\ \Pr[B_i]$. □

Independence is the formal statement that one event carries no information about another.

Definition 11.7 (Independence of events)**.** Two events A, B are *independent* if

$$\Pr[A \cap B] = \Pr[A]\ \Pr[B].$$

A family $\{A_i\}_{i \in I}$ of events is *mutually independent* if for every finite subset $J \subseteq I$,

$$\Pr\Big[\bigcap_{i \in J} A_i\Big] = \prod_{i \in J} \Pr[A_i].$$

Remark 11.8. When $\Pr[B] > 0$, independence of A and B is equivalent to $\Pr[A \mid B] = \Pr[A]$: conditioning on B does not change the probability of A. Mutual independence is strictly stronger than pairwise independence. The standard counterexample: toss two fair coins, and let A be "the first is heads", B "the second is heads", and C "the two coins differ". Each pair is independent, yet $\Pr[A \cap B \cap C] = 0 \neq \frac{1}{8} = \Pr[A]\ \Pr[B]\ \Pr[C]$, so the three are not mutually independent.

11.3 Random variables and expectation

A random variable assigns a value to each outcome; it summarises a random experiment by a number (or a vector, a string, a group element).

Definition 11.9 (Random variable)**.** Let $(\Omega, \Pr)$ be a finite probability space and T any set. A *(T-valued) random variable* is a function $X : \Omega \to T$. When $T \subseteq \mathbb{R}$, we call X *real-valued*. For $t \in T$,

$$\Pr[X = t] = \Pr[\{\omega \in \Omega : X(\omega) = t\}],$$

and more generally $\Pr[X \in B]$ denotes the probability that X takes a value in $B \subseteq T$. The function $t \mapsto \Pr[X = t]$ on the finite image of X is the *distribution* (or *law*) of X.

Definition 11.10 (Independent random variables)**.** Random variables $X_1, \dots, X_n$ (with values in sets $T_1, \dots, T_n$) are *mutually independent* if for all $t_1 \in T_1, \dots, t_n \in T_n$,

$$\Pr[X_1 = t_1, \dots, X_n = t_n] = \prod_{i=1}^{n} \Pr[X_i = t_i].$$

Definition 11.11 (Expectation)**.** Let $X : \Omega \to \mathbb{R}$ be a real-valued random variable on a finite space. Its *expectation* (or *mean*) is

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega)\ \Pr[\omega] = \sum_{t \in \mathrm{im}(X)} t \cdot \Pr[X = t],$$

where $\mathrm{im}(X)$ denotes the (finite) image of X. The two expressions agree by grouping outcomes according to their X-value.

A frequently used bridge between the two notions is the following: the expectation of the indicator of an event is exactly its probability.

Definition 11.12 (Indicator)**.** For an event A, its *indicator random variable* $1_A : \Omega \to \{0, 1\}$ takes the value $1_A(\omega) = 1$ if $\omega \in A$ and 0 otherwise.

Proposition 11.13 (Indicator expectation). *For any event A, $\mathbb{E}[1_A] = \Pr[A]$.*

Proof. By theorem 11.11, $\mathbb{E}[1_A] = \sum_\omega 1_A(\omega)\Pr[\omega] = \sum_{\omega\in A}\Pr[\omega] = \Pr[A]$, since the indicator vanishes off A. □

The principal property of expectation is its linearity, which holds with no independence assumption whatsoever; this absence of any independence hypothesis is what makes expectation so much more tractable than probability.

Theorem 11.14 (Linearity of expectation). *Let $X, Y : \Omega \to \mathbb{R}$ be random variables on a finite space and $a, b \in \mathbb{R}$. Then*

$$\mathbb{E}[aX + bY] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y].$$

More generally, for random variables $X_1, \dots, X_n$ and scalars $a_1, \dots, a_n$,

$$\mathbb{E}\Big[\sum_{i=1}^n a_i X_i\Big] = \sum_{i=1}^n a_i\,\mathbb{E}[X_i].$$

This holds whether or not the X_i are independent.

Proof. Directly from the definition,

$$\mathbb{E}[aX + bY] = \sum_{\omega\in\Omega}\big(aX(\omega) + bY(\omega)\big)\Pr[\omega] = a\sum_\omega X(\omega)\Pr[\omega] + b\sum_\omega Y(\omega)\Pr[\omega] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y],$$

splitting the finite sum and extracting the constants. The n-term statement follows by induction on n. □

A one-line consequence converts an expectation bound into a probability bound; it underlies every averaging ("heavy-row") argument in the series.

Proposition 11.15 (Markov's inequality). *Let $X \ge 0$ be a real-valued random variable and $a > 0$. Then $\Pr[X \ge a] \le \mathbb{E}[X]/a$.*

Proof. Every outcome contributes nonnegatively to $\mathbb{E}[X]$, and each outcome with $X(\omega) \ge a$ contributes at least $a\Pr[\omega]$; hence $\mathbb{E}[X] \ge a\Pr[X \ge a]$. □

11.4 The union bound and a birthday calculation

The payoff of this subsection, stated up front, is the square-root law that sets the output lengths of cryptographic hash functions: among roughly $\sqrt{N}$ uniform samples from a set of size N, two are likely to coincide, so an n-bit hash resists collision-finding only up to about $2^{n/2}$ work, and output lengths are chosen twice the desired security level. The engine of the calculation is the most frequently used inequality in cryptographic proofs: the union bound. It is crude, it requires no independence, and it almost always suffices.

Theorem 11.16 (Union bound / Boole's inequality). *For any events $A_1, \dots, A_n$ in a finite probability* *space,*

$$\Pr\Big[\bigcup_{i=1}^{n} A_i\Big] \le \sum_{i=1}^{n} \Pr[A_i].$$

Proof. Disjointify the union into "first occurrence" events: $B_1 = A_1$ and $B_i = A_i \setminus (A_1 \cup \cdots \cup A_{i-1})$ for $i \ge 2$. The B_i are pairwise disjoint, $B_i \subseteq A_i$, and $\bigcup_i B_i = \bigcup_i A_i$. By finite additivity and monotonicity (theorem 11.3),

$$\Pr\Big[\bigcup_i A_i\Big] = \Pr\Big[\bigcup_i B_i\Big] = \sum_i \Pr[B_i] \le \sum_i \Pr[A_i]. \qquad \square$$

The disjointification device — replacing events by pairwise disjoint pieces with the same union — recurs throughout the section: the "first occurrence" form above proves the union bound, and a telescoping form for increasing chains proves the continuity theorem of §11.7. In each case it turns a subadditivity or limit claim into an exact additivity computation. The complementary inequality for conjunctions is equally useful and follows by taking complements.

Corollary 11.17 (Union bound for the good event)**.** *If each event A_i holds except with probability at most ε_i (that is, $\Pr[\Omega \setminus A_i] \le \varepsilon_i$), then all of them hold simultaneously except with probability at most $\sum_i \varepsilon_i$:*

$$\Pr\Big[\bigcap_{i=1}^{n} A_i\Big] \ge 1 - \sum_{i=1}^{n} \varepsilon_i.$$

Proof. By De Morgan, $\Omega \setminus \bigcap_i A_i = \bigcup_i (\Omega \setminus A_i)$. Apply the union bound to the complements and take the complement of both sides. $\square$

We now derive the *birthday bound*, which governs how soon collisions appear when sampling uniformly.

Proposition 11.18 (Birthday bound)**.** *Sample $x_1, \dots, x_k$ independently and uniformly from a set of size N, and let C be the event that some two of them collide, i.e. $x_i = x_j$ for some $i < j$. Then*

$$\Pr[C] \le \binom{k}{2}\frac{1}{N} = \frac{k(k-1)}{2N},$$

and, on the other side,

$$\Pr[C] = 1 - \prod_{i=0}^{k-1}\Big(1 - \frac{i}{N}\Big) \ge 1 - e^{-k(k-1)/(2N)}.$$

In particular a collision becomes likely once $k = \Theta(\sqrt{N})$.

Proof. Upper bound. For $i < j$ let A_{ij} be the event $x_i = x_j$. Since x_i and x_j are independent and uniform, $\Pr[A_{ij}] = \sum_v \Pr[x_i = v]\,\Pr[x_j = v] = N \cdot (1/N)^2 = 1/N$. Now $C = \bigcup_{i<j} A_{ij}$, a union of $\binom{k}{2}$ events, so the union bound (theorem 11.16) gives $\Pr[C] \le \binom{k}{2}/N$.

Lower bound. The complement $\overline{C}$ is the event that all k samples are distinct. If $k > N$, the pigeonhole principle forces a collision, so $\Pr[\overline{C}] = 0$; the product also vanishes, through its factor $1 - N/N$, so the stated equality holds and the exponential bound is trivial. Let then $k \le N$, and for $0 \le i \le k$ let D_i be the event that the first i samples are pairwise distinct ($D_0 = D_1 = \Omega$, and $D_k = \overline{C}$). We show $\Pr[D_i] = \prod_{j=0}^{i-1}(1 - j/N)$ by induction on $i \le k$; every factor is positive, since $j \le k-1 \le N-1$, so in particular each D_i has positive probability and conditioning on it is legitimate (theorem 11.5). For the step, the events pinning the first i samples to specific distinct values partition D_i, each with probability $N^{-i} > 0$ by independence and uniformity; given any one of them, the next sample avoids the i values taken with probability $1-i/N$, again by independence and uniformity. Averaging over the cells (total probability, theorem 11.6) gives $\Pr[D_{i+1} \mid D_i] = 1 - i/N$, and $\Pr[D_{i+1}] = \Pr[D_{i+1} \mid D_i]\ \Pr[D_i]$ completes the induction. Taking $i = k$ yields the stated equality, in which every factor is nonnegative, so the inequality $1 - x \le e^{-x}$ applies factor by factor. (For $0 \le x < 1$ this inequality follows from Bernoulli's inequality, $(1 - x/n)^n \ge 1 - x$ for $n > x$, by letting $n \to \infty$ in the limit definition of e^{-x}; for $x \ge 1$ the left side is nonpositive.) Thus

$$\Pr[\overline{C}] = \prod_{i=0}^{k-1}\Big(1 - \frac{i}{N}\Big) \le \prod_{i=0}^{k-1} e^{-i/N} = \exp\Big(-\frac{1}{N}\sum_{i=0}^{k-1} i\Big) = e^{-k(k-1)/(2N)},$$

summing the exponents via $\sum_{i=0}^{k-1} i = k(k-1)/2$. Taking complements gives $\Pr[C] = 1 - \Pr[\overline{C}] \ge 1 - e^{-k(k-1)/(2N)}$. □

Example 11.19. With $N = 365$ and $k = 23$ — the classical birthday party — the exact product gives $\Pr[C] = 0.5073$, already a majority chance, and the proposition brackets it as $0.5000 \le \Pr[C] \le 0.6932$. The threshold is visible at small scale too: with $N = 16$, the fifth sample (just past $\sqrt{16} = 4$) tips the odds, $\Pr[C] = 0.5001$ at $k = 5$.

Remark 11.20. The threshold $k \approx \sqrt{N}$ is the reason an n-bit hash function (so $N = 2^n$) offers only about $2^{n/2}$ collision resistance: an adversary expects a collision after roughly $2^{n/2}$ random evaluations — for $N = 2^{256}$, that is $\sqrt{2^{256}} = 2^{128}$. For this reason one chooses output lengths twice the desired security level. The same square-root phenomenon recurs in random-walk discrete-log attacks (Pollard's rho, theorem 10.27), so designers size the curves used in Halo 2 and Orchard so that $\sqrt{N}$ is itself astronomically large.

11.5 Statistical distance

Comparing distributions — for example a real key-generation procedure against an idealised uniform one — requires a notion of how far apart two distributions are. The appropriate notion for "no statistical test can distinguish them well" is statistical distance.

Definition 11.21 (Statistical distance)**.** Let P and Q be probability distributions on the same finite set S (i.e. probability mass functions $P, Q : S \to [0, 1]$). Their *statistical distance* (or *total variation distance*) is

$$\Delta(P, Q) = \frac{1}{2}\sum_{s\in S}\big|P(s) - Q(s)\big|.$$

For random variables X, Y with values in S, $\Delta(X, Y)$ denotes the statistical distance of their distributions.

The factor $\frac{1}{2}$ makes Δ range over $[0, 1]$ and gives it a clean interpretation as the maximum advantage of a distinguisher, which part (3) of the next theorem establishes.

Theorem 11.22 (Properties of statistical distance)**.** *Let P, Q, R be distributions on a finite set S. Then:*

1. $0 \le \Delta(P, Q) \le 1$*; and* $\Delta(P, Q) = 0$ *if and only if* $P = Q$.
2. (Metric.) Δ *is symmetric and satisfies the triangle inequality* $\Delta(P, R) \le \Delta(P, Q) + \Delta(Q, R)$.

3. (Variational characterisation.) $\Delta(P, Q) = \max_{A \subseteq S}\big(P(A) - Q(A)\big) = \max_{A \subseteq S}\big|P(A) - Q(A)\big|$, *where* $P(A) = \sum_{s \in A} P(s)$.

4. (Data processing / post-processing.) *For any (possibly randomised) function f applied to a sample, $\Delta(f(P), f(Q)) \le \Delta(P, Q)$. In particular no algorithm can increase statistical distance.*

Proof. (1) Nonnegativity is clear. For the upper bound, split S into $S^+ = \{s : P(s) \ge Q(s)\}$ and $S^- = S \setminus S^+$. Since $\sum_s P(s) = \sum_s Q(s) = 1$, we have $\sum_s (P(s) - Q(s)) = 0$, so $\sum_{s \in S^+}(P(s) - Q(s)) = \sum_{s \in S^-}(Q(s) - P(s)) =: D \ge 0$. Hence $\Delta(P, Q) = \frac{1}{2}\big(\sum_{S^+}(P - Q) + \sum_{S^-}(Q - P)\big) = D$. As $D = \sum_{S^+}(P(s) - Q(s)) \le \sum_{S^+} P(s) \le 1$, we get $\Delta \le 1$. If $\Delta = 0$ then every $|P(s) - Q(s)| = 0$, so $P = Q$; the converse is immediate.

(2) Symmetry is evident. For the triangle inequality, $|P(s) - R(s)| \le |P(s) - Q(s)| + |Q(s) - R(s)|$ termwise by the triangle inequality on $\mathbb{R}$; summing over s and halving gives the claim.

(3) With D as above, take $A = S^+$: then $P(A) - Q(A) = \sum_{S^+}(P - Q) = D = \Delta(P, Q)$, so the maximum is at least Δ. Conversely, for *any* $A \subseteq S$, $P(A) - Q(A) = \sum_{s \in A}(P(s) - Q(s)) \le \sum_{s \in A \cap S^+}(P(s) - Q(s)) \le D = \Delta$, since dropping the negative terms only increases the sum and extending from $A \cap S^+$ to all of S^+ increases it further. Thus the maximum over A of $P(A) - Q(A)$ is exactly Δ; replacing A by its complement swaps the sign, giving the same value for $\max_A |P(A) - Q(A)|$.

(4) First take f deterministic, $f : S \to U$. For $u \in U$, $f(P)(u) = P(f^{-1}(u))$. Then

$$\Delta(f(P), f(Q)) = \frac{1}{2}\sum_u \big|P(f^{-1}(u)) - Q(f^{-1}(u))\big| \le \frac{1}{2}\sum_u \sum_{s \in f^{-1}(u)} |P(s) - Q(s)| = \Delta(P, Q),$$

using $|\sum_s a_s| \le \sum_s |a_s|$ on each fibre, and that the fibres $f^{-1}(u)$ partition S. A randomised f is a deterministic function of the sample s together with independent fresh coins r; apply the deterministic case to the joint variable (s, r), noting that the two joint distributions $P \times U_r$ and $Q \times U_r$ have the same statistical distance $\Delta(P, Q)$, the shared coin distribution contributing nothing. □

Remark 11.23 (Distinguishing interpretation). Part (3) states that $\Delta(P, Q)$ is exactly the best advantage of any test — even a computationally unbounded one — that, given a single sample, must guess whether it came from P or Q: the optimal test outputs "P" on the set $A = S^+$, and its advantage $\Pr_P[A] - \Pr_Q[A]$ equals $\Delta(P, Q)$. Part (4) is what makes statistical distance compose in proofs: applying the same procedure to two close inputs keeps the outputs close. If $\Delta(P, Q) \le \varepsilon$, then P and Q are interchangeable in any analysis at the cost of an additive ε in every probability. When ε is negligible (theorem 11.42 below), we call P and Q *statistically indistinguishable*.

11.6 Uniform sampling and the bias of modular reduction

Cryptography requires uniform elements of finite sets throughout: a uniform scalar in $\mathbb{Z}/q\mathbb{Z}$ (section 2), a uniform field element of $\mathbb{F}_p$ (section 6), a uniform point in a group. The most tractable source of randomness, however, is a stream of independent uniform bits. We must therefore convert uniform bitstrings into uniform elements of a target set S, and we must understand the bias the conversion introduces.

If $|S| = 2^\ell$ is a power of two, the conversion is exact: read ℓ uniform bits as a number in $\{0, \dots, 2^\ell - 1\}$ and index into S. The nontrivial case arises when $|S|$ is not a power of two — most importantly when $S = \mathbb{Z}/q\mathbb{Z}$ for a prime q, the scalar field of an elliptic curve (section 10). A common and elementary method is *reduce-modulo*: sample a long uniform integer and reduce it mod q. The result is not perfectly uniform — some residues receive one more preimage than others — but the bias is negligible provided the integer is long enough. The next proposition makes this precise.

Proposition 11.24 (Bias of modular reduction). *Let $q \ge 2$ be an integer and let $K = 2^L$ for a nonnegative integer L. Sample U uniformly from $\{0, 1, \dots, K - 1\}$ (equivalently, read L uniform bits* *as an integer), and set* $R = U \bmod q \in \{0, 1, \dots, q-1\}$*. Let* $\mathcal{U}_q$ *denote the uniform distribution on* $\{0, \dots, q-1\}$*. Then the statistical distance of R from uniform satisfies*

$$\Delta(R, \mathcal{U}_q) \le \frac{q}{K} = \frac{q}{2^L}.$$

Consequently, if $q < 2^n$ *and one takes* $L = n + s$ *bits, then* $\Delta(R, \mathcal{U}_q) < 2^{-s}$*.*

Proof. Write $K = mq + t$ with $m = \lfloor K/q \rfloor$ and remainder $t = K \bmod q$, so $0 \le t < q$. For a residue $r \in \{0, \dots, q-1\}$, the number of integers in $\{0, \dots, K-1\}$ congruent to r modulo q is $m+1$ if $r < t$ and m if $r \ge t$. Hence

$$\Pr[R = r] = \begin{cases} (m+1)/K, & r < t, \\ m/K, & r \ge t. \end{cases}$$

Comparing with the target probability $1/q$ and using $m/K \le 1/q \le (m+1)/K$ (since $mq \le K \le (m+1)q$),

$$\left|\Pr[R = r] - \tfrac{1}{q}\right| \le \frac{m+1}{K} - \frac{m}{K} = \frac{1}{K} \qquad \text{for every } r.$$

Therefore

$$\Delta(R, \mathcal{U}_q) = \frac{1}{2} \sum_{r=0}^{q-1} \left|\Pr[R = r] - \tfrac{1}{q}\right| \le \frac{1}{2} \cdot q \cdot \frac{1}{K} = \frac{q}{2K} \le \frac{q}{K}.$$

(The computation actually yields the sharper bound $q/(2K)$; q/K is the usual stated form.) For the final claim, $q < 2^n$ and $K = 2^{n+s}$ give $q/K < 2^n/2^{n+s} = 2^{-s}$. □

Example 11.25. Reduce $L = 6$ uniform bits modulo $q = 5$. Here $K = 64 = 12 \cdot 5 + 4$, so the four residues $r < 4$ receive 13 preimages each and the residue $r = 4$ receives 12; the statistical distance from uniform works out to exactly $1/80$, within the guarantee 2^{-3} of the proposition (as $q < 2^3$ and $s = 6 - 3 = 3$).

Remark 11.26 (Sampling scalars in practice). The consequence to remember: to sample a scalar mod a ∼256-bit prime q with statistical distance below 2^{-128} from uniform, draw $256 + 128 = 384$ uniform bits and reduce. The bias 2^{-s}, with the slack s set to the security parameter, is negligible (theorem 11.42), so the reduce-modulo sample is statistically indistinguishable from a perfectly uniform scalar, and by the data-processing inequality (theorem 11.22(4)) substituting one for the other anywhere costs at most 2^{-128} per use. An alternative, *rejection sampling* — draw $\lceil \log_2 q \rceil$ bits, accept if the result is $< q$, else retry — gives *exactly* uniform output at the cost of a variable but (with overwhelming probability) small number of retries; for this series' purposes the negligible bias of reduce-modulo is acceptable and simpler to analyse. The Orchard specification employs exactly this reduce-modulo sampling when deriving scalars from hash outputs: its ToScalar reduces the full 512-bit $\mathsf{PRF^{expand}}$ output modulo the 255-bit group order — a slack of 257 bits, comfortably beyond the 128 of the worked example (Zcash protocol specification §4.2.3 and §5.4.2; implemented as `to_scalar` in the `orchard` crate, `src/spec.rs`).

11.7 Beyond finite spaces: countable additivity, limits, and densities

Everything so far lives on a finite sample space, and for the cryptographic material of the upper volumes that is enough: keys, challenges, and transcripts are finite strings. Two uses need more. A process watched for as long as it takes — a random walk pursued until it first hits a barrier — has no finite horizon, and a waiting time measured in continuous units has no finite sample space at all. This subsection extends the finite theory by exactly the rungs those uses require: countable additivity, continuity along monotone events, a toolkit for waiting times under stopping rules, and laws defined by densities.

Definition 11.27 (Countable probability space). A *countable probability space* is a pair $(\Omega, \Pr)$ in which Ω is a countable set and $\Pr : \Omega \to [0,1]$ satisfies $\sum_{\omega\in\Omega} \Pr[\omega] = 1$; the sum is independent of the enumeration order because its terms are nonnegative. Events are arbitrary subsets $A \subseteq \Omega$, with $\Pr[A] = \sum_{\omega\in A} \Pr[\omega]$, and Definitions 11.2–11.11 carry over with "finite" read as "countable" (a random variable's image may now be countably infinite), with one exception: theorem 11.4 does not — equal weights cannot sum to 1 on a countably infinite set, so the uniform distribution remains a strictly finite notion. Expectation additionally requires its defining sum to converge absolutely.

Proposition 11.28 (Countable additivity). *Let* $(\Omega, \Pr)$ *be a countable probability space and* $A_1, A_2, \ldots$ *pairwise disjoint events. Then*

$$\Pr\Big[\bigcup_{n\geq 1} A_n\Big] = \sum_{n\geq 1} \Pr[A_n],$$

and in particular the basic axioms of theorem 11.3 hold unchanged.

Proof. Every outcome of the union lies in exactly one A_n, so both sides sum the same nonnegative terms $\Pr[\omega]$, grouped differently; a series of nonnegative terms may be summed in any order, or in groups, without changing its value. □

The free regrouping of nonnegative series invoked here — reorder, regroup, exchange double sums, recount by multiplicity — is the workhorse of every infinite-space argument in this subsection: it proves countable additivity above, the tail-sum formula below (where each term is counted by its multiplicity), and the exchange step in Wald's identity.

Theorem 11.29 (Continuity along monotone events). *Let* $\Pr$ *be a probability assignment, on any sample space, defined for a family of events closed under complement and countable union, and countably additive on it. If* $A_1 \subseteq A_2 \subseteq \cdots$ *is an increasing chain of events, then*

$$\Pr\Big[\bigcup_{n\geq 1} A_n\Big] = \lim_{n\to\infty} \Pr[A_n],$$

and dually $\Pr[\bigcap_n B_n] = \lim_n \Pr[B_n]$ *for a decreasing chain* $B_1 \supseteq B_2 \supseteq \cdots$*. On a countable space the hypothesis holds automatically, by theorem 11.28.*

Proof. Disjointify by telescoping: set $D_1 = A_1$ and $D_n = A_n \setminus A_{n-1}$ for $n \geq 2$. The D_n are pairwise disjoint, $\bigcup_{k\leq n} D_k = A_n$, and $\bigcup_n D_n = \bigcup_n A_n$. Countable additivity gives

$$\Pr\Big[\bigcup_n A_n\Big] = \sum_{n\geq 1} \Pr[D_n] = \lim_{n\to\infty} \sum_{k=1}^{n} \Pr[D_k] = \lim_{n\to\infty} \Pr[A_n],$$

the final step by finite additivity. The decreasing case follows by complementation. □

Definition 11.30 (Infinite sequence of independent trials). Fix a finite outcome set T and a distribution p on T. An *infinite sequence of independent trials* with law p is a sequence of random variables $X_1, X_2, \ldots$ such that every prefix $(X_1, \ldots, X_n)$ consists of n mutually independent p-distributed trials in the sense of theorem 11.10 — equivalently, the prefix's law is that of the finite probability space T^n carrying the product weights $\Pr[(t_1, \ldots, t_n)] = p(t_1) \cdots p(t_n)$. An event is *finitely determined* if membership in it is decided by some fixed finite prefix; its probability is the finite-space value. An event of the form "some prefix satisfies P" is the union of an increasing chain of finitely determined events, and theorem 11.29 assigns it the limit of the prefix probabilities.

Remark 11.31 (What is taken as given). Theorem 11.30 presumes that a single, countably additive probability assignment on infinite sequences exists that restricts correctly to every finite prefix at once. That existence statement is the Kolmogorov extension theorem, and its proof belongs to measure theory, which lies off this series' critical path: the space of infinite sequences over at least two possible outcomes is uncountable, so it escapes theorem 11.27, and the extension specifies the admissible events. We therefore take it — with its countable additivity, hence theorem 11.29 — as a foundation stone rather than a theorem. On infinite-sequence spaces, only finitely determined events and their monotone limits appear in this series, and for those the values are forced by the finite theory together with continuity; the density-defined laws of theorem 11.32 carry the analogous debt, their interval and joint-event probabilities being taken to cohere as the stated integrals.

Definition 11.32 (Law defined by a density). A real-valued random quantity X has *density* $f : \mathbb{R} \to \mathbb{R}_{\geq 0}$ if $\int_{-\infty}^{\infty} f(t)\,dt = 1$ and

$$\Pr[a \leq X \leq b] = \int_a^b f(t)\,dt \qquad \text{for all } a \leq b;$$

improper Riemann integrals suffice throughout this series. The *distribution function* is $F(t) = \Pr[X \leq t] = \int_{-\infty}^{t} f$, so $\Pr[X > t] = 1 - F(t)$, and single points carry probability zero. The expectation is $\mathbb{E}[X] = \int_{-\infty}^{\infty} t\, f(t)\,dt$ when this integral converges absolutely. Two density-defined quantities X, Y are *independent* if $\Pr[X \in I,\ Y \in J] = \Pr[X \in I]\Pr[Y \in J]$ for all intervals I, J; probabilities of joint events are then taken to factor through iterated integrals of the product $f_X(s)\, f_Y(t)$.

11.7.1 The waiting-time toolkit: tail sums, geometric trials, and Wald's identity

Waiting times are the extension's chief dividend. The three short results of this toolkit — used by later volumes' extractor and race analyses — follow from countable expectation and finitely determined events alone. One convention first: for a variable with values in $\mathbb{N} \cup \{\infty\}$, the defining sum of the expectation has nonnegative terms, and we permit the value $+\infty$ when it diverges.

Lemma 11.33 (Tail-sum formula). *Let T be a random variable with values in $\mathbb{N} \cup \{\infty\}$ and $\Pr[T = \infty] = 0$. Then $\mathbb{E}[T] = \sum_{k \geq 1} \Pr[T \geq k]$, either side being infinite exactly when the other is.*

Proof. Expanding $\Pr[T \geq k] = \sum_{m \geq k} \Pr[T = m]$ and regrouping the doubly indexed nonnegative series, each term $\Pr[T = m]$ is counted once for every $k \leq m$, that is, m times; nonnegative series may be regrouped freely. □

Proposition 11.34 (Geometric waiting time). *In an infinite sequence of independent trials, each succeeding with probability $\varepsilon > 0$, let T be the index of the first success. Then $\Pr[T \geq k] = (1 - \varepsilon)^{k-1}$ and $\mathbb{E}[T] = 1/\varepsilon$.*

Proof. The event $T \geq k$ says the first $k - 1$ trials all fail — a finitely determined event of probability $(1 - \varepsilon)^{k-1}$; in particular $\Pr[T = \infty] = \lim_k (1 - \varepsilon)^{k-1} = 0$ by continuity (theorem 11.29). The tail-sum formula then gives $\mathbb{E}[T] = \sum_{k \geq 1} (1 - \varepsilon)^{k-1} = 1/\varepsilon$, a geometric series. □

Proposition 11.35 (Wald's identity). *Let $X_1, X_2, \ldots$ be an infinite sequence of independent trials with common law, each $X_i \geq 0$ with finite mean, and let T be a* stopping rule*: a random index such that, for every k, whether $T = k$ is decided by the first k trials alone. If $\mathbb{E}[T] < \infty$, then*

$$\mathbb{E}\Big[\sum_{i=1}^{T} X_i\Big] = \mathbb{E}[T] \cdot \mathbb{E}[X_1].$$

Proof. Write $S = \sum_{i=1}^{T} X_i$ and $S_k = \sum_{i=1}^{k} X_i$, so that $S = S_k$ on the event $T = k$. The hypothesis $\mathbb{E}[T] < \infty$ forces $\Pr[T = \infty] = 0$, by the convention preceding theorem 11.33. The variable S takes countably many values — on $\{T = k\}$ it agrees with S_k, whose values run over a finite set, the trials having finitely many outcomes — and each event $\{S = s\}$ is the disjoint countable union $\bigcup_k \{T = k,\ S_k = s\}$ of finitely determined events, so its probability is assigned by countable additivity (theorem 11.31). By $\mathbb{E}[S]$ we mean $\sum_s s \Pr[S = s]$, the expectation of this countably supported distribution, a sum of nonnegative terms.

Expanding each $\Pr[S = s]$ and regrouping the doubly indexed nonnegative series,

$$\mathbb{E}[S] = \sum_s s \sum_{k \geq 1} \Pr[T = k,\ S_k = s] = \sum_{k \geq 1} \mathbb{E}[S_k 1_{T=k}],$$

each inner expectation living on the finite prefix space of the first k trials (theorem 11.30), where linearity (theorem 11.14) expands it as $\sum_{i=1}^{k} \mathbb{E}[X_i 1_{T=k}]$. Regrouping the nonnegative double series once more,

$$\mathbb{E}[S] = \sum_{i \geq 1} \sum_{k \geq i} \mathbb{E}[X_i 1_{T=k}].$$

Fix i and let $n \geq i$. The events $T = k$ (for $k \leq n$), $T > n$, and $T \geq i$ are all decided by the first n trials, and on that prefix space the pointwise identity $1_{T \geq i} = \sum_{k=i}^{n} 1_{T=k} + 1_{T>n}$ holds, so linearity gives

$$\sum_{k=i}^{n} \mathbb{E}[X_i 1_{T=k}] = \mathbb{E}[X_i 1_{T \geq i}] - \mathbb{E}[X_i 1_{T>n}].$$

The trials take finitely many values, so $X_i \leq M$ for a constant M, and $0 \leq \mathbb{E}[X_i 1_{T>n}] \leq M \Pr[T > n] \to 0$ as $n \to \infty$: the terms $\Pr[T \geq k]$ of the convergent series $\mathbb{E}[T]$ tend to zero. Hence $\sum_{k \geq i} \mathbb{E}[X_i 1_{T=k}] = \mathbb{E}[X_i 1_{T \geq i}]$.

Now the crux factorisation. The event $T \geq i$ — the negation of $T \leq i-1$ — is decided by the first $i-1$ trials, so its indicator is a function $g(t_1, \ldots, t_{i-1})$ of the first $i-1$ trial values, and on the prefix space of the first i trials the expectation splits along the product weights of theorem 11.30: with p the common law,

$$\mathbb{E}[X_i 1_{T \geq i}] = \sum_{t_1, \ldots, t_i} g(t_1, \ldots, t_{i-1})\ t_i\ p(t_1) \cdots p(t_i) = \Pr[T \geq i] \cdot \mathbb{E}[X_1],$$

the double sum factoring into the sum over $(t_1, \ldots, t_{i-1})$, which totals $\Pr[T \geq i]$, times $\sum_{t_i} t_i\, p(t_i) = \mathbb{E}[X_1]$. Assembling the pieces and applying the tail-sum formula (theorem 11.33),

$$\mathbb{E}[S] = \sum_{i \geq 1} \mathbb{E}[X_1] \Pr[T \geq i] = \mathbb{E}[X_1] \cdot \mathbb{E}[T].$$ □

Later volumes build block-discovery and extraction analyses directly on this base: exponential clocks are density-defined laws, block-attribution sequences are infinite sequences of independent trials, extractor running times are the waiting times above (the geometric waiting time and Wald's identity), and the limit arguments instantiate theorem 11.29.

11.8 Asymptotic notation

The discussion now turns from probability to growth rates. To address efficiency and hardness, we must compare how functions of an integer parameter grow as that parameter tends to infinity. The Bachmann–Landau notation makes such comparisons precise. Throughout, functions map $\mathbb{N}$ (or a cofinite subset of it) to the nonnegative reals $\mathbb{R}_{\geq 0}$, and "for sufficiently large n" means "for all n beyond some threshold n_0".

Definition 11.36 (Big-O, Omega, Theta, little-o, little-omega)**.** Let $f, g : \mathbb{N} \to \mathbb{R}_{\geq 0}$ with g eventually positive.

- $f = O(g)$ (read "f is big-O of g", an asymptotic upper bound) if there exist constants $c > 0$ and n_0 with $f(n) \leq c\, g(n)$ for all $n \geq n_0$.
- $f = \Omega(g)$ (asymptotic lower bound) if there exist $c > 0$ and n_0 with $f(n) \geq c\, g(n)$ for all $n \geq n_0$; equivalently, $f = \Omega(g)$ if and only if $g = O(f)$.
- $f = \Theta(g)$ (tight bound) if $f = O(g)$ and $f = \Omega(g)$; i.e. there are $c_1, c_2 > 0$ and n_0 with $c_1 g(n) \leq f(n) \leq c_2 g(n)$ for $n \geq n_0$.
- $f = o(g)$ (read "little-o", strictly smaller order) if for every $c > 0$ there is n_0 with $f(n) \leq c\, g(n)$ for all $n \geq n_0$; equivalently $\lim_{n\to\infty} f(n)/g(n) = 0$.
- $f = \omega(g)$ (strictly larger order) if $g = o(f)$; equivalently $\lim_{n\to\infty} f(n)/g(n) = \infty$.

Remark 11.37. The "=" in "$f = O(g)$" is traditional but denotes set membership: $O(g)$ is the class of all functions bounded above by a constant multiple of g, and $f = O(g)$ means $f \in O(g)$. One never reads such equations right to left. We follow the customary abuse of notation.

Example 11.38. The polynomial $3n^2 + 10n + 7$ is $\Theta(n^2)$: for $n \geq 1$ one has $3n^2 \leq 3n^2 + 10n + 7 \leq 20n^2$, with equality on the right at $n = 1$, so the constant 20 is exact there. Also $n^2 = o(n^3)$, since $n^2/n^3 = 1/n \to 0$; $\log_2 n = o(n)$; $n^{100} = o(2^n)$ (any polynomial is little-o of any exponential with base > 1); and $n! = \omega(2^n)$. A cautionary pair: $2^n = o(2^{n+1})$ is *false*, but $2^n = \Theta(2^{n+1})$ is true, since $2^{n+1} = 2 \cdot 2^n$ is within a constant factor.

Proposition 11.39 (Useful closure facts)**.** *Let $f_1 = O(g_1)$ and $f_2 = O(g_2)$. Then $f_1 + f_2 = O(\max(g_1, g_2)) = O(g_1 + g_2)$ and $f_1 f_2 = O(g_1 g_2)$. The same closure statements hold with O replaced uniformly by Ω or Θ. Moreover $O(\cdot)$ is transitive: $f = O(g)$ and $g = O(h)$ imply $f = O(h)$.*

Proof. Choose witnesses (c_1, n_1) and (c_2, n_2) for the two hypotheses and let $n_0 = \max(n_1, n_2)$. For $n \geq n_0$, $f_1(n) + f_2(n) \leq c_1 g_1(n) + c_2 g_2(n) \leq (c_1 + c_2)\max(g_1(n), g_2(n))$, and $\max(g_1, g_2) \leq g_1 + g_2 \leq 2\max(g_1, g_2)$, so the two right-hand forms agree up to a constant; this proves the sum rule. For products, $f_1(n) f_2(n) \leq c_1 c_2\, g_1(n) g_2(n)$. Transitivity: $f \leq c\, g$ and $g \leq c' h$ eventually give $f \leq cc'\, h$ eventually. The Ω statements follow by the equivalence $f = \Omega(g) \Leftrightarrow g = O(f)$, and Θ by combining the two. □

11.9 Polynomial, exponential, and negligible functions

The payoff of this subsection is a closure theorem (theorem 11.44): the class of *negligible* functions — those shrinking faster than the reciprocal of every polynomial — is stable under exactly the operations that security proofs perform, addition and multiplication by polynomials, which is why "negligible" is the right formalisation of "ignorable". We first classify the growth rates relevant to cryptography. Fix a distinguished variable $\lambda \in \mathbb{N}$, the *security parameter* (discussed in detail in §11.11); informally, larger λ means more security and larger keys. Resources (running time, key length) and success probabilities are functions of λ.

Definition 11.40 (Polynomial and super-polynomial)**.** A function $f : \mathbb{N} \to \mathbb{R}_{\geq 0}$ is *polynomially bounded*, written $f = \mathrm{poly}(\lambda)$, if there is a constant c (and a threshold) with $f(\lambda) \leq \lambda^c + c$ for all large λ; equivalently, $f(\lambda) = O(\lambda^c)$ for some constant c. It is *super-polynomial* if it grows faster than every polynomial, i.e. $f(\lambda) = \omega(\lambda^c)$ for every constant c (equivalently, $\lambda^c = o(f(\lambda))$ for all c).

Definition 11.41 (Exponential and sub-exponential)**.** A function f is *exponential* if $f(\lambda) = 2^{\Omega(\lambda)}$; the prototypical case is $f(\lambda) = 2^{c\lambda}$ for a constant $c > 0$. A super-polynomial function with $f(\lambda) = 2^{o(\lambda)}$ is *sub-exponential*; examples are $\lambda^{\log_2 \lambda} = 2^{(\log_2 \lambda)^2}$ and the index-calculus running time $L_p[1/3] = 2^{\Theta(\lambda^{1/3}(\log \lambda)^{2/3})}$ in the bit length $\lambda = \log_2 p$ (theorem 7.21). Every exponential function is super-polynomial; the converse fails by these examples.

The central cryptographic notion is that of a negligible function. A negligible probability may be ignored because no efficient (polynomially many) repetition of an experiment can amplify it to a noticeable chance — the closure theorem below makes this precise.

Definition 11.42 (Negligible function)**.** A function $\nu : \mathbb{N} \to \mathbb{R}_{\geq 0}$ is *negligible*, written $\nu = \mathrm{negl}(\lambda)$, if for every positive constant c there exists λ_0 such that

$$\nu(\lambda) < \frac{1}{\lambda^c} \qquad \text{for all } \lambda \geq \lambda_0.$$

Equivalently, $\nu(\lambda) = o(\lambda^{-c})$ for every constant $c > 0$; equivalently, $\lambda^c\, \nu(\lambda) \to 0$ as $\lambda \to \infty$ for every constant c. A function is *non-negligible* if it is not negligible, i.e. $f(\lambda) \geq \lambda^{-c}$ for infinitely many λ, for some constant c; it is *noticeable* if there exist c and λ_0 with $f(\lambda) \geq \lambda^{-c}$ for all $\lambda \geq \lambda_0$ (noticeable implies non-negligible, but not conversely); it is *overwhelming* if $1 - f$ is negligible.

Example 11.43. The functions $2^{-\lambda}$, $2^{-\sqrt{\lambda}}$, and $\lambda^{-\log \lambda}$ are negligible. For $2^{-\lambda}$: given any c, the product $2^{-\lambda}\lambda^c = \lambda^c/2^\lambda$ tends to 0 (a polynomial over an exponential), meeting the definition. By contrast, $1/\lambda^{100}$ is *not* negligible — taking $c = 101$ shows $1/\lambda^{100} > 1/\lambda^{101}$ for all $\lambda \geq 2$ — it is noticeable, even though it tends to 0. This is the key subtlety: tending to zero does not suffice; a negligible function must beat every inverse polynomial.

Proposition 11.44 (Closure properties of negligible functions)**.** *Let ν, μ be negligible and let p be a polynomially bounded function. Then:*

1. *$\nu + \mu$ is negligible (the class is closed under addition, hence under any fixed finite sum).*
2. *$p \cdot \nu$ is negligible (a polynomial times a negligible function is negligible).*
3. *If $f(\lambda) \leq \nu(\lambda)$ for all large λ, then f is negligible (domination by a negligible function).*
4. *Summing $p(\lambda)$ functions, each bounded by a single negligible function ν, yields a negligible function; in particular a union bound over polynomially many negligible events is negligible.*

Proof. The mechanism throughout is an exponent shift: to beat λ^{-c}, invoke negligibility at a strictly larger exponent, so that the surplus absorbs the combining factor. The same shift is the standard mechanism behind hybrid and union-bound steps in security proofs.

(1) Fix c. Applying the definition at the exponent $c+1$, there is a threshold beyond which $\nu(\lambda), \mu(\lambda) < \lambda^{-(c+1)} \leq \frac{1}{2}\lambda^{-c}$ (the latter once $\lambda \geq 2$); adding the two bounds gives $\nu(\lambda) + \mu(\lambda) < \lambda^{-c}$ for all large λ.

(2) Suppose $p(\lambda) \leq \lambda^d$ for large λ (and some constant d). Fix c. By negligibility applied at the exponent $c+d$, $\nu(\lambda) < \lambda^{-(c+d)}$ for large λ, whence $p(\lambda)\nu(\lambda) < \lambda^d \lambda^{-(c+d)} = \lambda^{-c}$ for large λ. Hence $p\nu$ is negligible.

(3) Immediate: if $f \leq \nu$ eventually and $\nu < \lambda^{-c}$ eventually, then $f < \lambda^{-c}$ eventually, for every c.

(4) Write the sum $\Sigma(\lambda) = \sum_{i=1}^{p(\lambda)} \nu_i(\lambda)$, where each $\nu_i \leq \nu$ by hypothesis (in security proofs all the ν_i arise from one bound). Then $\Sigma(\lambda) \leq p(\lambda)\,\nu(\lambda)$, which is negligible by (2); conclude by (3). □

Remark 11.45 (Why negligible is the threshold of security). Suppose an adversary wins a game with negligible probability $\nu(\lambda)$. If it repeats the attack $p(\lambda)$ times (polynomially many, the most an efficient adversary can afford), the probability that some repetition succeeds is, by the union bound, at most $p(\lambda)\nu(\lambda)$ — still negligible by theorem 11.44(2). Thus negligible success probability remains negligible under any efficient amplification: it is genuinely ignorable. Conversely, a noticeable success probability $\geq\lambda^{-c}$ can be boosted to a constant by λ^c repetitions. The negligible/non-negligible split is the formal backbone of the phrase "the scheme is secure except with negligible probability": security asserts a negligible advantage, and an attack means a non-negligible one. (A merely non-negligible — not noticeable — success probability is boosted to a constant only on the infinitely many λ where the inverse-polynomial bound holds, which already suffices to contradict a security definition.)

11.10 Algorithms, running time, and PPT

It remains to make "efficient algorithm" and "infeasible" precise. We adopt the standard model: algorithms are Turing machines (equivalently, up to polynomial factors, programs in any reasonable model of computation), inputs and outputs are finite binary strings, and the measured resource is the number of elementary steps as a function of input length. Write $\{0,1\}^*$ for the set of all finite binary strings and $|x|$ for the length of $x \in \{0,1\}^*$.

Definition 11.46 (Deterministic polynomial-time algorithm)**.** An algorithm M runs in *(deterministic) polynomial time* if there is a polynomial p such that, on every input $x \in \{0,1\}^*$, M halts within at most $p(|x|)$ steps and produces an output. The class of decision problems solvable by such machines is P.

Determinism is insufficient for cryptography: key generation and sampling require randomness. We model randomness by granting the machine an auxiliary tape of independent uniform random bits (the "coins").

Definition 11.47 (Probabilistic algorithm and PPT)**.** A *probabilistic* (or *randomised*) algorithm M is a Turing machine with an additional read-only *random tape* initialised with independent uniformly random bits $r \in \{0,1\}^{\mathbb{N}}$. For a fixed input x, the output $M(x)$ is a random variable over the choice of r; write $M(x;r)$ to display the coins explicitly, and then $M(x;\cdot)$ is a deterministic function. The algorithm runs in *probabilistic polynomial time* (it is *PPT*) if there is a polynomial p such that, for every input x *and every choice of coins* r, M halts within $p(|x|)$ steps. (Bounding the time for every r, not merely in expectation, is the standard "strict" notion and ensures M ever reads only $p(|x|)$ coins.)

Remark 11.48 (Non-uniformity and circuits). There are two standard ways to formalise an efficient adversary. The *uniform* model is a single PPT machine that works for all input lengths, as above. The *non-uniform* model allows a separate algorithm (equivalently, a Boolean circuit) of size $\mathrm{poly}(\lambda)$ for each λ, which may be hard-wired with an arbitrary polynomial-length "advice" string depending on

λ. Non-uniform adversaries are at least as powerful as uniform ones and are often more convenient in reduction proofs, because the advice can encode a "best" attack. The asymptotic theory of this section is identical for both models; security definitions in the Halo 2/Orchard setting typically state security against non-uniform PPT adversaries, the more conservative choice.

With efficiency pinned down, we can state the strong hardness notion used in cryptography. The guiding principle is the Cobham–Edmonds thesis: "efficiently computable" means "computable in polynomial time". Feasibility and the infeasibility notion below are useful opposite regimes, but they are not logical negations: intermediate success probabilities can fall into neither class.

Definition 11.49 (Computational infeasibility)**.** A computational task parameterised by the security parameter λ is *(computationally) feasible* if there is a PPT algorithm solving it with overwhelming success probability. It is *infeasible* if for *every* PPT algorithm $\mathcal{A}$ the probability that $\mathcal{A}$ solves it is negligible in λ:

$$\Pr\big[\mathcal{A} \text{ solves the task on parameter } \lambda\big] = \operatorname{negl}(\lambda),$$

where the probability is over the task's randomness and $\mathcal{A}$'s coins. Equivalently: no efficient algorithm succeeds with non-negligible probability.

Remark 11.50 (Reading a hardness assumption)*.* Cryptographic assumptions are infeasibility statements. For instance, the discrete-logarithm assumption in a family of groups $\{\mathbb{G}_\lambda\}$ of order $q(\lambda)$ asserts: for every PPT $\mathcal{A}$,

$$\Pr\big[\mathcal{A}(g, g^x) = x\big] = \operatorname{negl}(\lambda),$$

the probability taken over a uniformly chosen generator g of $\mathbb{G}_\lambda$, a uniform $x \leftarrow \mathbb{Z}/q\mathbb{Z}$, and $\mathcal{A}$'s coins. The entire edifice of Halo 2 and Orchard rests on such asymptotic infeasibility assumptions; choosing a parameter size does not prove them. Separately, concrete analysis asks how much work the best known attacks need at one fixed size. Generic discrete-log attacks take $\Theta(\sqrt{q})$ group operations by the birthday/rho phenomenon of theorem 11.20; absent a faster structure-specific attack, an order near $2^{2\lambda}$ therefore targets about λ bits of generic-group security. The Pasta orders are just above 2^{254}, giving roughly 2^{127} generic work. The asymptotic language permits the terms "negligible" and "polynomial"; concrete estimates remain conditional on the attacks presently known.

11.11 The security parameter

The section closes by making explicit the parameter that threads through every preceding definition, and by paying its opening debt.

Definition 11.51 (Security parameter)**.** The *security parameter* is a positive integer $\lambda \in \mathbb{N}$ supplied to all algorithms of a cryptographic scheme, conventionally in unary as the string 1^λ (so that "polynomial in the input length" coincides with "polynomial in λ"). It governs simultaneously:

- the *sizes* of objects — key lengths and encodings, each $\operatorname{poly}(\lambda)$; in a generic discrete-log instantiation one commonly chooses a group order near $2^{2\lambda}$;
- the *running time* of honest algorithms, which must be $\operatorname{poly}(\lambda)$; and
- the *security level*: every PPT adversary's advantage must be $\operatorname{negl}(\lambda)$.

Remark 11.52 (Why unary, and what λ buys)*.* Passing 1^λ (a string of λ ones, of length λ) rather than the integer λ in binary (length $\log_2 \lambda$) is a deliberate convention: it forces "polynomial time in the input length" to mean polynomial in λ itself, so an honest key generator may take time $\operatorname{poly}(\lambda)$ to produce λ-bit keys. In a scheme satisfying the security definition, increasing λ raises honest cost only polynomially while every PPT adversary's success remains negligible; in common concrete families, the best known attack cost grows exponentially. Asymptotic security is the statement that this gap holds for all sufficiently large λ; the practitioner then fixes a single value (commonly $\lambda = 128$, targeting 2^{128}-operation attacks) and checks the concrete estimates against that target — the just-over-2^{254} Pallas/Vesta group orders give roughly 2^{127} generic group operations (section 10; Zcash protocol specification §5.4.9.6; the `pasta_curves` crate, `src/fields/`), hash outputs of ≥ 256 bits per the birthday bound of theorem 11.18, and sampling slack of at least 128 extra bits per theorem 11.24 (the deployed ToScalar of theorem 11.26 reduces 512 bits, a slack of 257). Every "negligible" in the remainder of the series is implicitly a function of this λ.

The opening cheque can now be cashed, with its two layers kept separate. Asymptotic infeasibility (theorem 11.49) quantifies over every probabilistic polynomial-time forger — uniform or hard-wired with advice (theorem 11.48) — and requires its success to be negligible in λ, a bound that no polynomial number of repetitions amplifies to a noticeable chance (theorem 11.45). A concrete "128-bit" estimate instead says that the best known applicable attack at a fixed parameter size costs on the order of 2^{128} operations; it is an attack estimate, not a theorem about every possible algorithm. Pasta's just-over-2^{254} group orders give roughly 2^{127} generic group operations and are conventionally placed near that target. The forger of the opening paragraph may still guess a key and win with astronomical luck; the formal promise is negligible success against every PPT forger, conditional on the stated cryptographic assumptions.